

Appendix

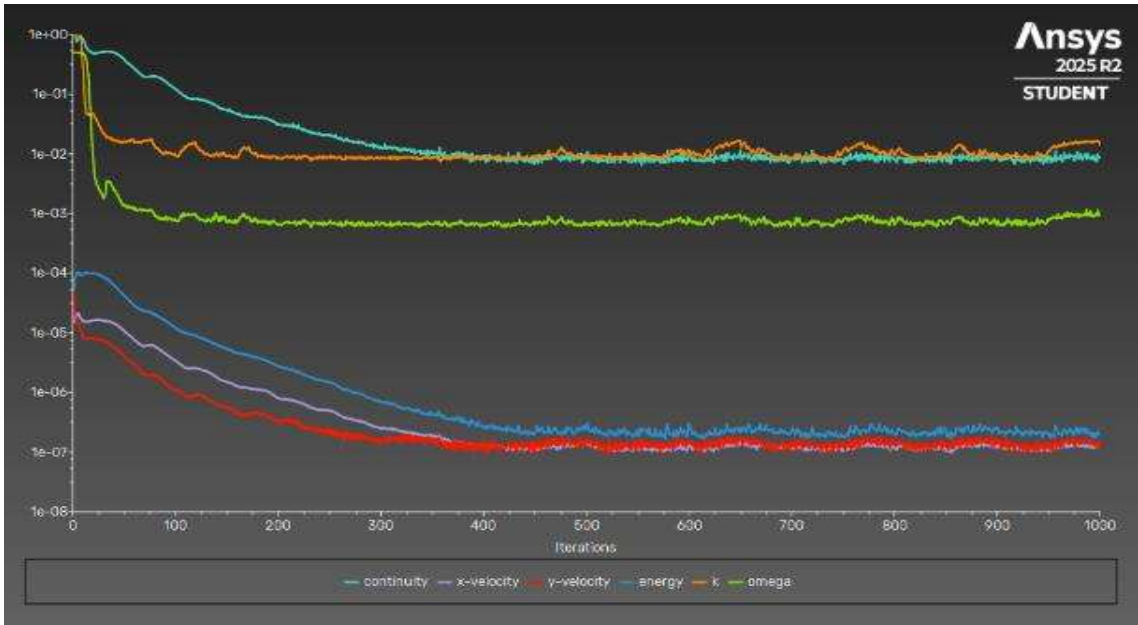


Image 1: Scaled Residual of Double-Wedge.

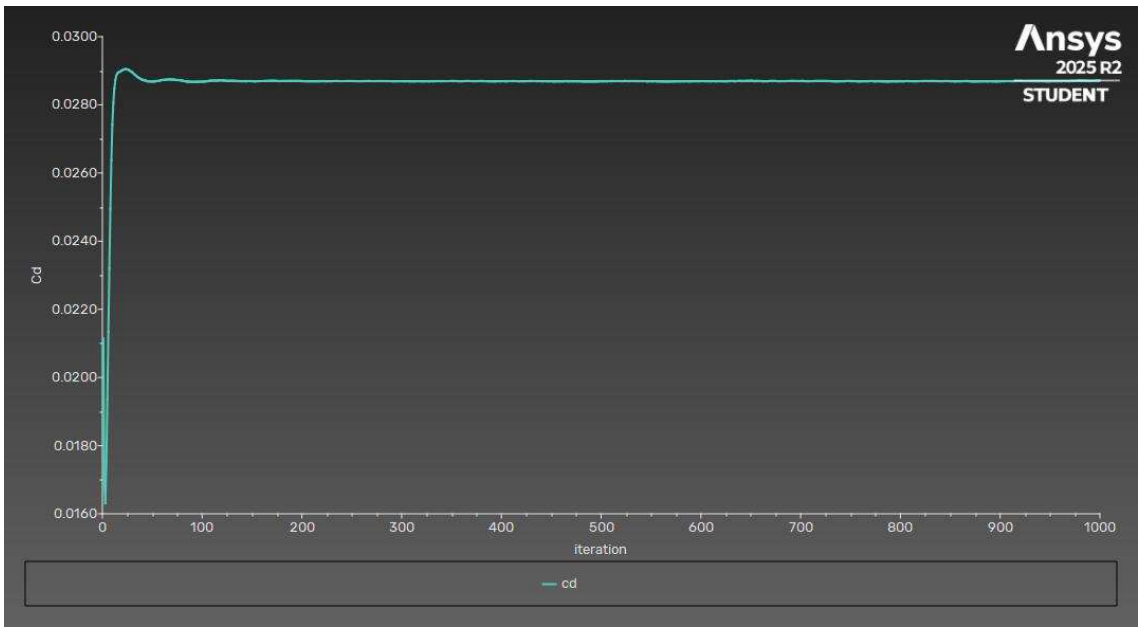


Image 2: Coefficient of Drag of Double-Wedge, Stable at 0.0287.

Stable Value: -0.00057

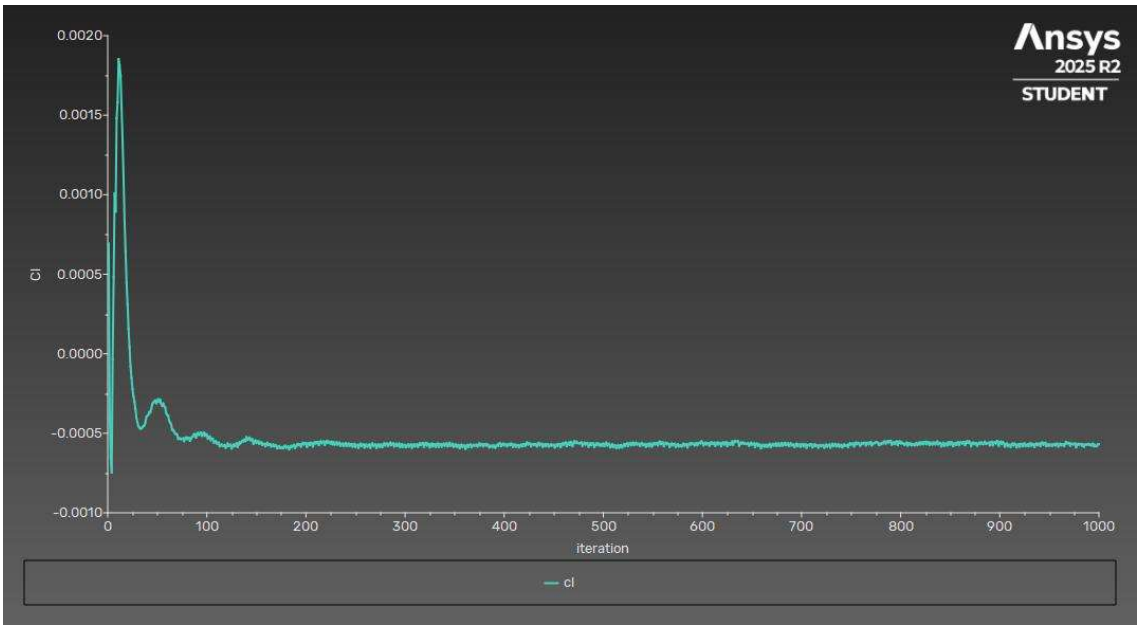


Image 3: Coefficient of Lift of Double-Wedge, Stable at -0.00057.

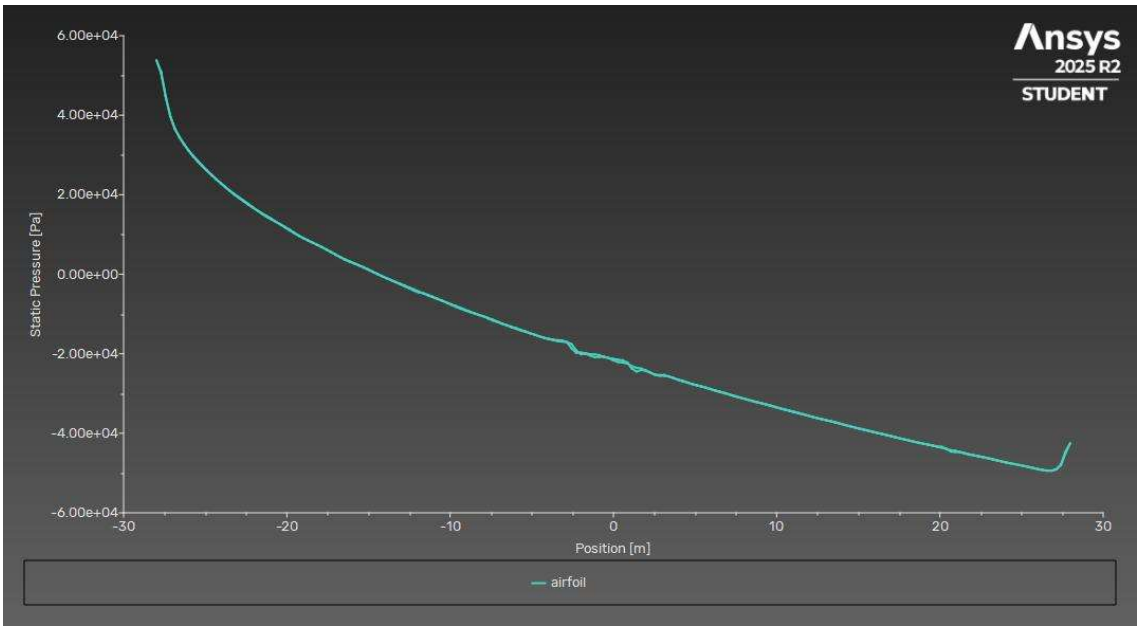


Image 4: Pressure plot for Bi-Convex around Airfoil.

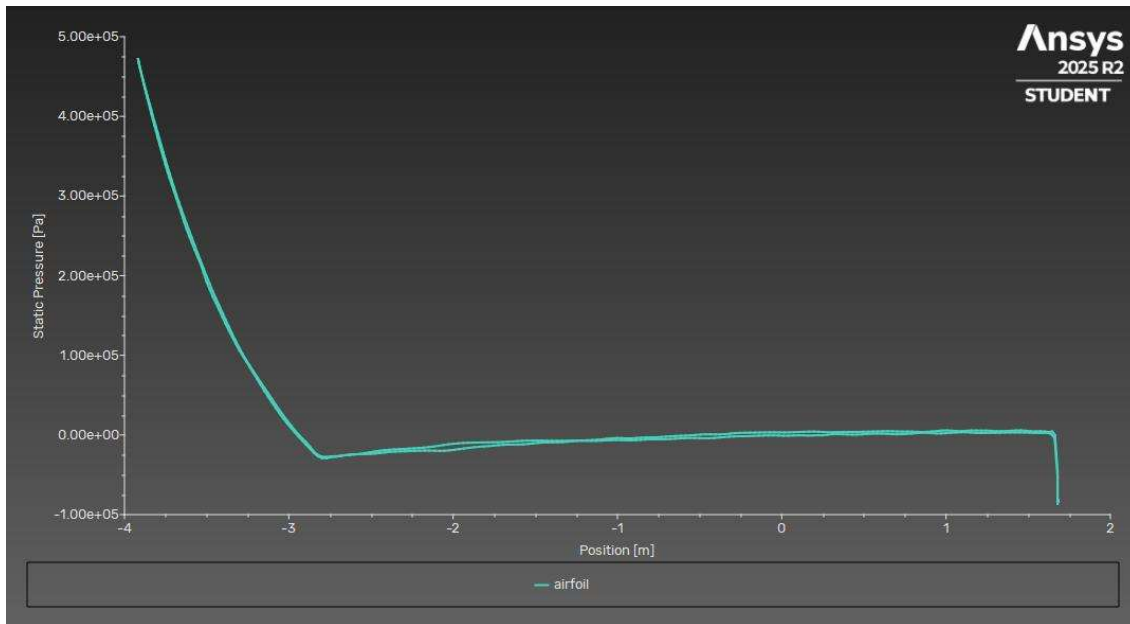


Image 5 Pressure plot for the dome around the airfoil.

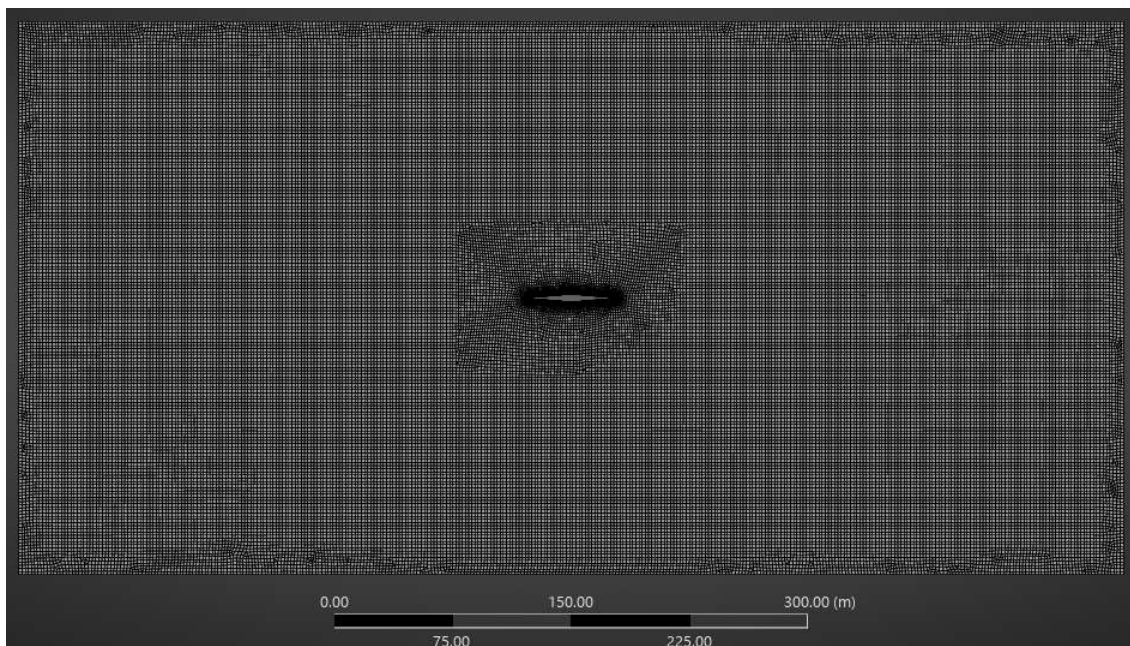
DOUBLE WEDGE:

Mesh:

Average Mesh Quality: 0.987

No. of Nodes: 78,336

No. of Elements: 77,547



MACH 2:

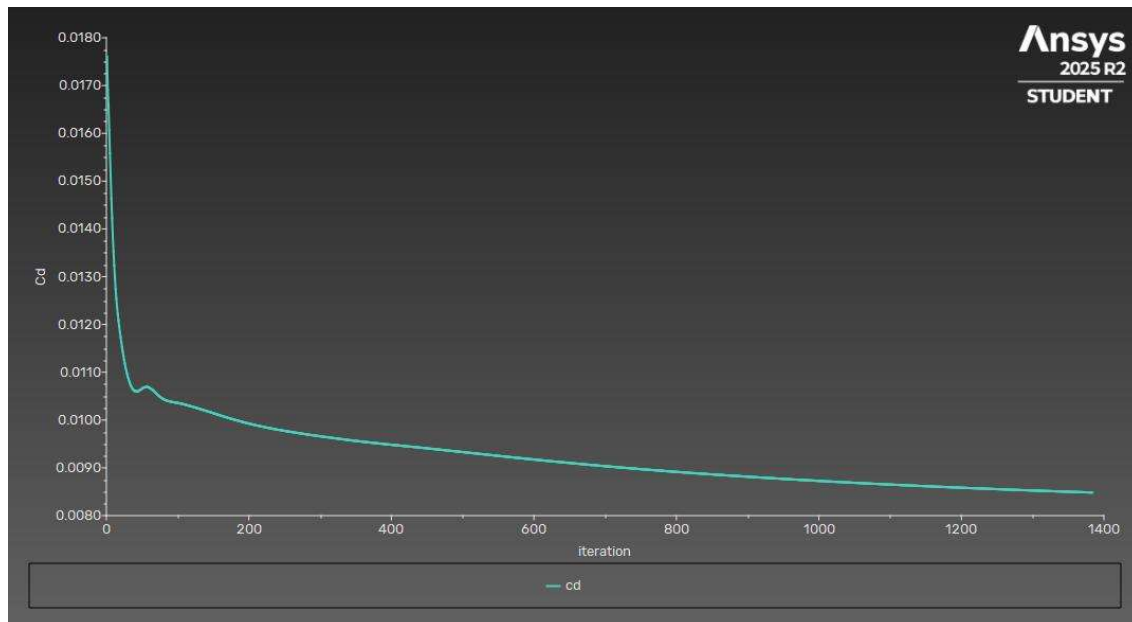
The following images below list the results for the double wedge run on the Density-based solver. Iteration 1384 solution has converged.

Scaled Residual:



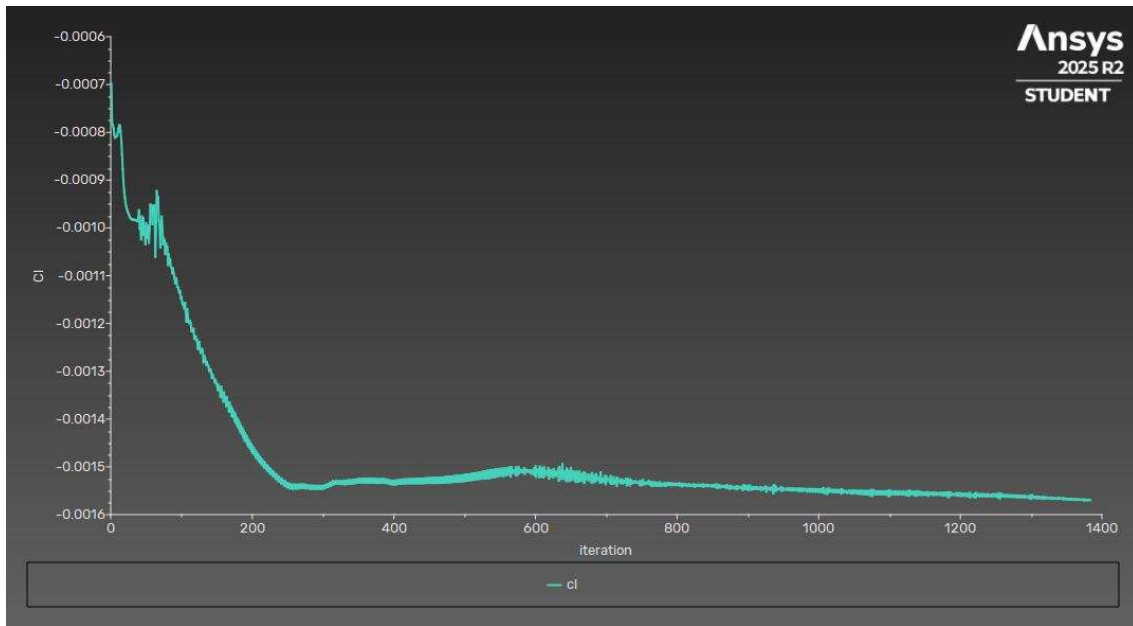
Coefficient of Drag:

Stable Value: 0.0084

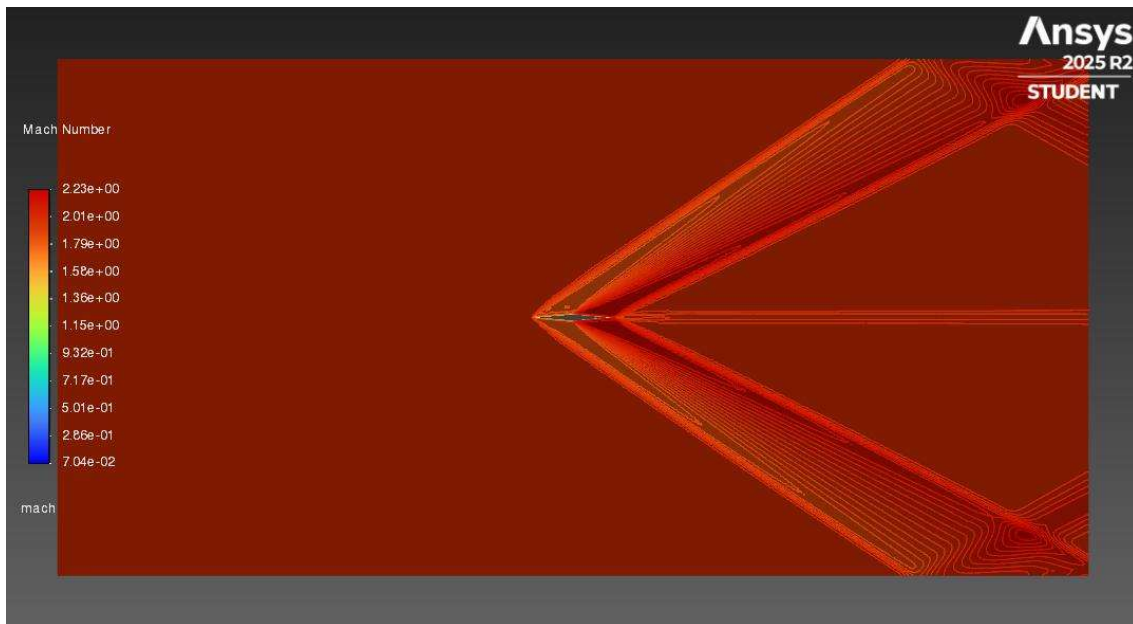


Coefficient of Lift:

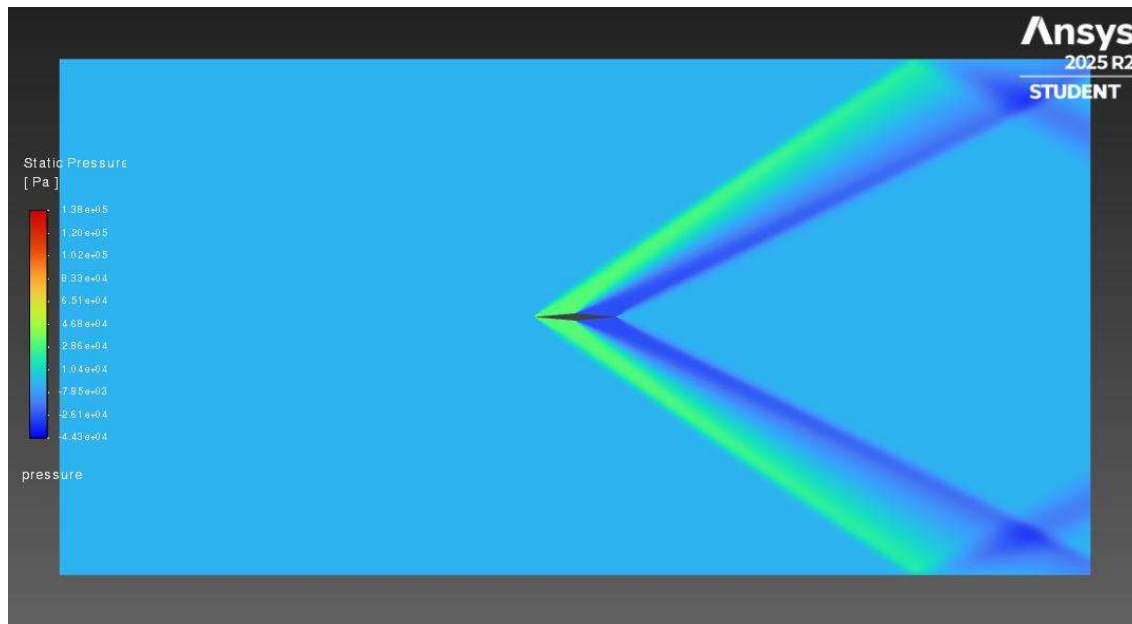
Stable Value: -0.0015



Mach Contours:



Pressure Contours:



MACH 3:

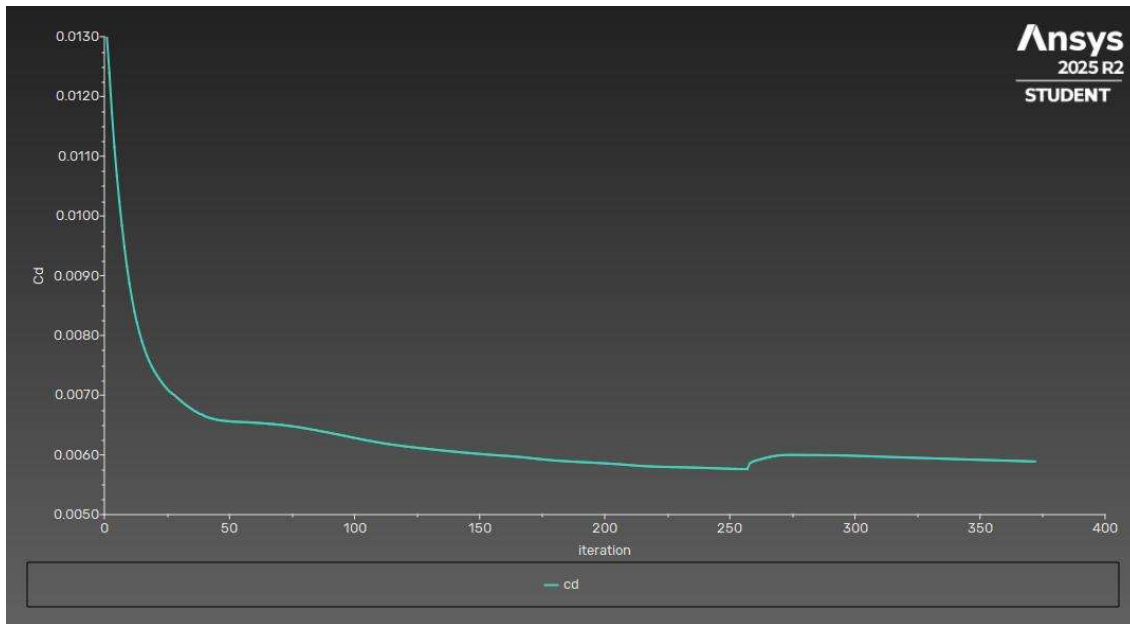
The following images below list the results for the double wedge run on the density-based solver. Iteration 372 solution has converged. Contour lines to clearly depict the flow.

Scaled Residual:



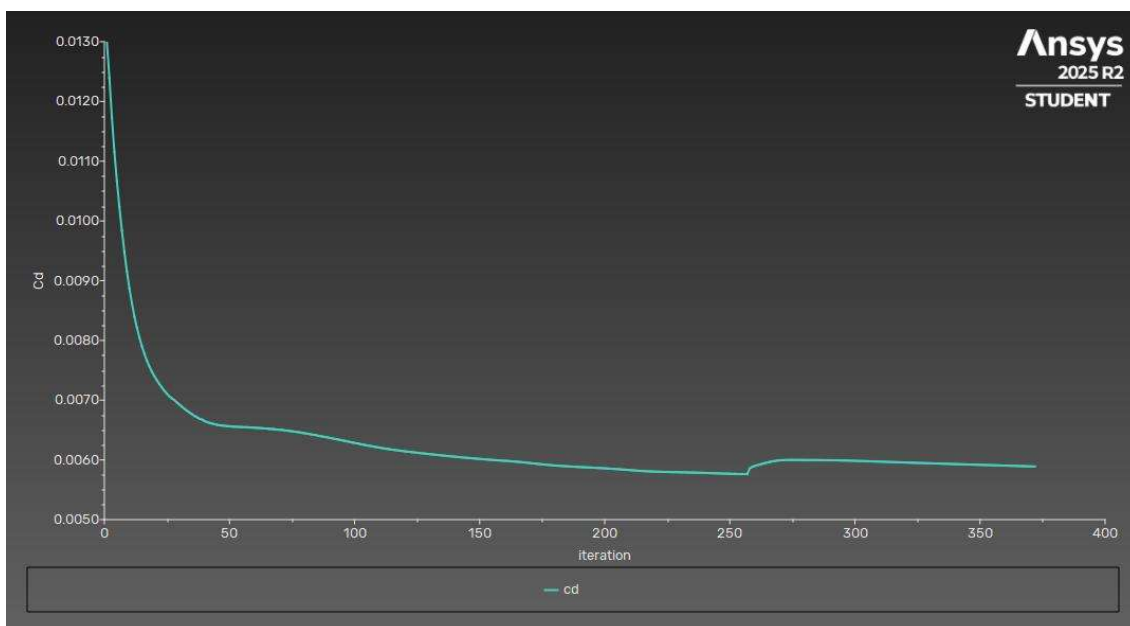
Coefficient of Drag:

Stable Value: 0.0059

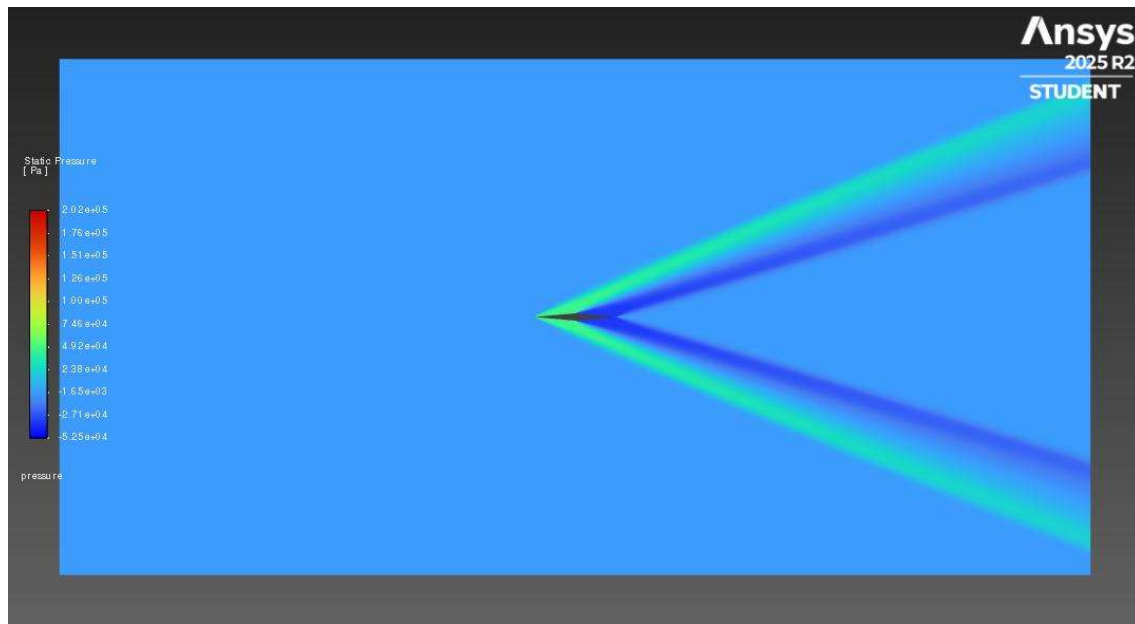


Coefficient of Lift:

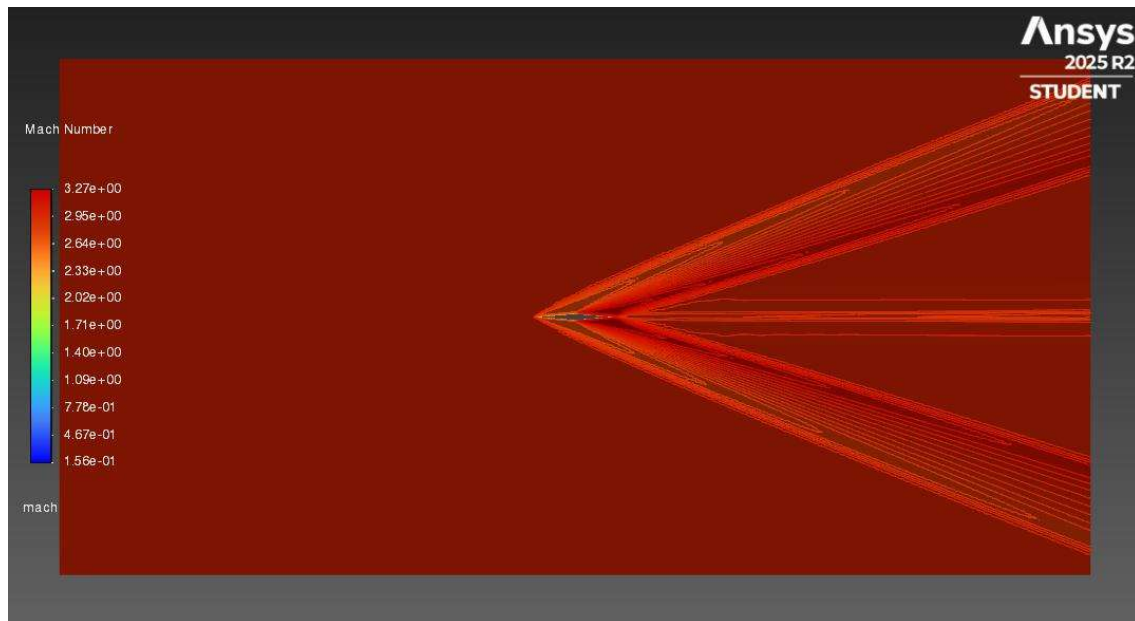
Stable Value: -0.0009



Mach Contours:



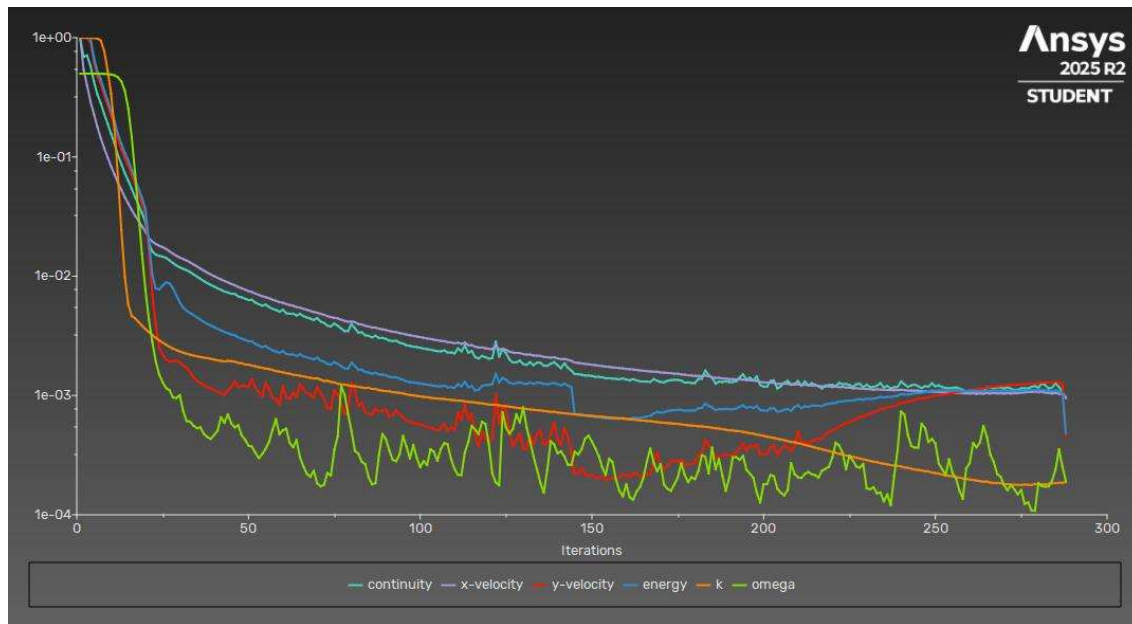
Pressure Contours:



MACH 4:

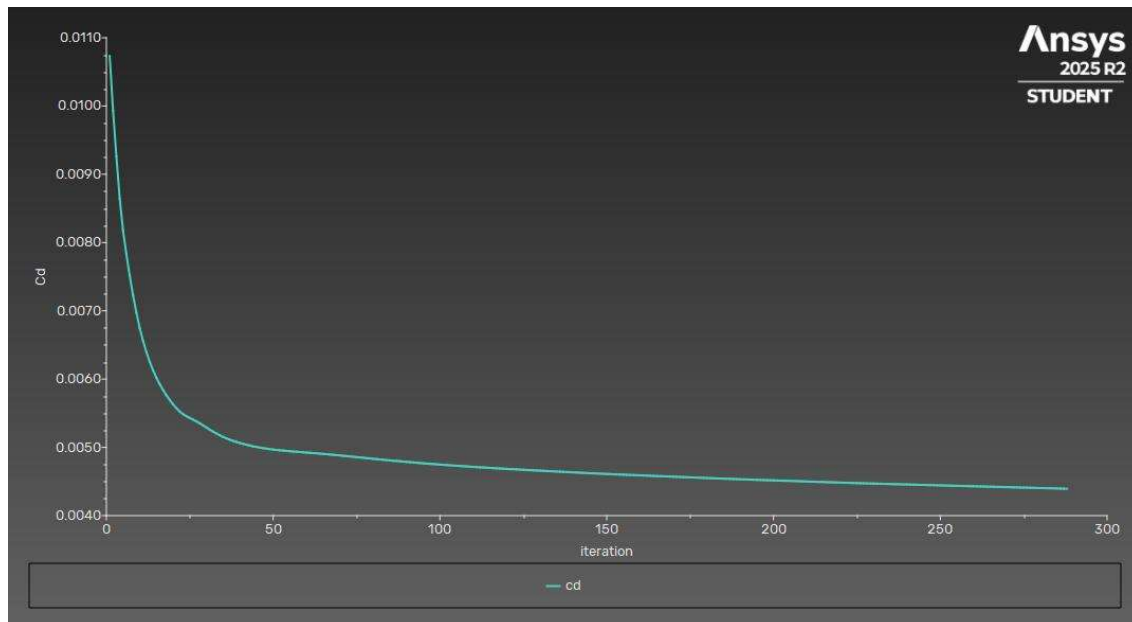
The following images below list the results for the double wedge run on the Density-based solver. Iteration 288 solution has converged.

Scaled Residual:



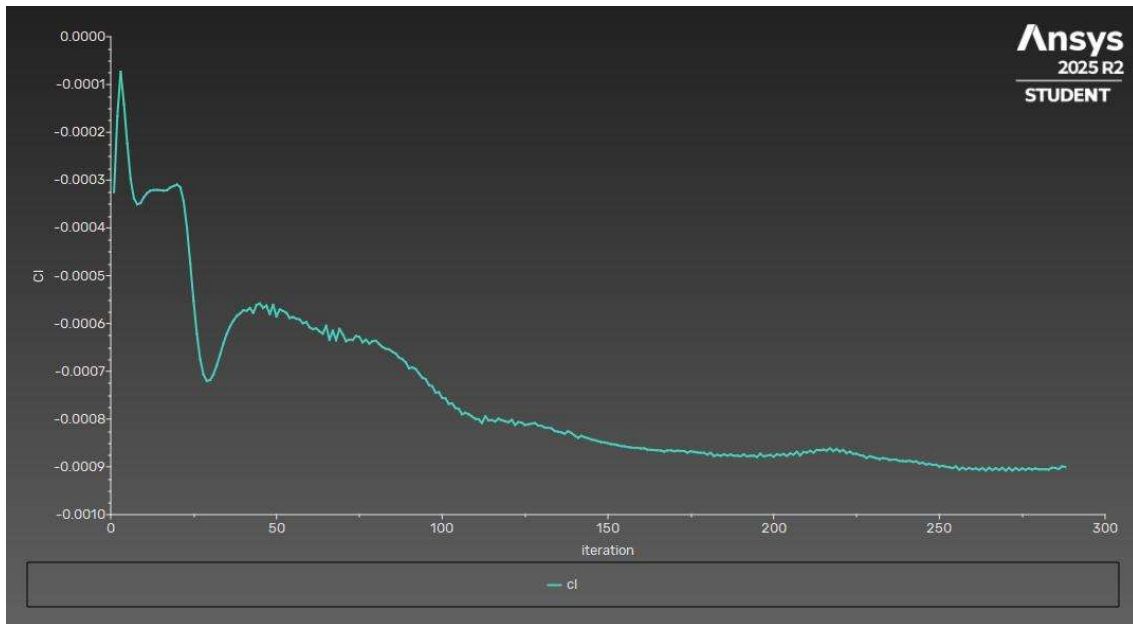
Coefficient of Drag:

Stable Value: 0.00439

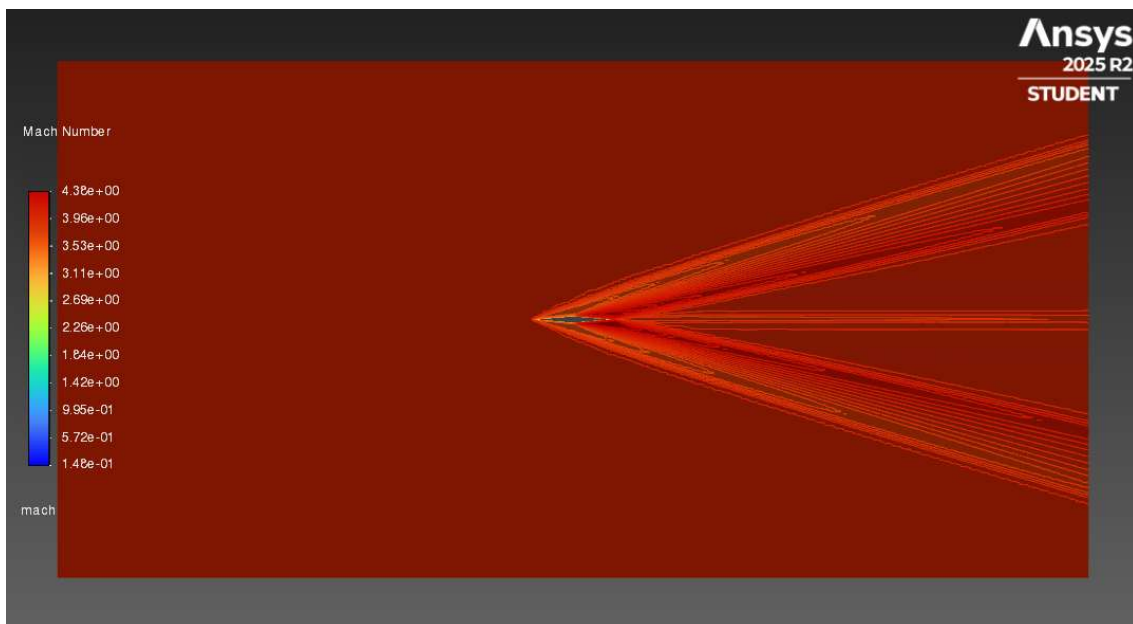


Coefficient of Lift:

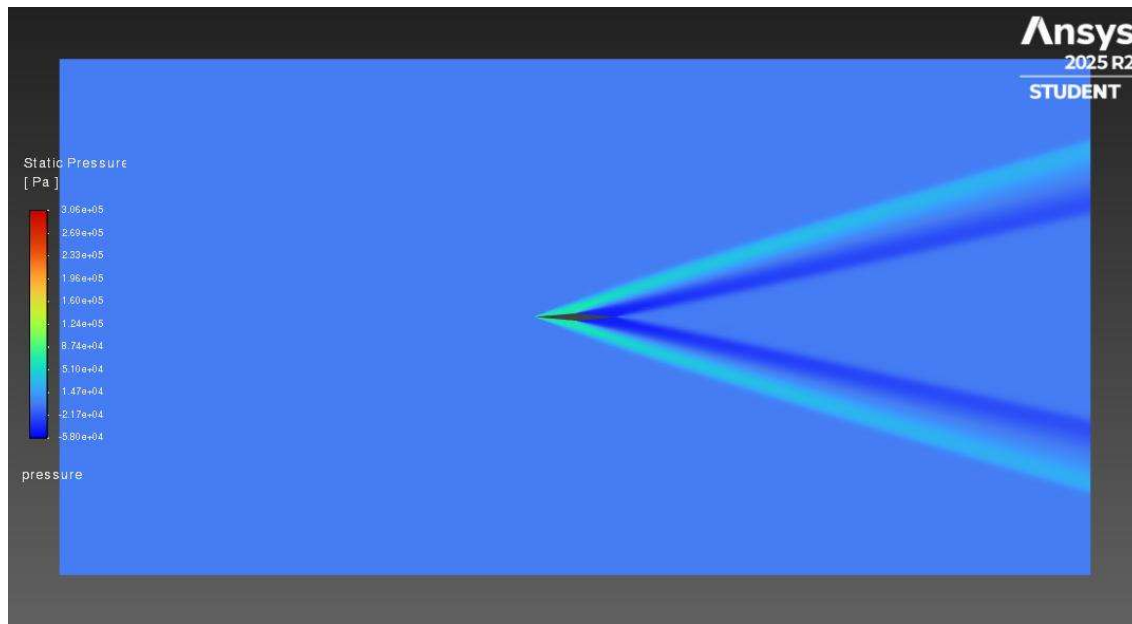
Stable Value: -0.0009



Mach Contours:



Pressure Contours:



MACH 5:

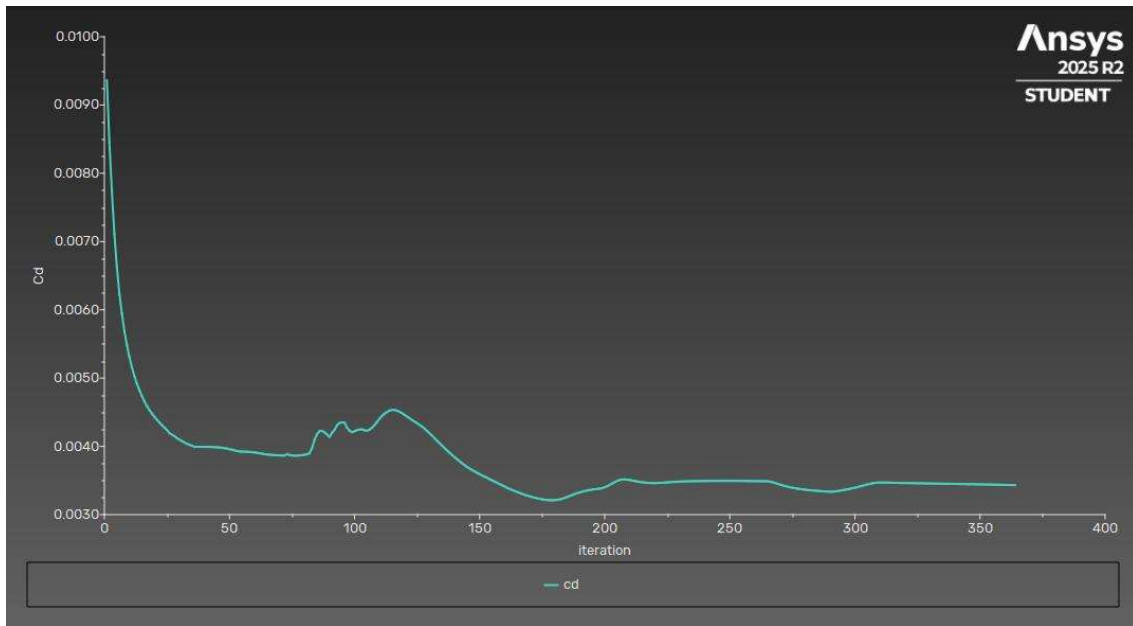
The following images below list the results for the double wedge run on the Density-based solver. Iteration 364 solution has converged.

Scaled Residual:



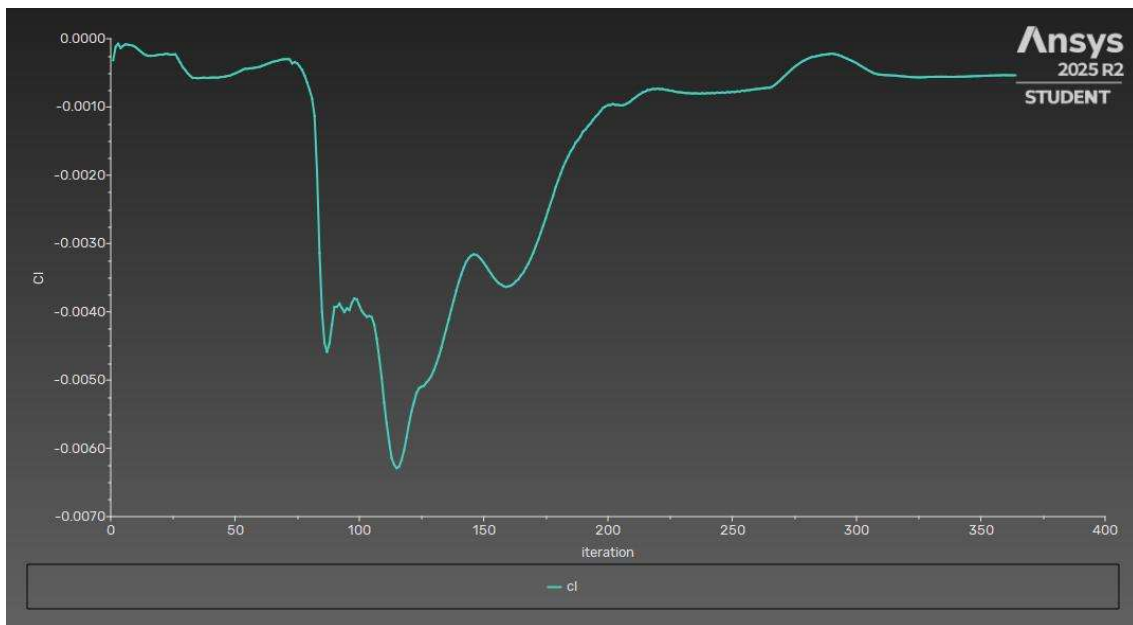
Coefficient of Drag:

Stable Value: 0.00343

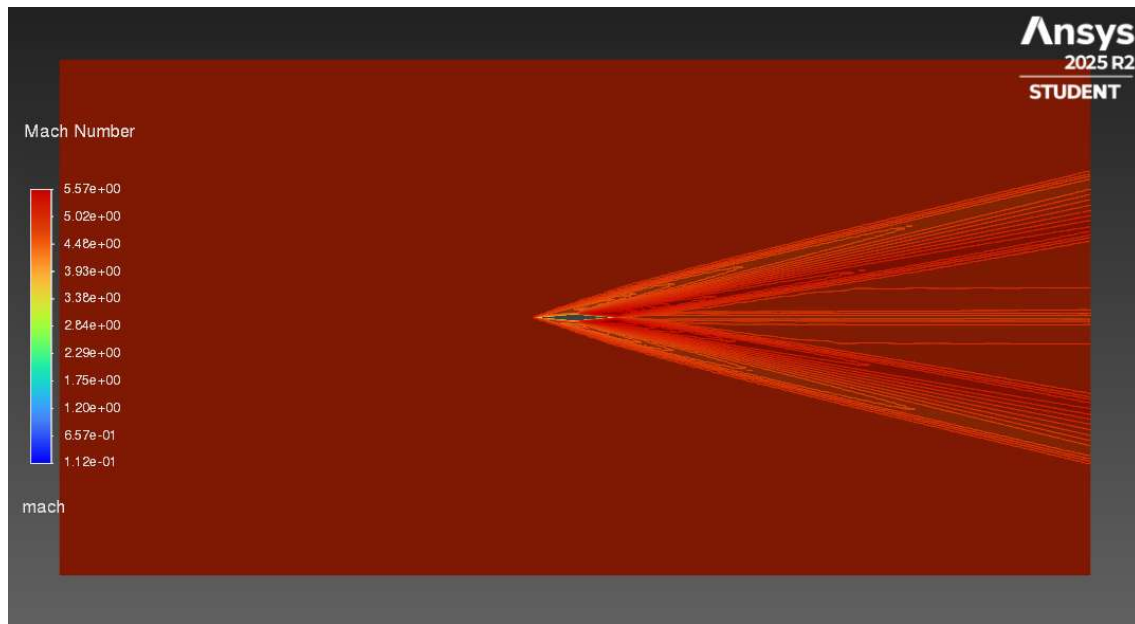


Coefficient of Lift:

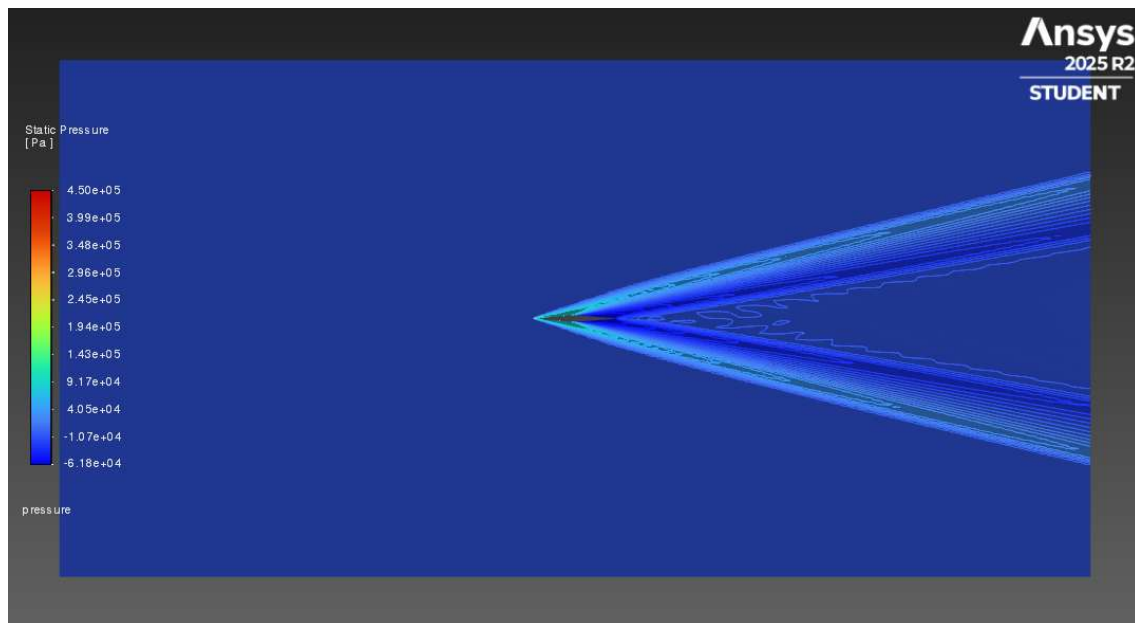
Stable Value: -0.000535



Mach Contours:



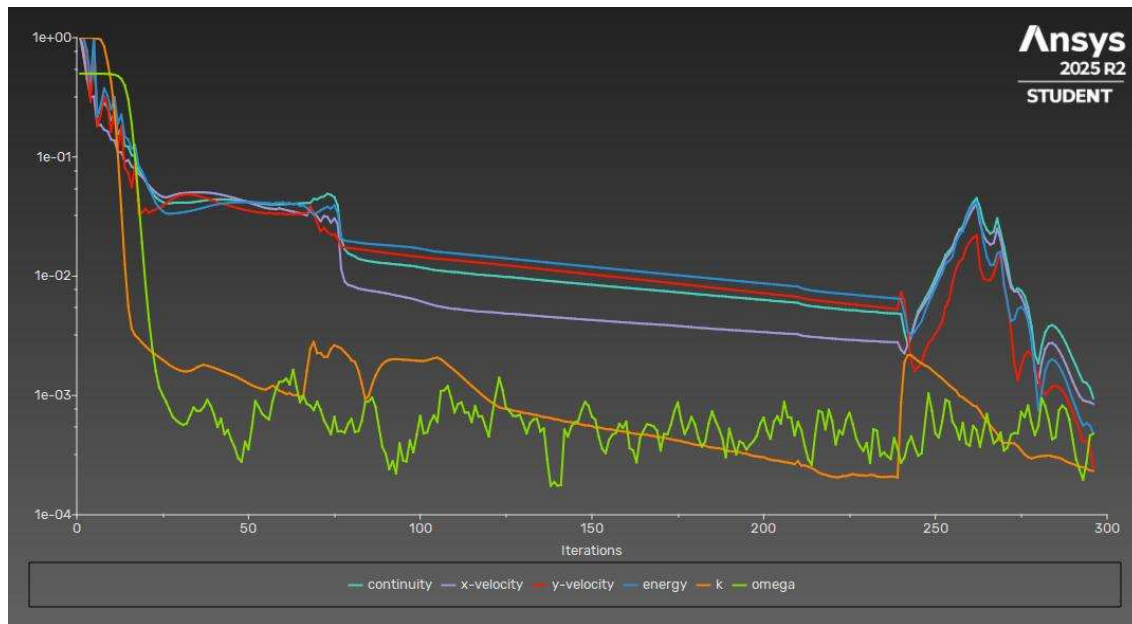
Pressure Contours:



MACH 6:

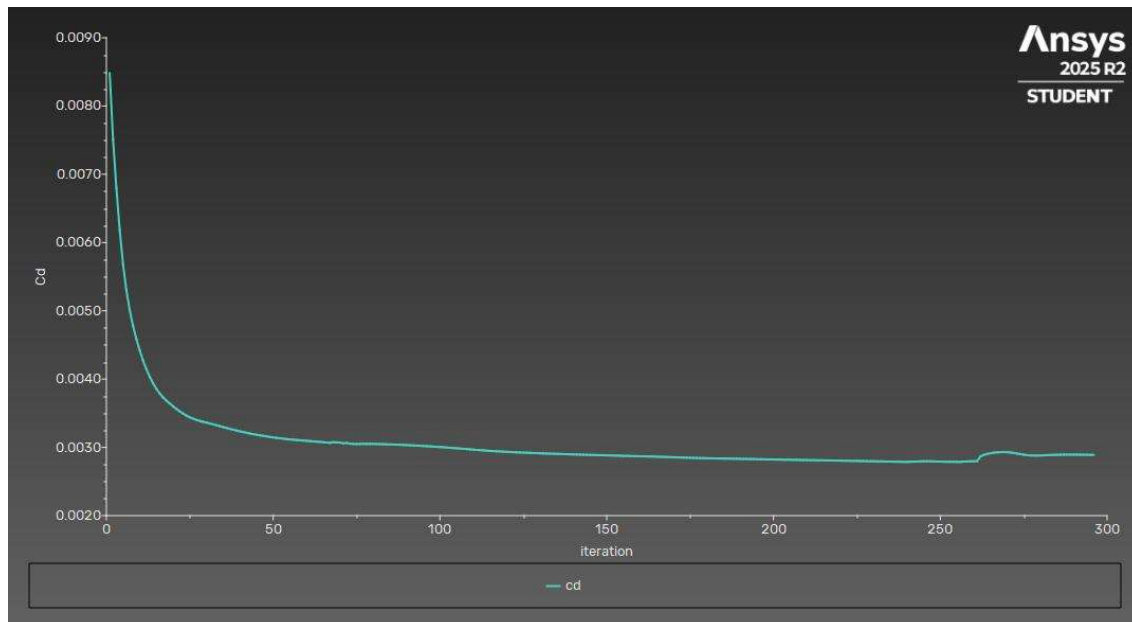
The following images below list the results for the double wedge run on the Density-based solver. Iteration 296 solution has converged.

Scaled Residual:



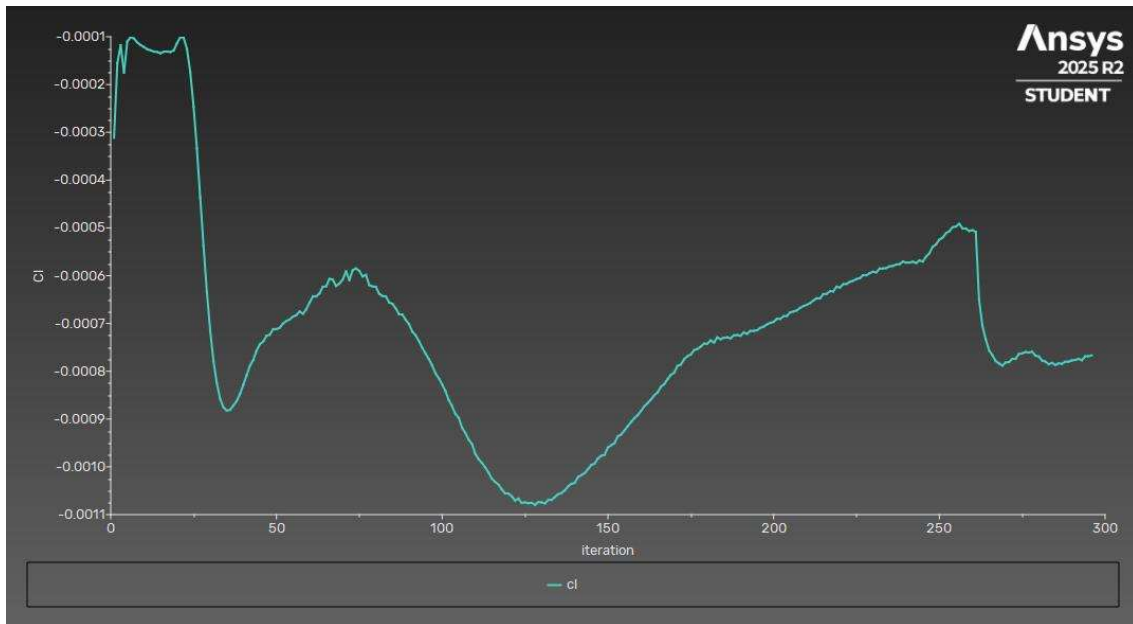
Coefficient of Drag:

Stable Value: 0.00288

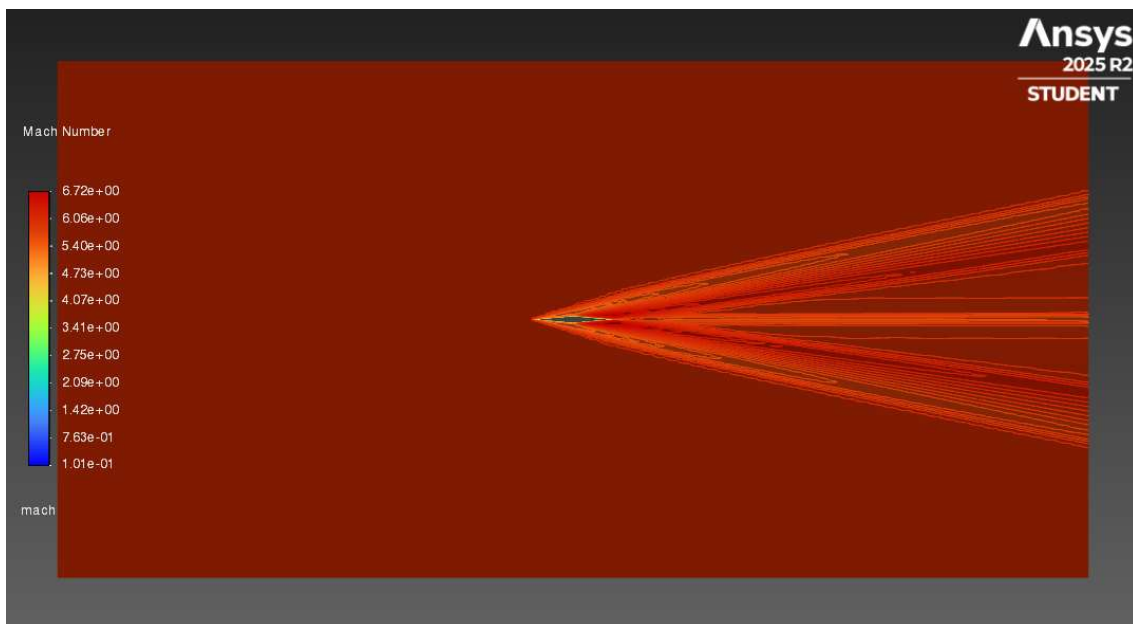


Coefficient of Lift:

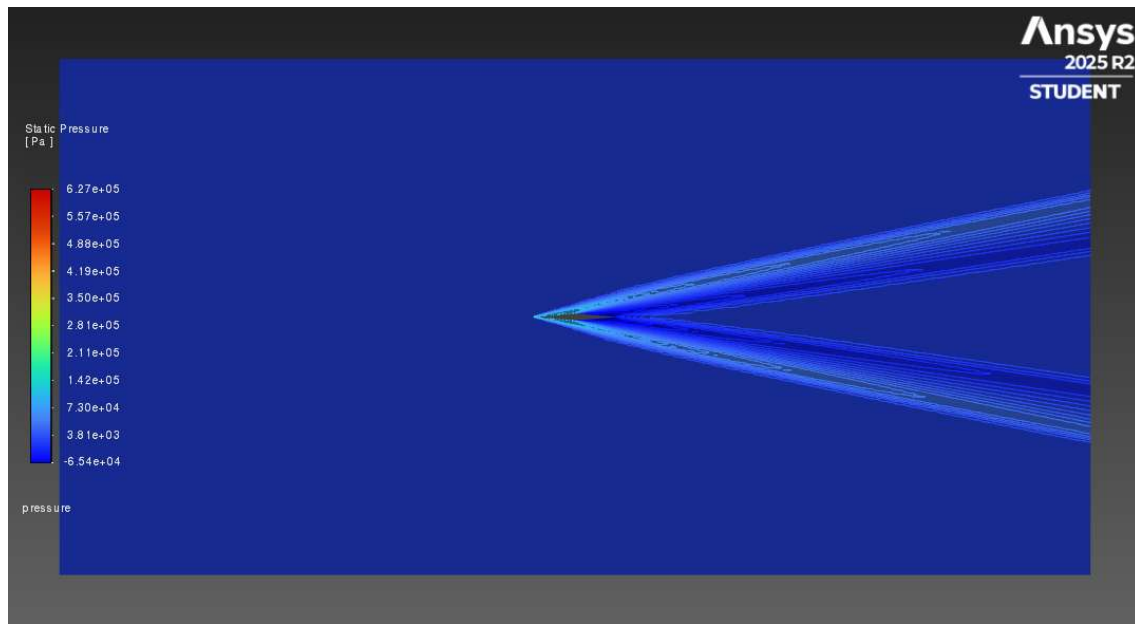
Stable Value: -0.0007



Mach Contours:



Pressure Contours:



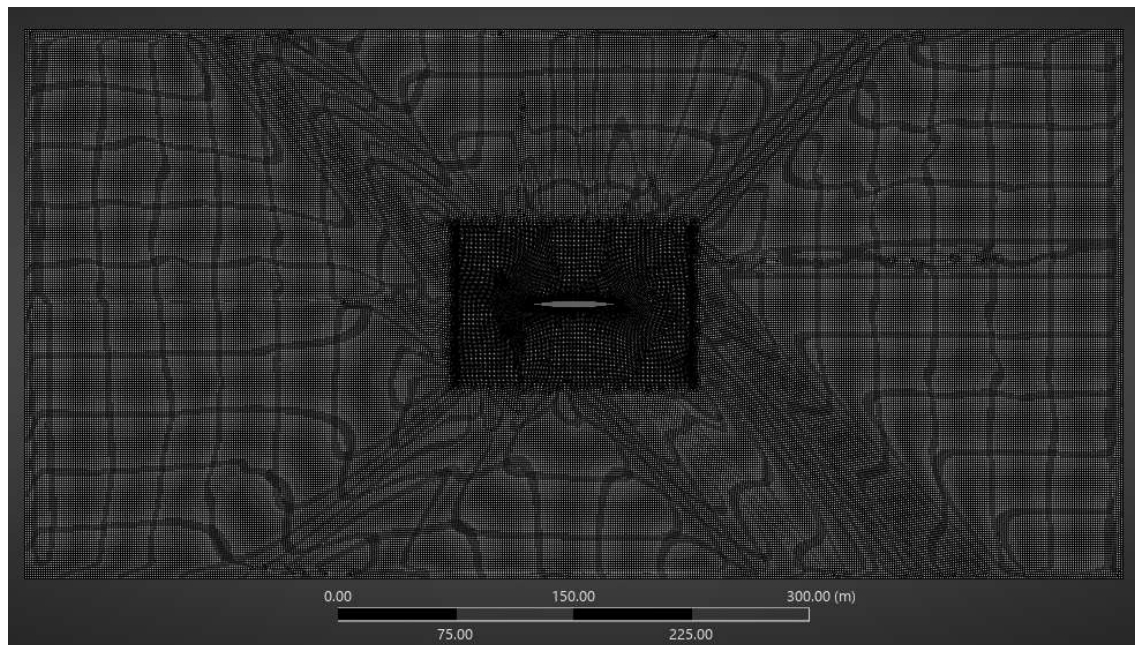
BI-CONVEX:

Mesh:

Average Mesh Quality: 0.985

No. of Nodes: 131,254

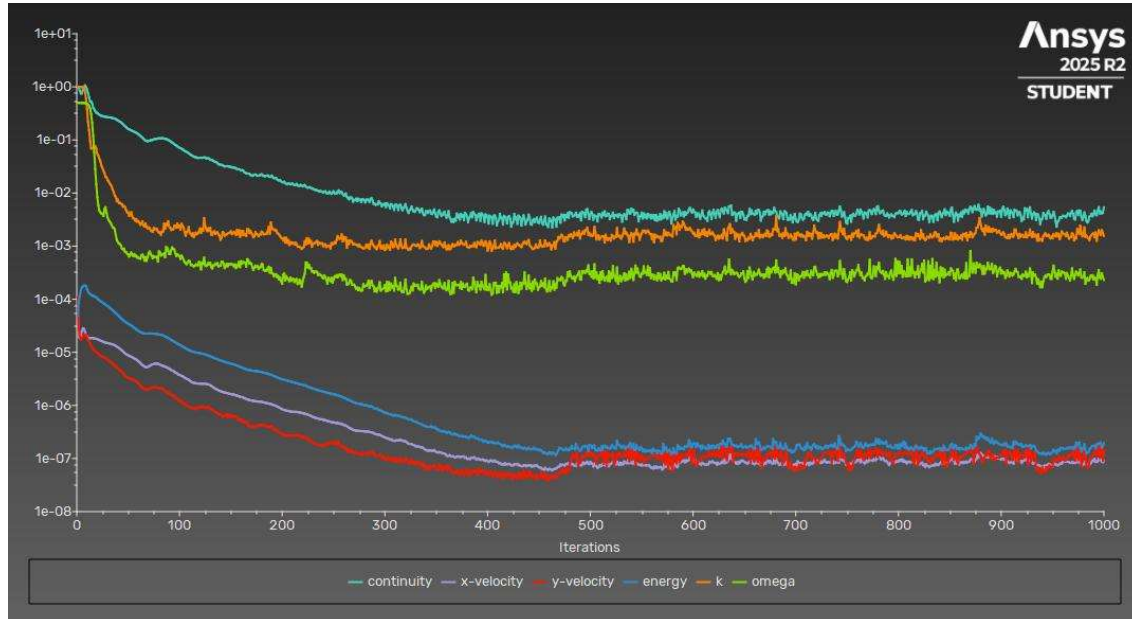
No. of Elements: 130,416



MACH 1:

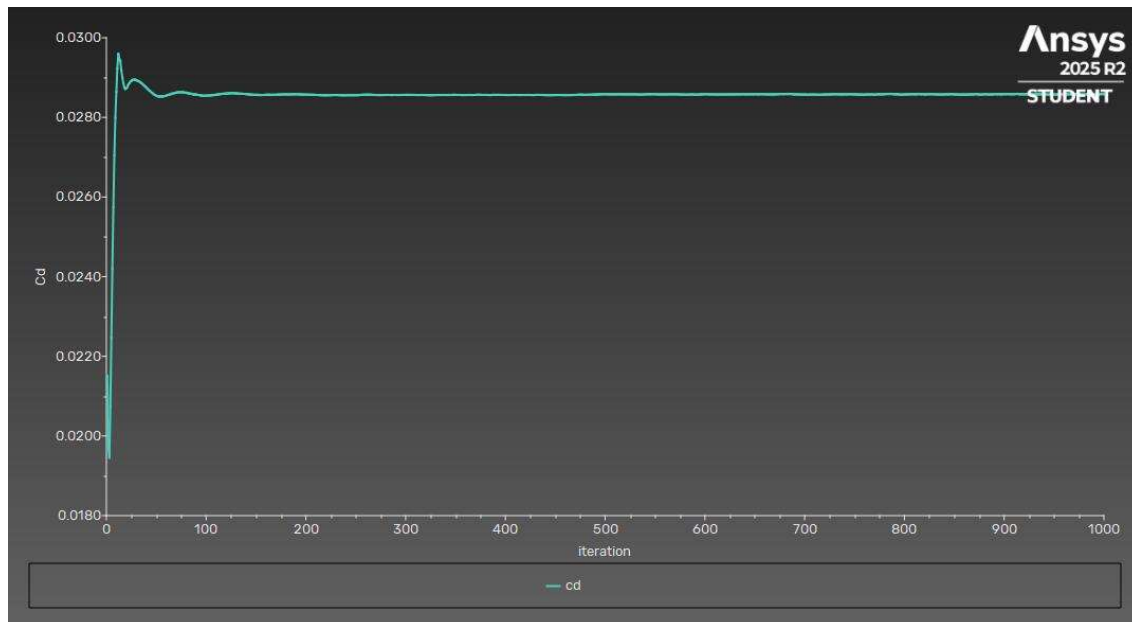
The following images below list the results for the double wedge run on the pressure-based solver.

Scaled Residual:



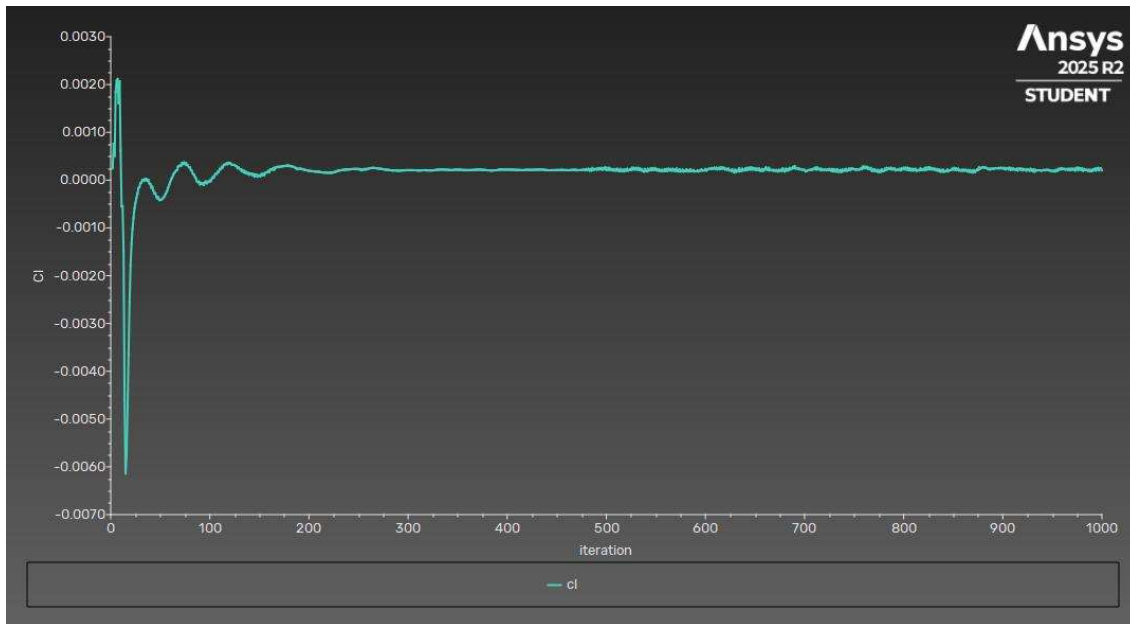
Coefficient of Drag:

Stable Value: 0.0285

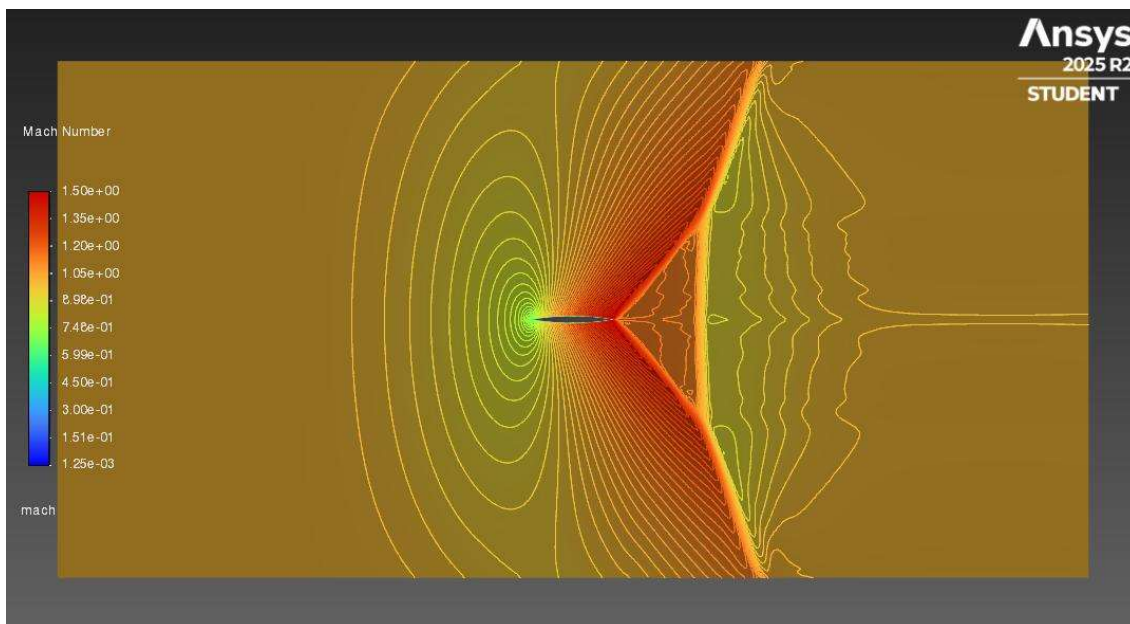


Coefficient of Lift:

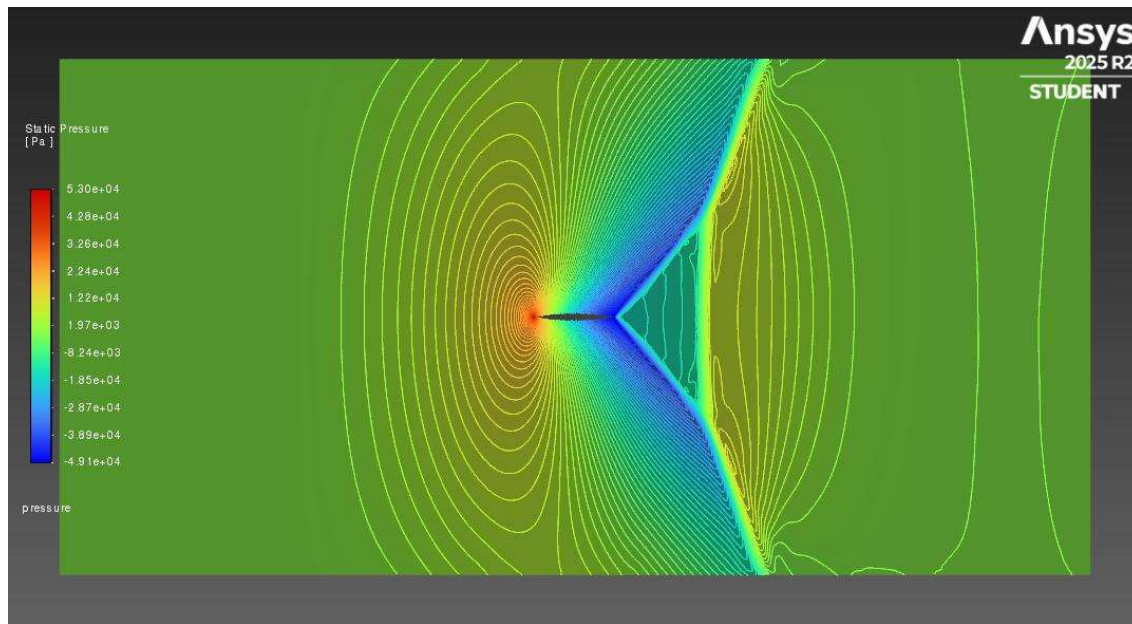
Stable Value: 0.00022



Mach Contours:



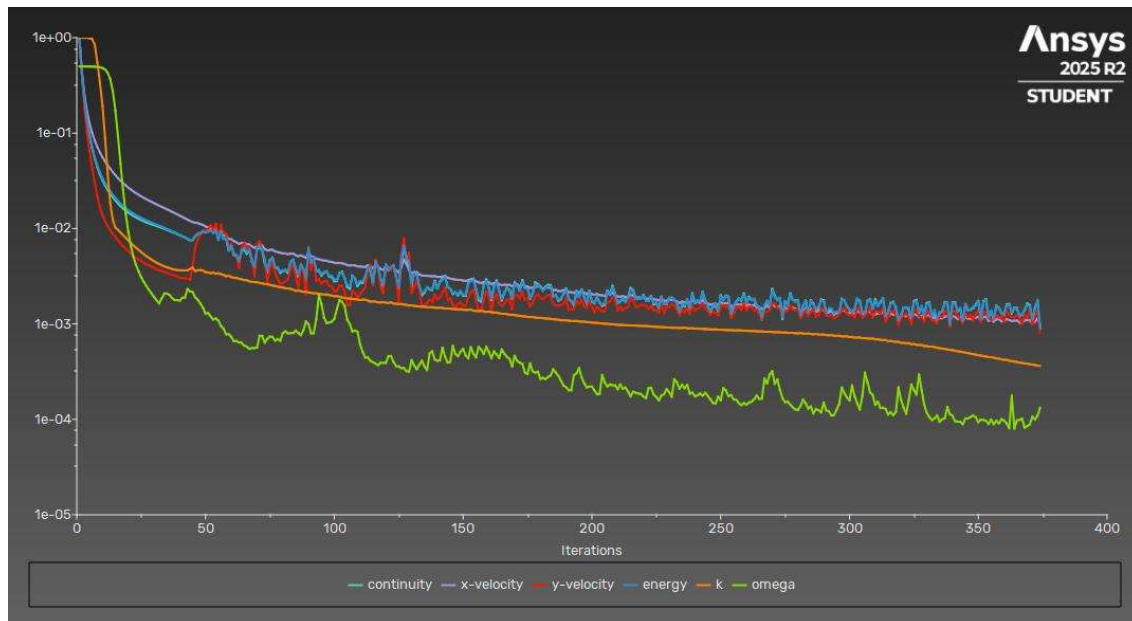
Pressure Contours:



MACH 2:

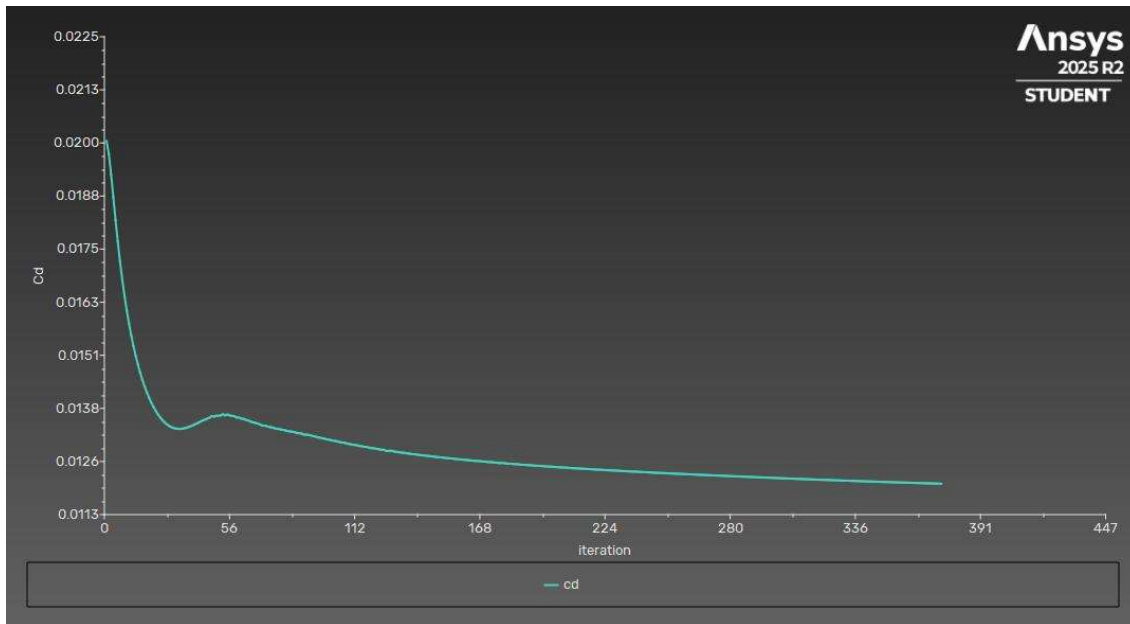
The following images below list the results for the double wedge run on the Density-based solver. Iteration 374 solution has converged.

Scaled Residual:



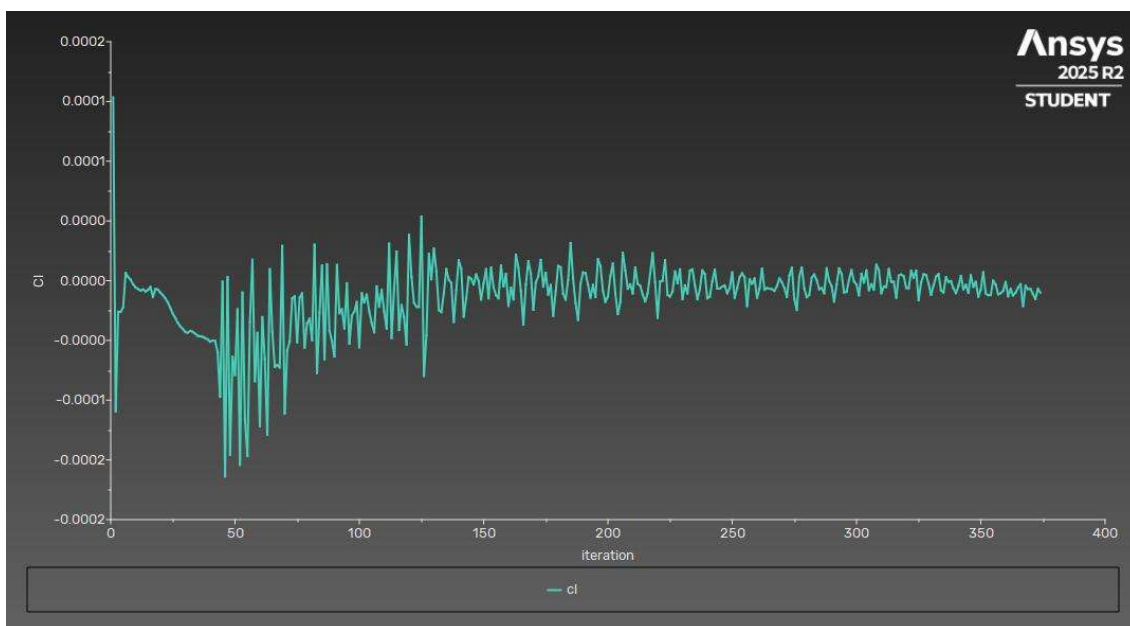
Coefficient of Drag:

Stable Value: 0.012

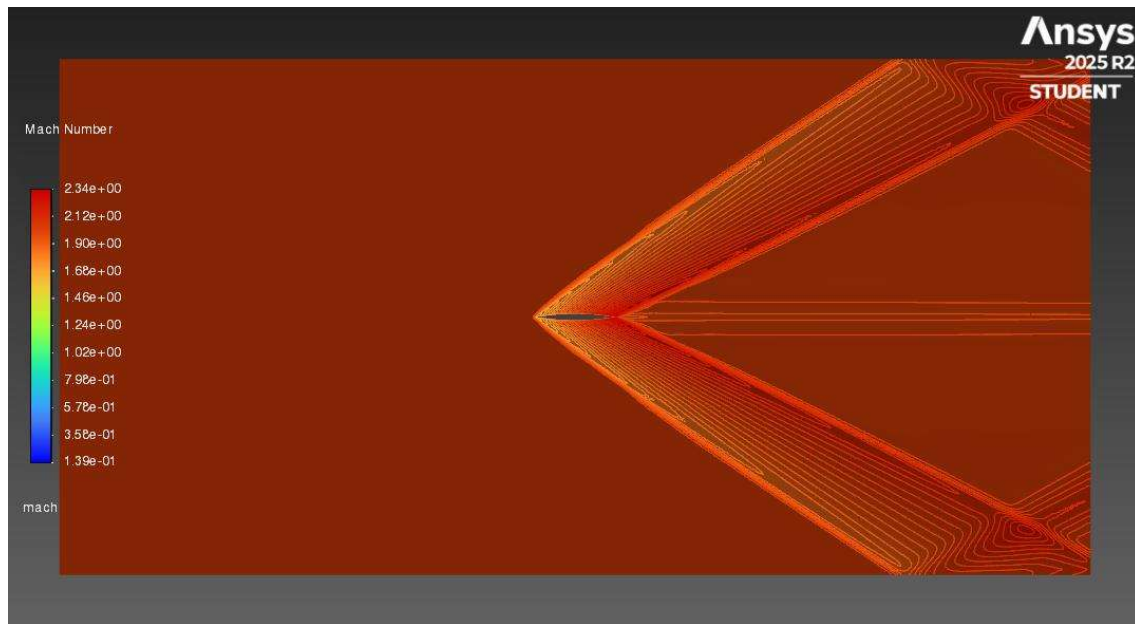


Coefficient of Lift:

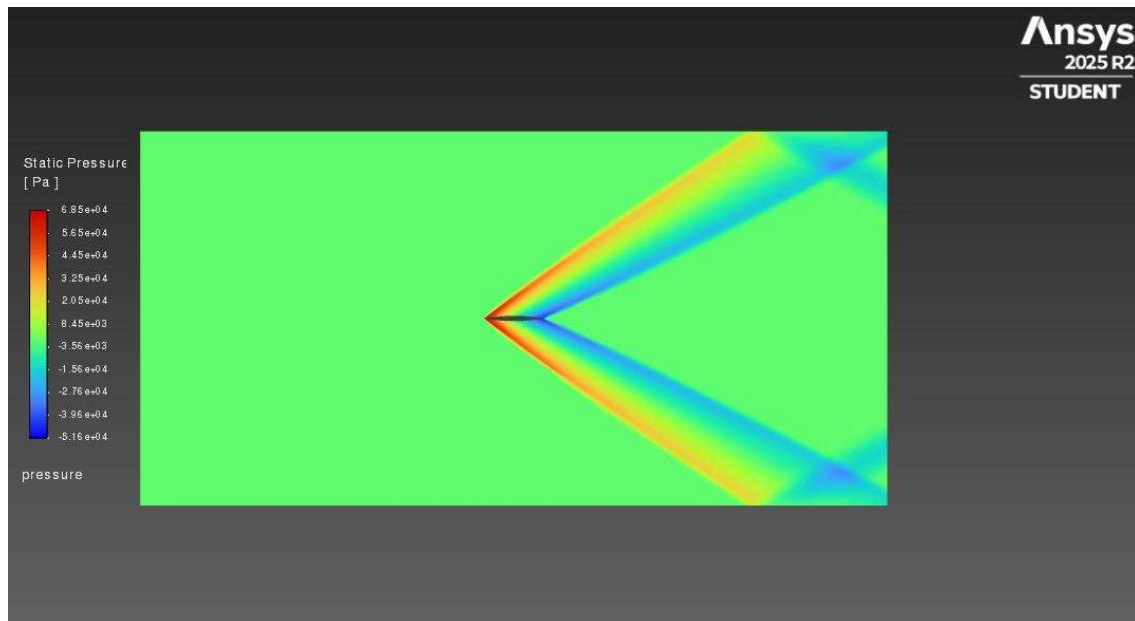
Stable Value: -0.00001



Mach Contours:



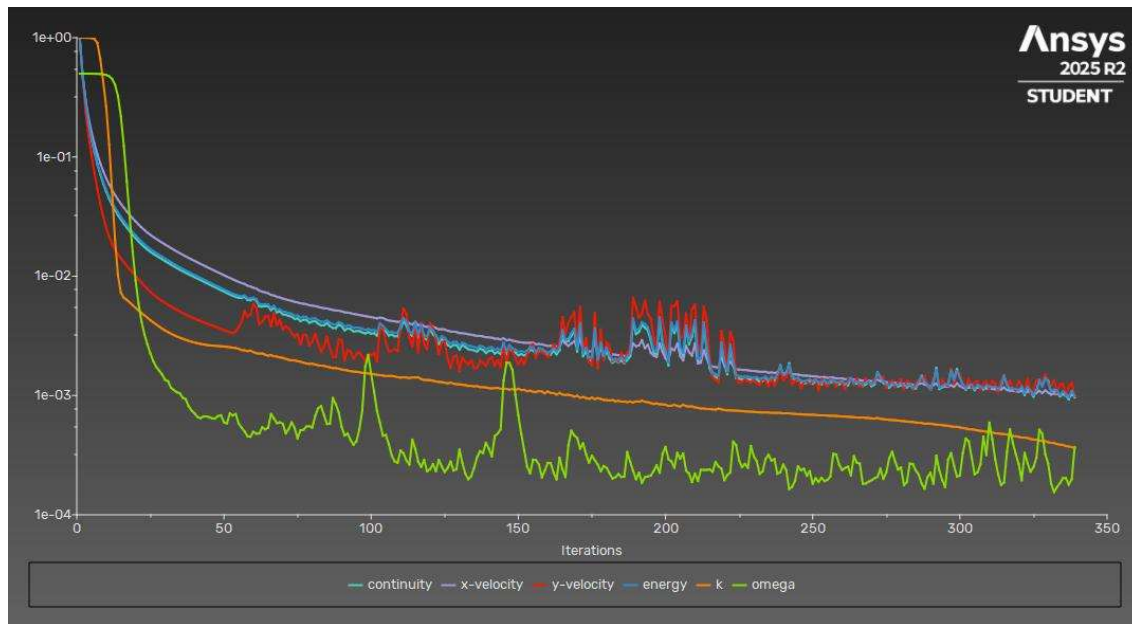
Pressure Contours:



MACH 3:

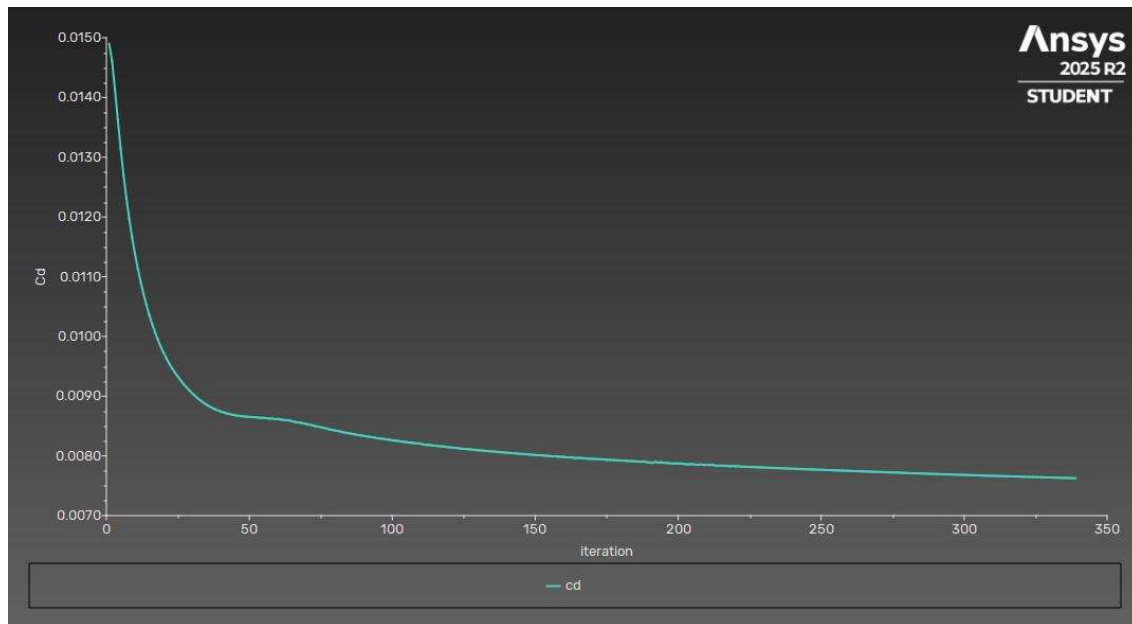
The following images below list the results for the double wedge run on the Density-based solver. Iteration 339 solution has converged.

Scaled Residual:



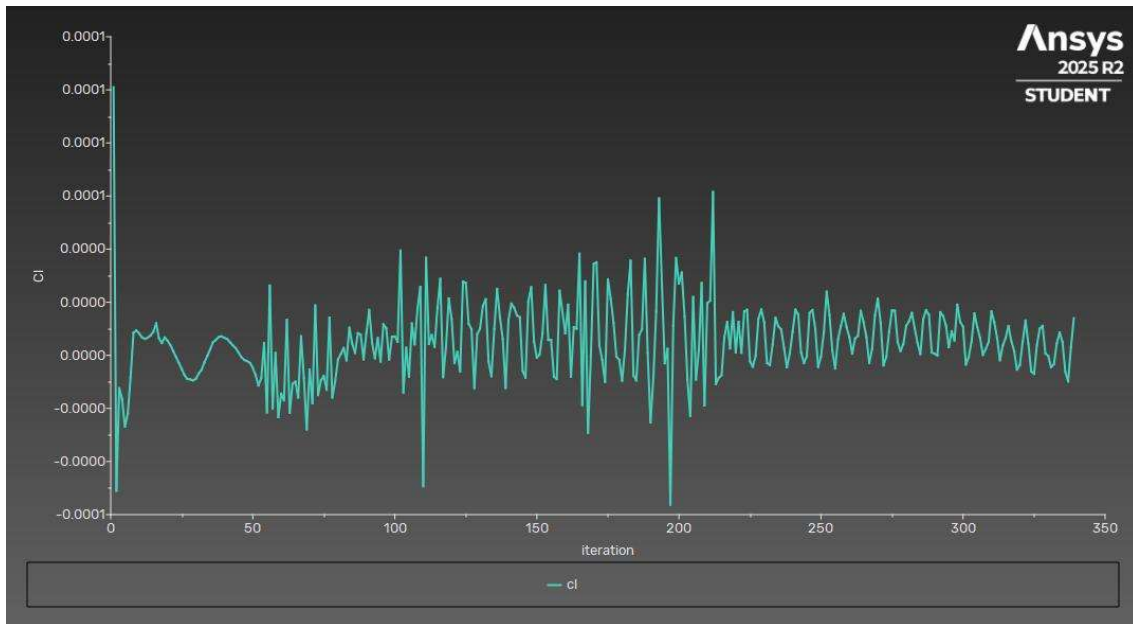
Coefficient of Drag:

Stable Value: 0.00762

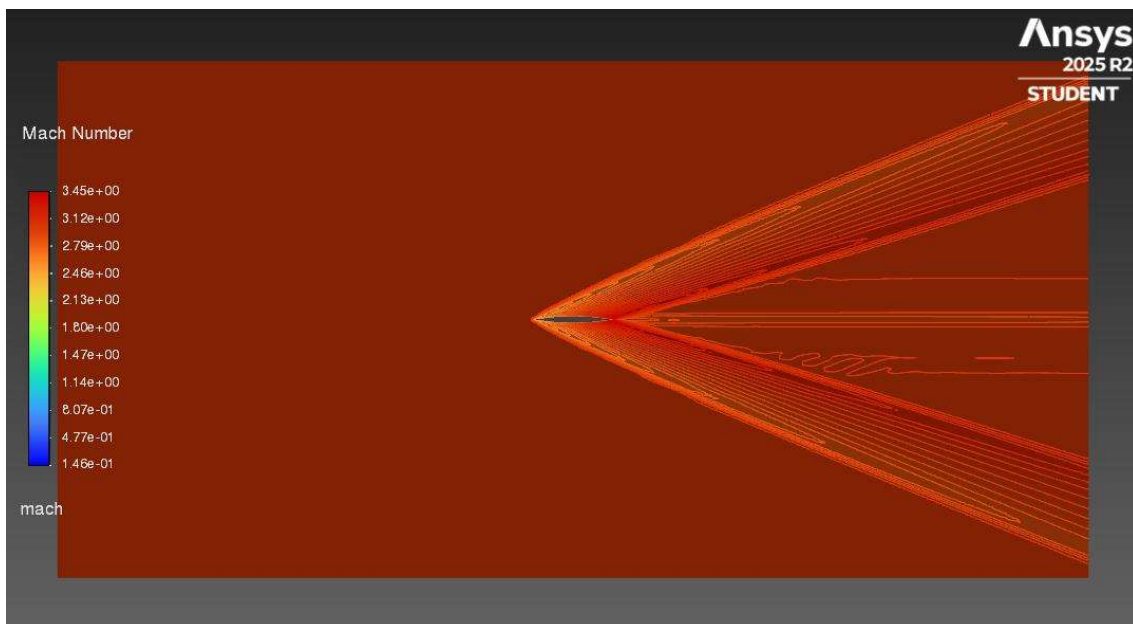


Coefficient of Lift:

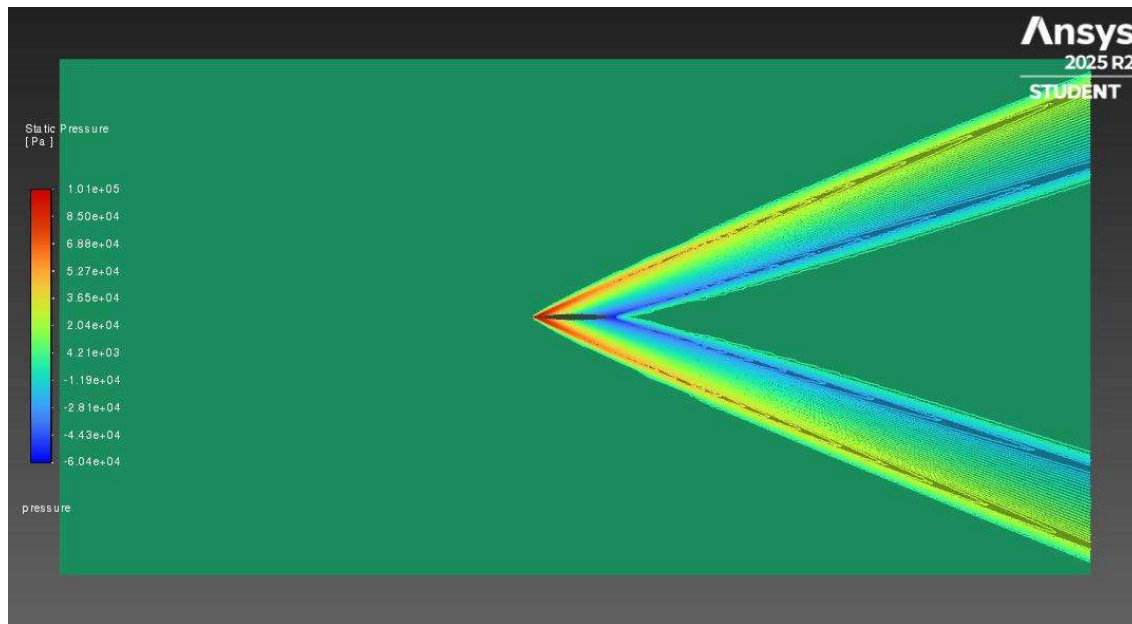
Stable Value: 0.00001



Mach Contours:



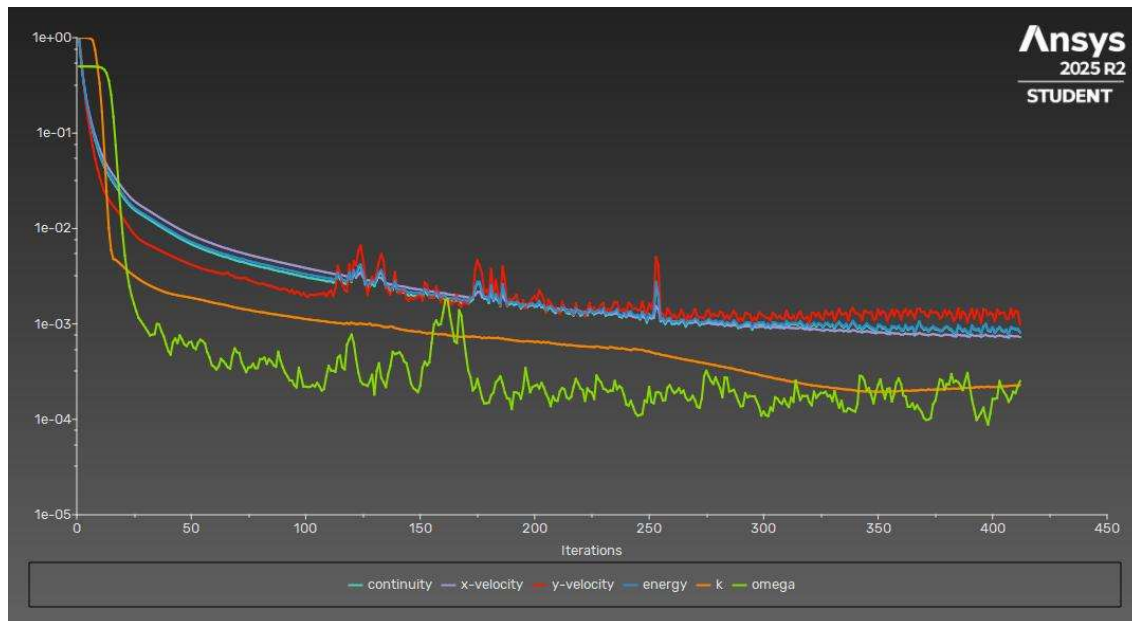
Pressure Contours:



MACH 4:

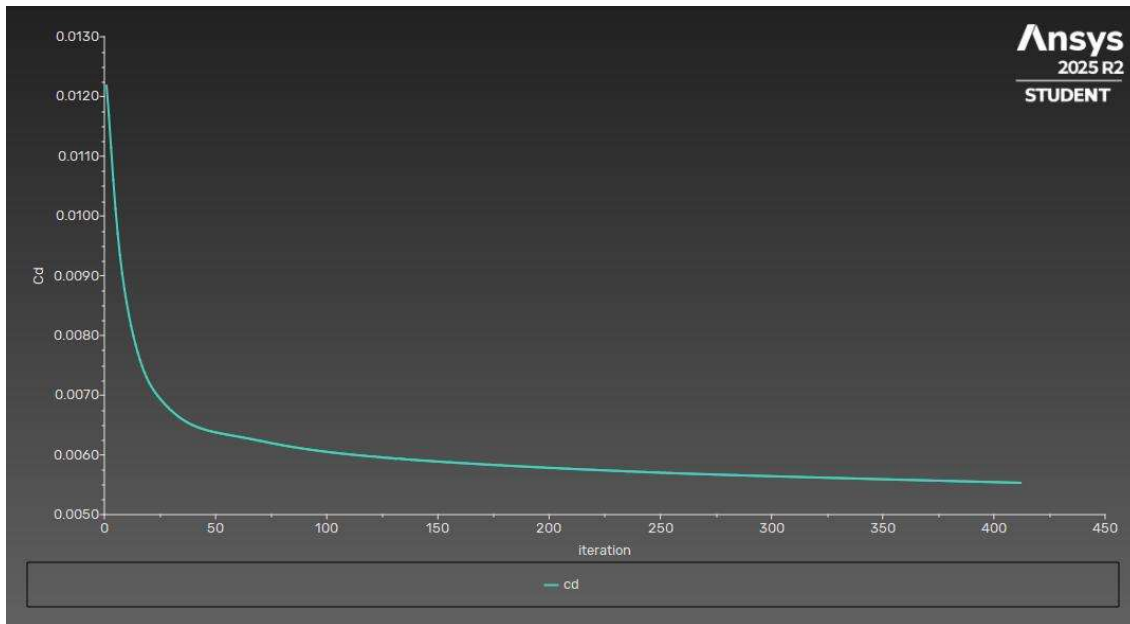
The following images below list the results for the double wedge run on the Density-based solver. Iteration 412 solution has converged.

Scaled Residual:



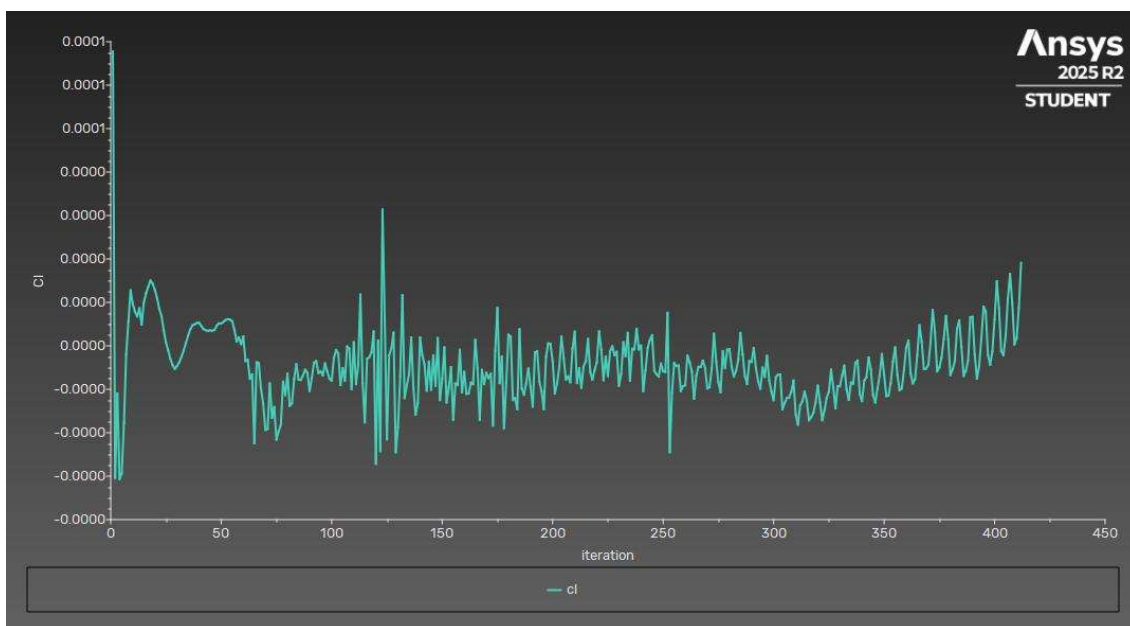
Coefficient of Drag:

Stable Value: 0.0055

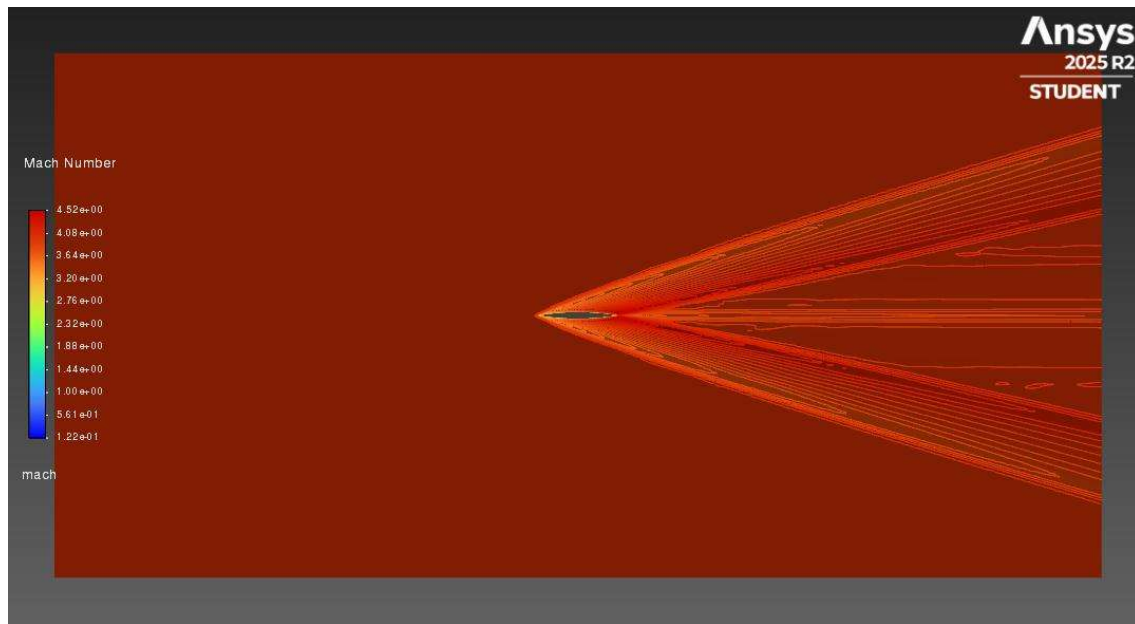


Coefficient of Lift:

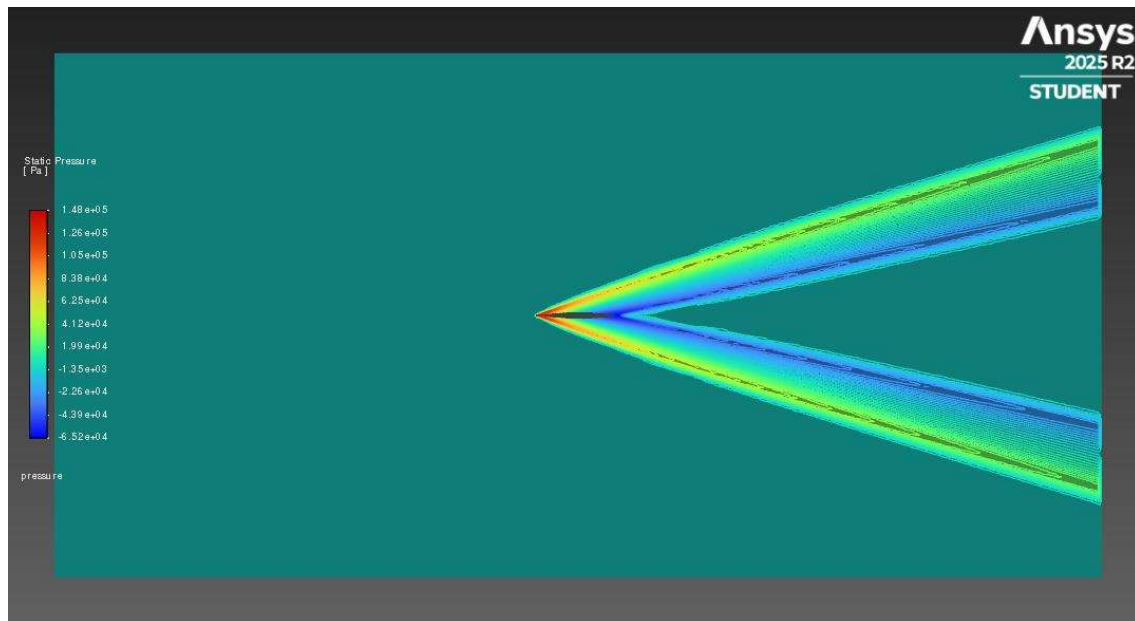
Stable Value: 0.000019



Mach Contours:



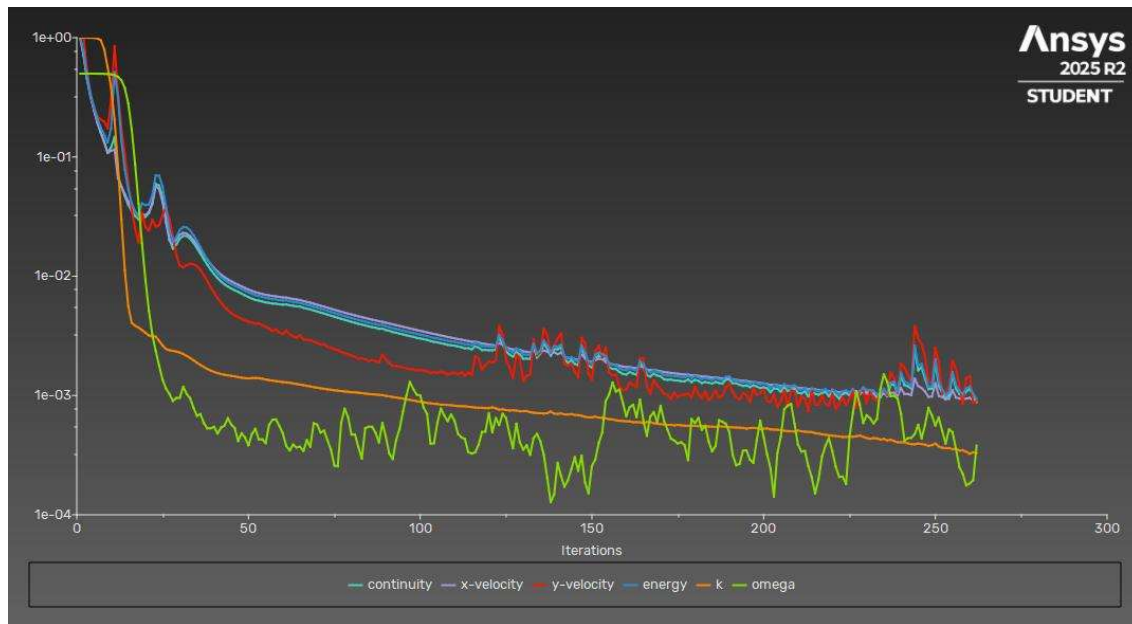
Pressure Contours:



MACH 5:

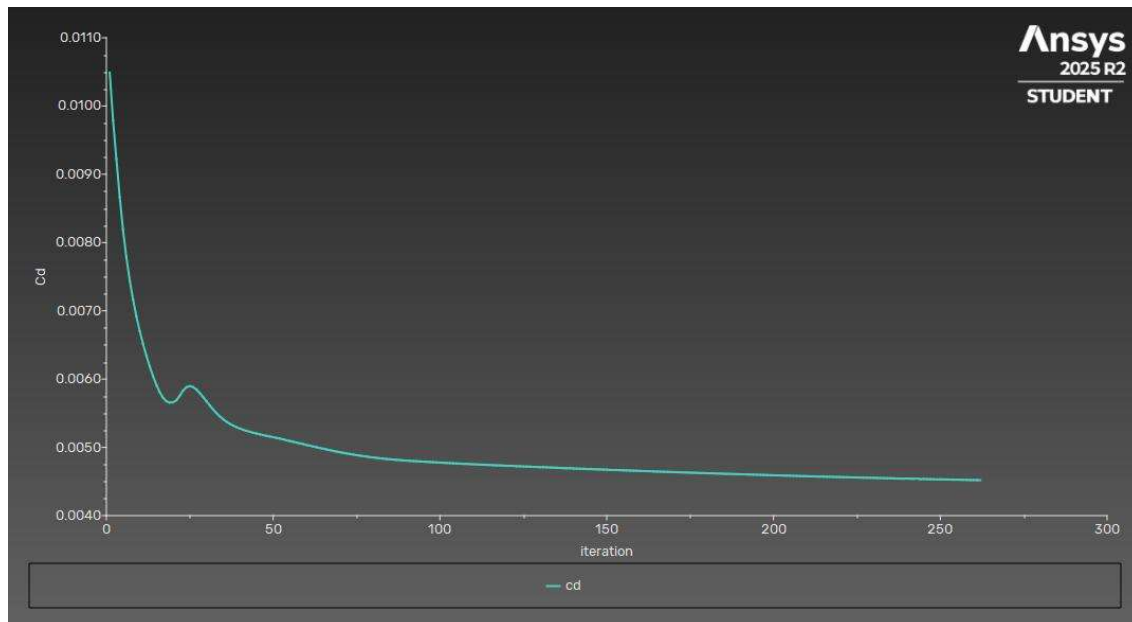
The following images below list the results for the double wedge run on the Density-based solver. Iteration 262 solution has converged.

Scaled Residual:



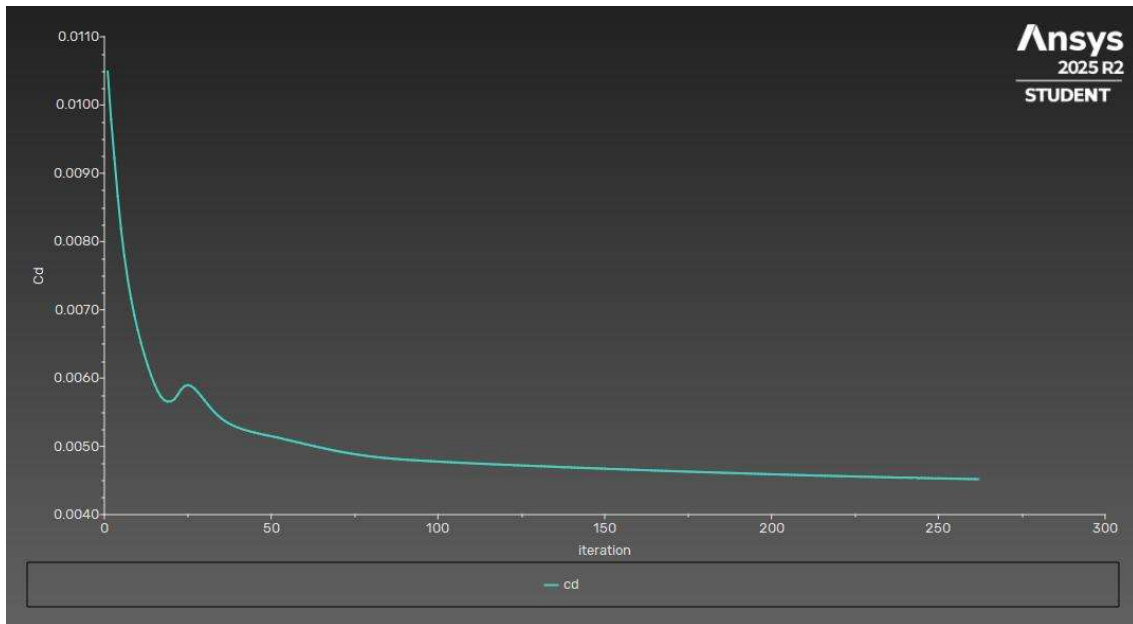
Coefficient of Drag:

Stable Value: 0.0045

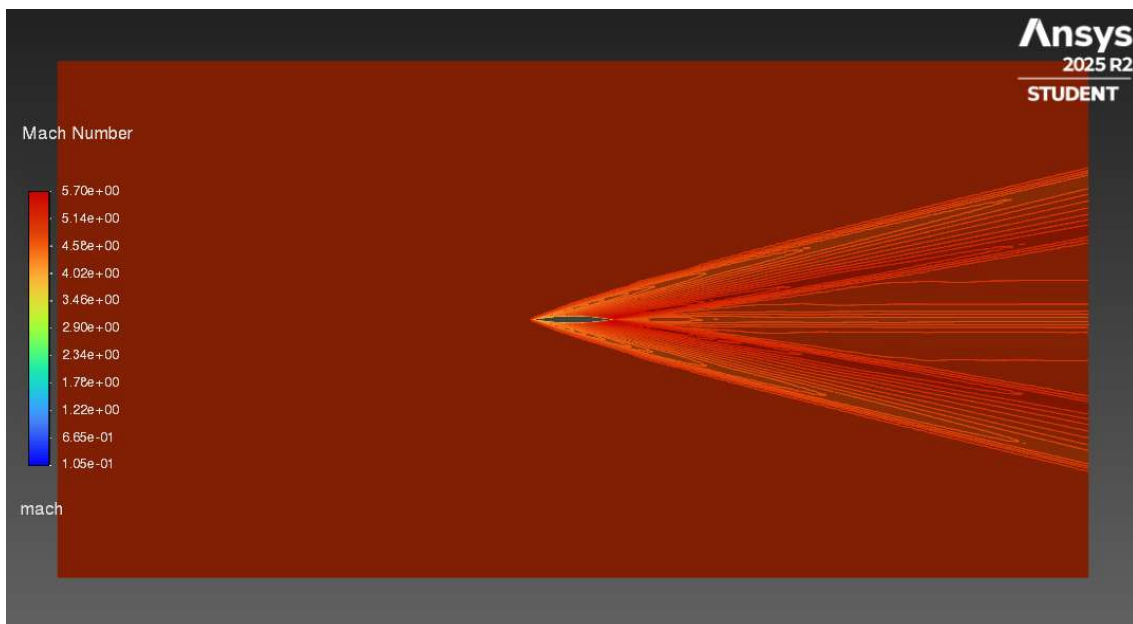


Coefficient of Lift:

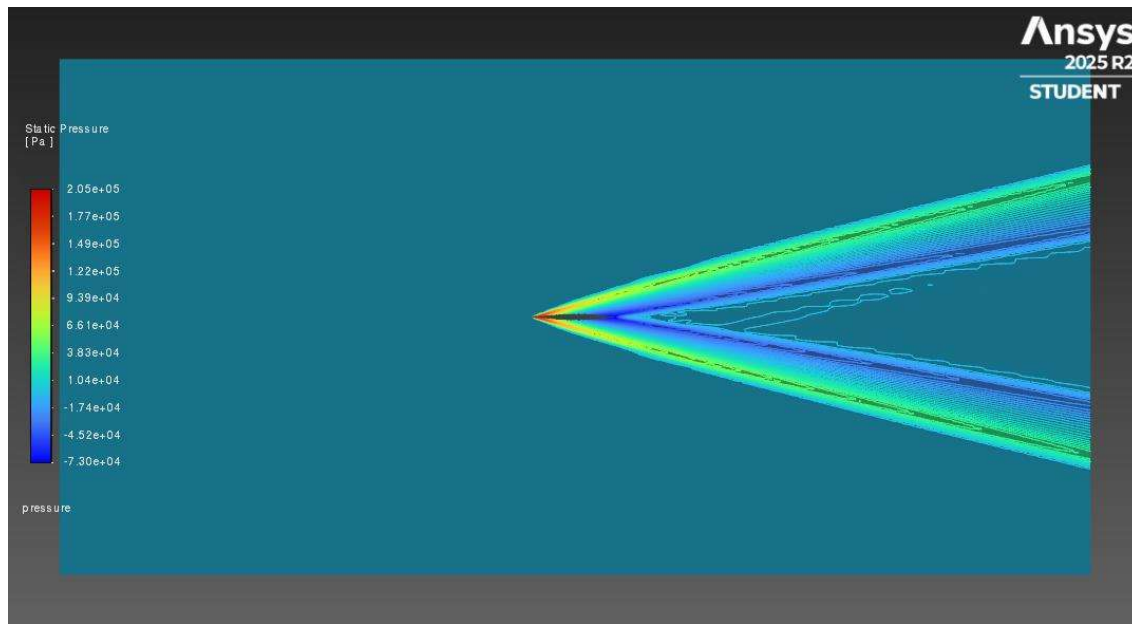
Stable Value: -0.000024



Mach Contours:



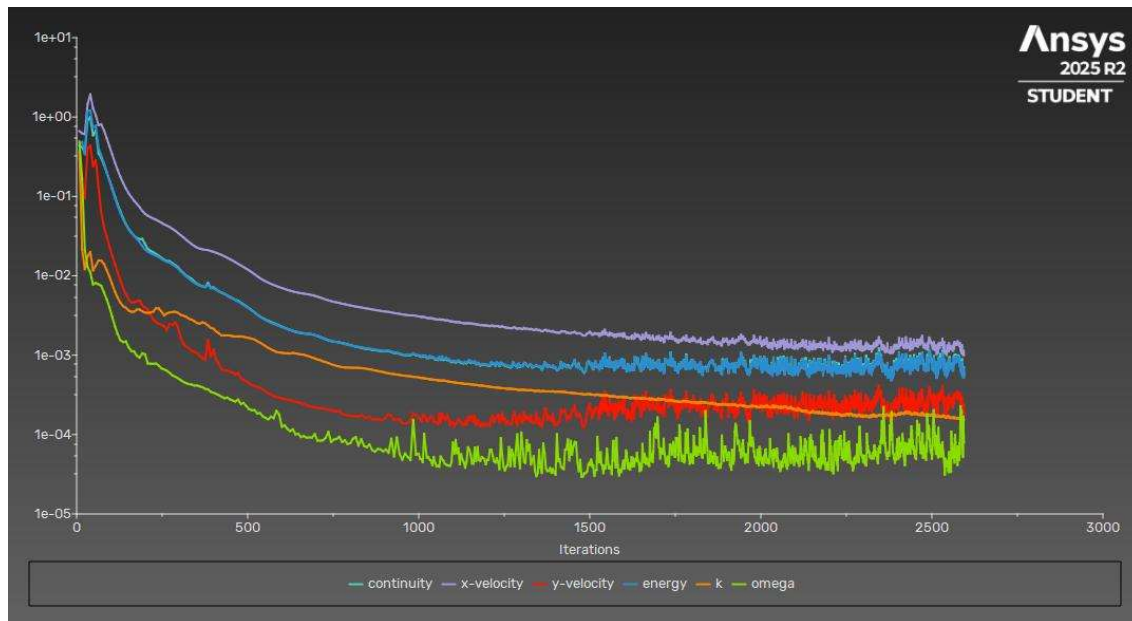
Pressure Contours:



MACH 6:

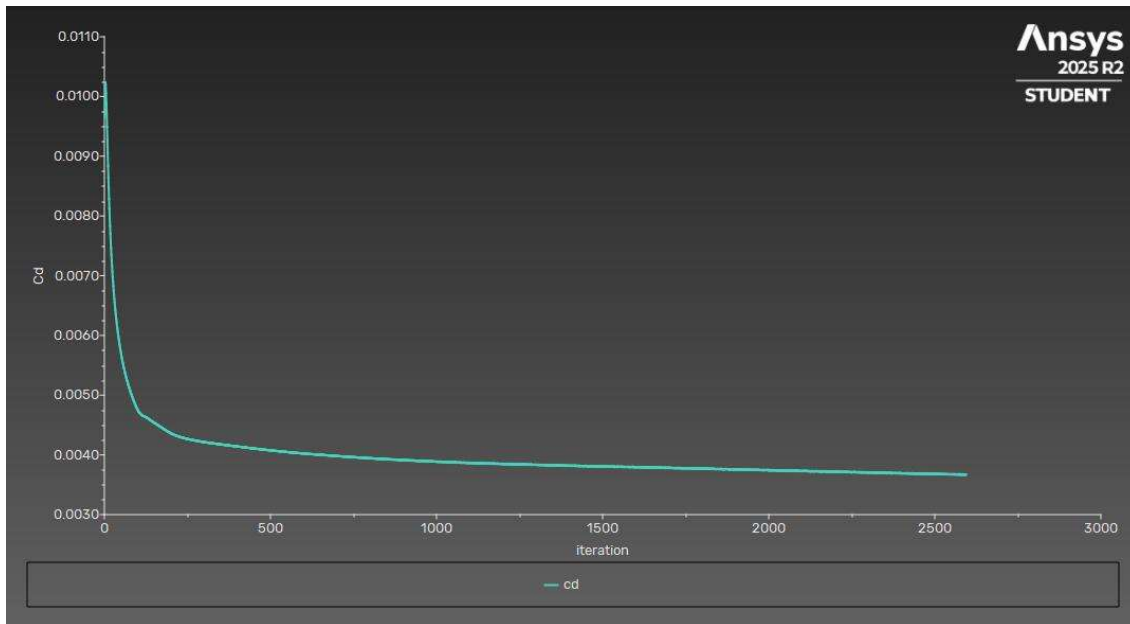
The following images below list the results for the double wedge run on the Density-based solver. Iteration 2594 solution has converged. Courant Number 0.75.

Scaled Residual:



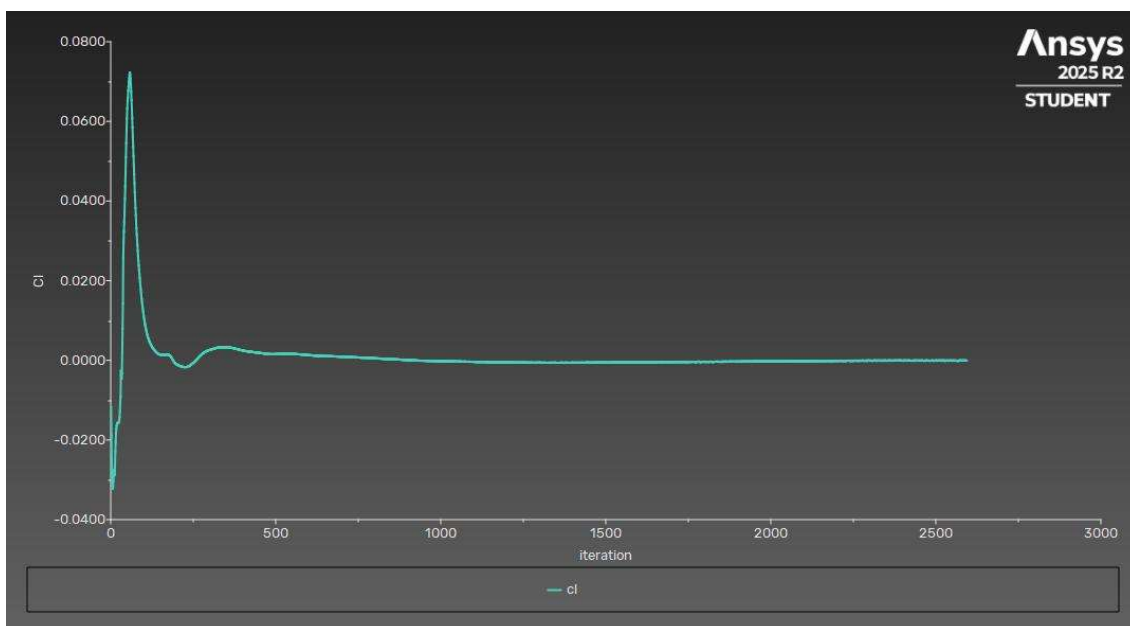
Coefficient of Drag:

Stable Value: 0.0036

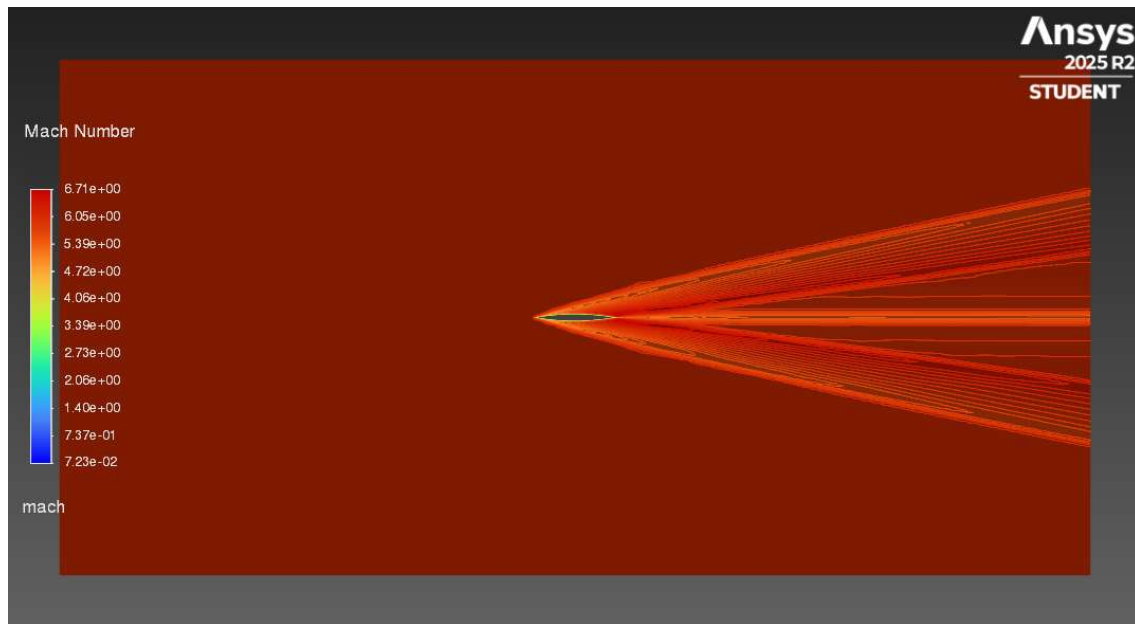


Coefficient of Lift:

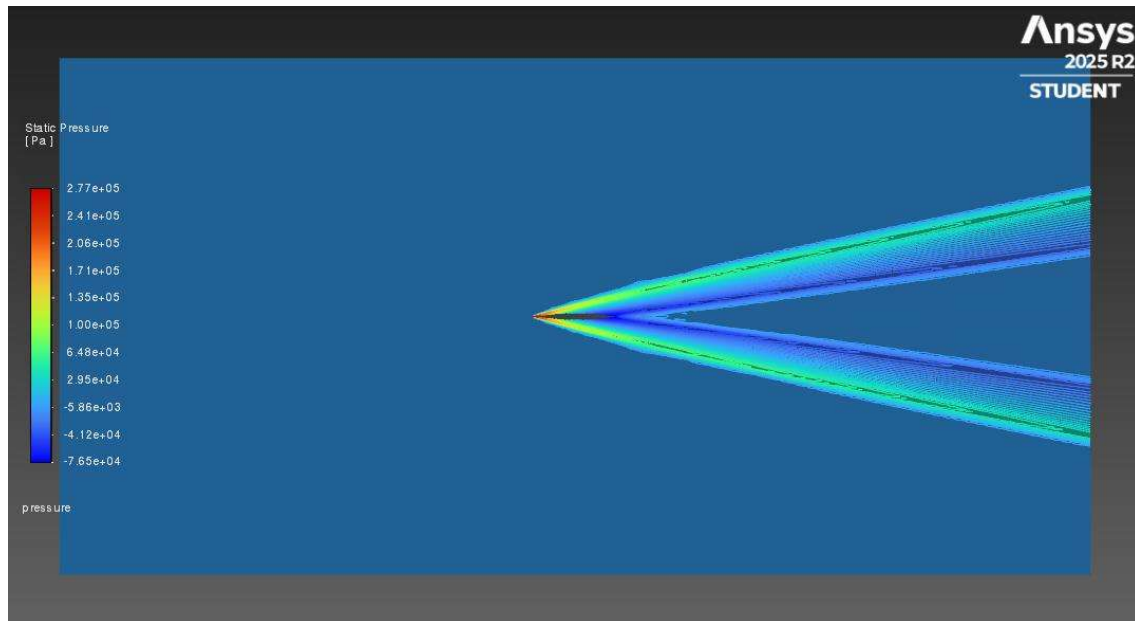
Stable Value: -0.00009



Mach Contours:



Pressure Contours:



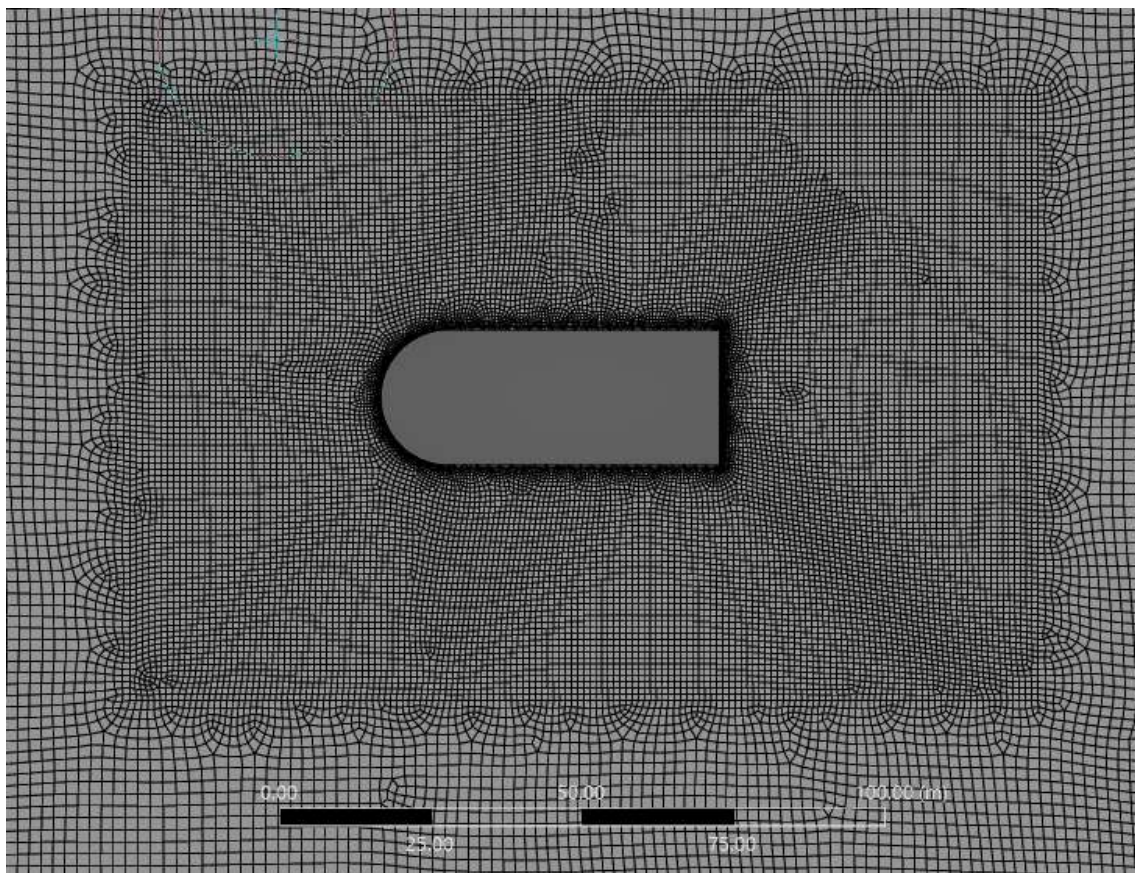
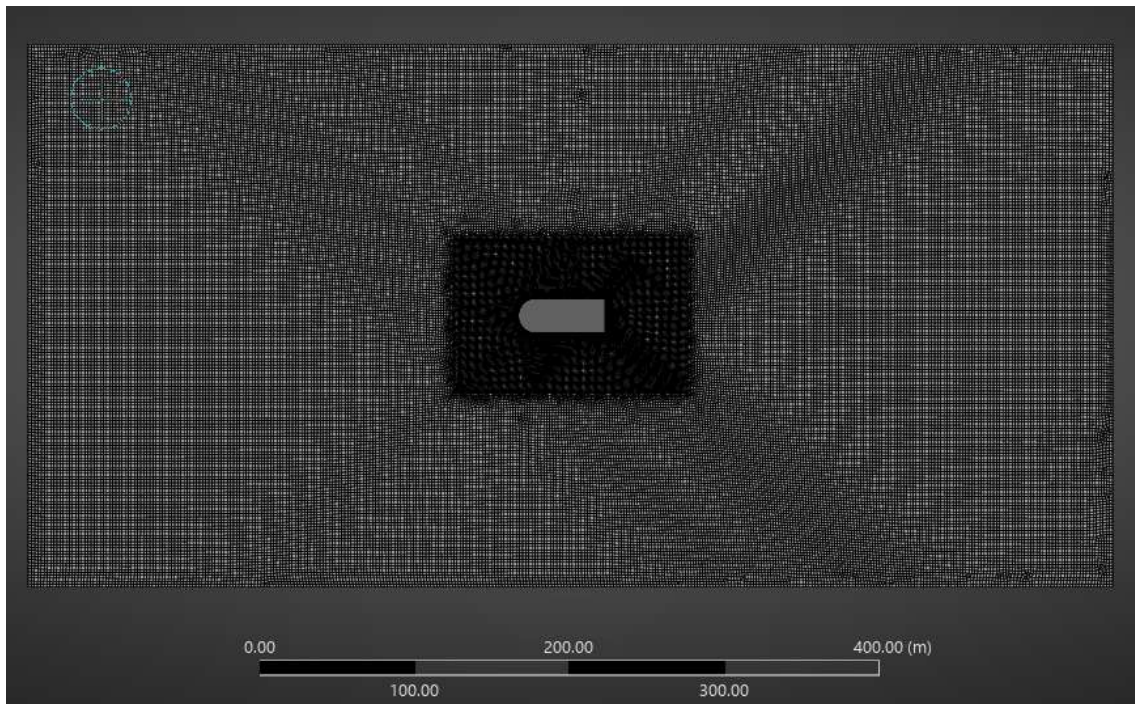
DOME SHAPE:

Mesh:

Average Mesh Quality: 0.976

No. of Nodes: 91,458

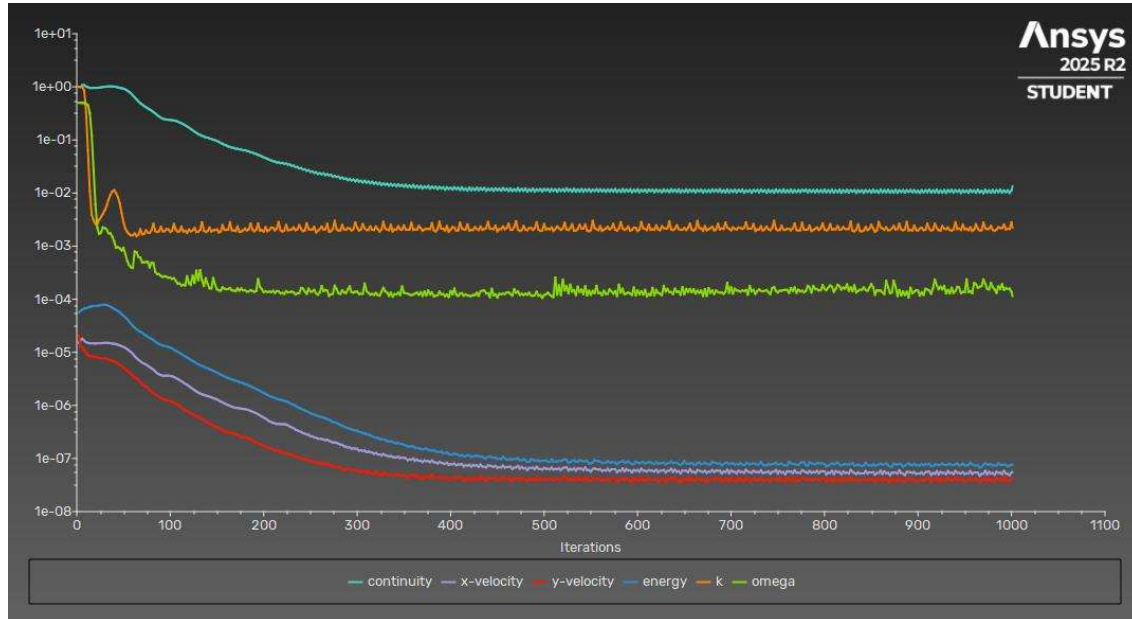
No. of Elements: 90,654



MACH 1:

The following images below list the results for the double wedge run on the pressure-based solver.

Scaled Residual:



Coefficient of Drag:

Stable Value:0.686

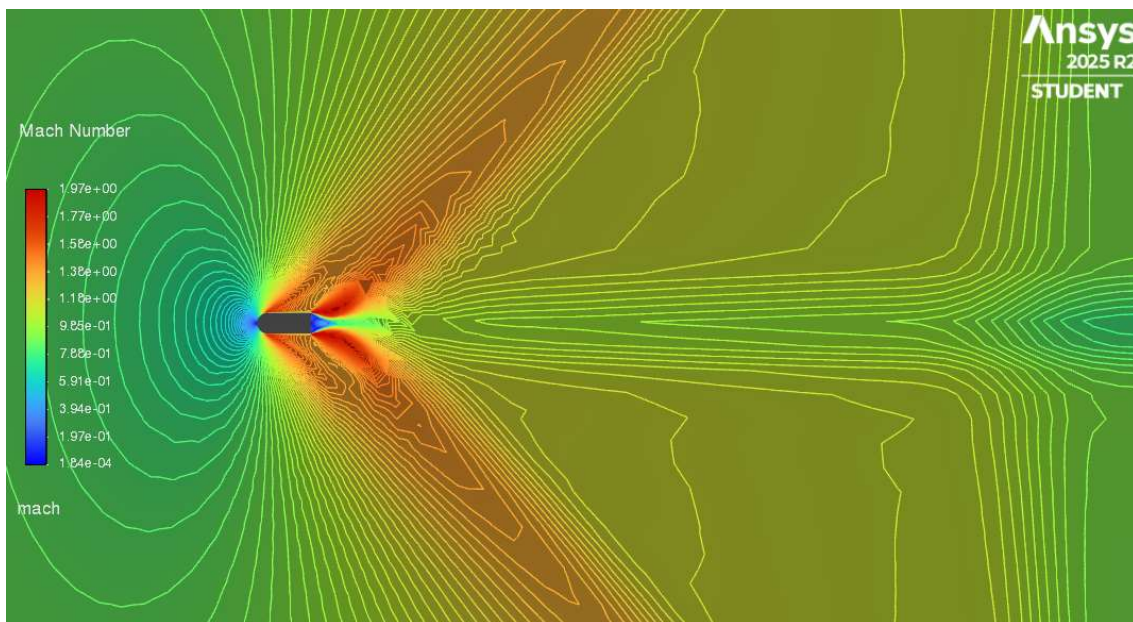


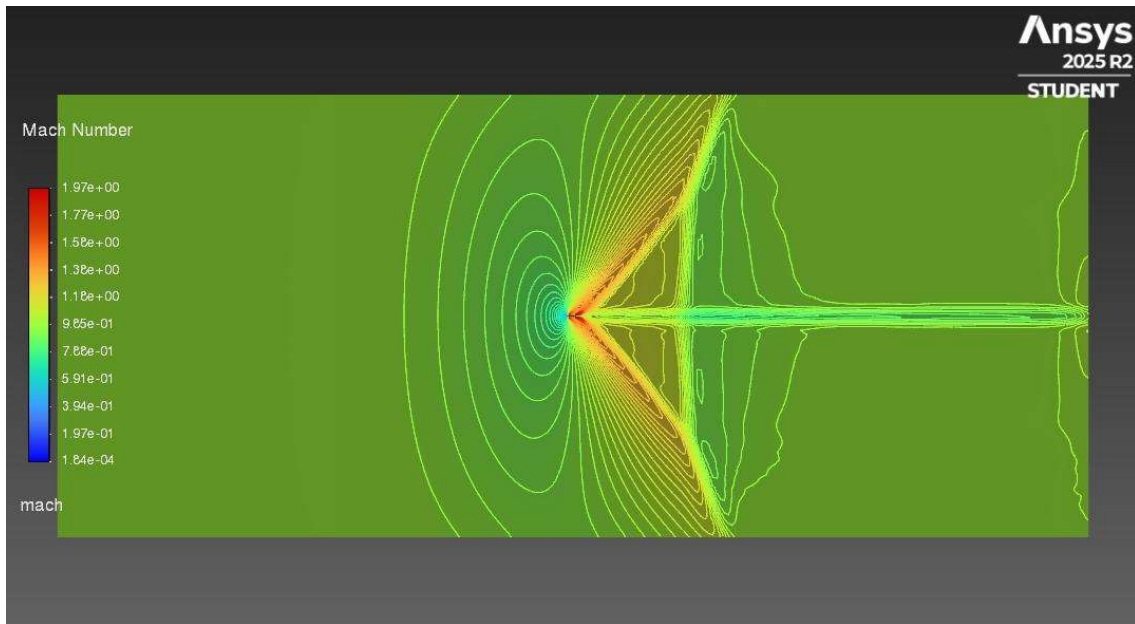
Coefficient of Lift:

Stable Value:0.00618

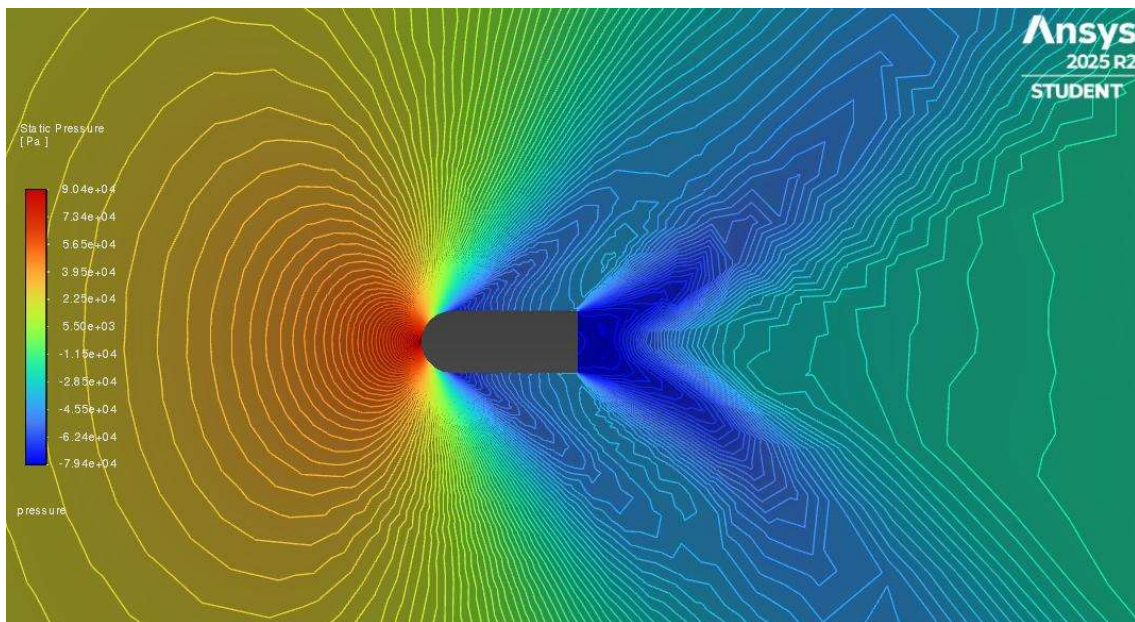


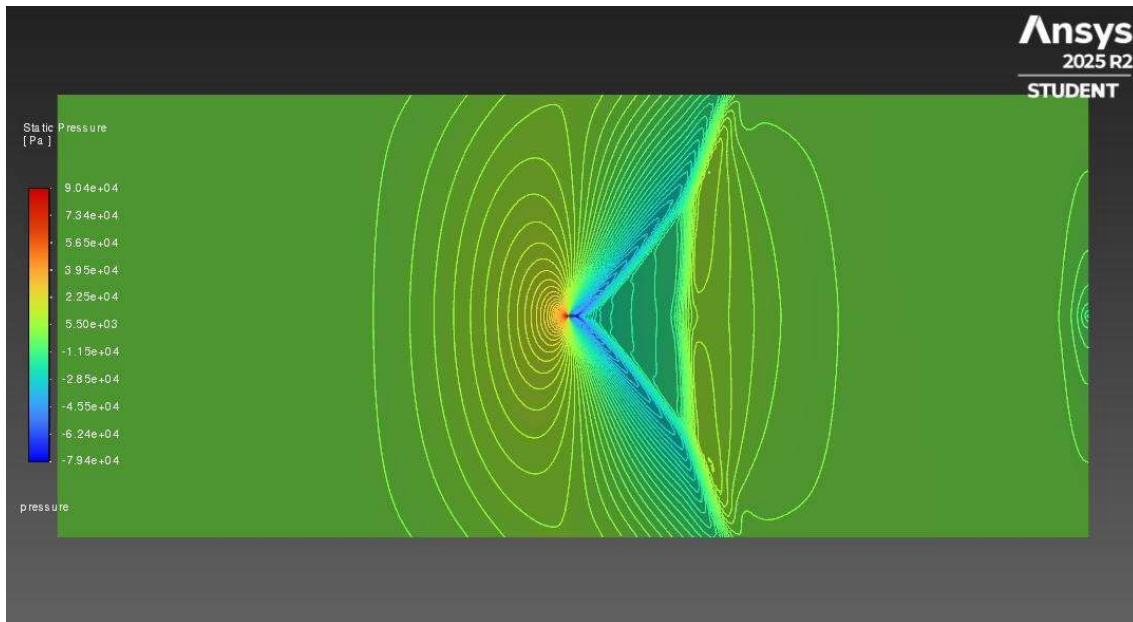
Mach Contours:





Pressure Contours:





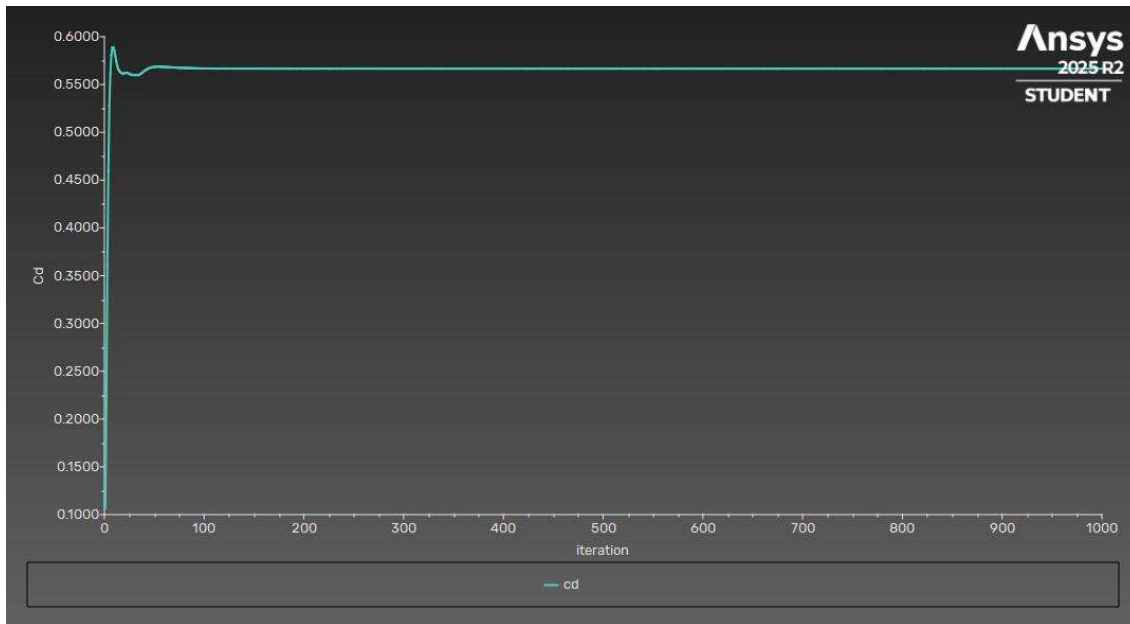
MACH 2:

Scaled Residual:



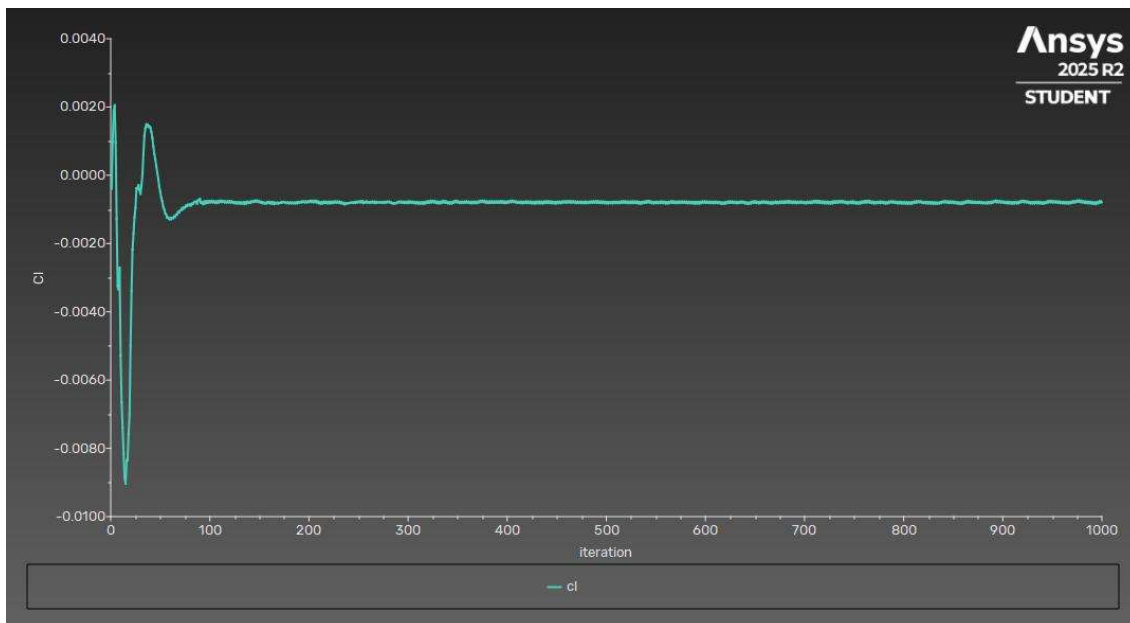
Coefficient of Drag:

Stable Value:0.566

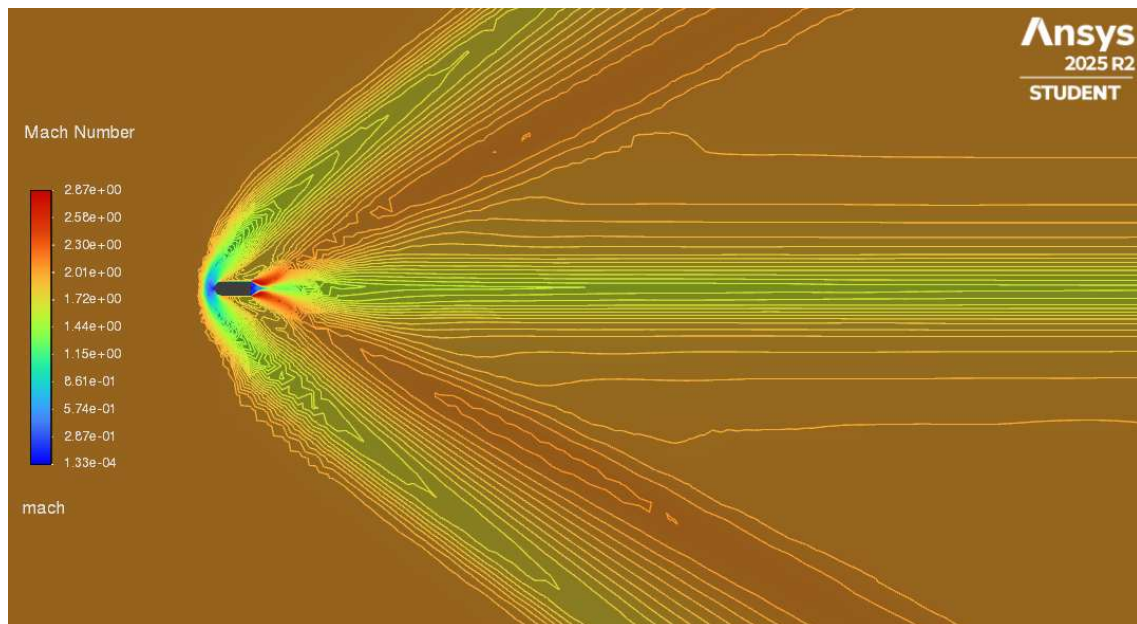
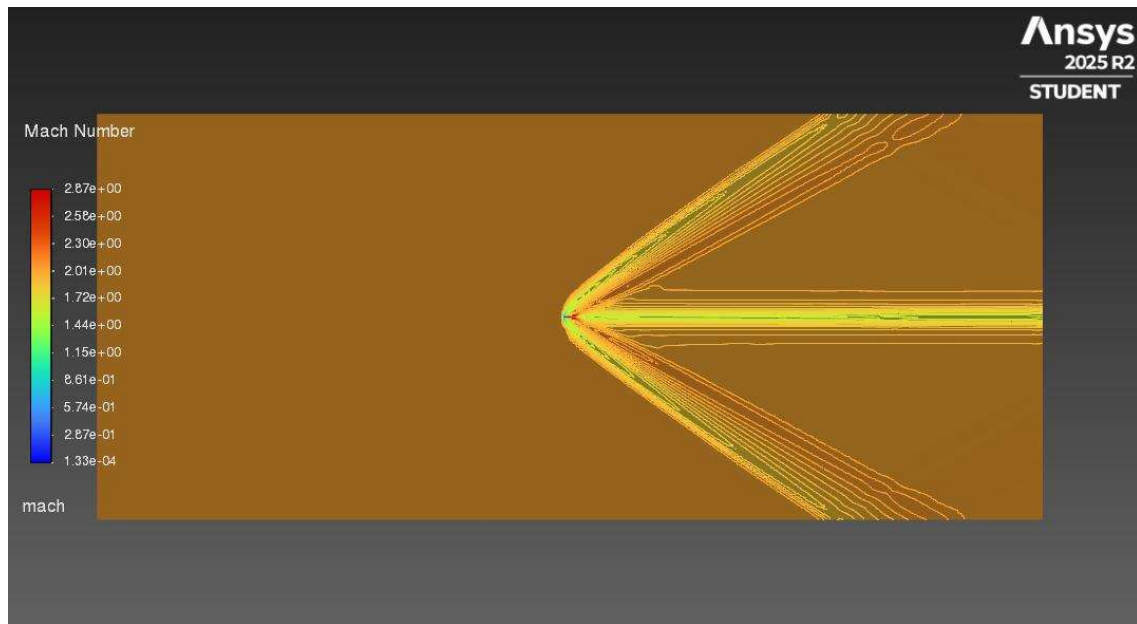


Coefficient of Lift:

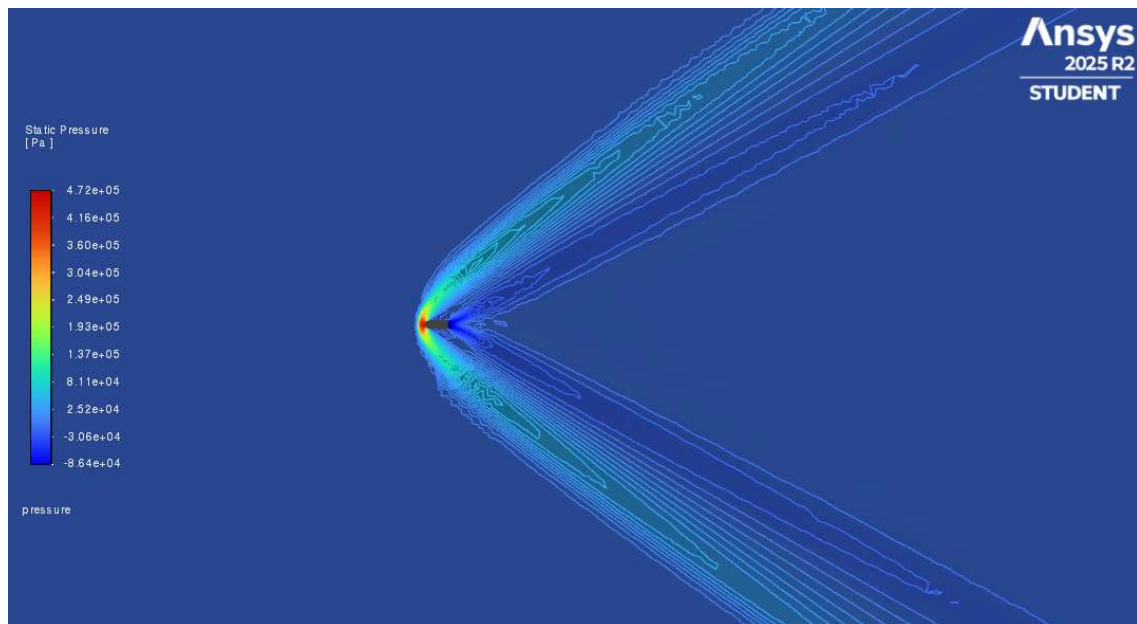
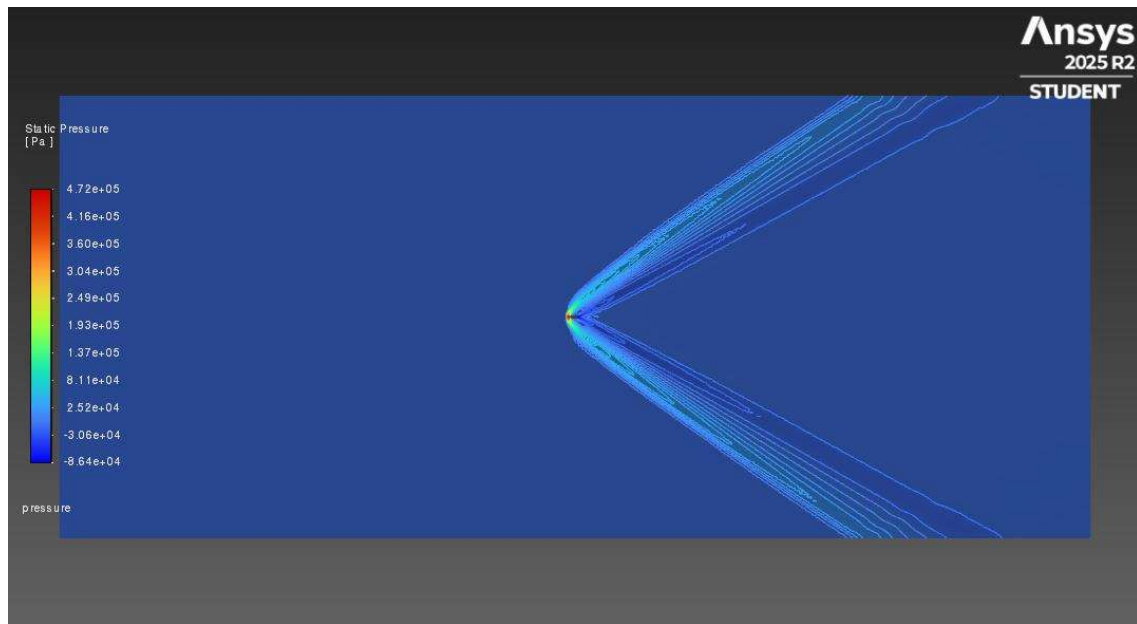
Stable Value: -0.000758



Mach Contours:

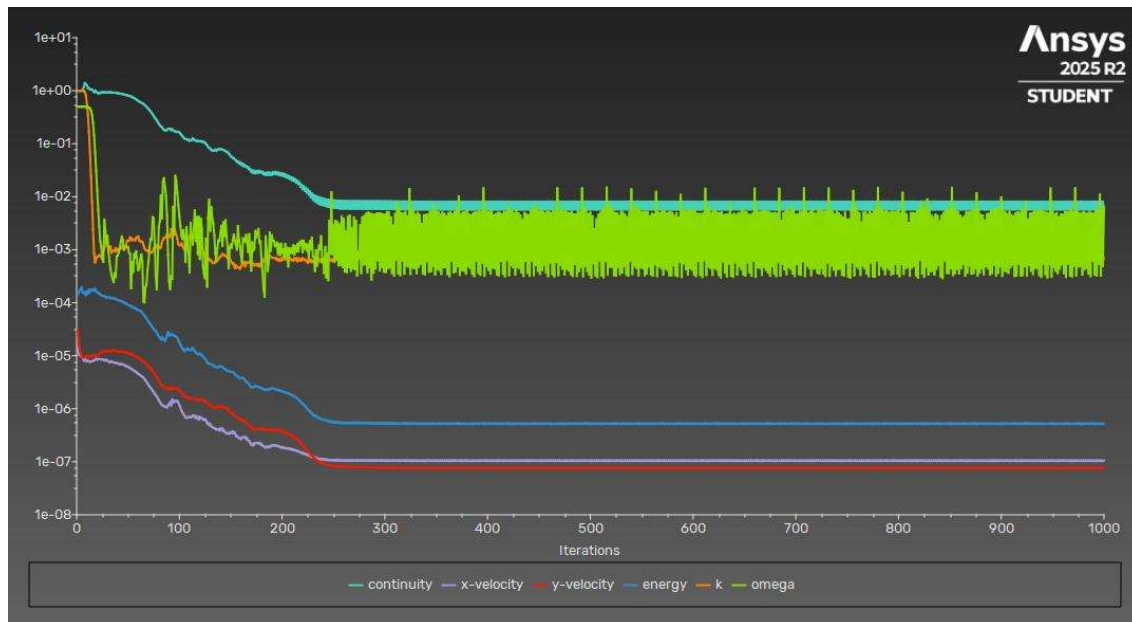


Pressure Contours:



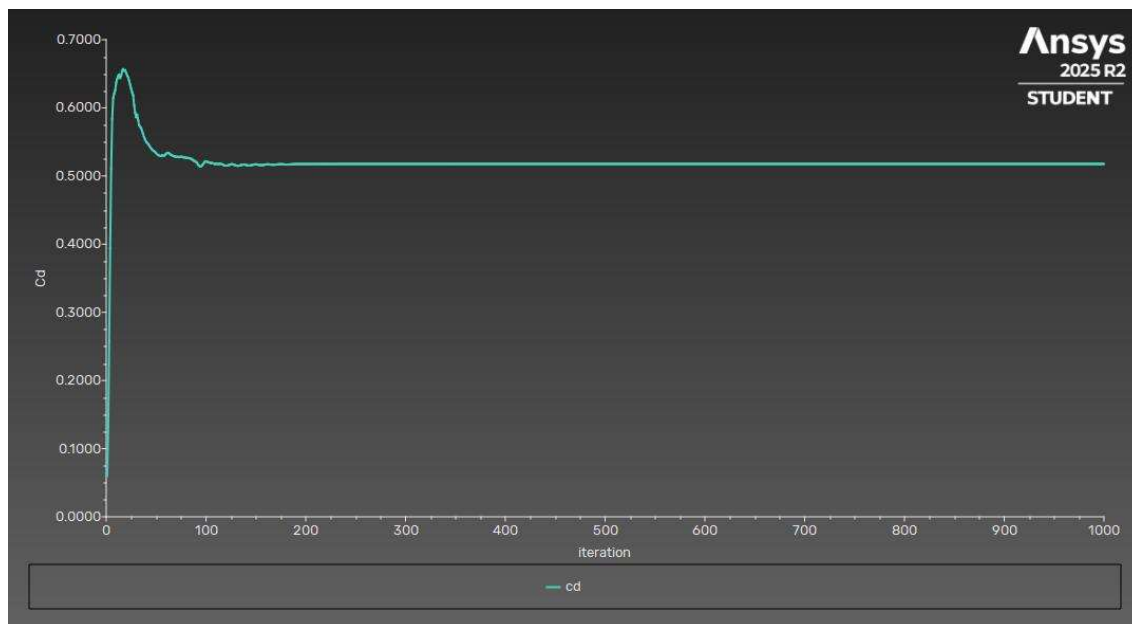
MACH 3:

Scaled Residual:



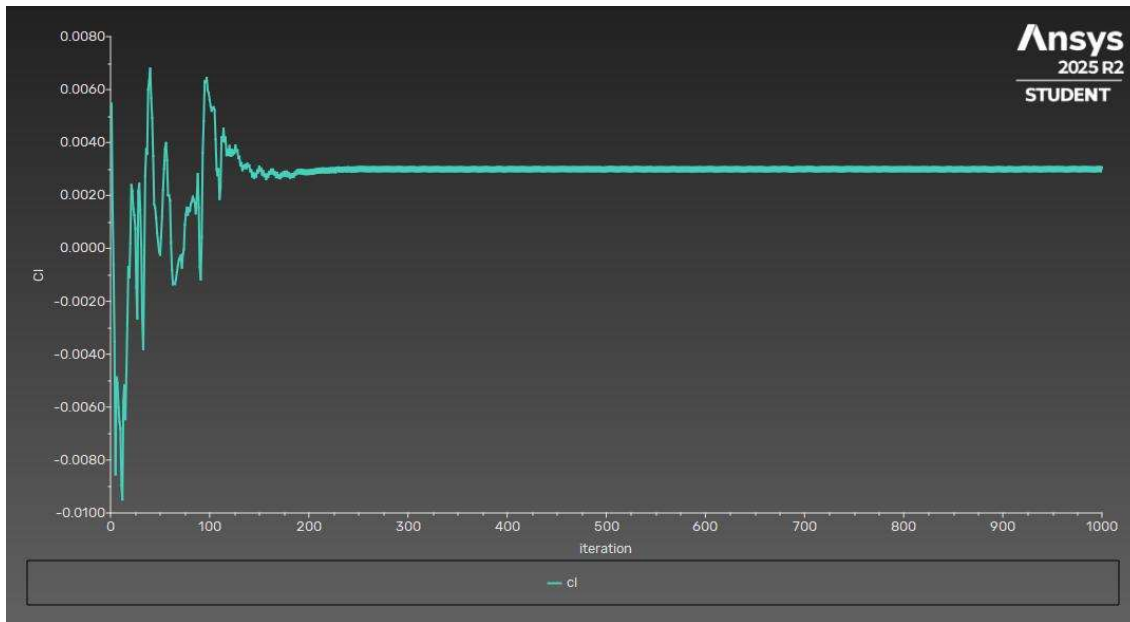
Coefficient of Drag:

Stable Value:0.517

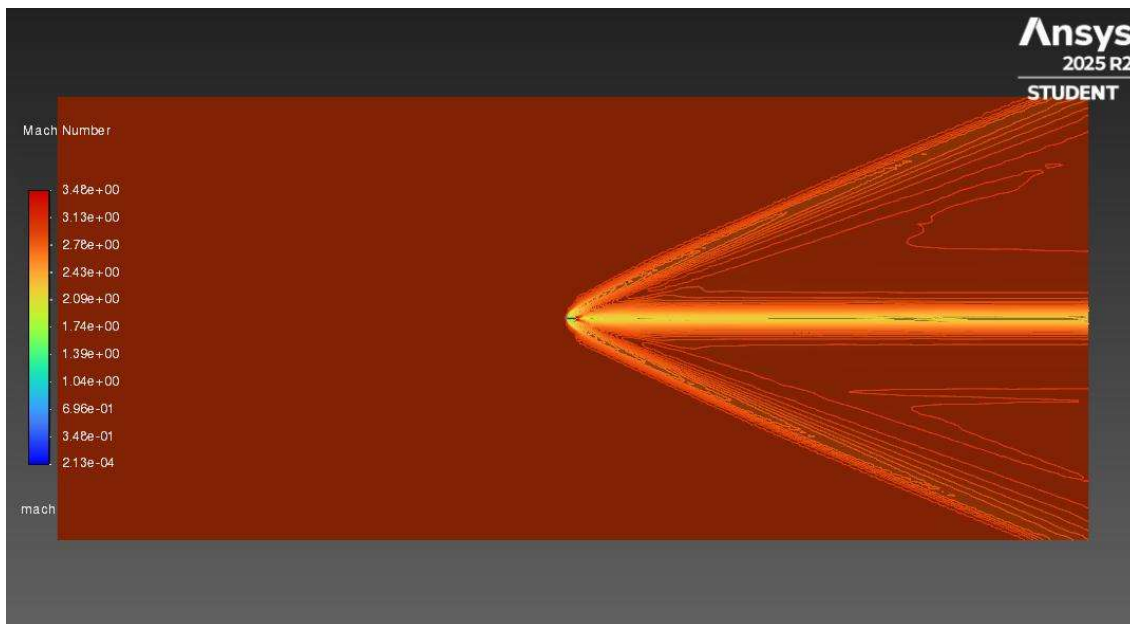


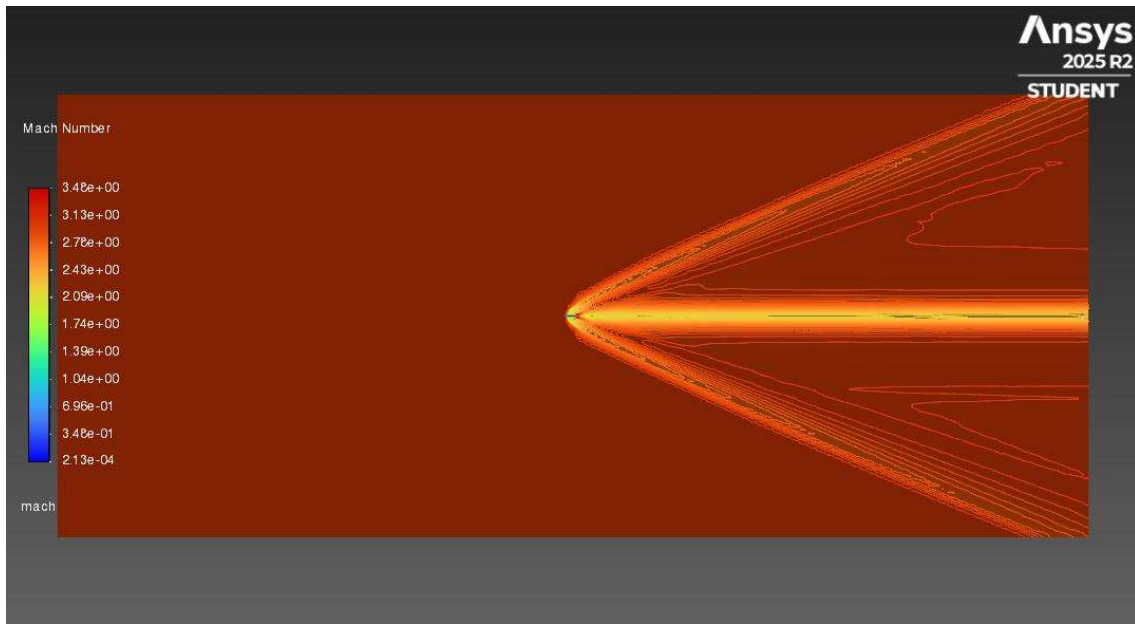
Coefficient of Lift:

Stable Value:0.003

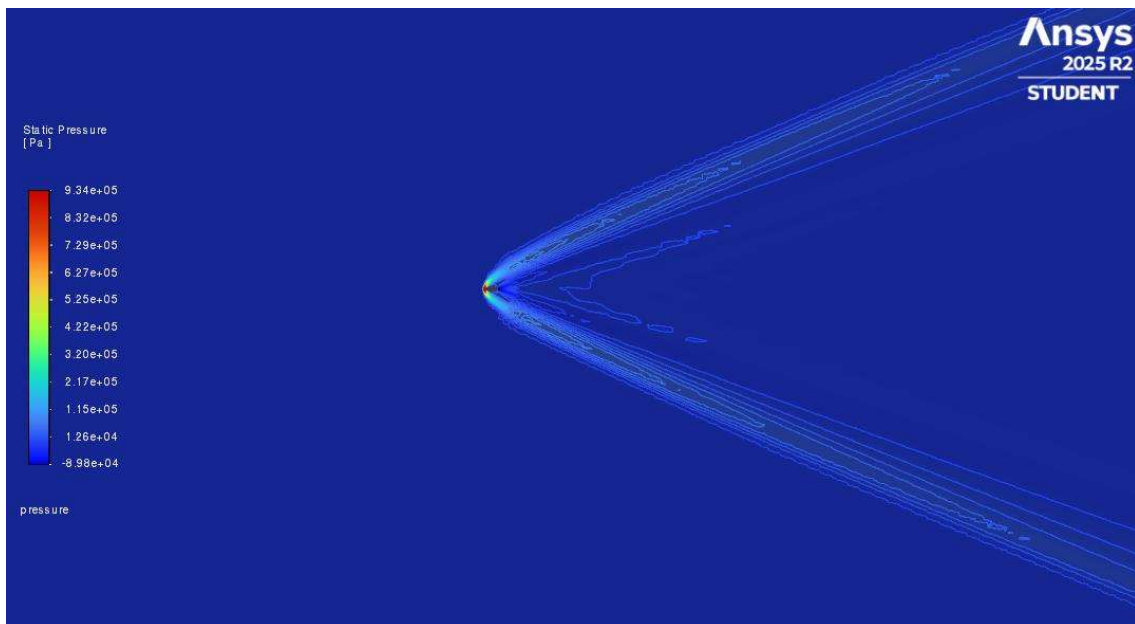


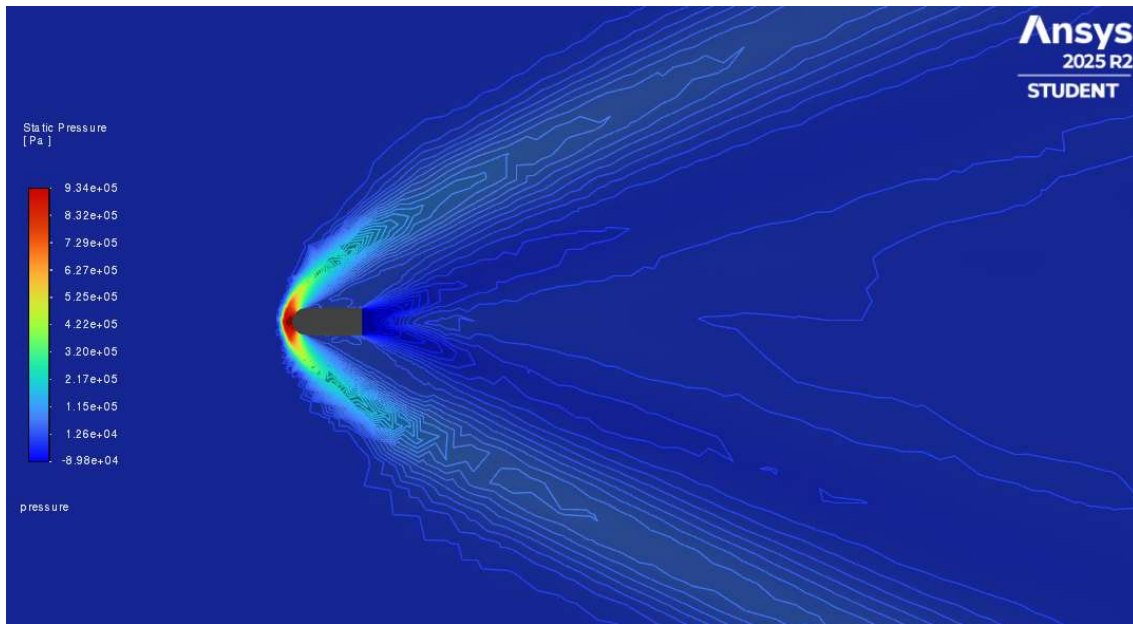
Mach Contours:





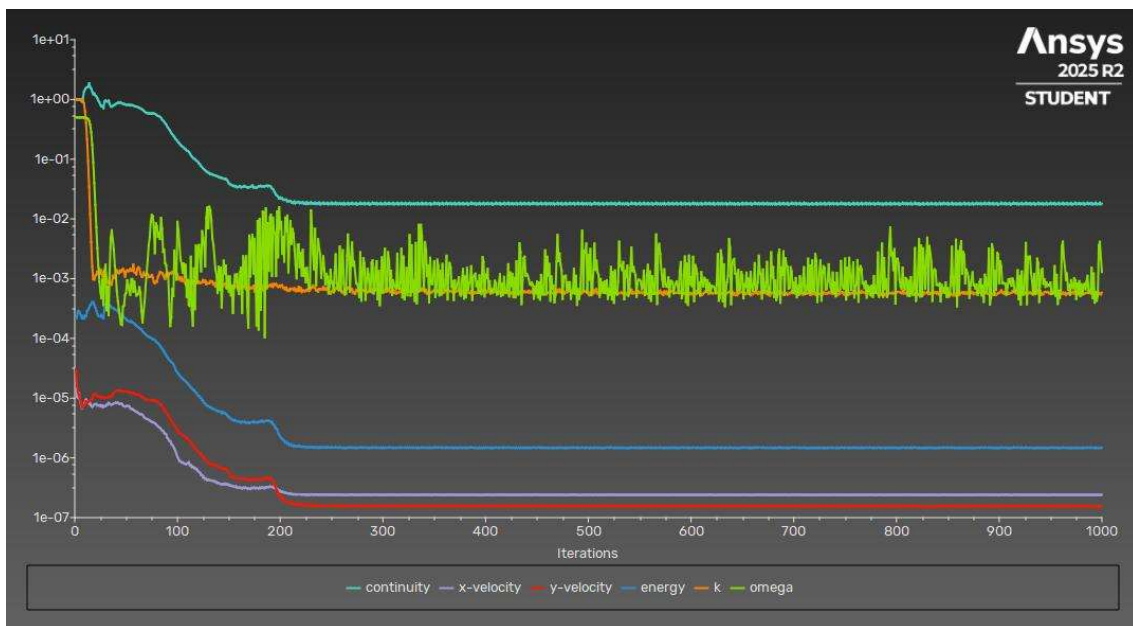
Pressure Contours:





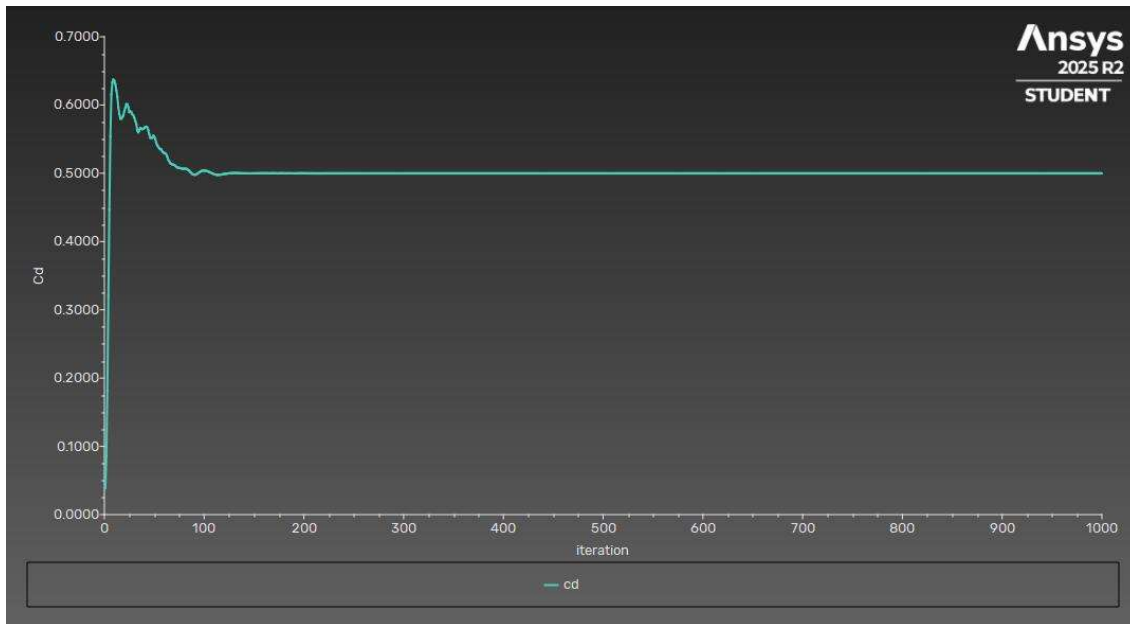
MACH 4:

Scaled Residual:



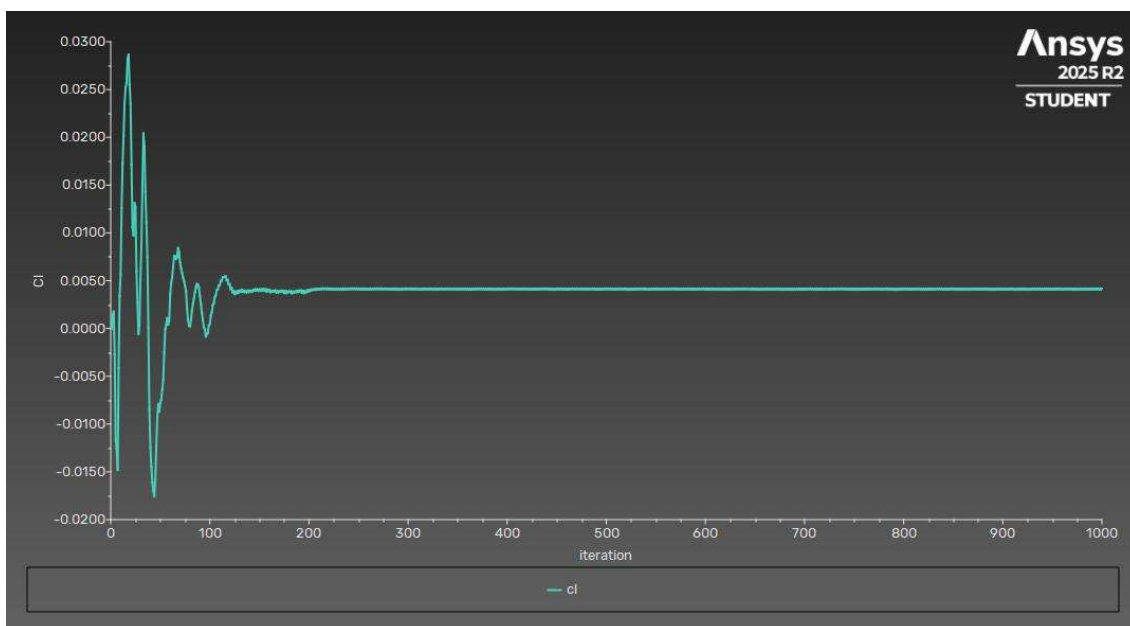
Coefficient of Drag:

Stable Value:0.5

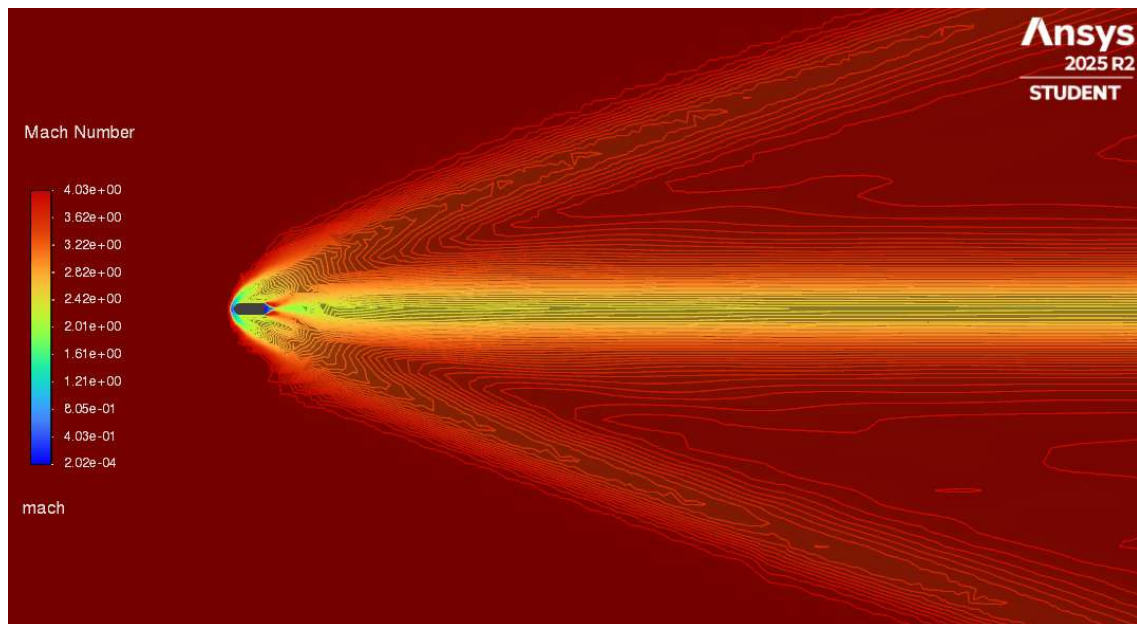
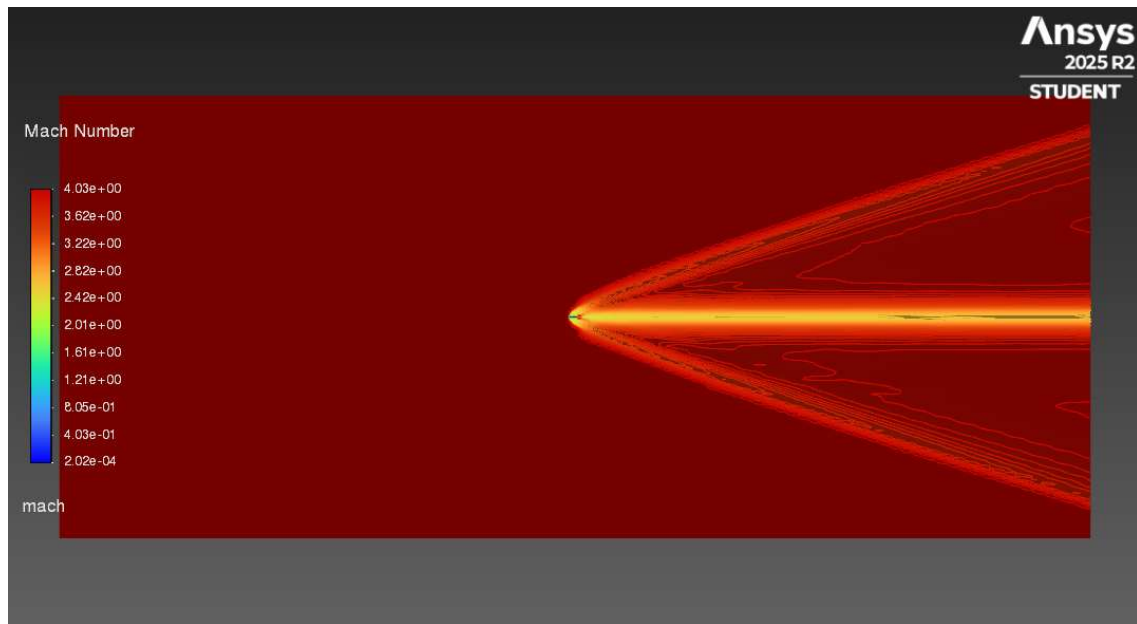


Coefficient of Lift:

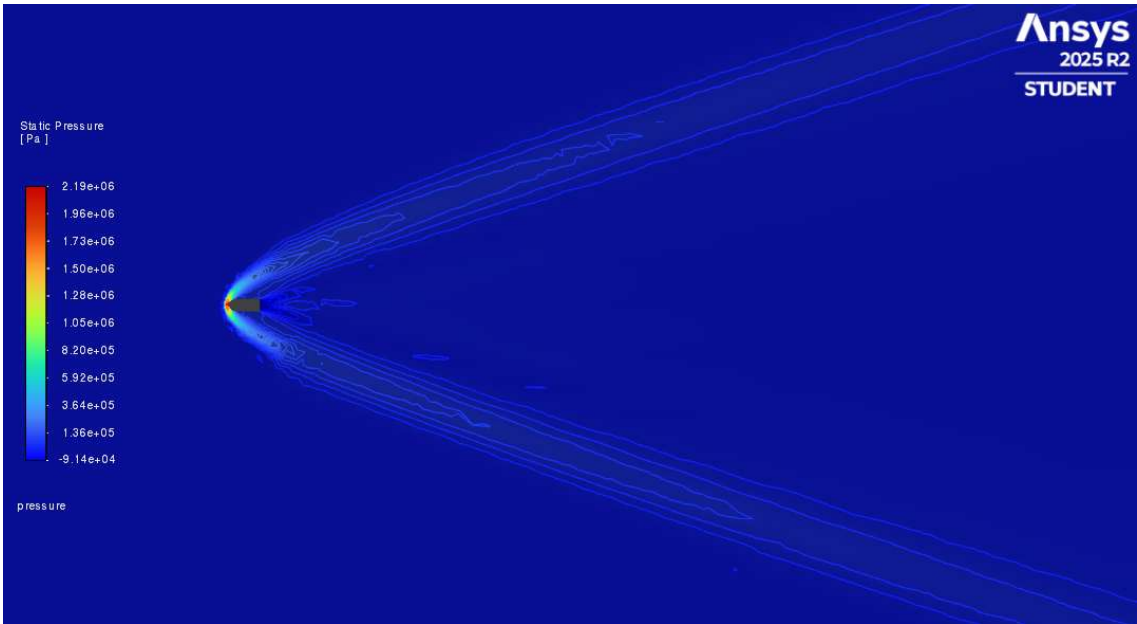
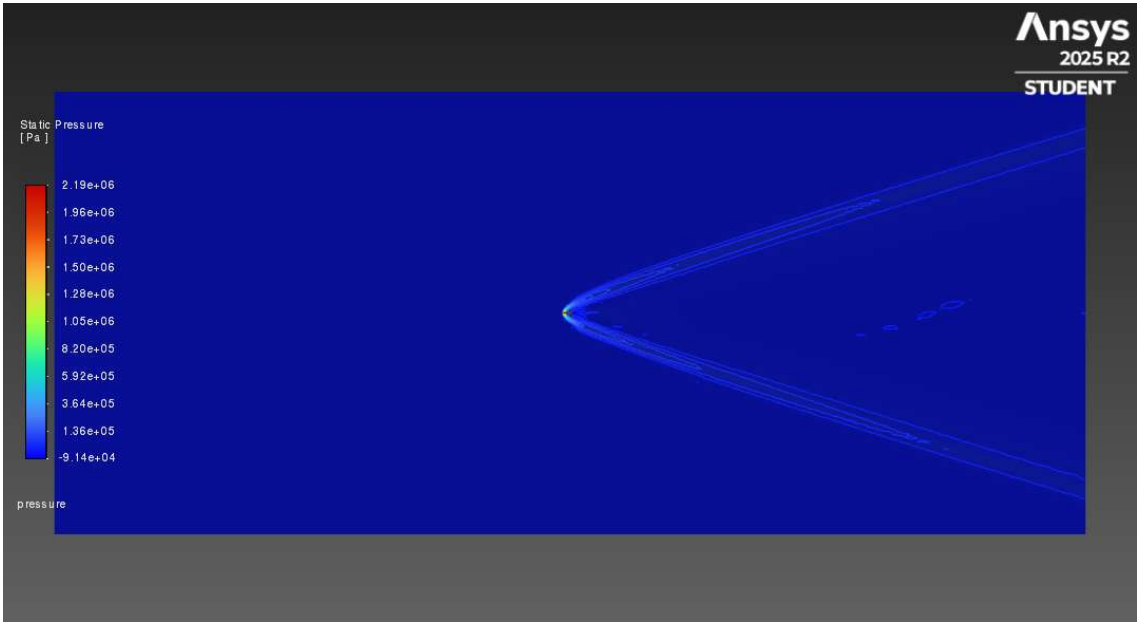
Stable Value:0.004



Mach Contours:



Pressure Contours:



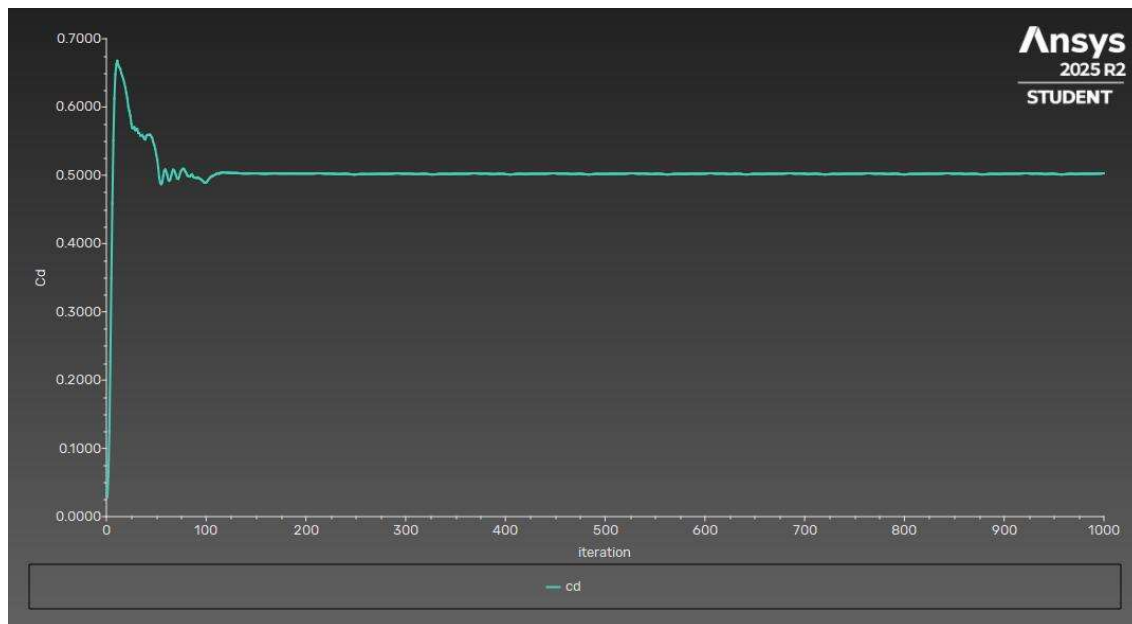
MACH 5:

Scaled Residual:



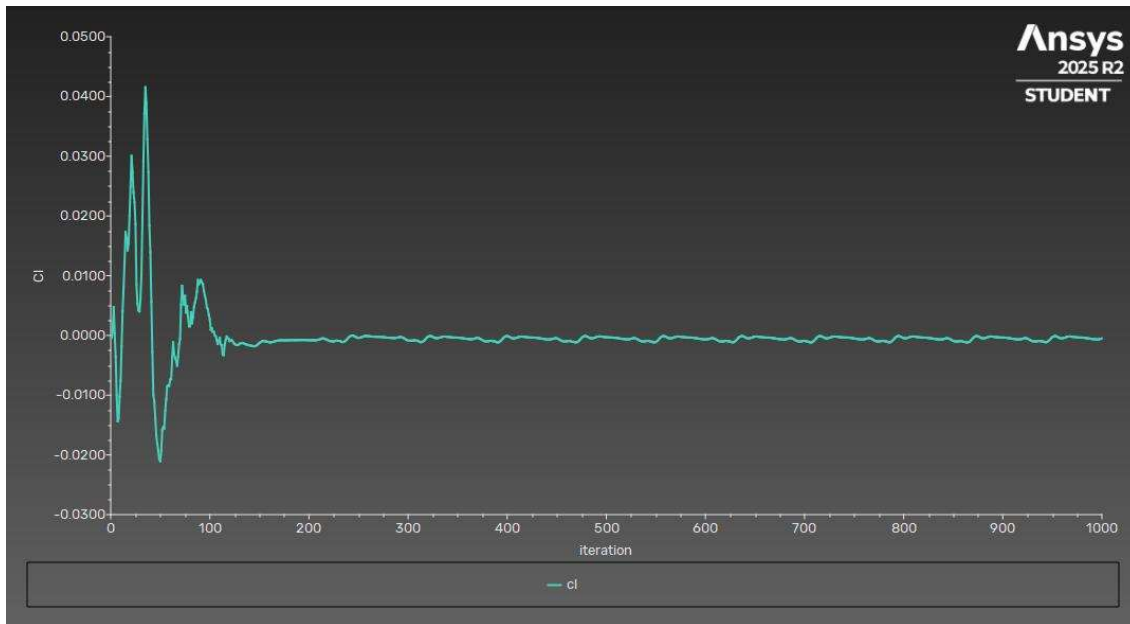
Coefficient of Drag:

Stable Value:0.497

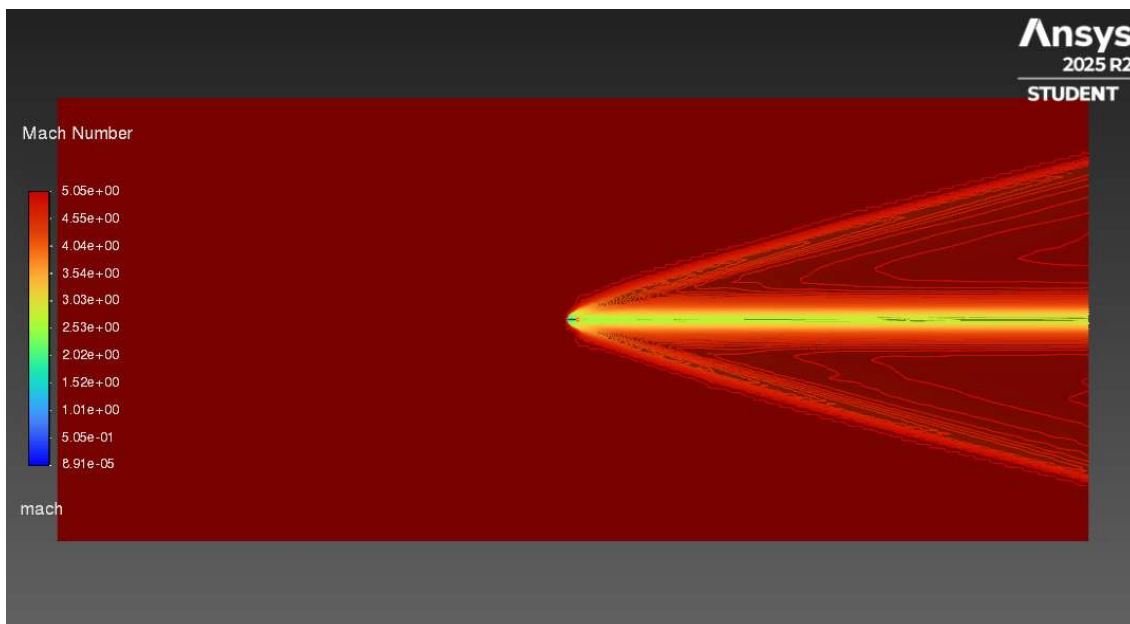


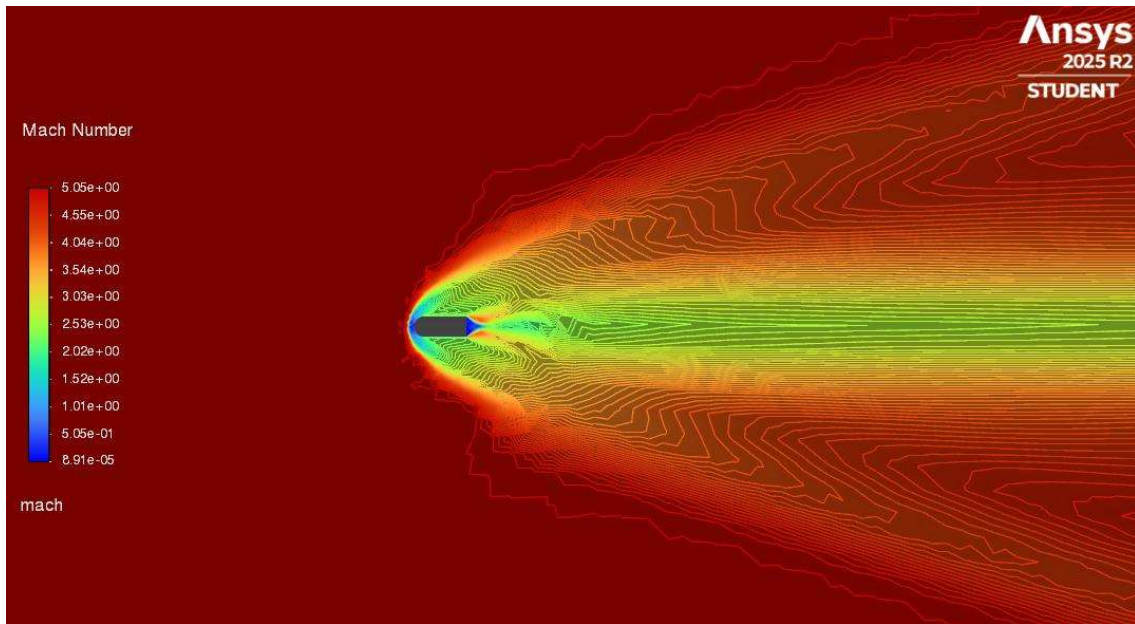
Coefficient of Lift:

Stable Value: -0.0005

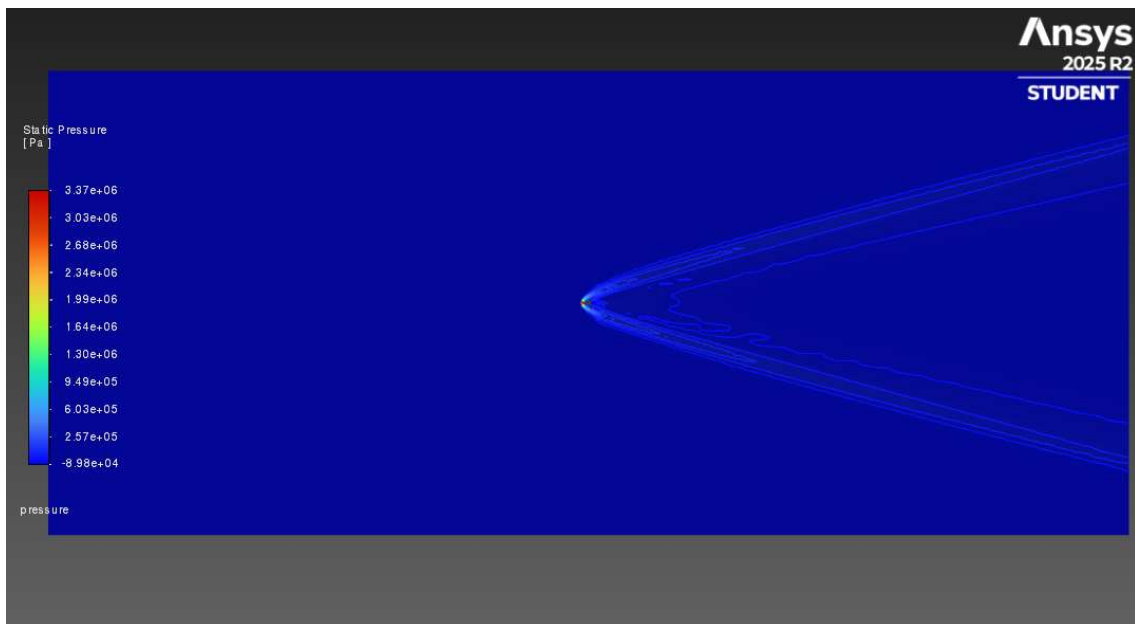


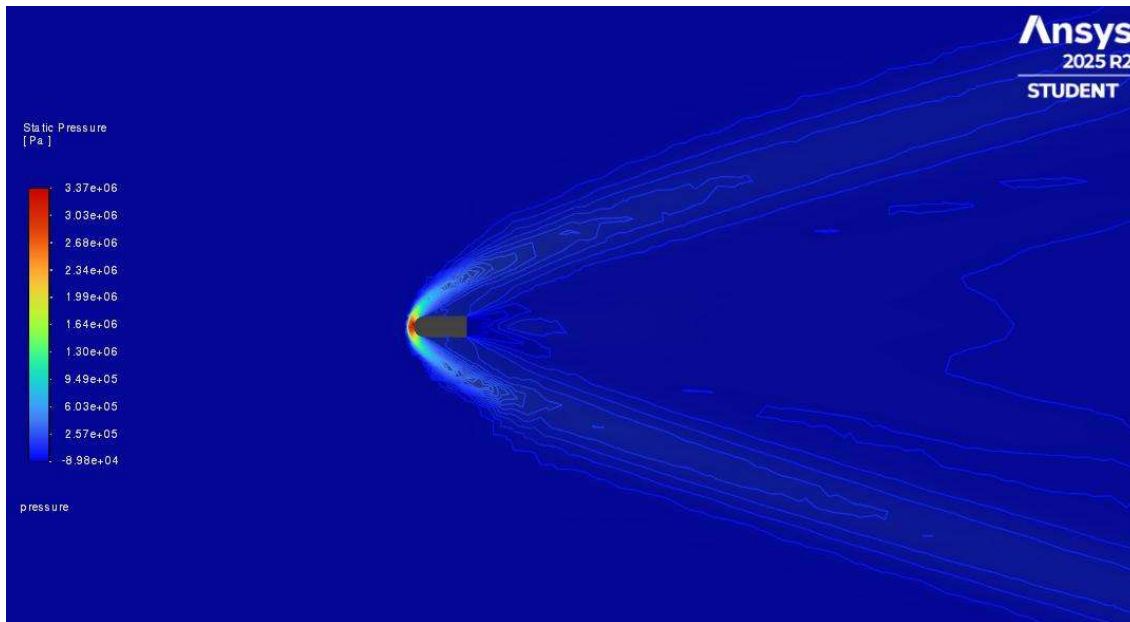
Mach Contours:





Pressure Contours:

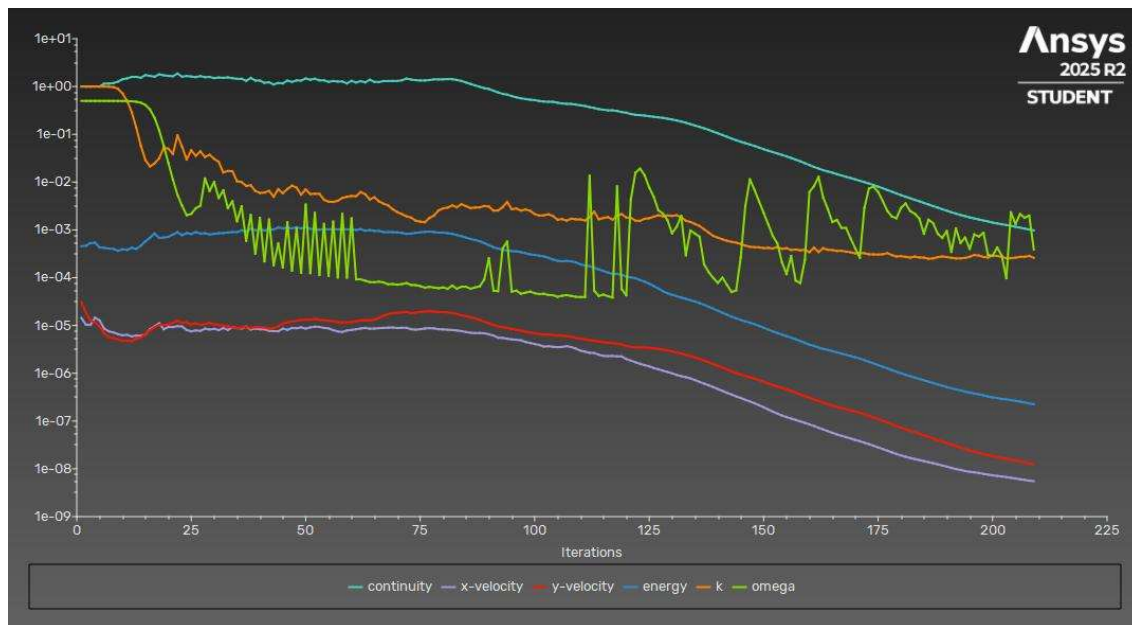




MACH 6:

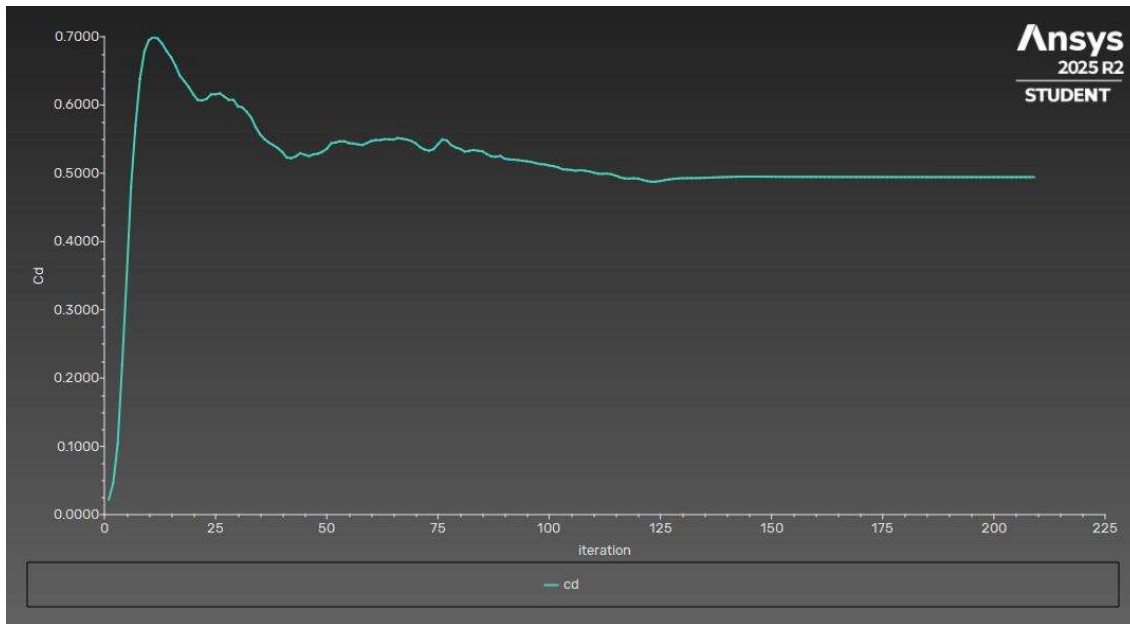
Iteration 209 solution has converged.

Scaled Residual:



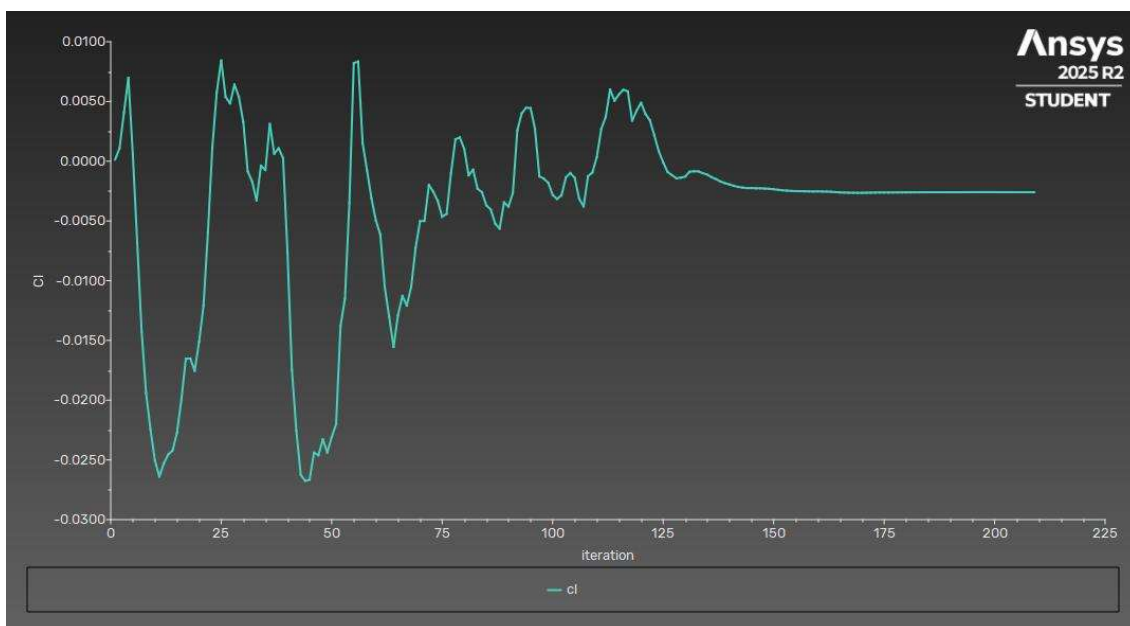
Coefficient of Drag:

Stable Value:0.485

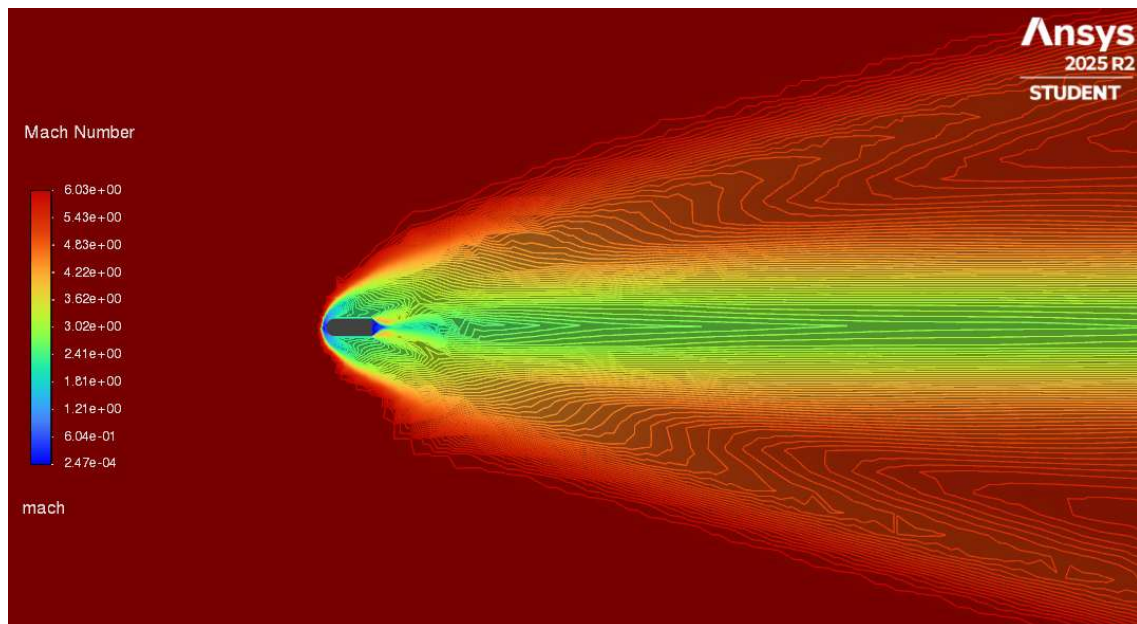
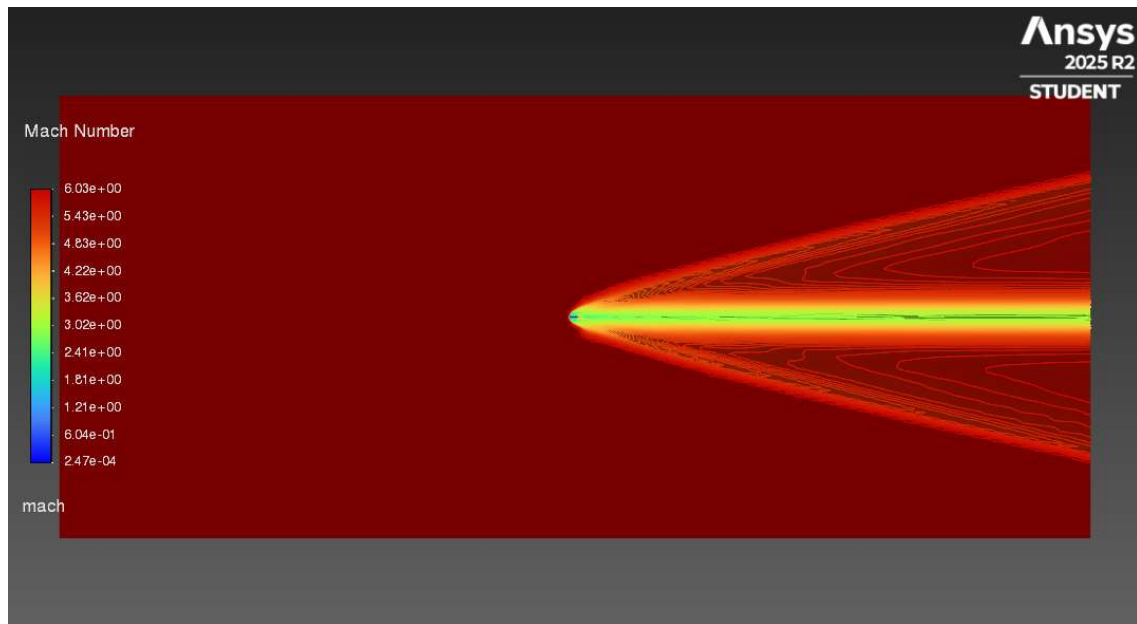


Coefficient of Lift:

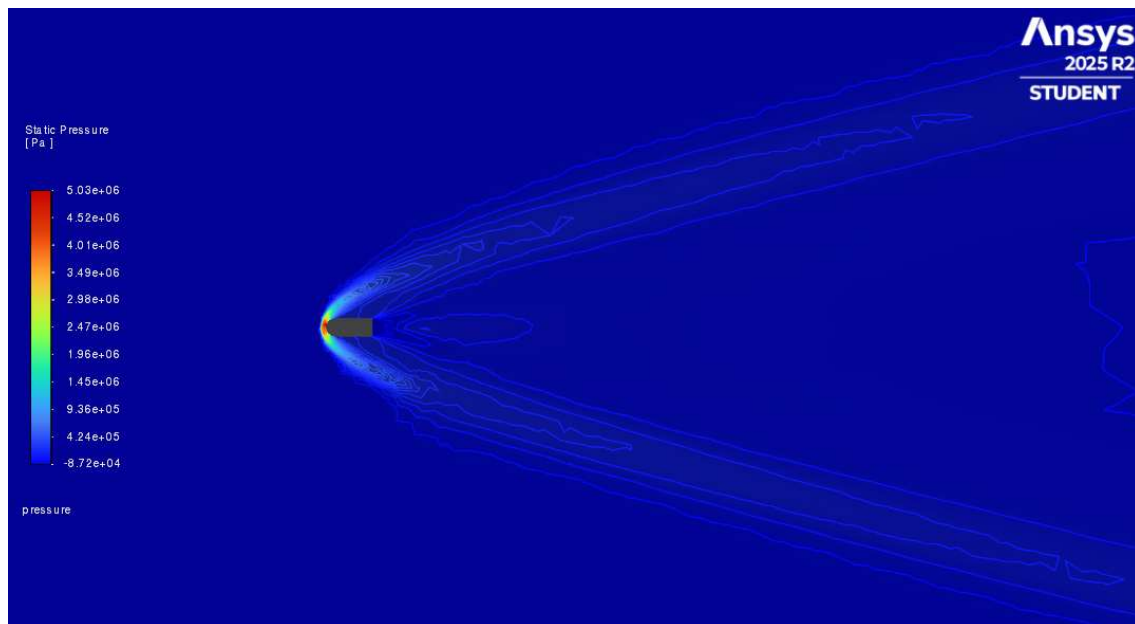
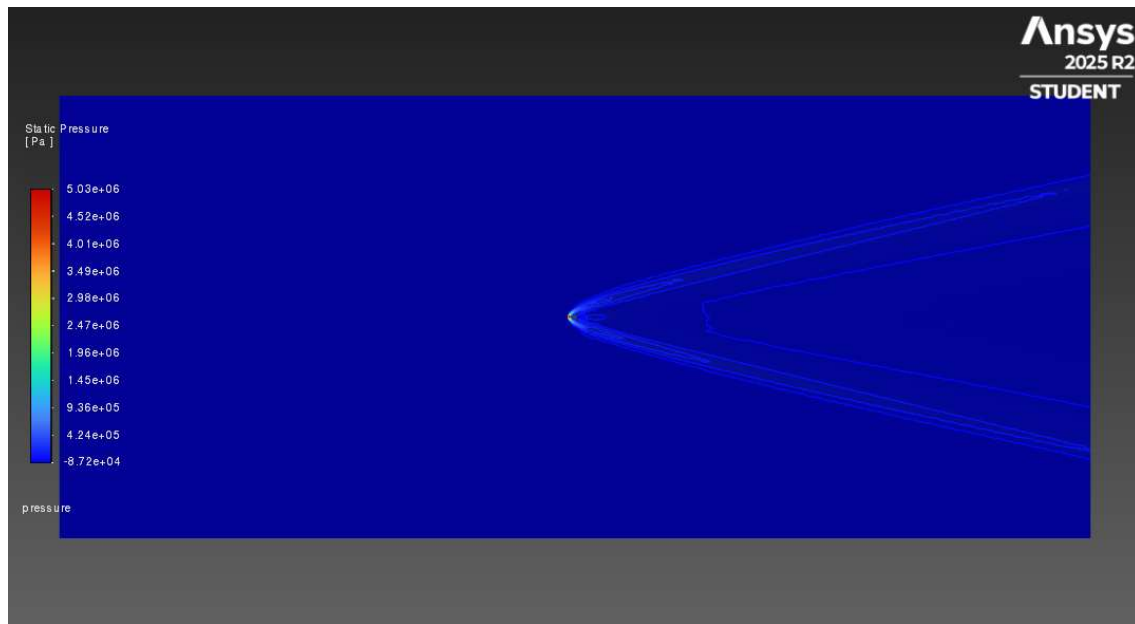
Stable Value: -0.002



Mach Contours:



Pressure Contours:



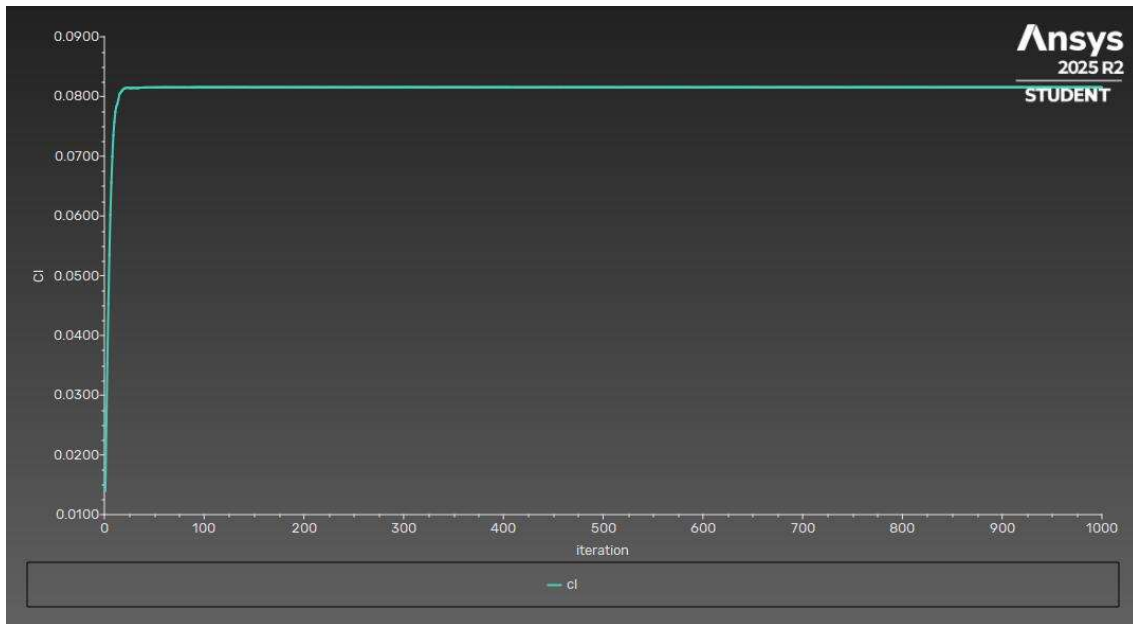
Question C

DOUBLE-WEDGE ANGLE TESTS:

5 Degree:

Coefficient of Lift:

Stability: 0.0815



Coefficient of Drag:

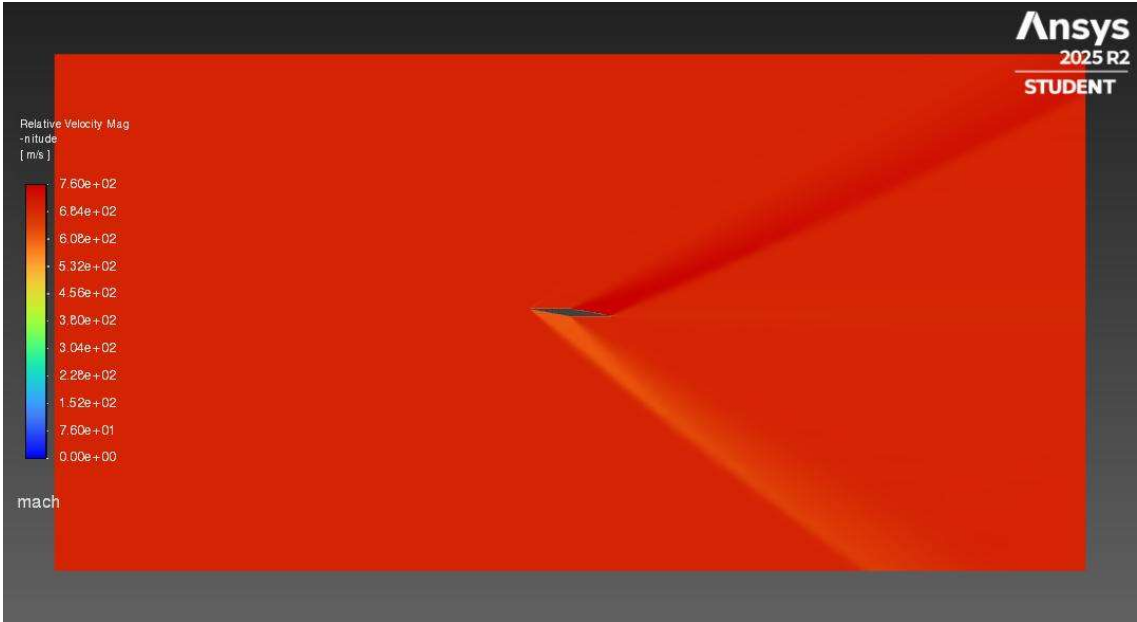
Stability: 0.015



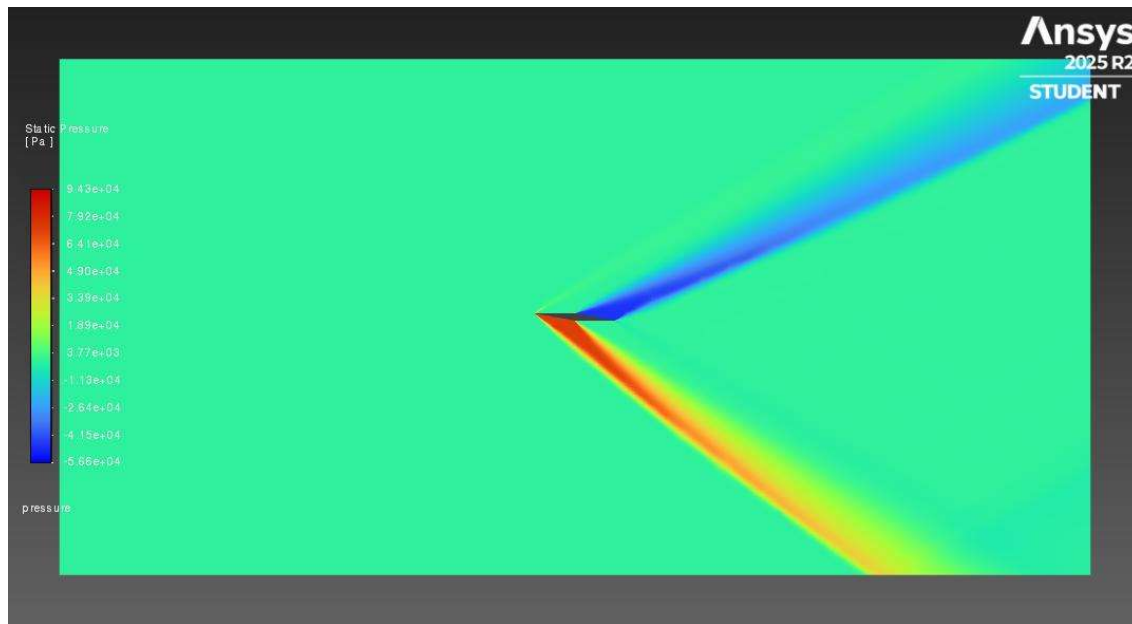
Scaled residuals:



Mach:



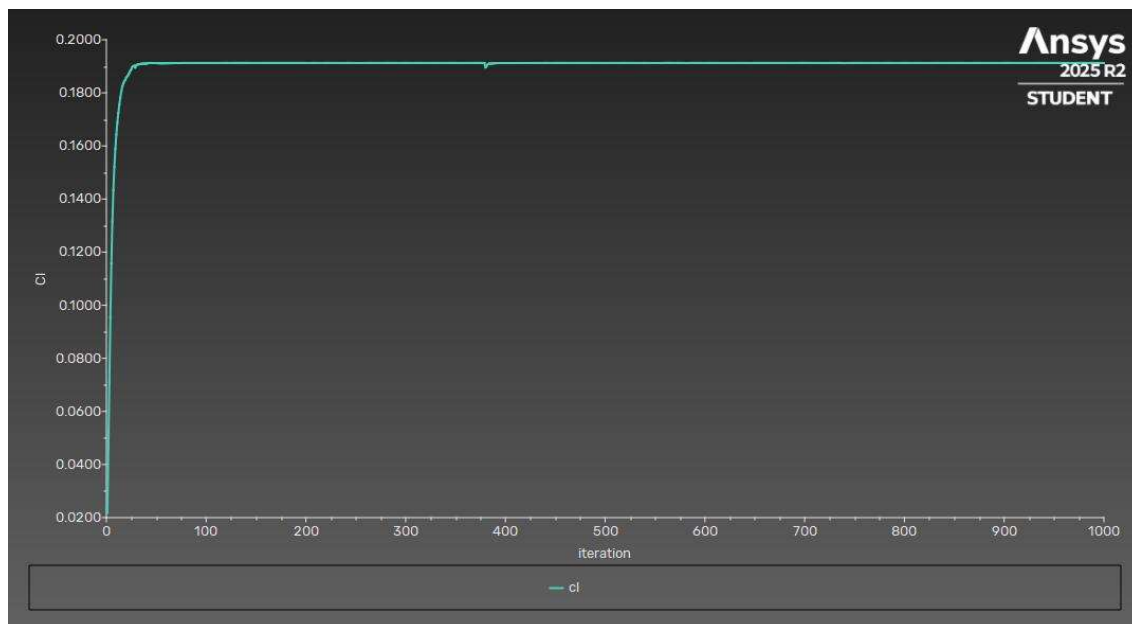
Pressure:



10 Degree:

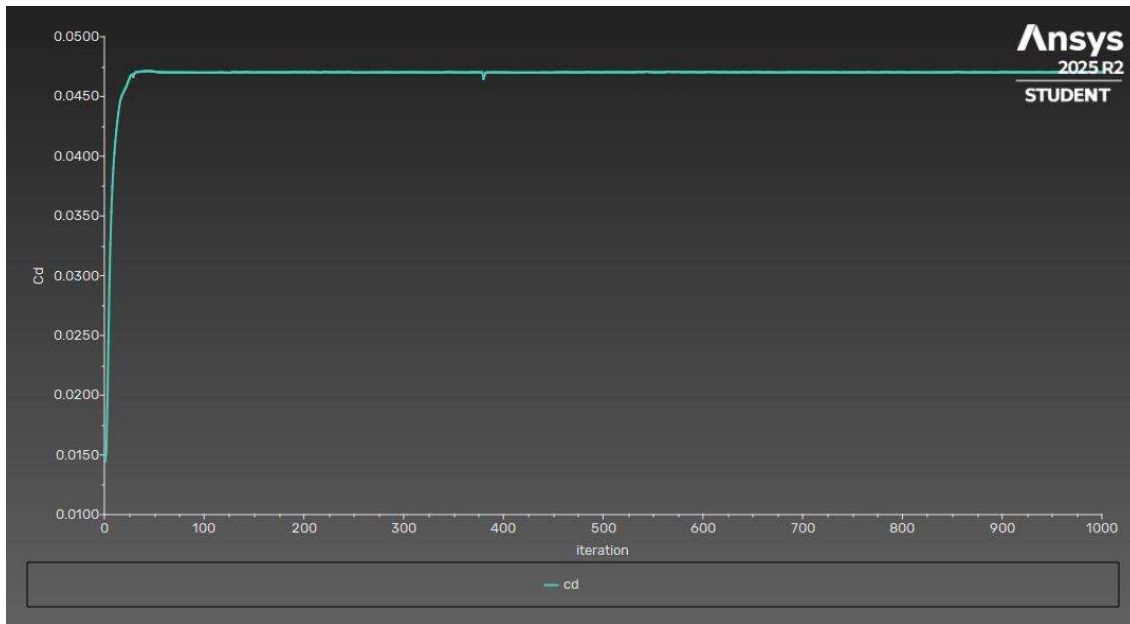
Coefficient of Lift:

Stability: 0.191



Coefficient of Drag:

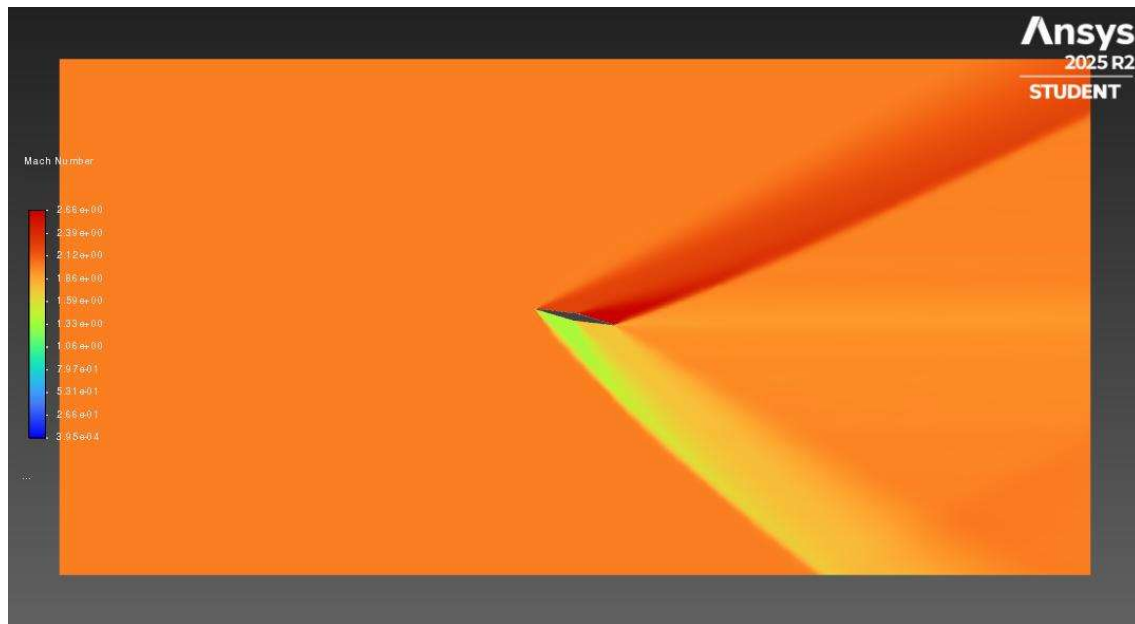
Stability: 0.047



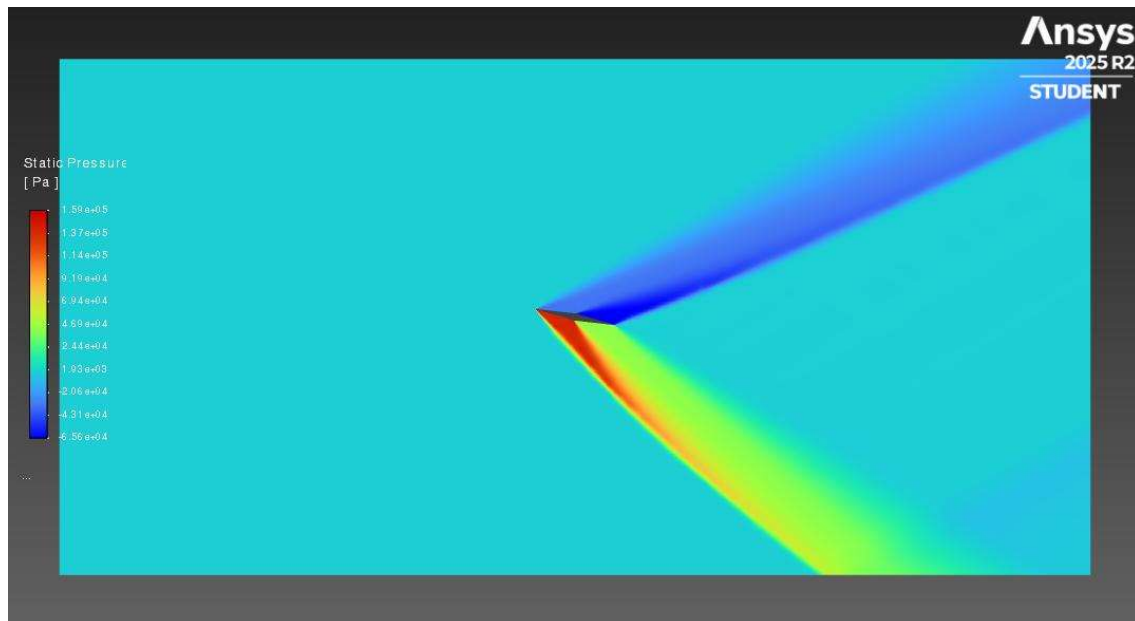
Scaled residuals:



Mach:



Pressure:



15 Degree:

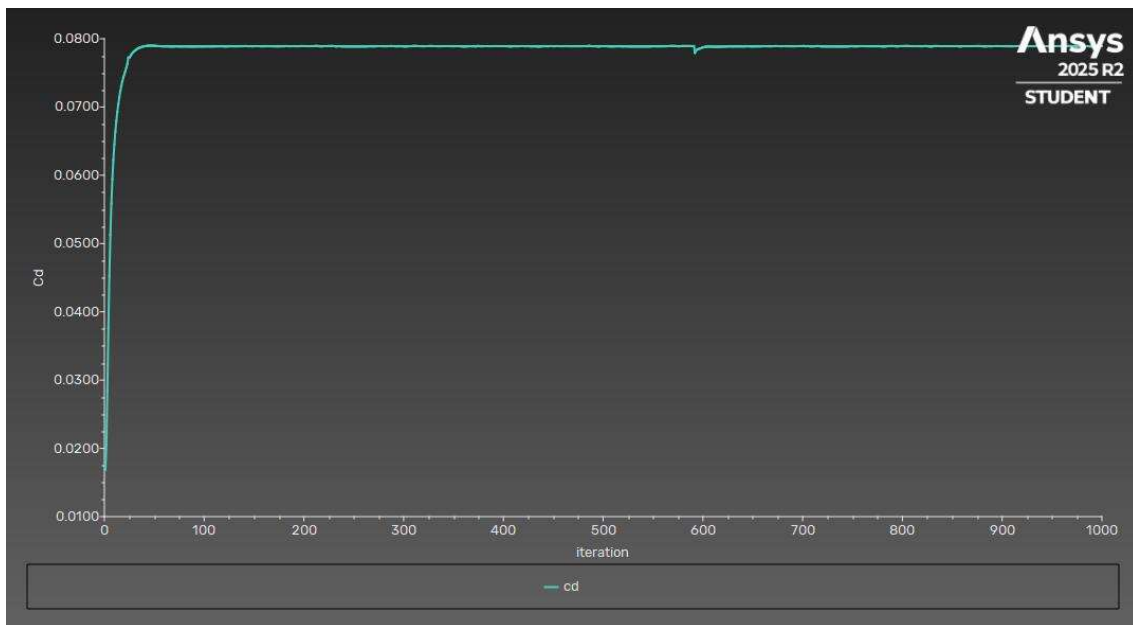
Coefficient of Lift:

Stability: 0.259



Coefficient of Drag:

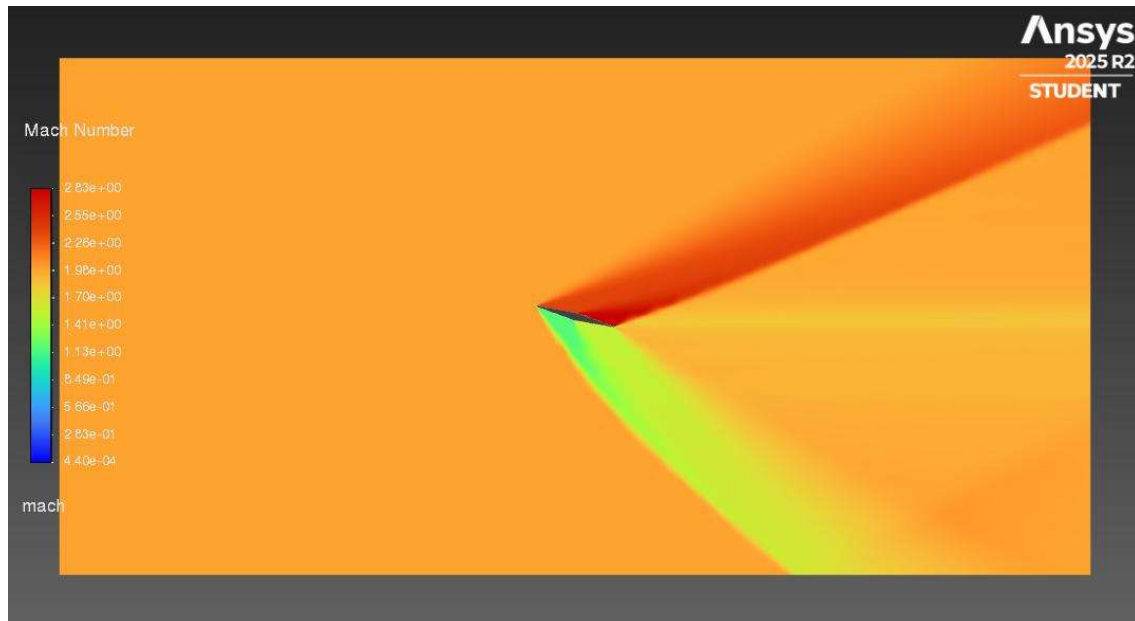
Stability:0.0789



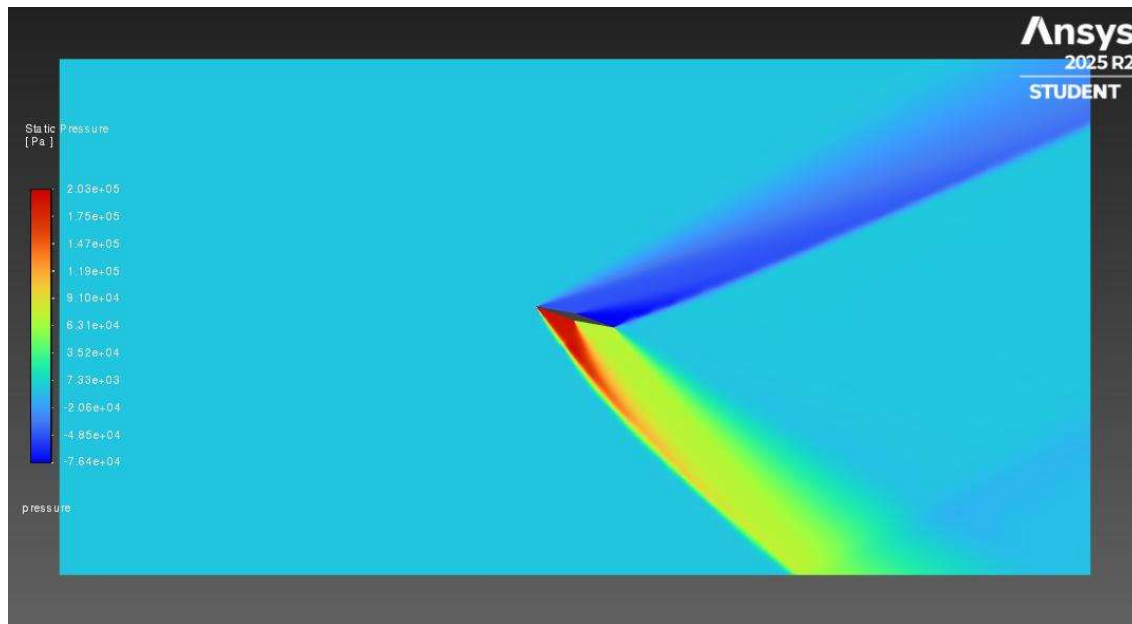
Scaled residuals:



Mach:



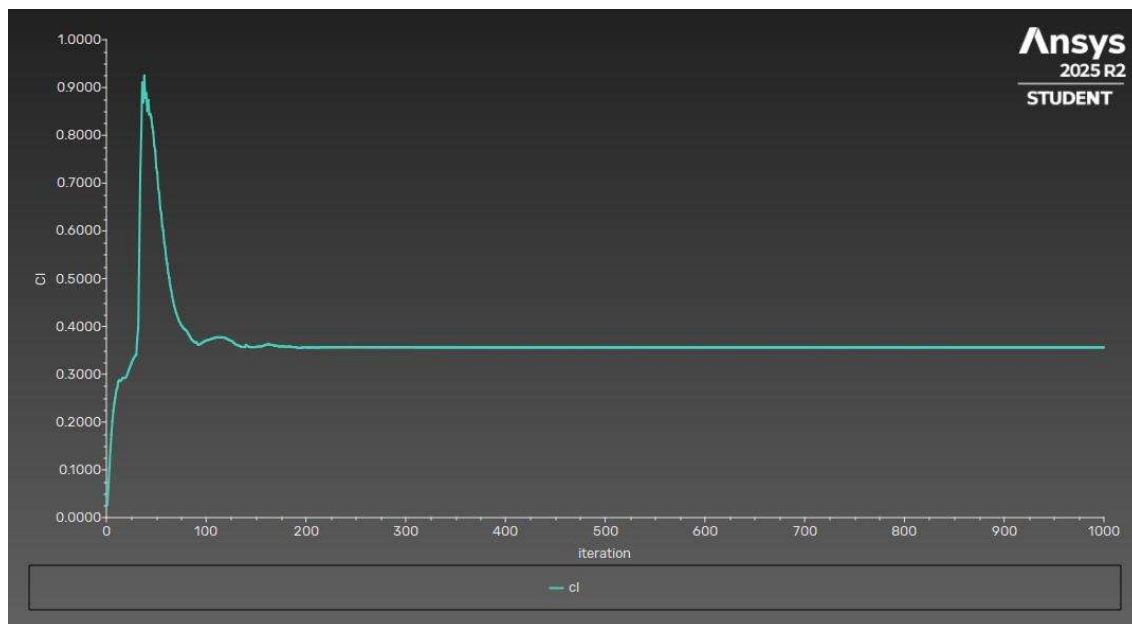
Pressure:



20 Degree:

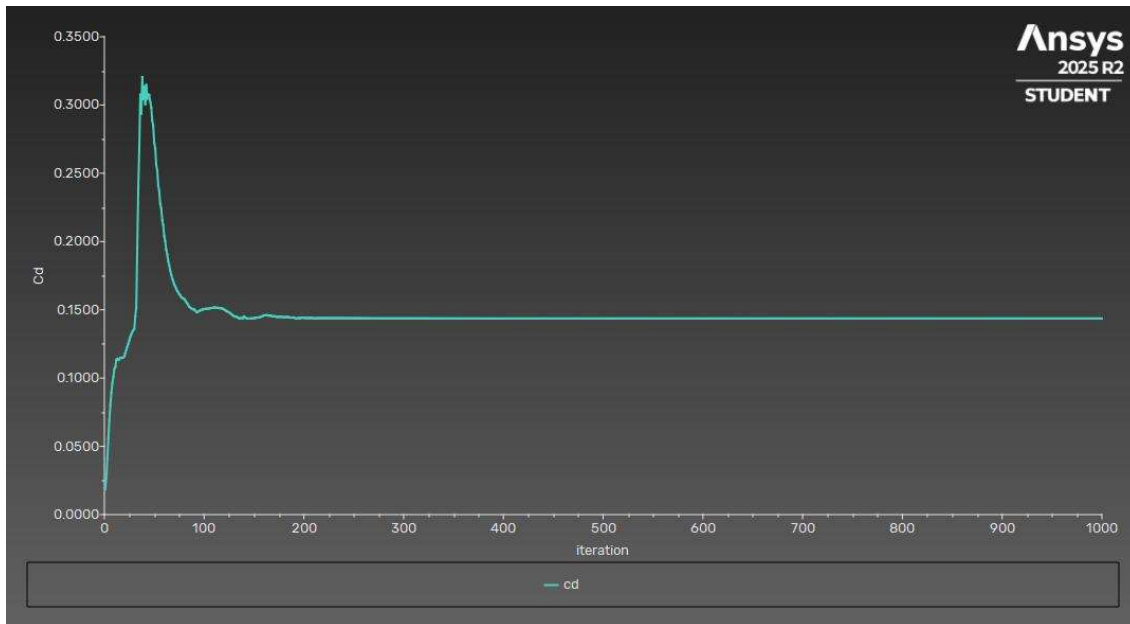
Coefficient of Lift:

Stability:0.356

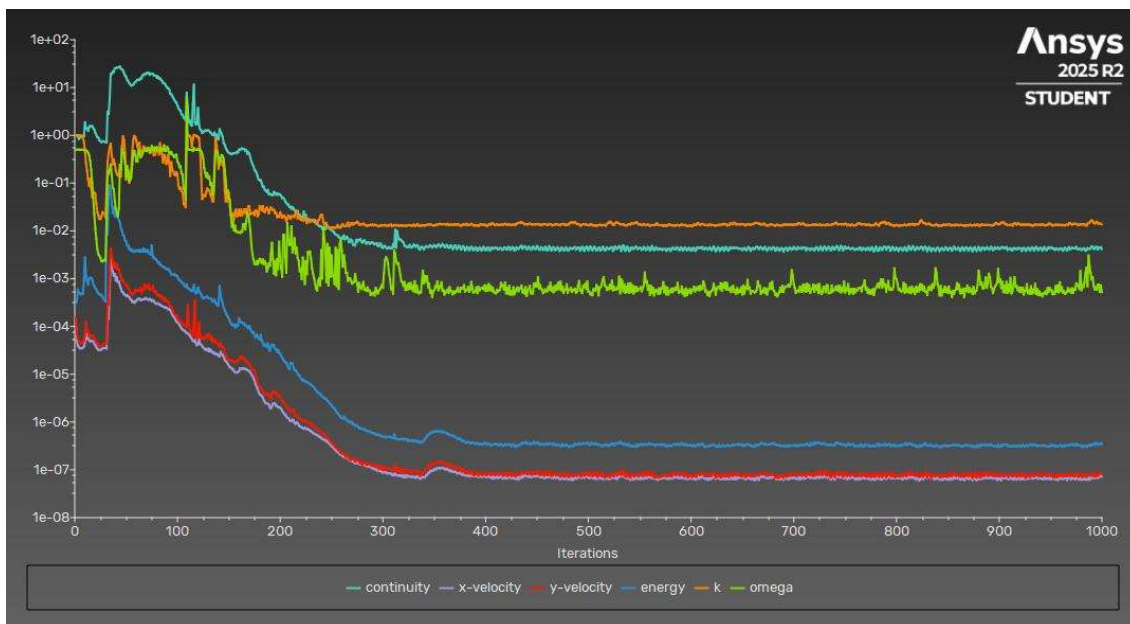


Coefficient of Drag:

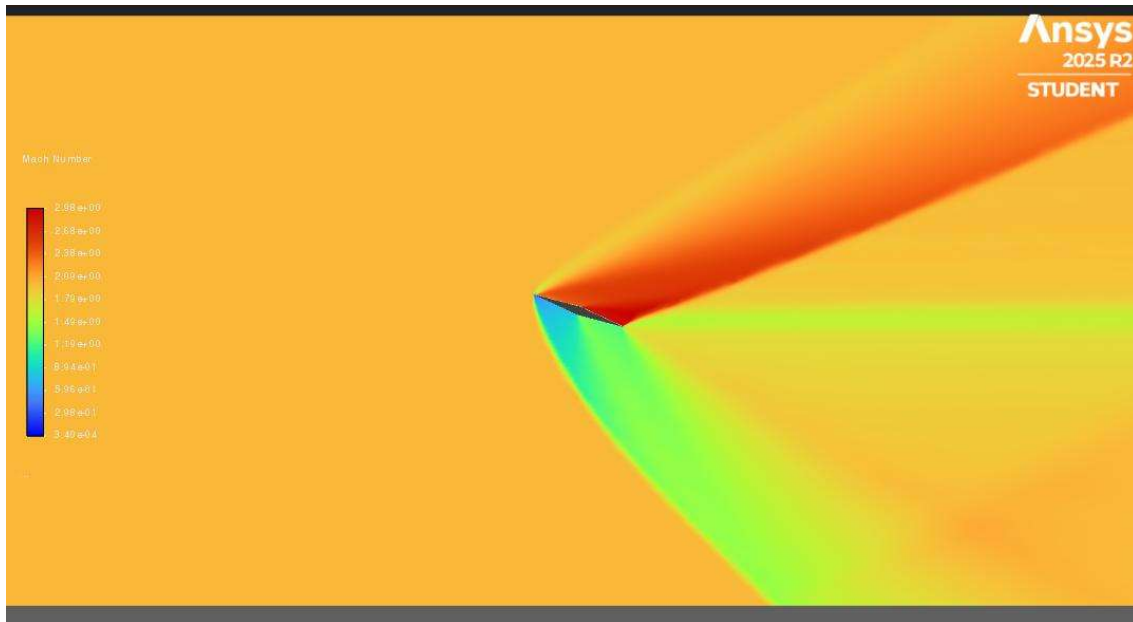
Stability:0.143



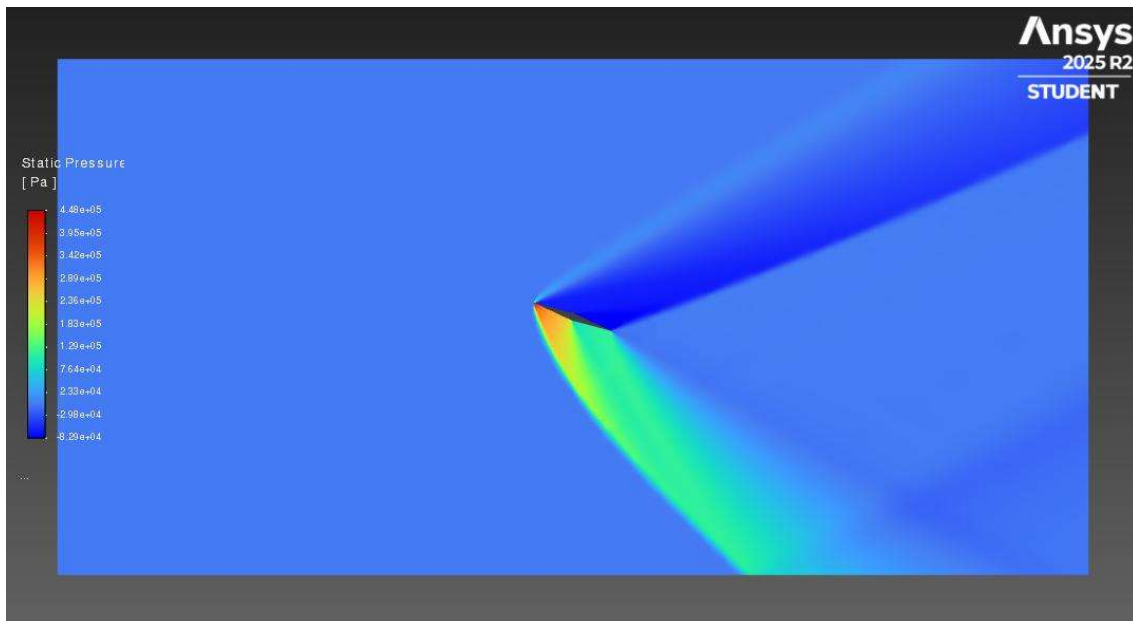
Scaled residuals:



Mach:



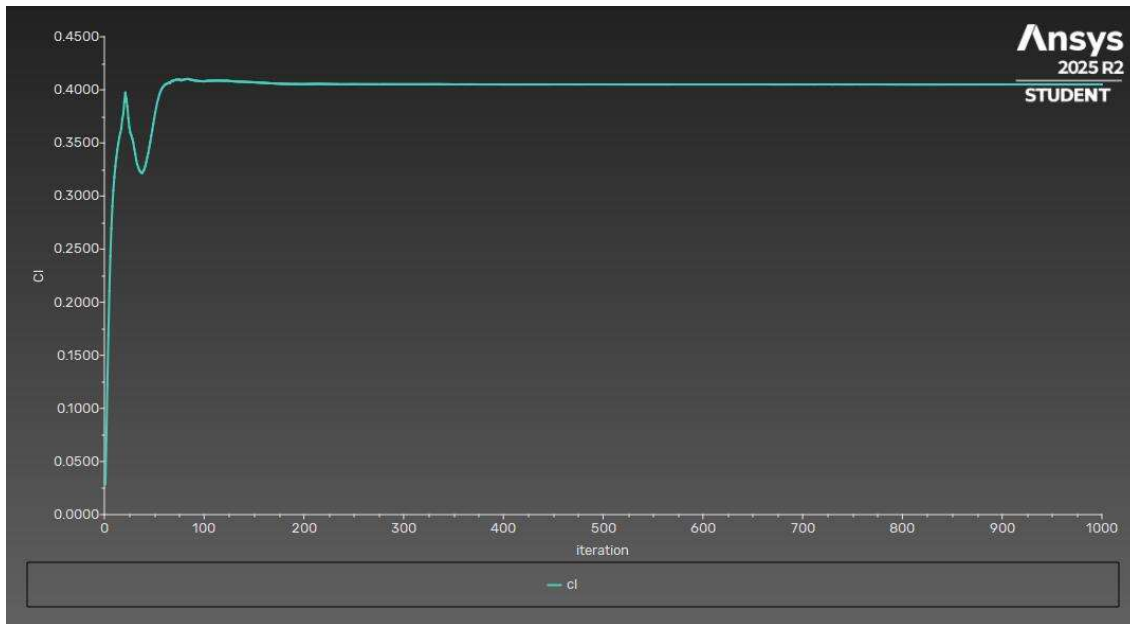
Pressure:



25 Degree:

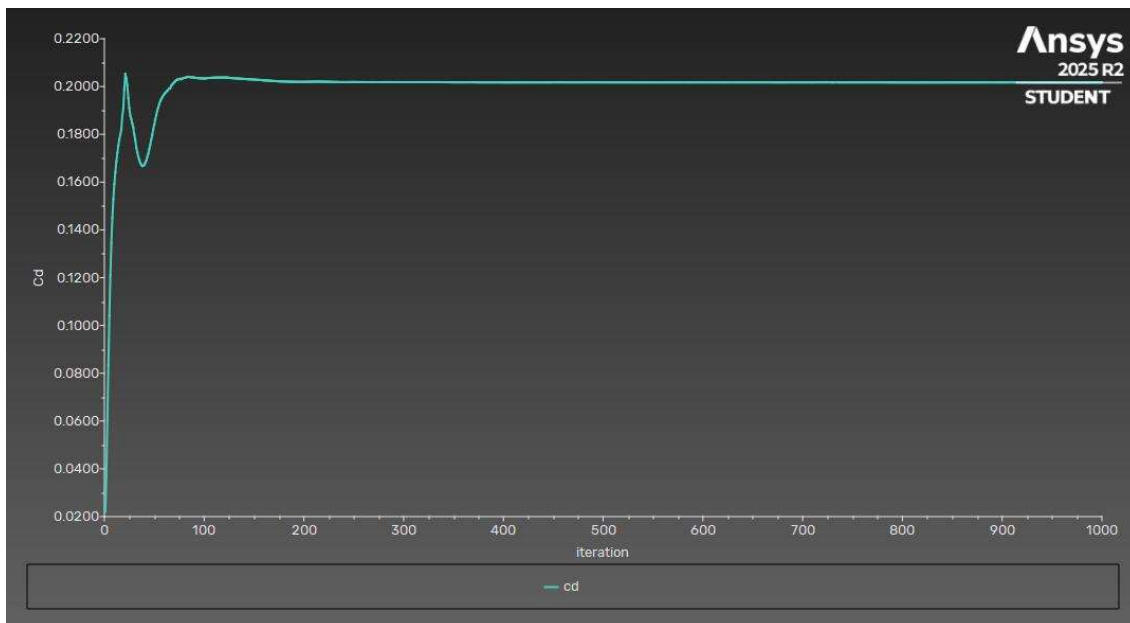
Coefficient of Lift:

Stability:0.405



Coefficient of Drag:

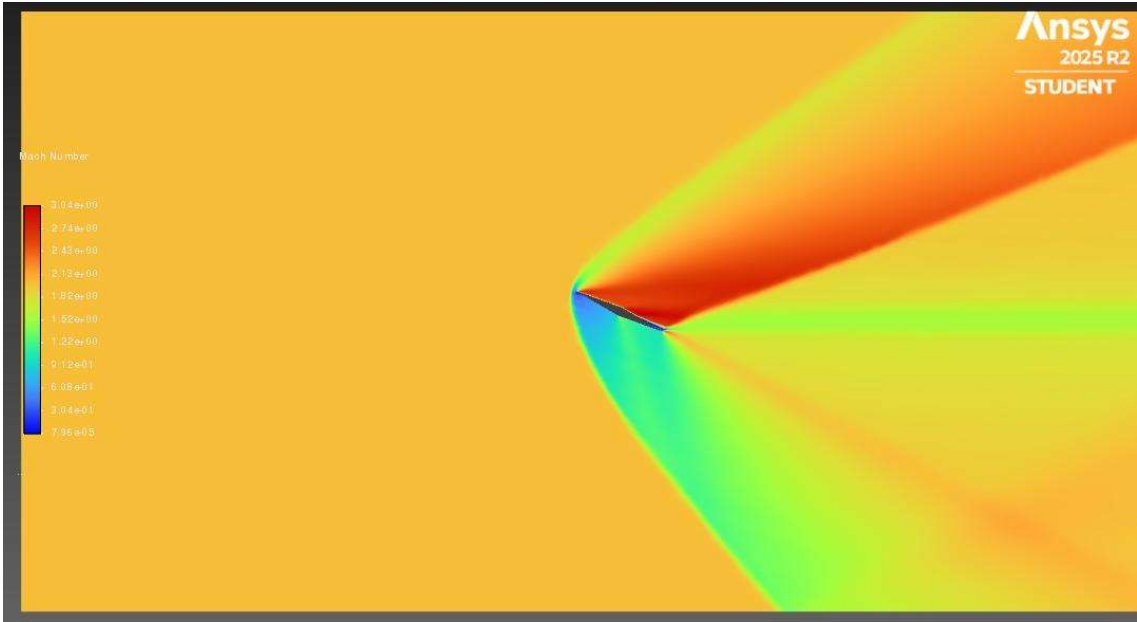
Stability: 0.201



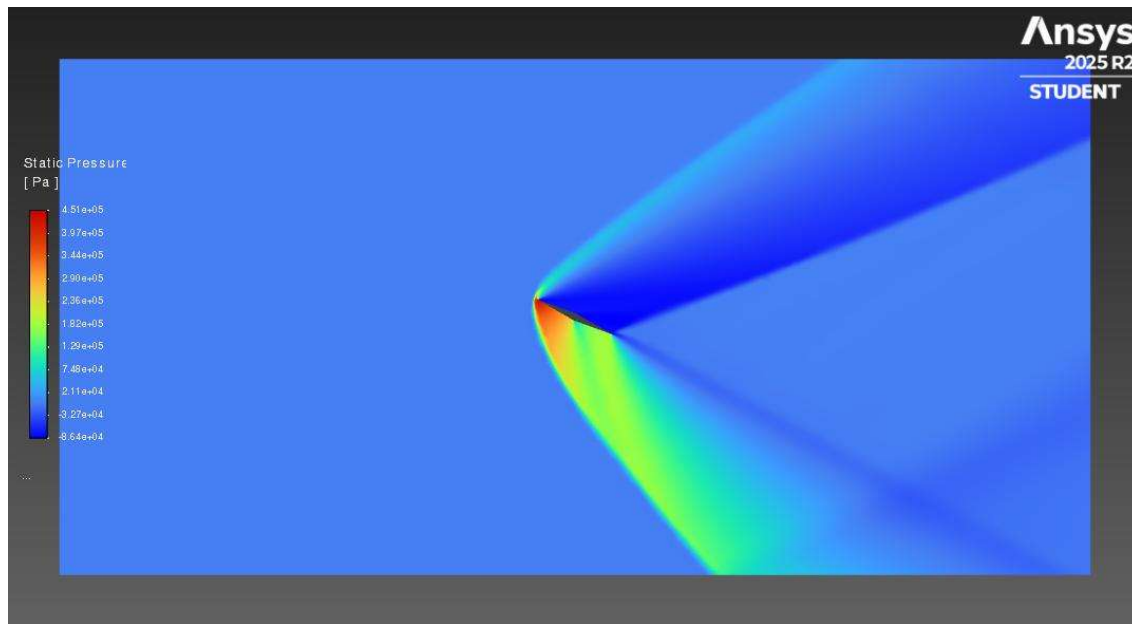
Scaled residuals:



Mach:



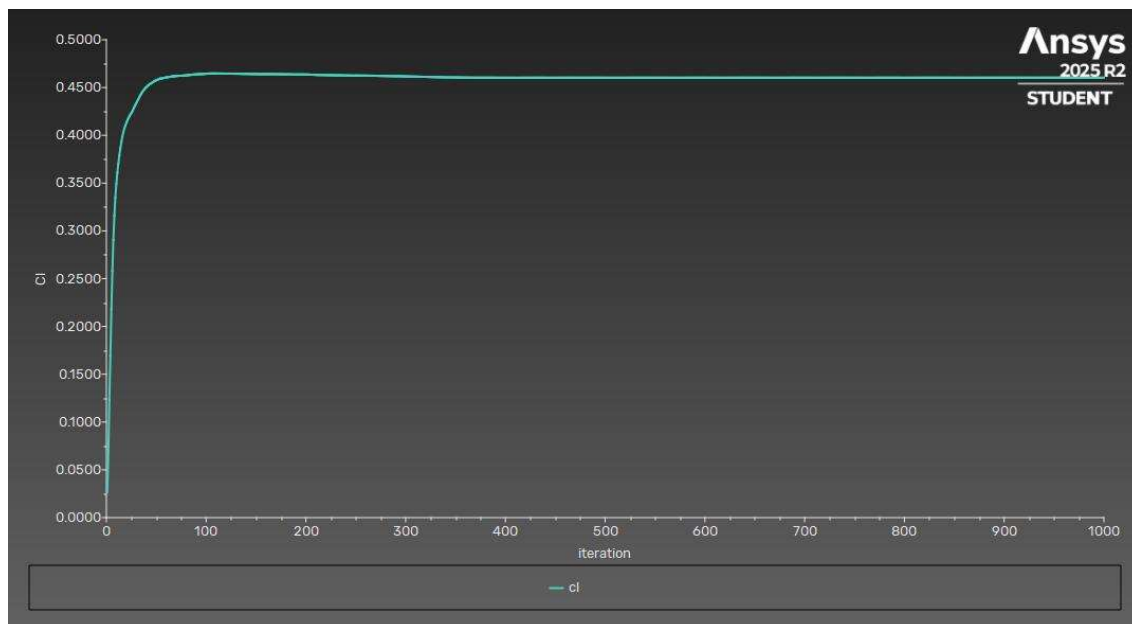
Pressure:



30 Degree:

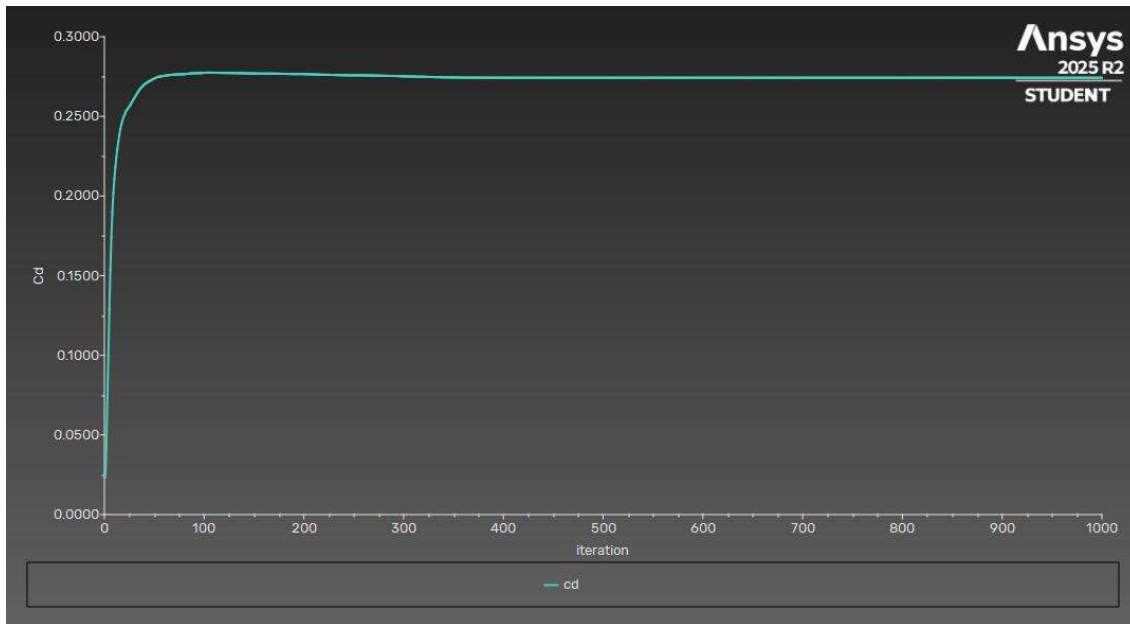
Coefficient of Lift:

Stability:0.46



Coefficient of Drag:

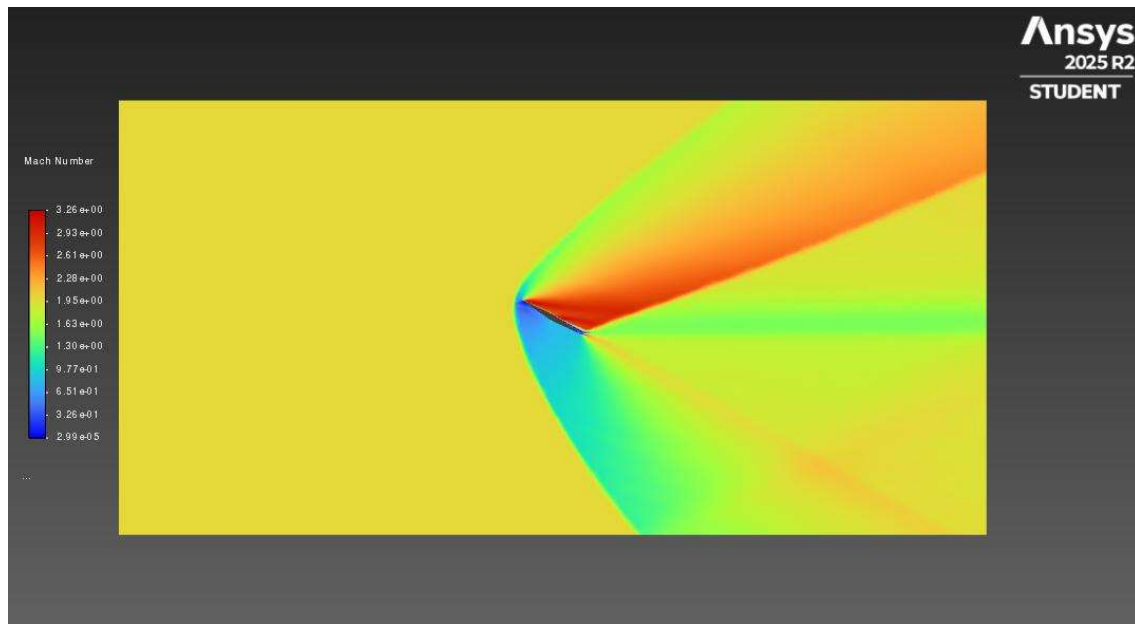
Stability:0.274



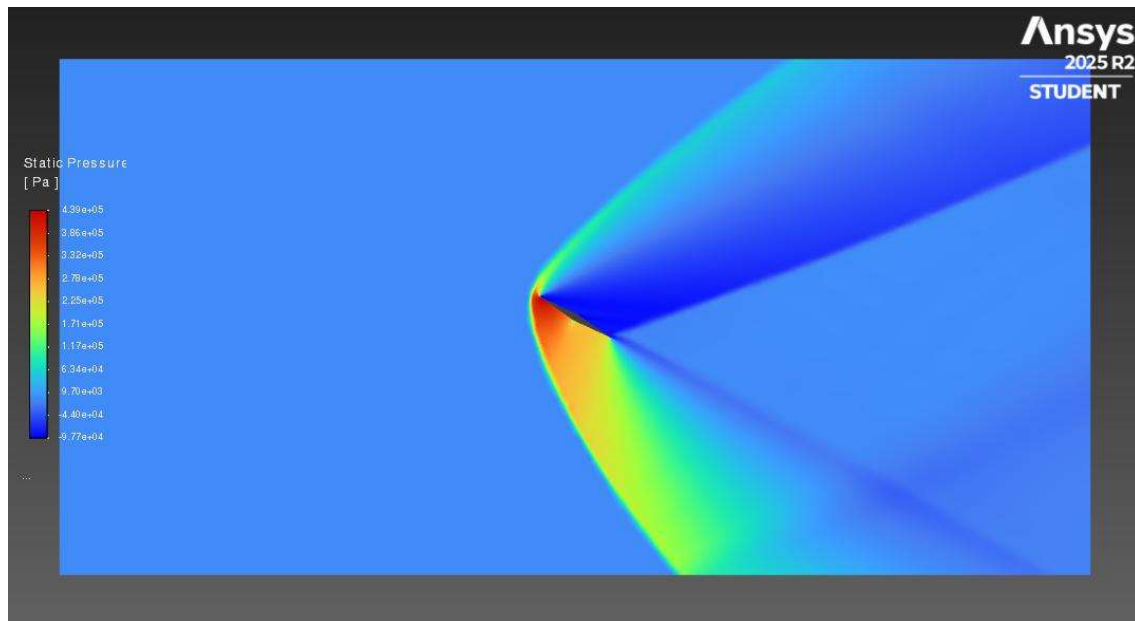
Scaled residuals:



Mach:



Pressure:

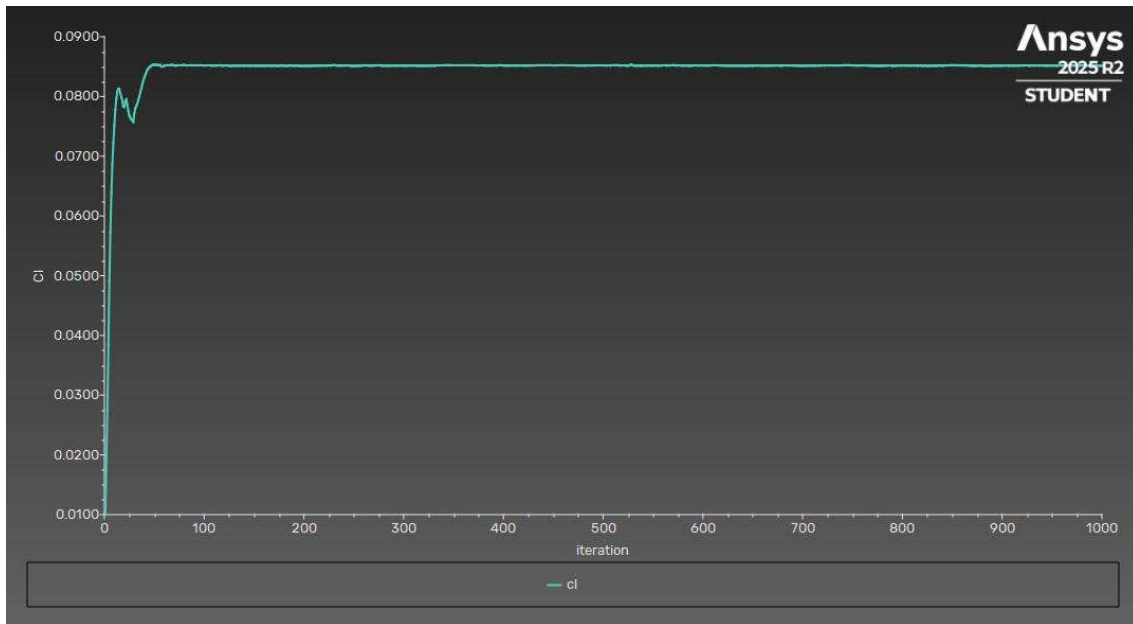


BI-CONVEX ANGLE TESTS:

5 Degree:

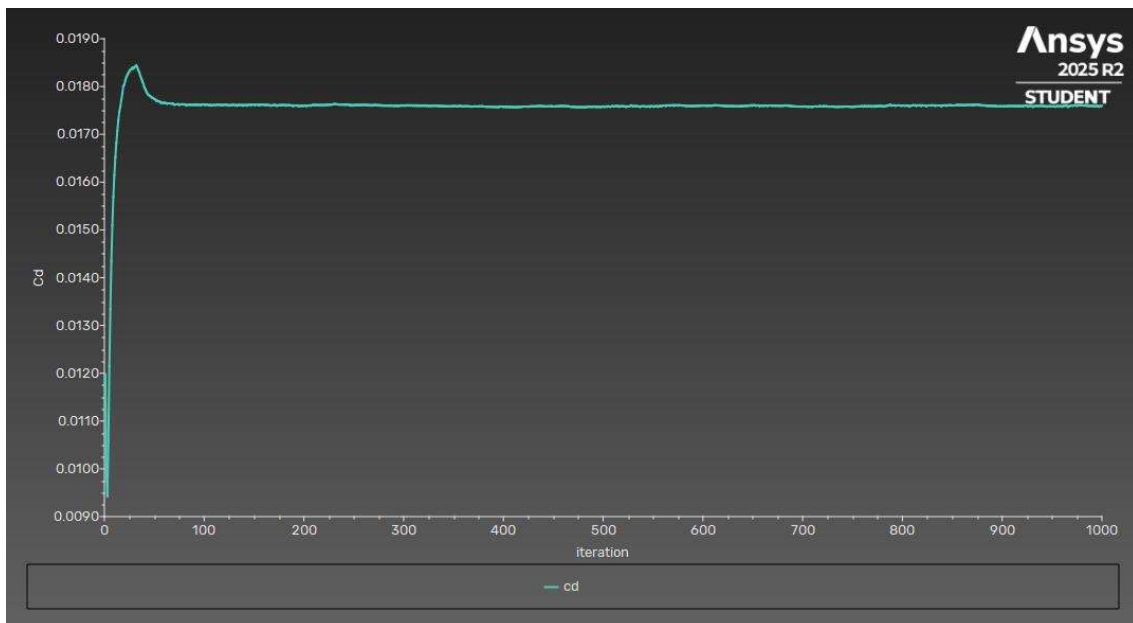
Coefficient of Lift:

Stability: 0.0852

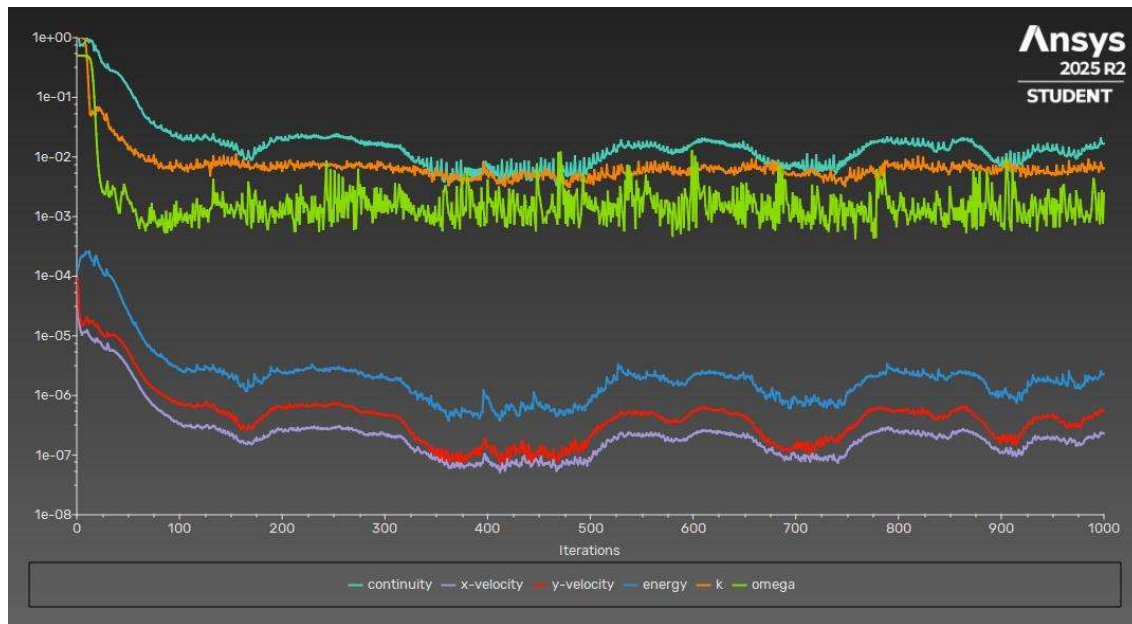


Coefficient of Drag:

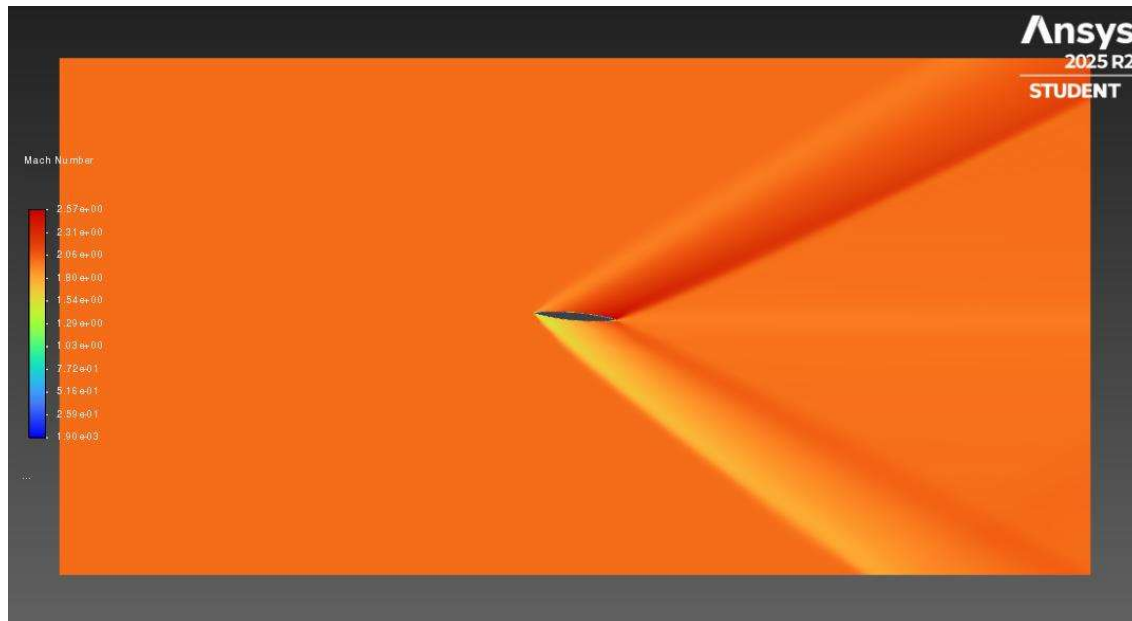
Stability: 0.0175



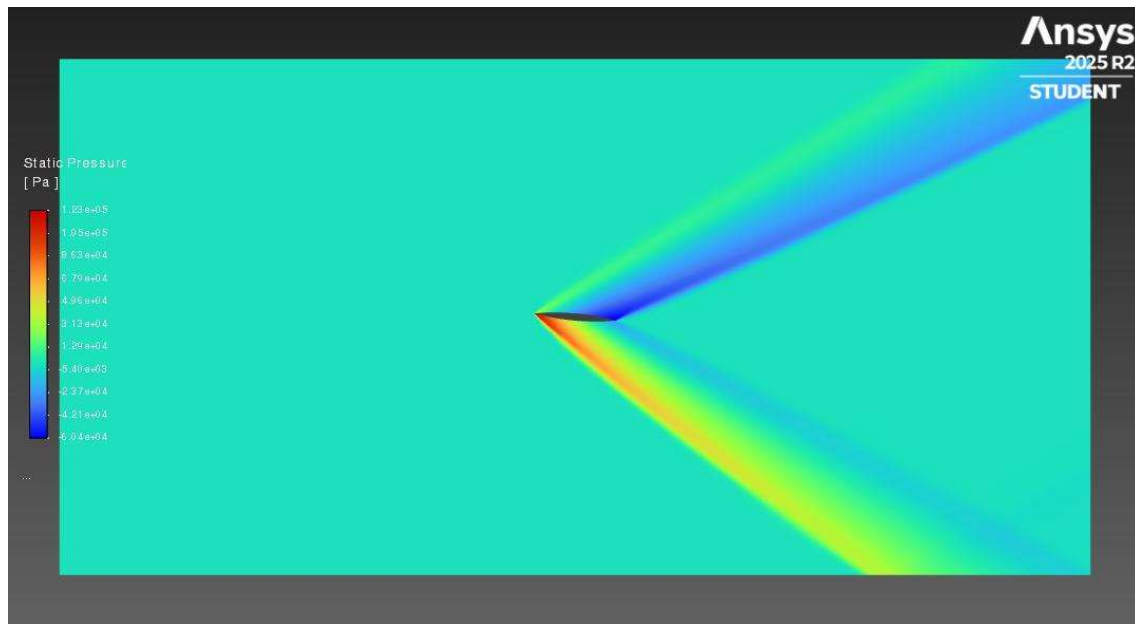
Scaled residuals:



Mach:



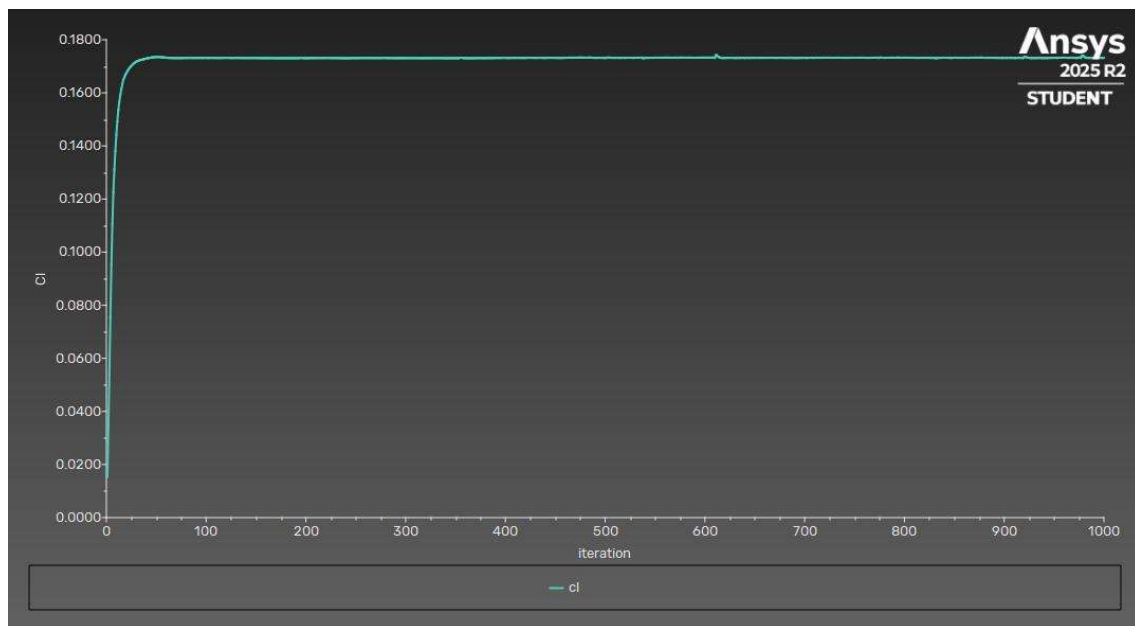
Pressure:



10 Degree:

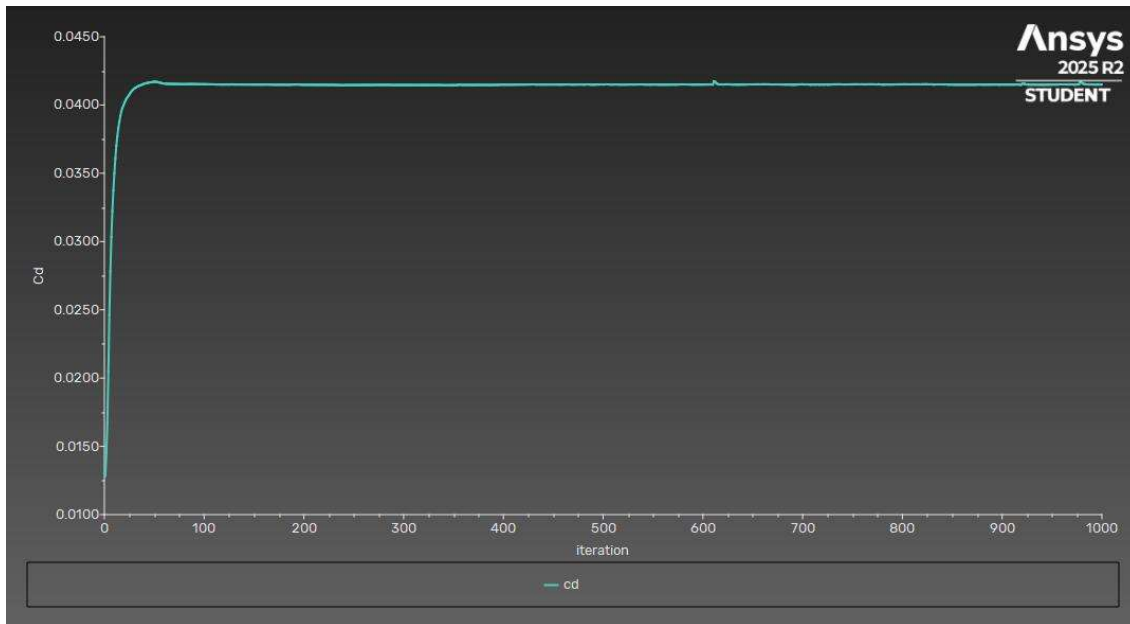
Coefficient of Lift:

Stability: 0.173

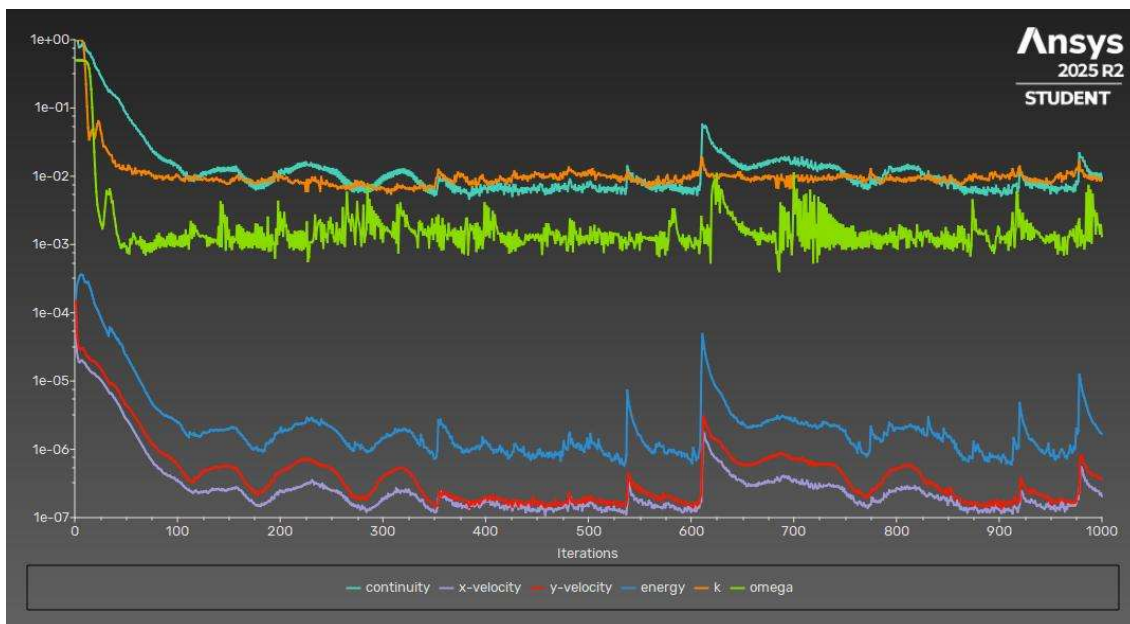


Coefficient of Drag:

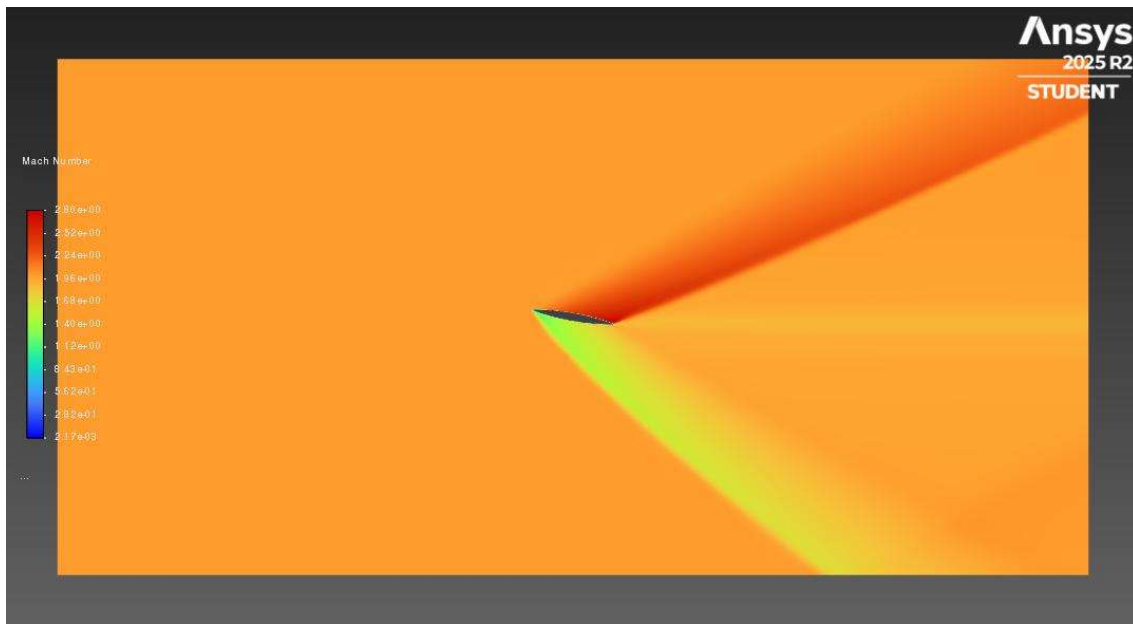
Stability: 0.0414



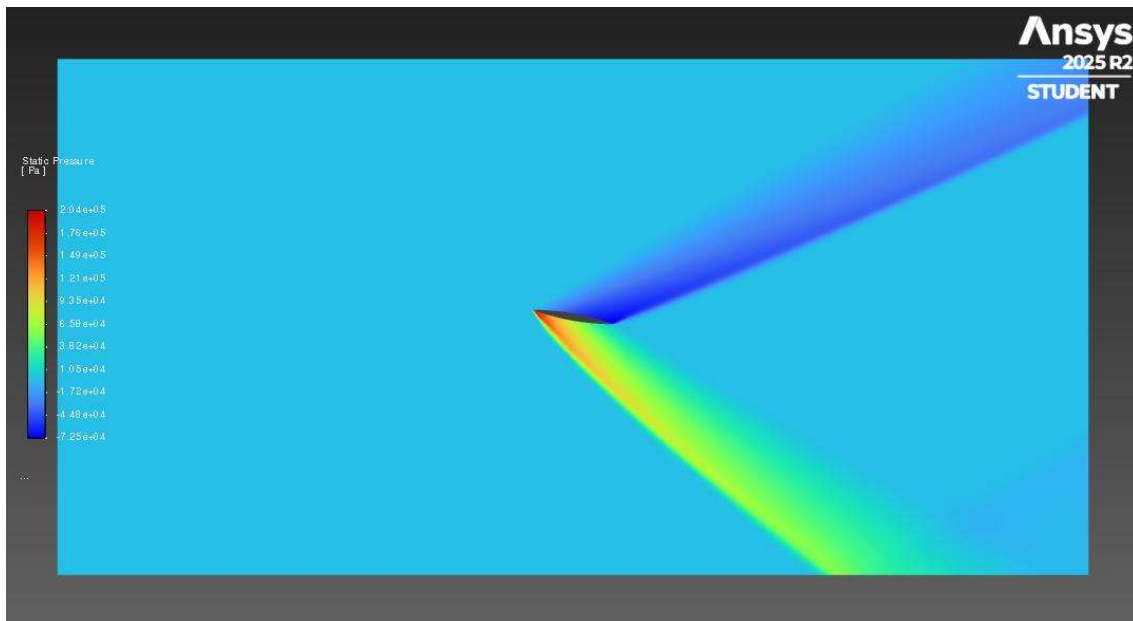
Scaled residuals:



Mach:



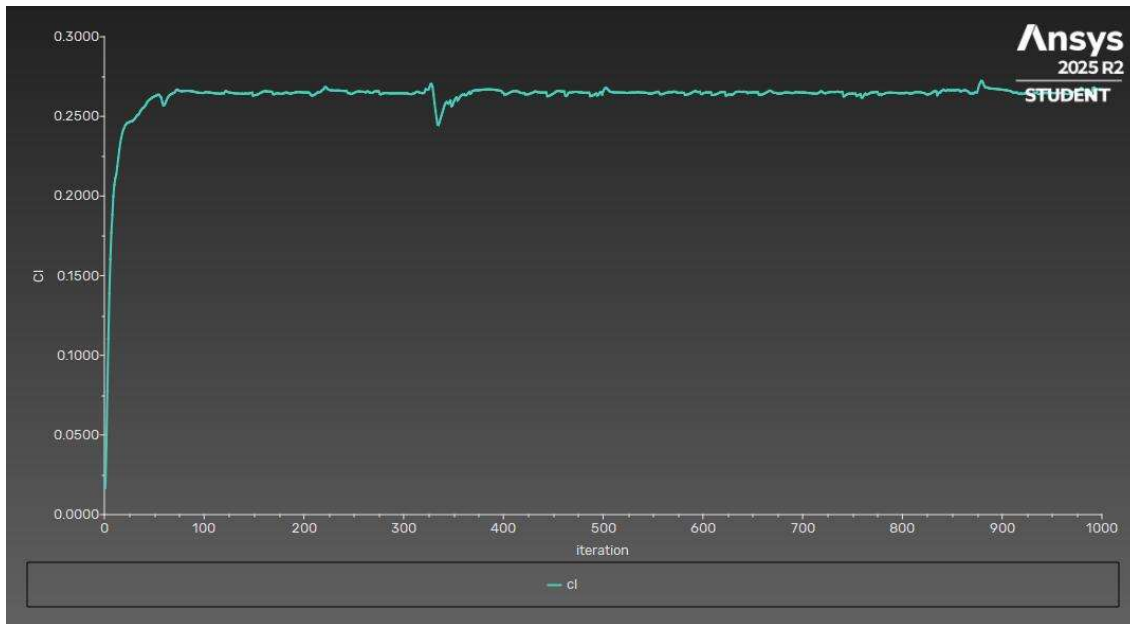
Pressure:



15 Degree:

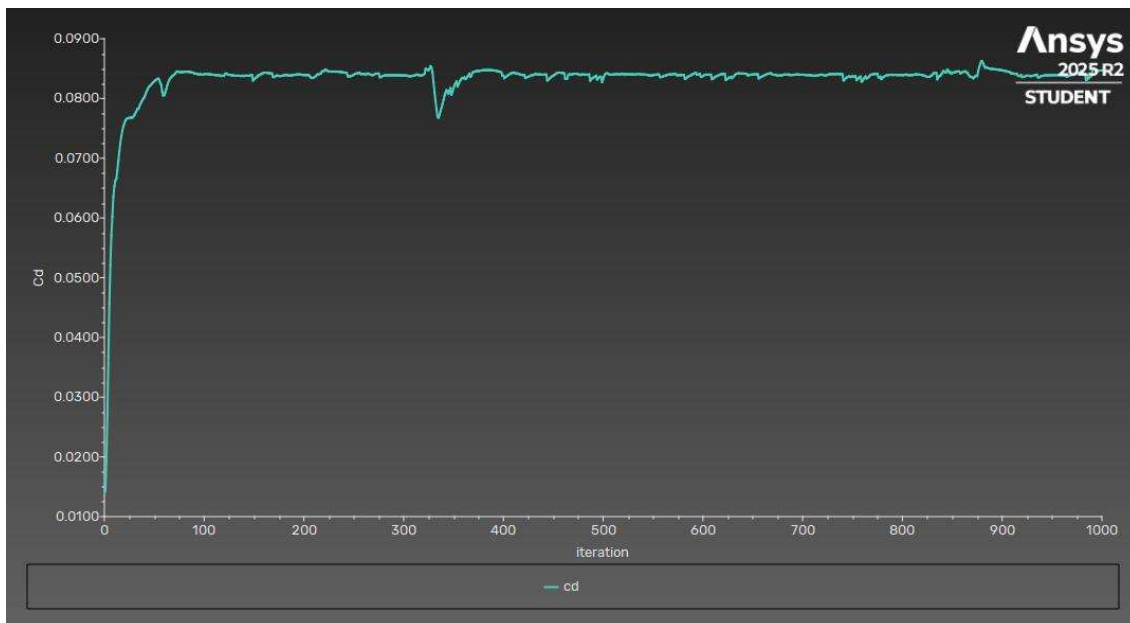
Coefficient of Lift:

Stability: 0.267



Coefficient of Drag:

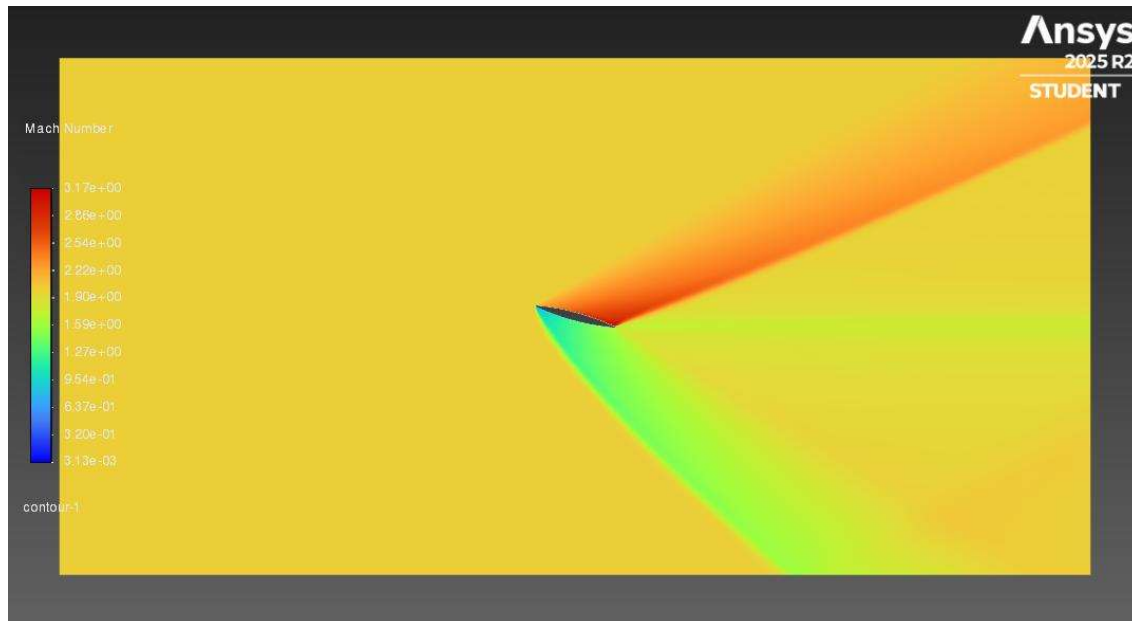
Stability: 0.0849



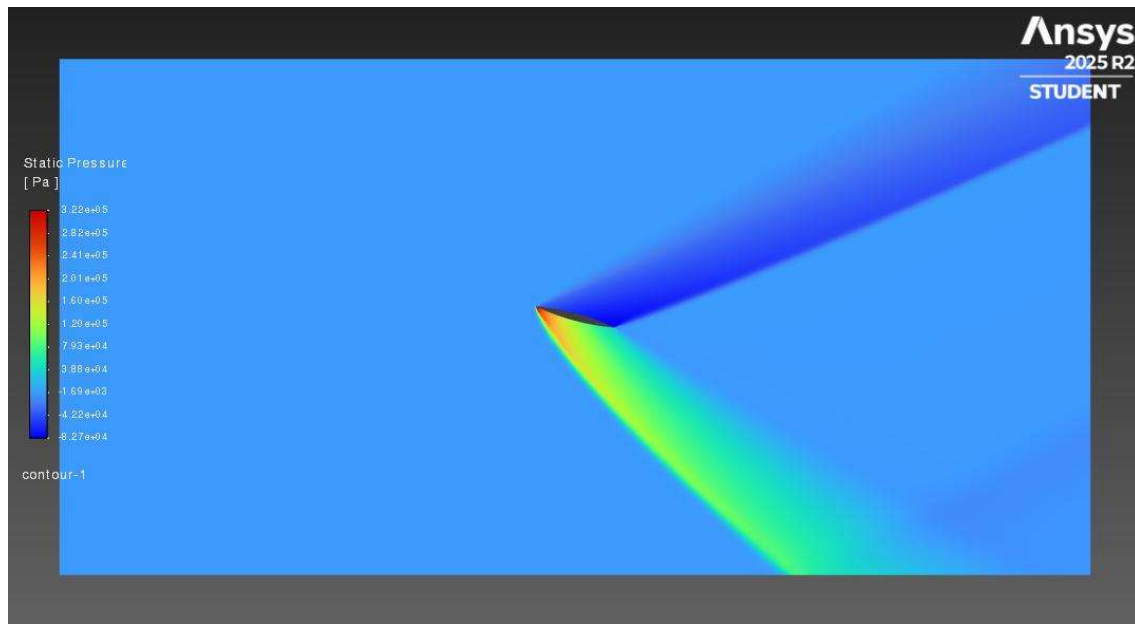
Scaled residuals:



Mach:



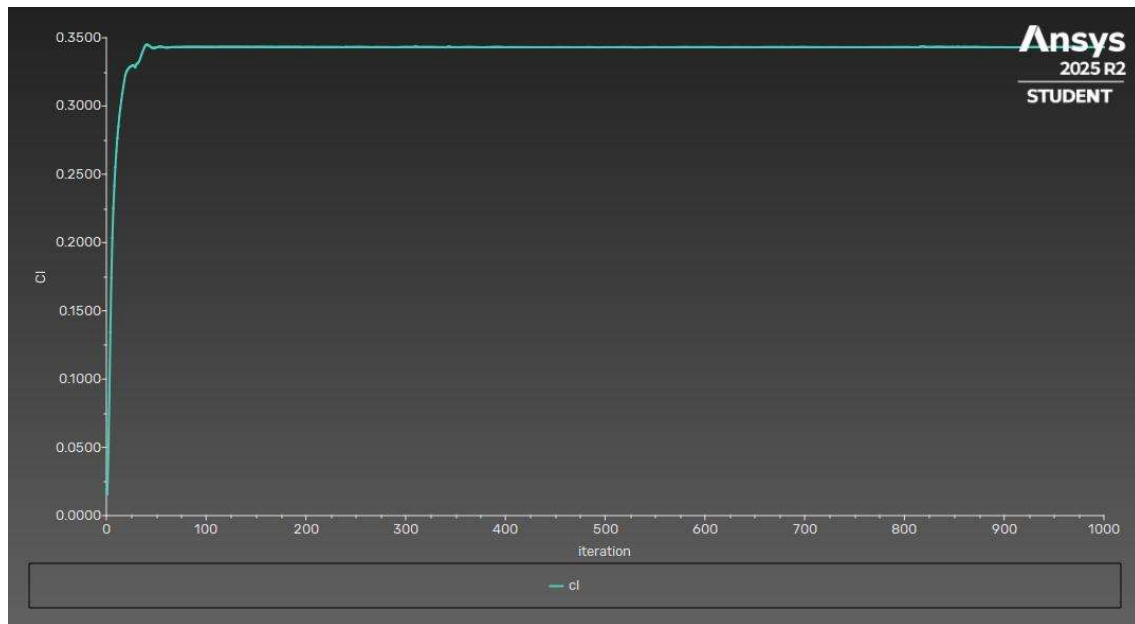
Pressure:



20 Degree:

Coefficient of Lift:

Stability: 0.343

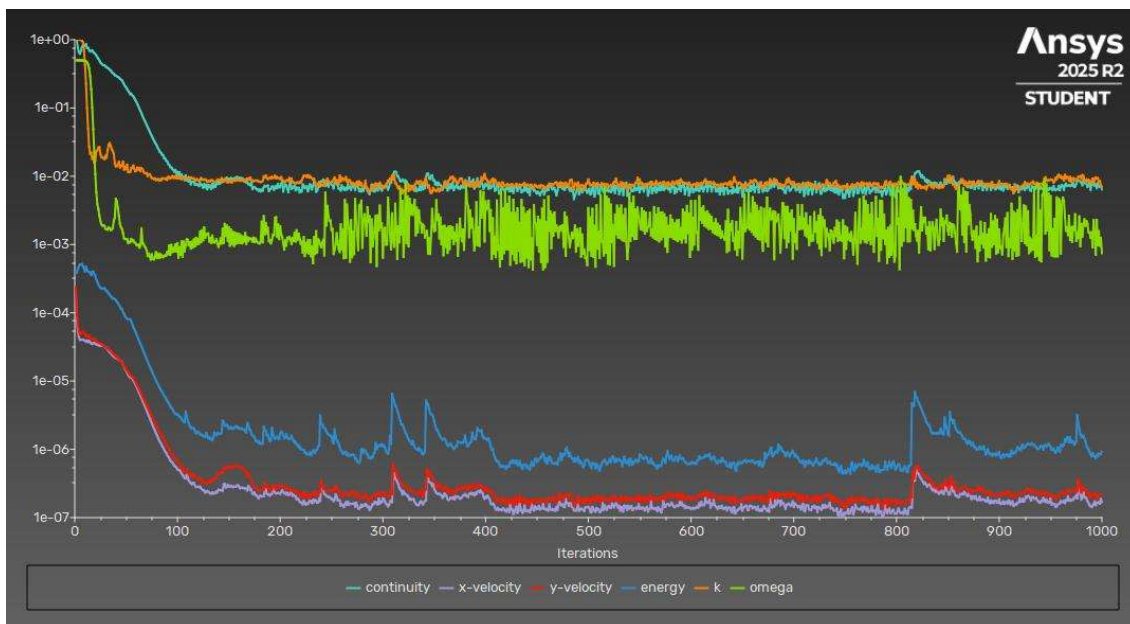


Coefficient of Drag:

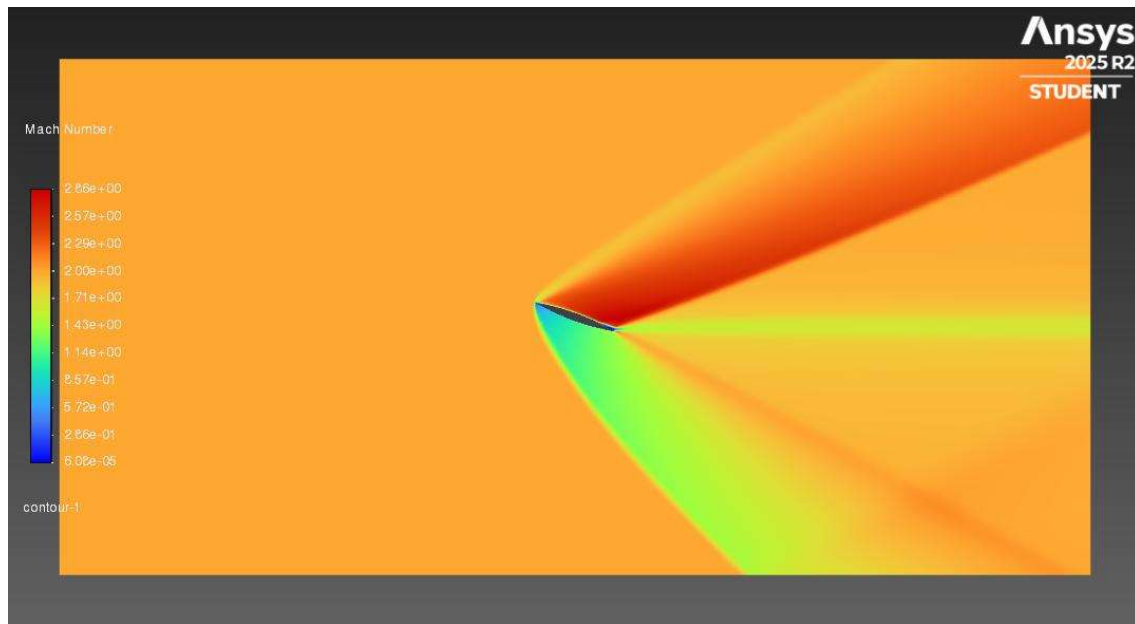
Stability: 0.139



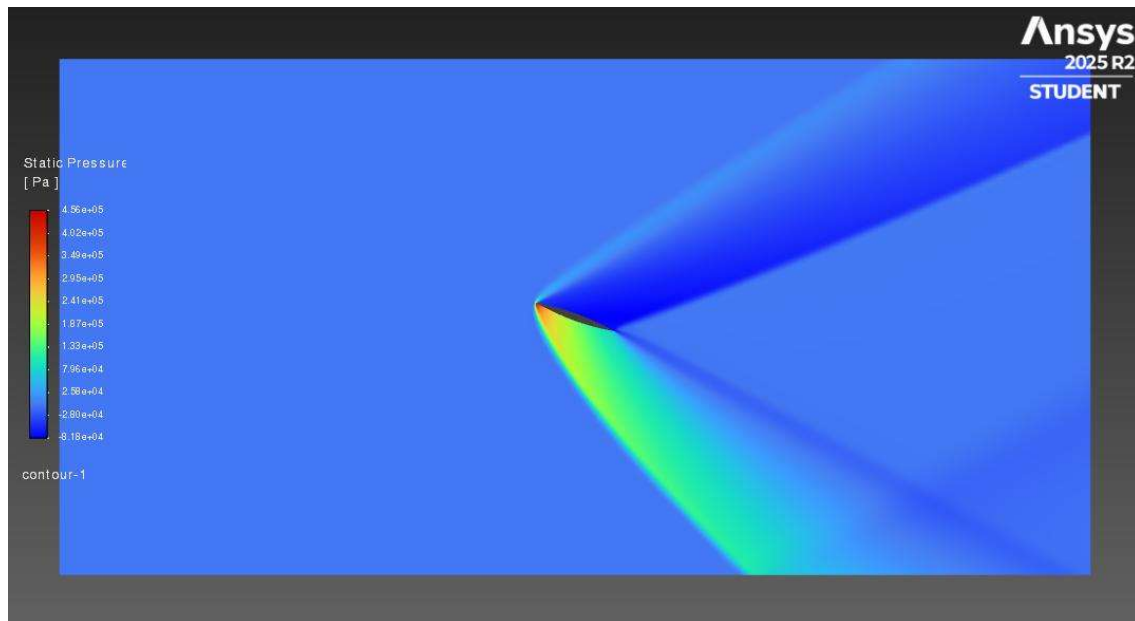
Scaled residuals:



Mach:



Pressure:



25 Degree:

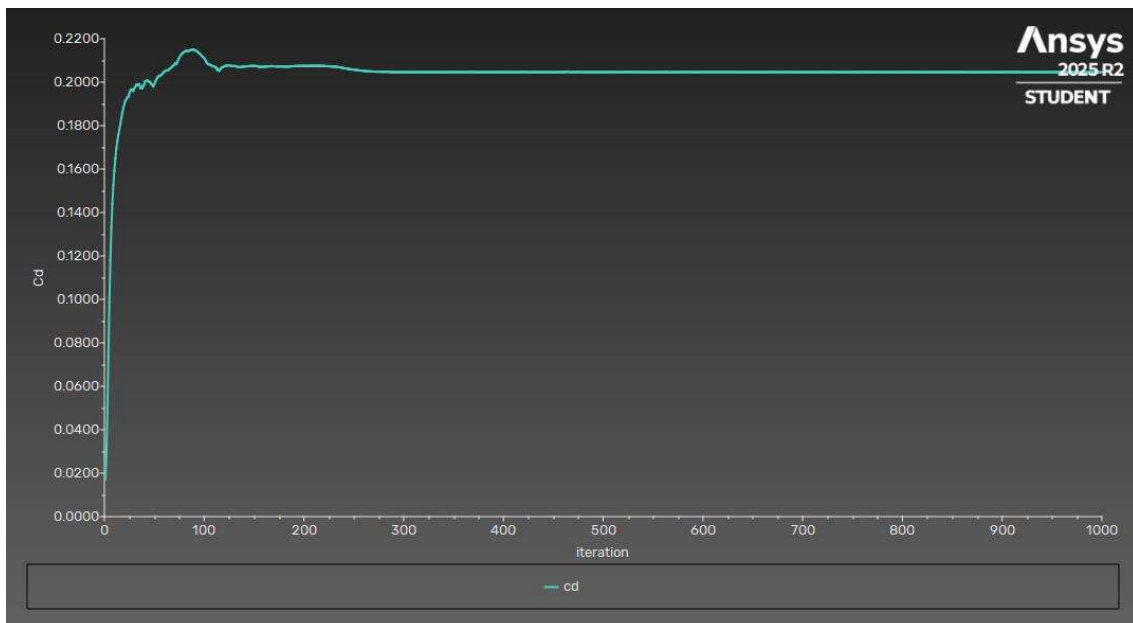
Coefficient of Lift:

Stability: 0.406



Coefficient of Drag:

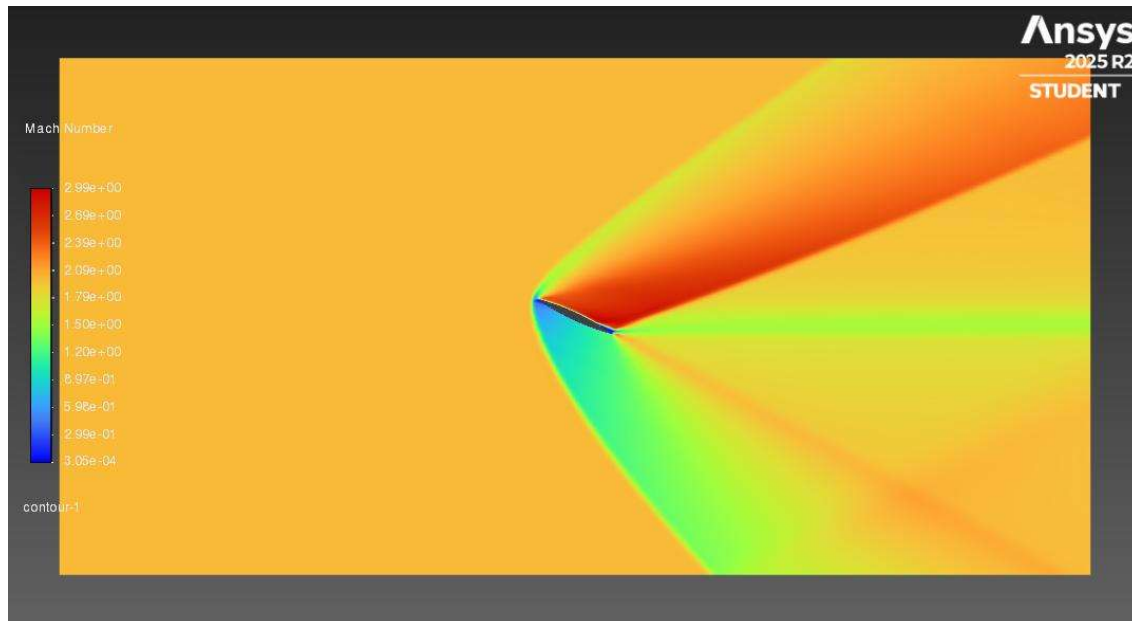
Stability: 0.204



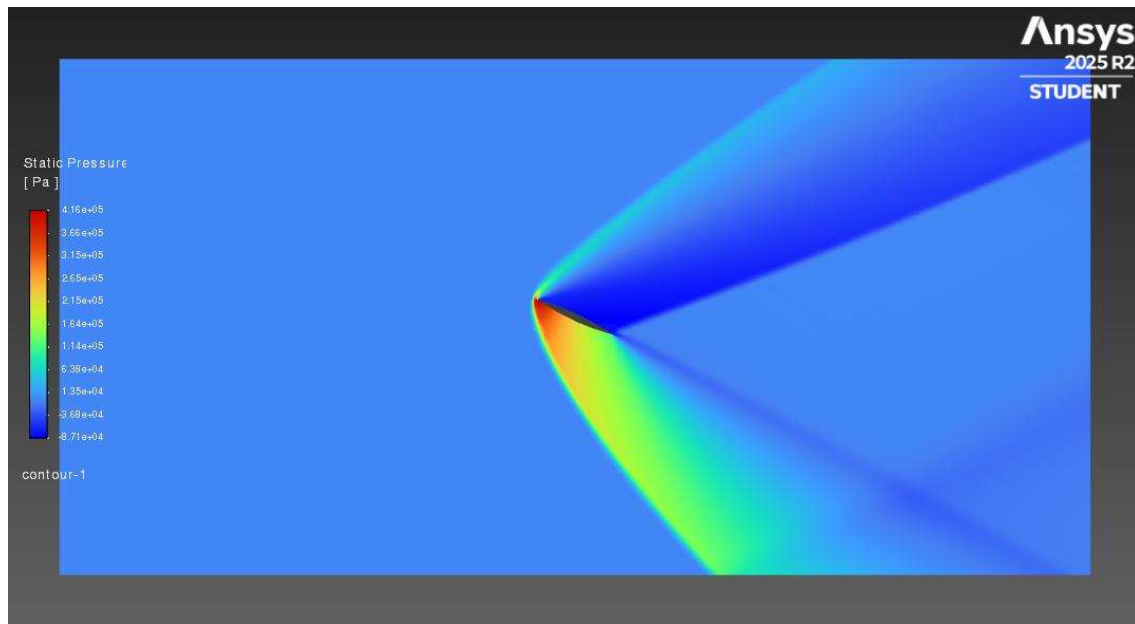
Scaled residuals:



Mach:



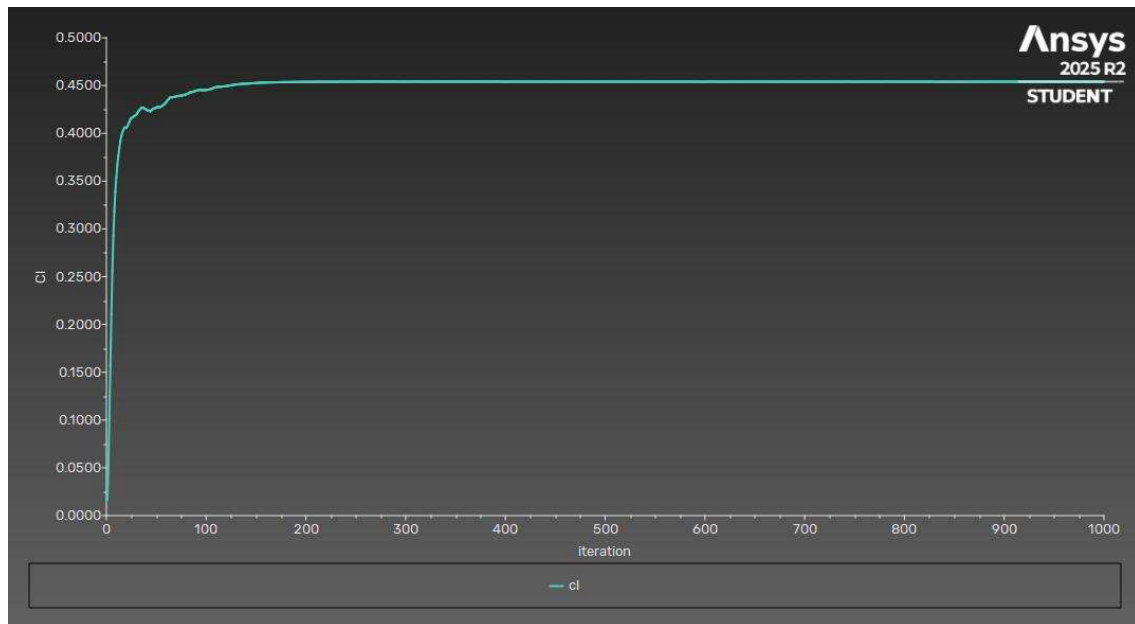
Pressure:



30 Degree:

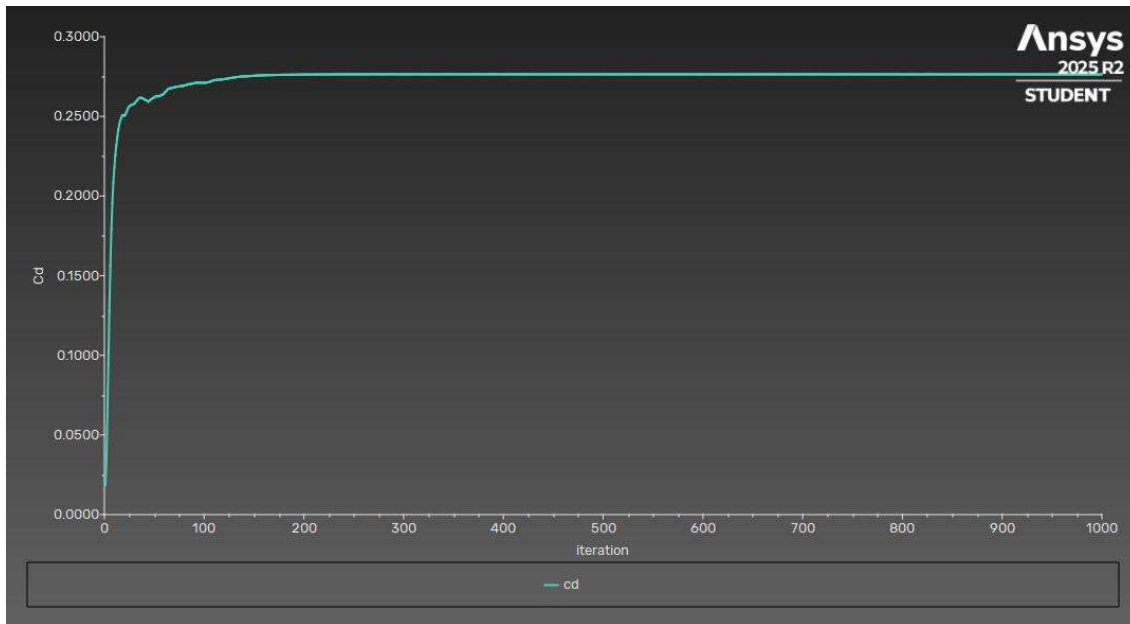
Coefficient of Lift:

Stability: 0.453



Coefficient of Drag:

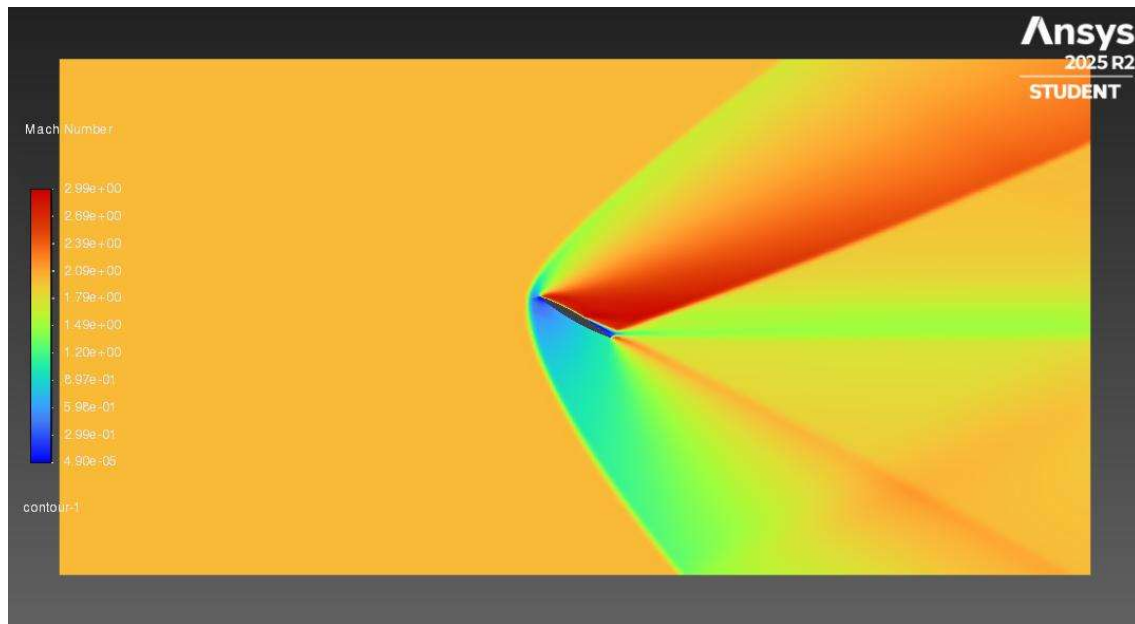
Stability: 0.276



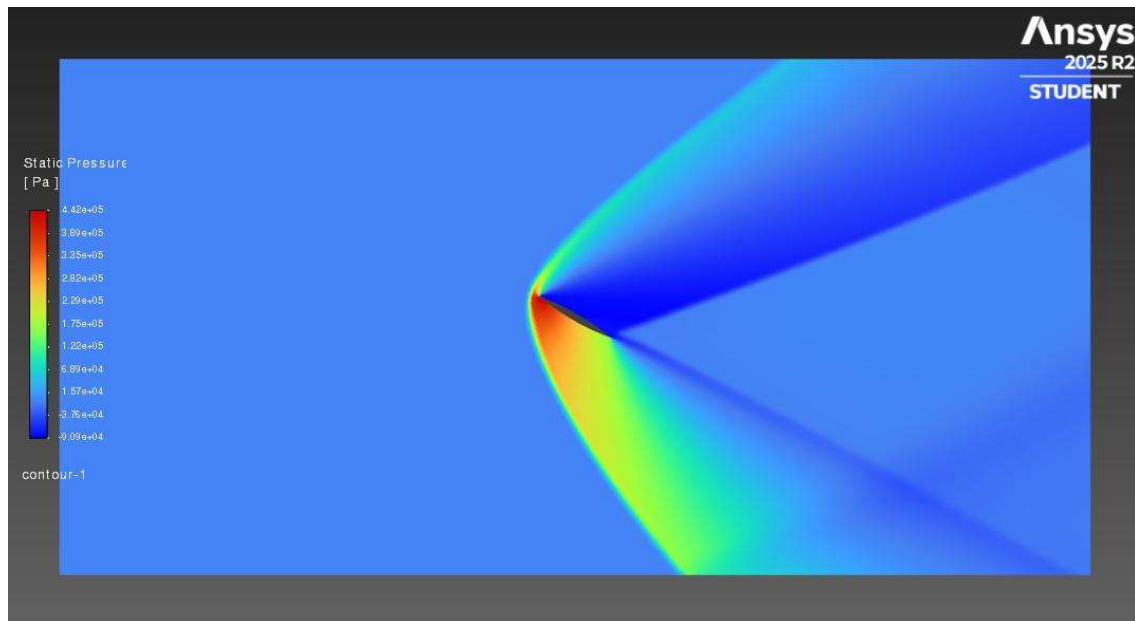
Scaled residuals:



Mach:



Pressure:

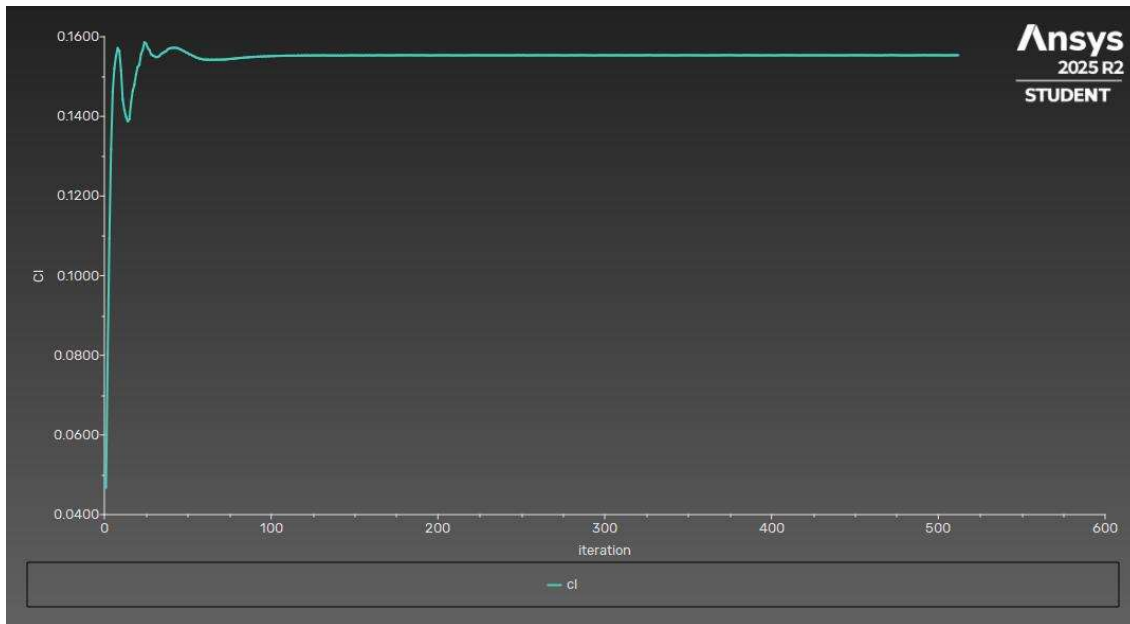


DOMES ANGLE TESTS:

5 Degree:

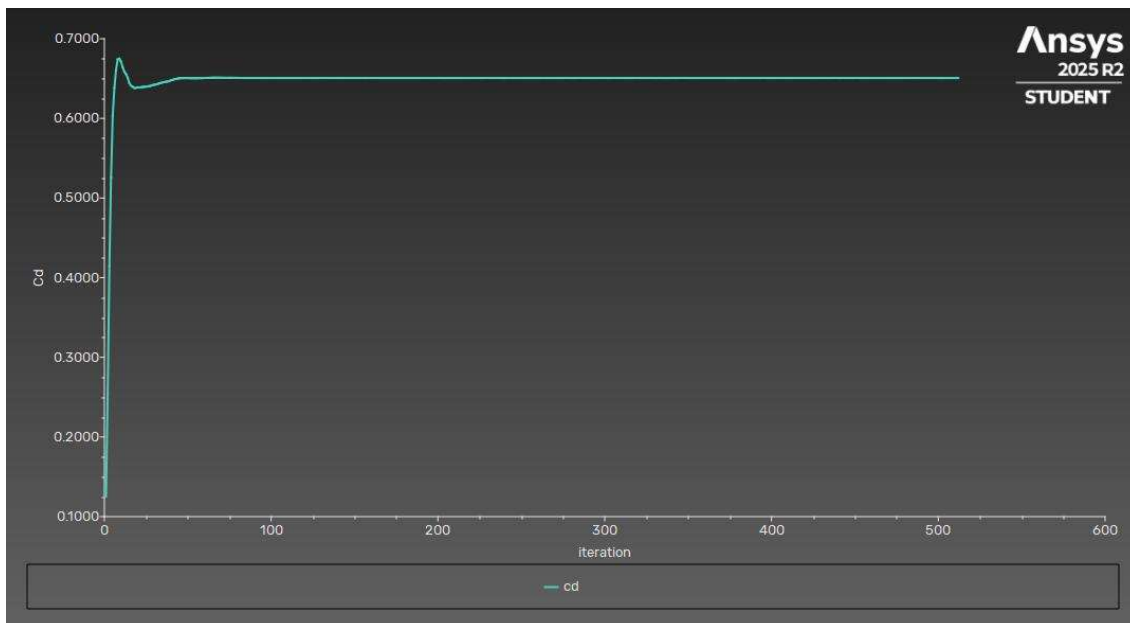
Coefficient of Lift:

Stability: 0.155

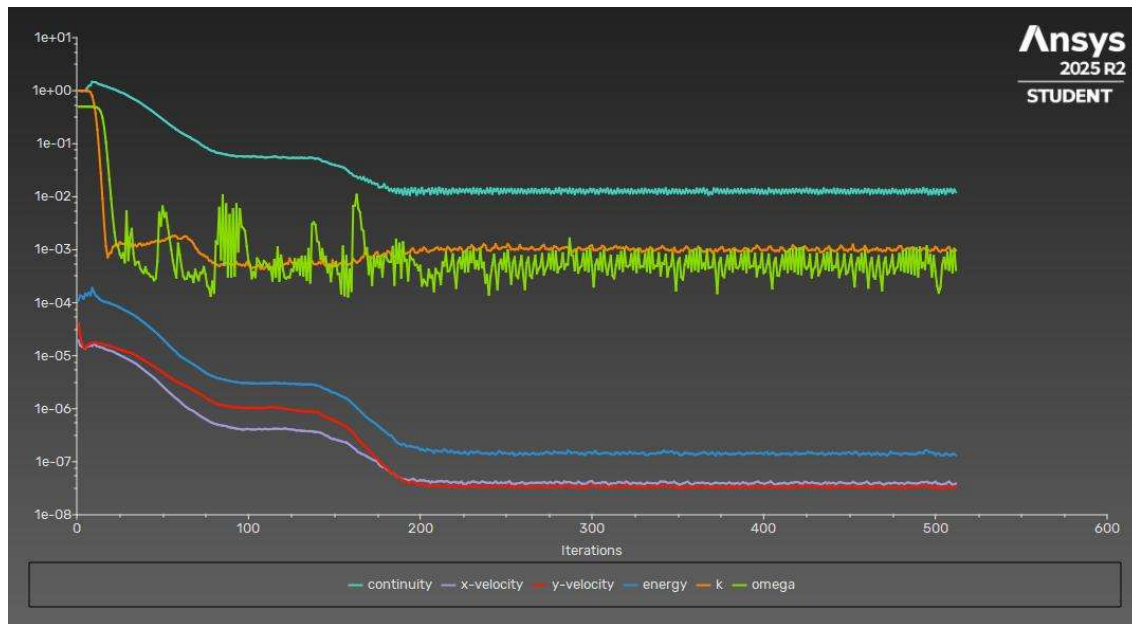


Coefficient of Drag:

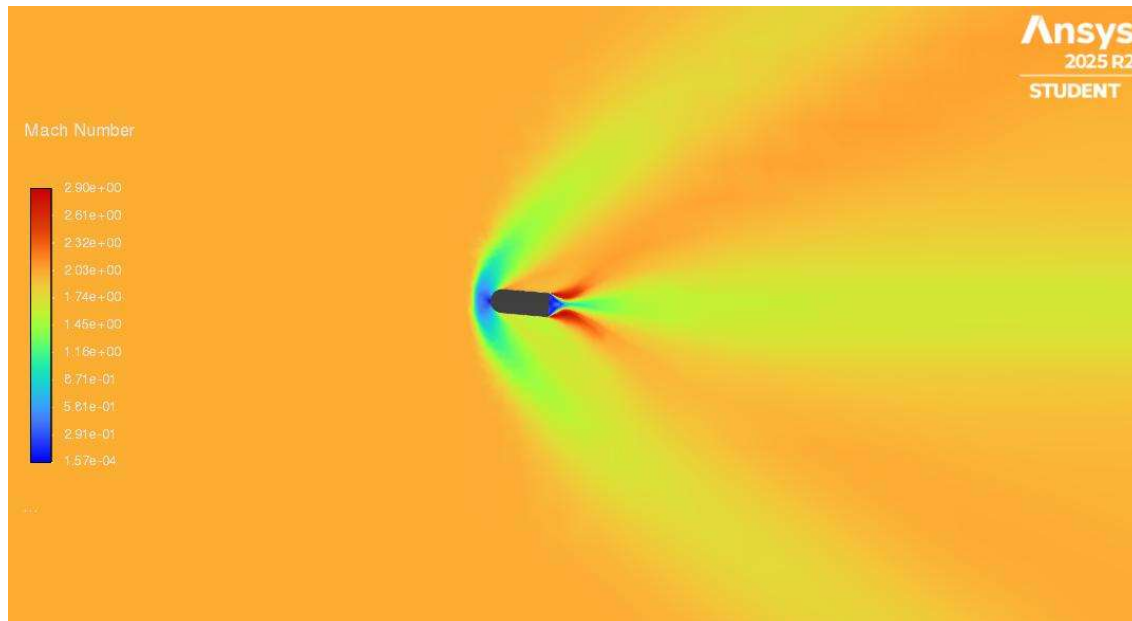
Stability: 0.65



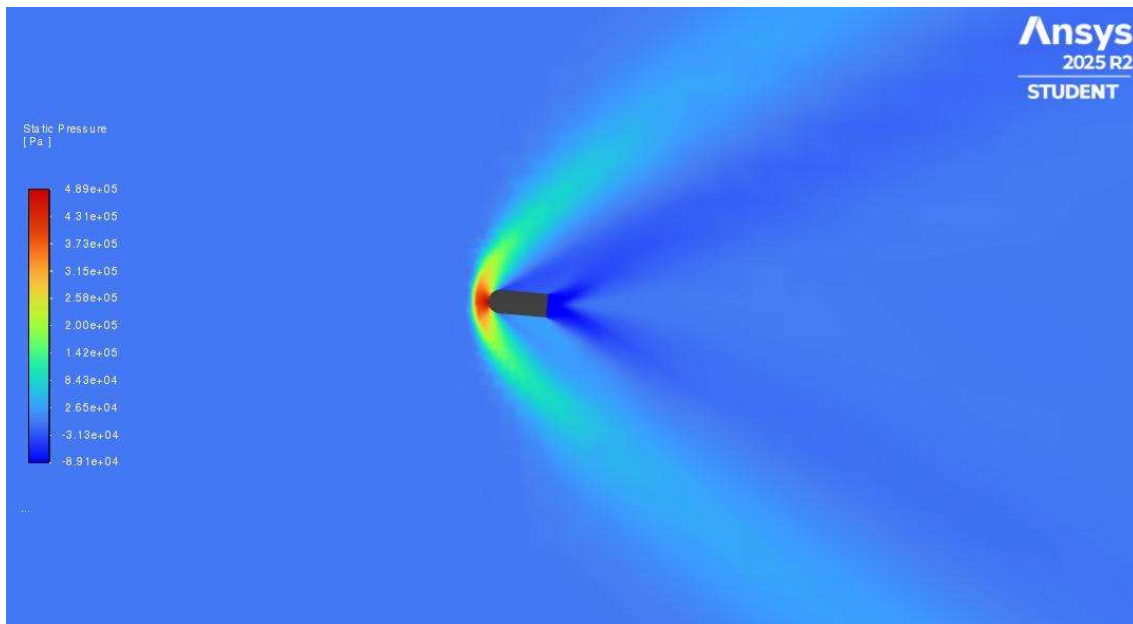
Scaled residuals:



Mach:



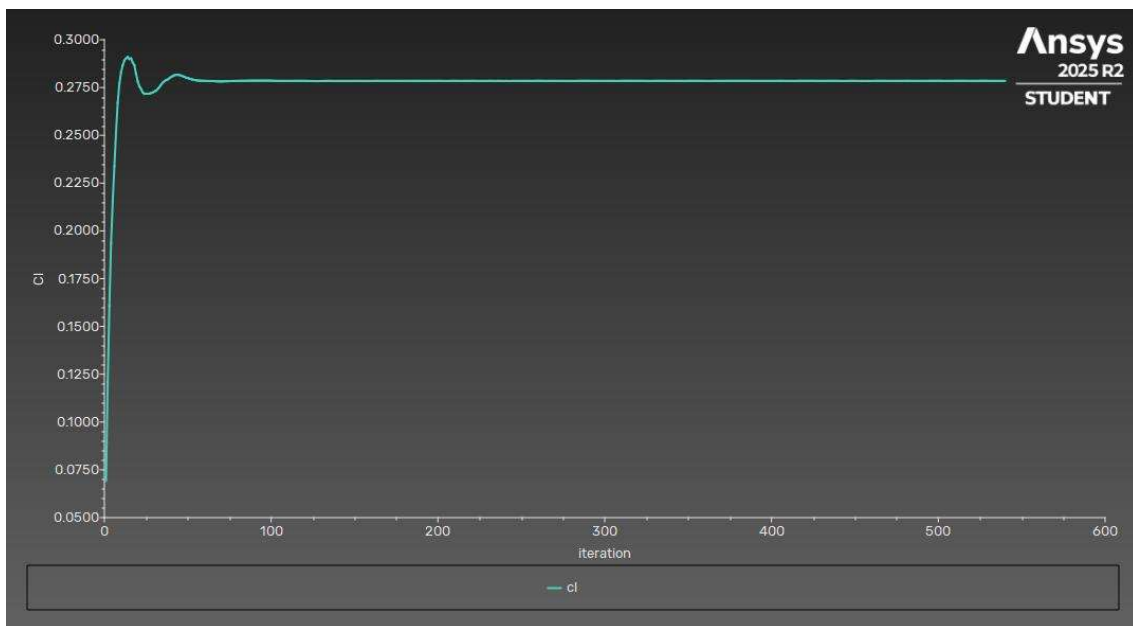
Pressure:



10 Degree:

Coefficient of Lift:

Stability: 0.278

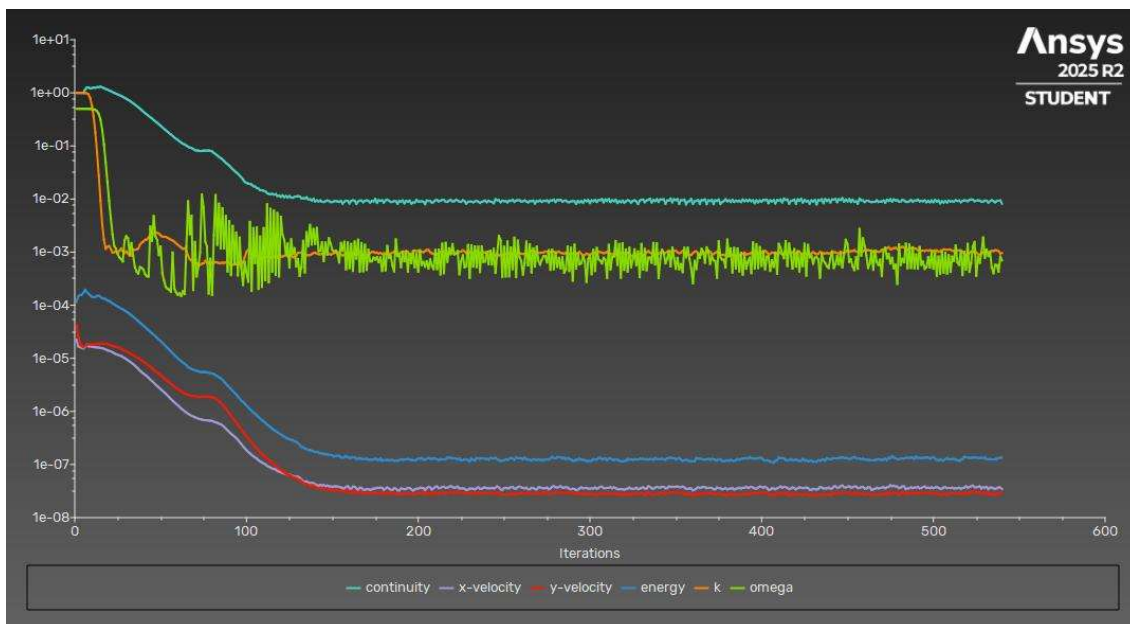


Coefficient of Drag:

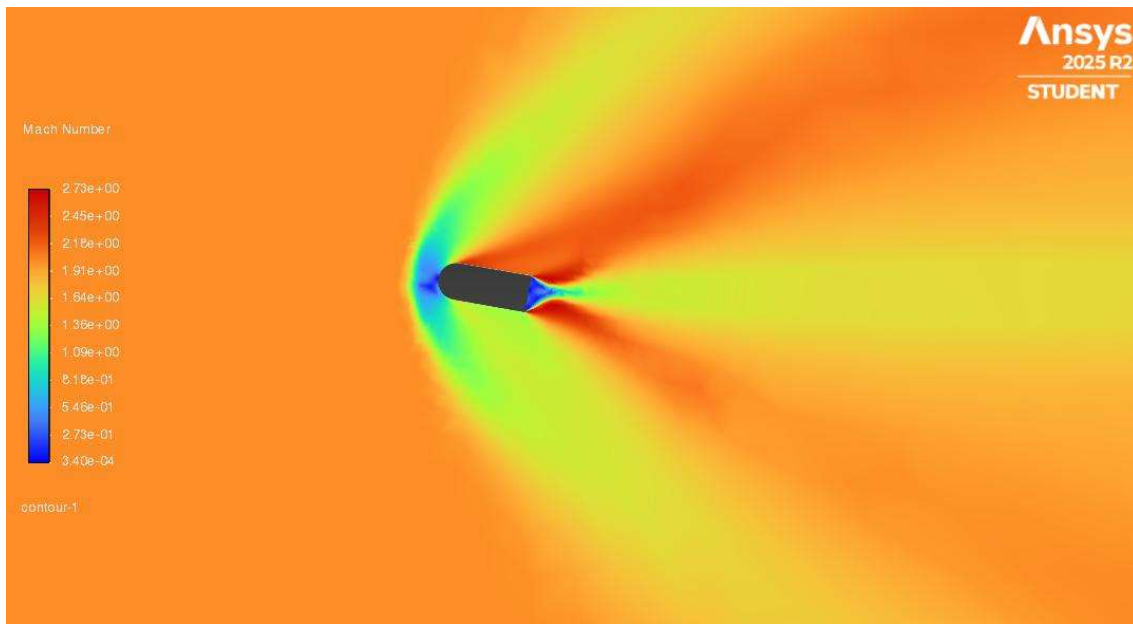
Stability: 0.611



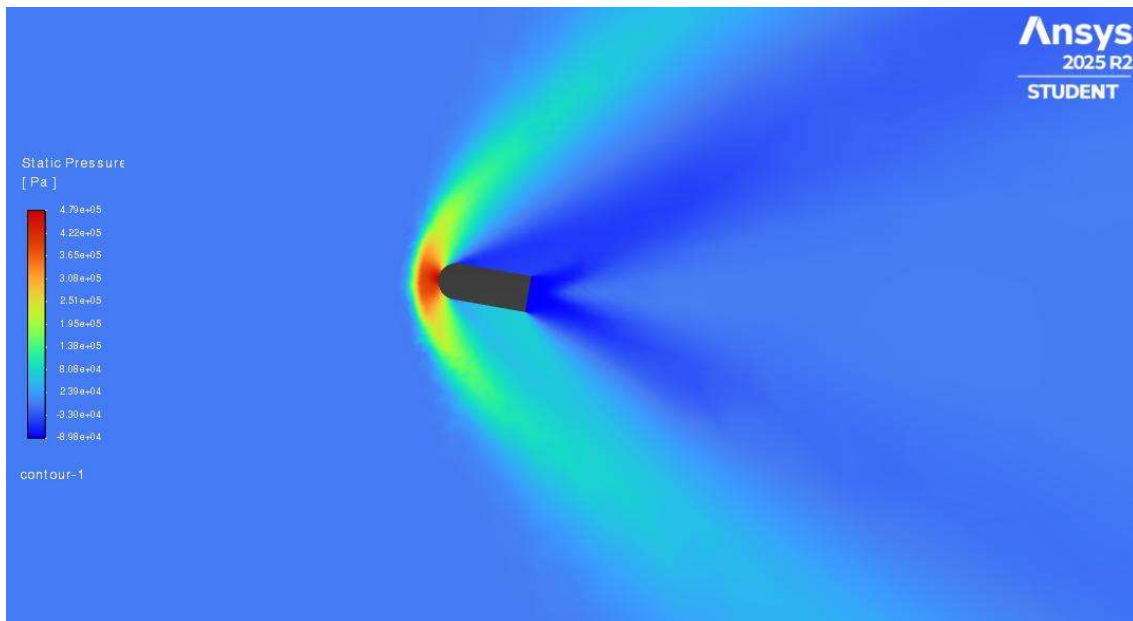
Scaled residuals:



Mach:



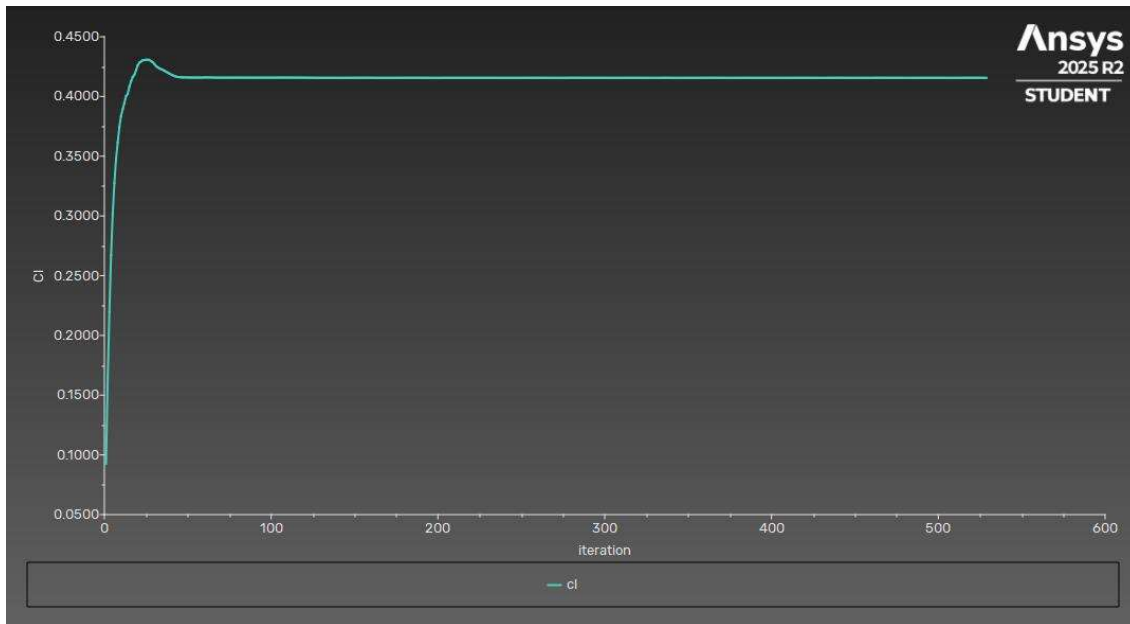
Pressure:



15 Degree:

Coefficient of Lift:

Stability: 0.415



Coefficient of Drag:

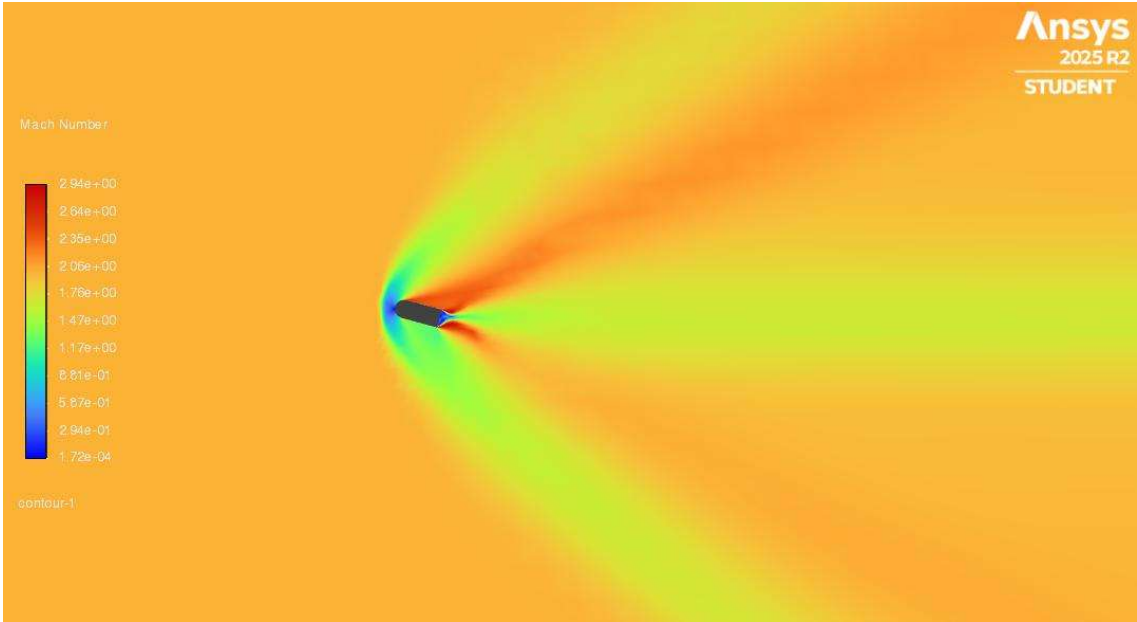
Stability: 0.686



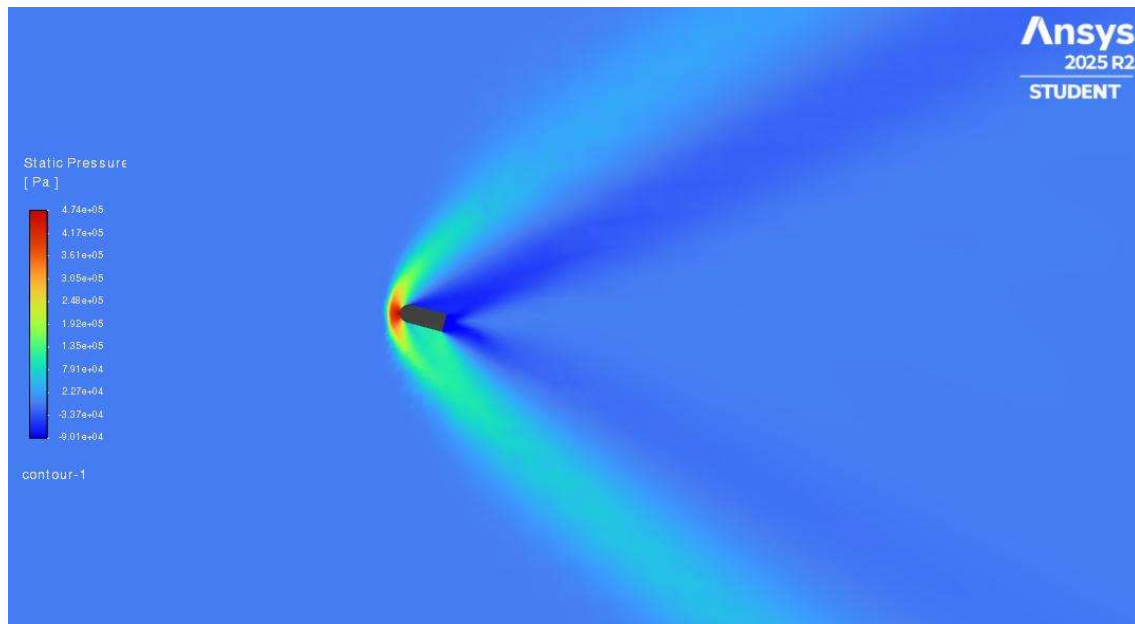
Scaled residuals:



Mach:



Pressure:



20 Degree:

Coefficient of Lift:

Stability: 0.622

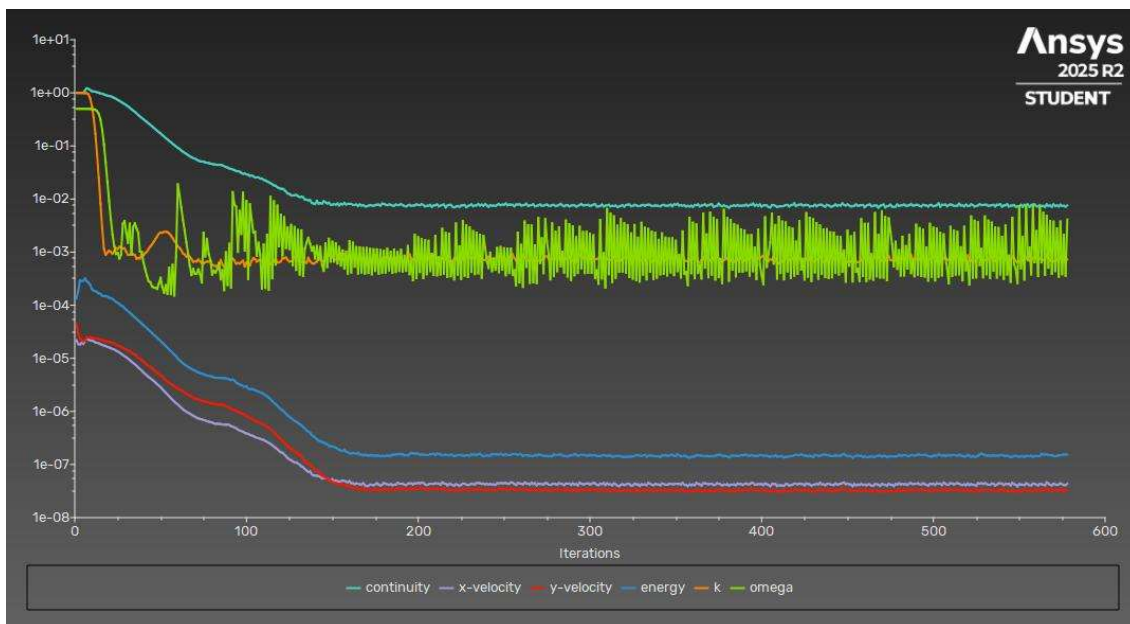


Coefficient of Drag:

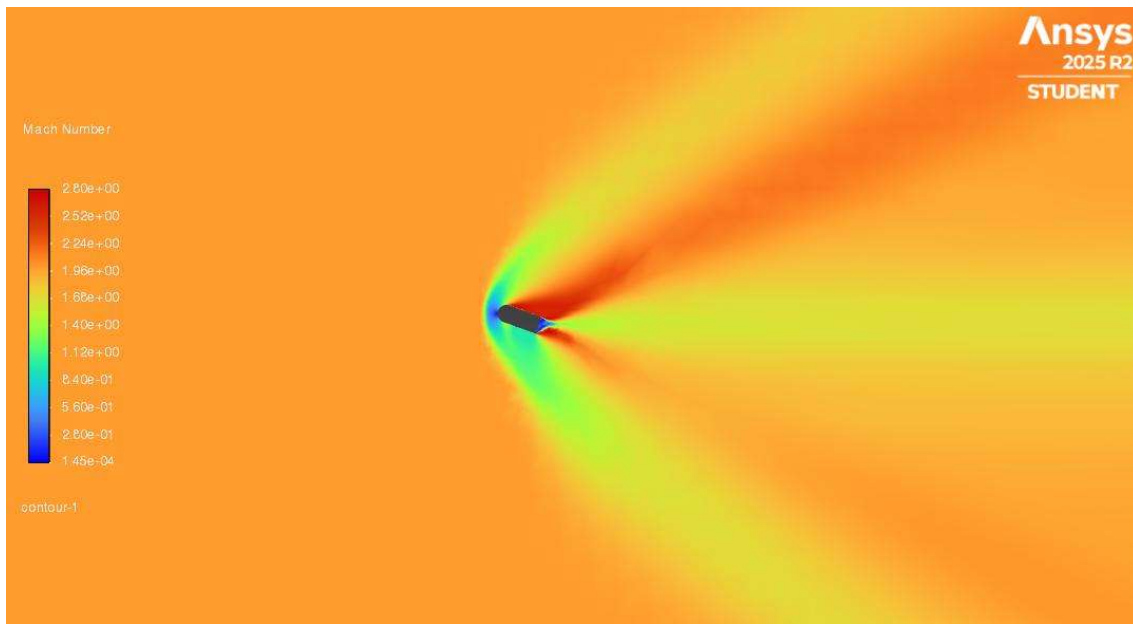
Stability: 0.797



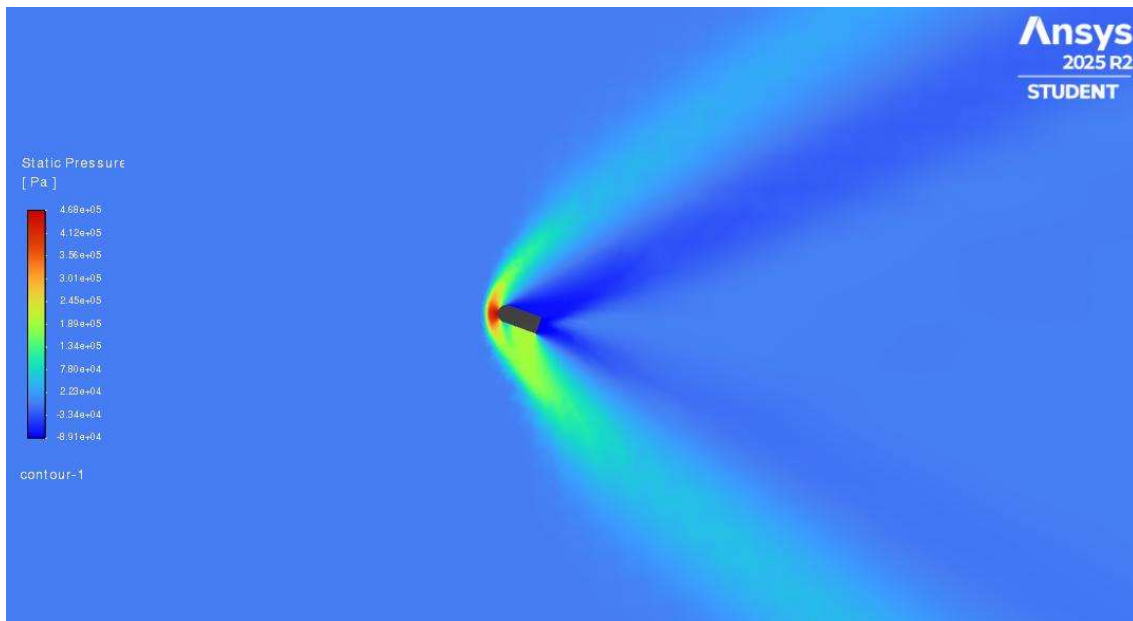
Scaled residuals:



Mach:



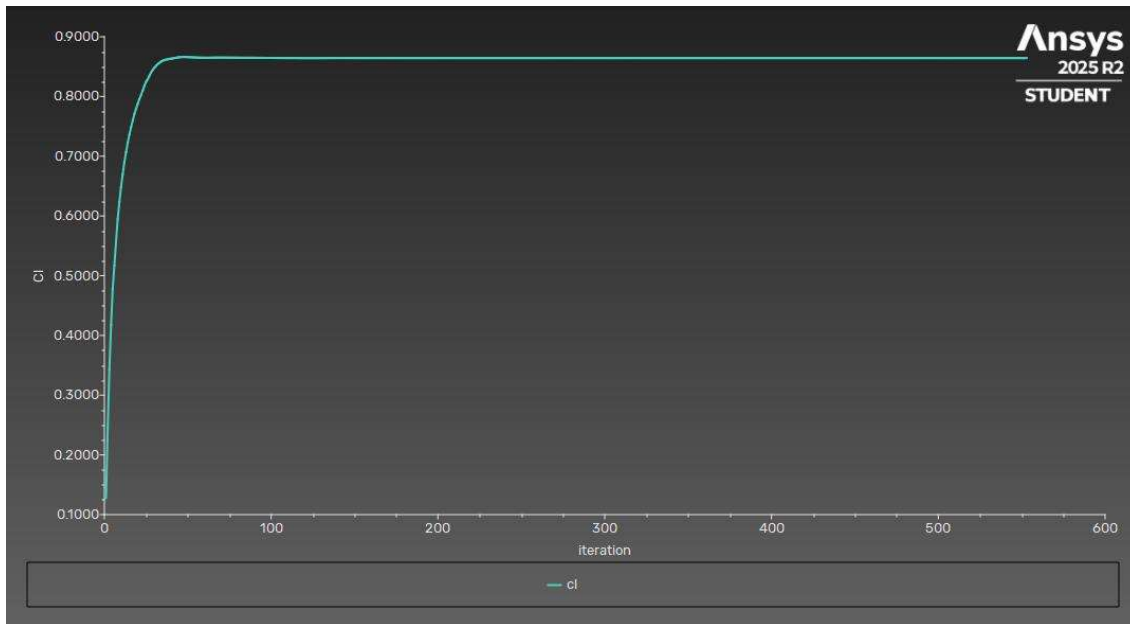
Pressure:



25 Degree:

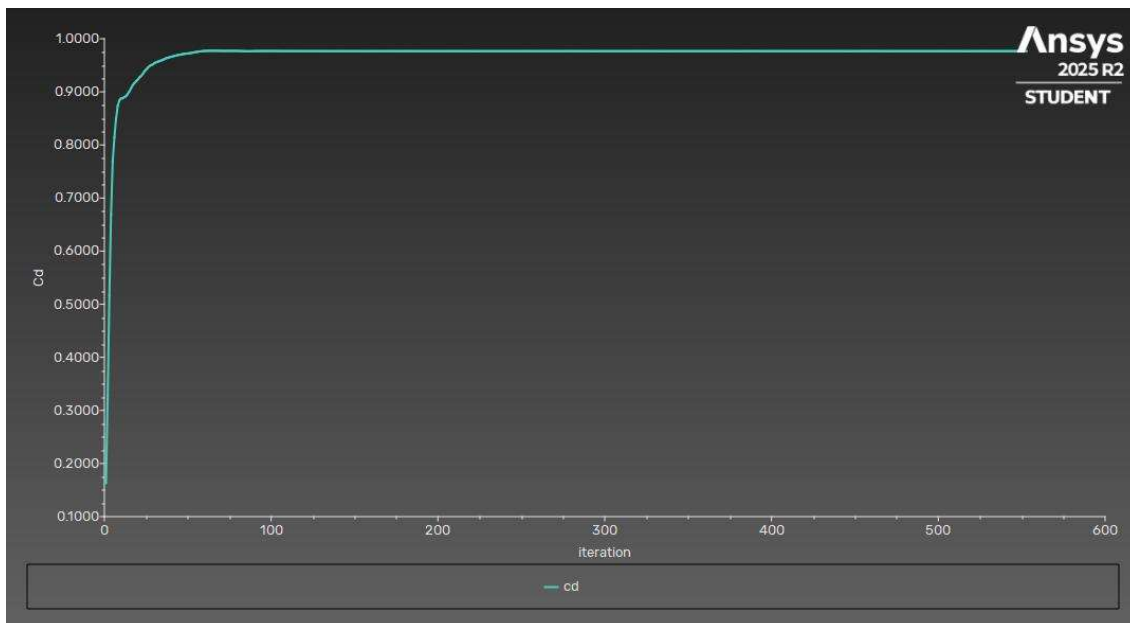
Coefficient of Lift:

Stability: 0.864



Coefficient of Drag:

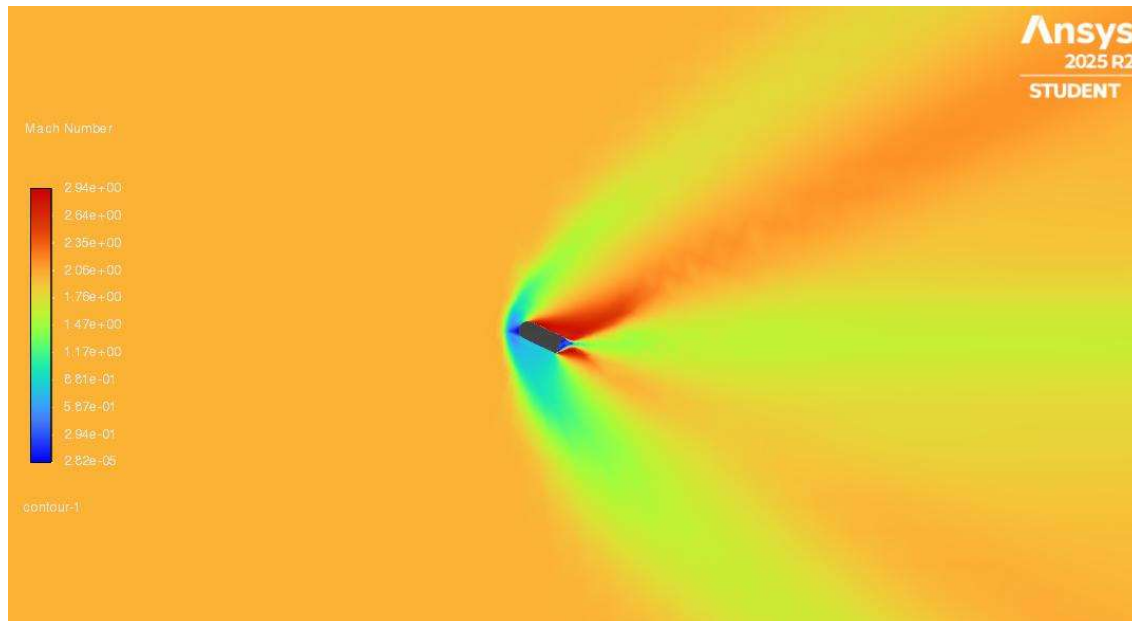
Stability: 0.976



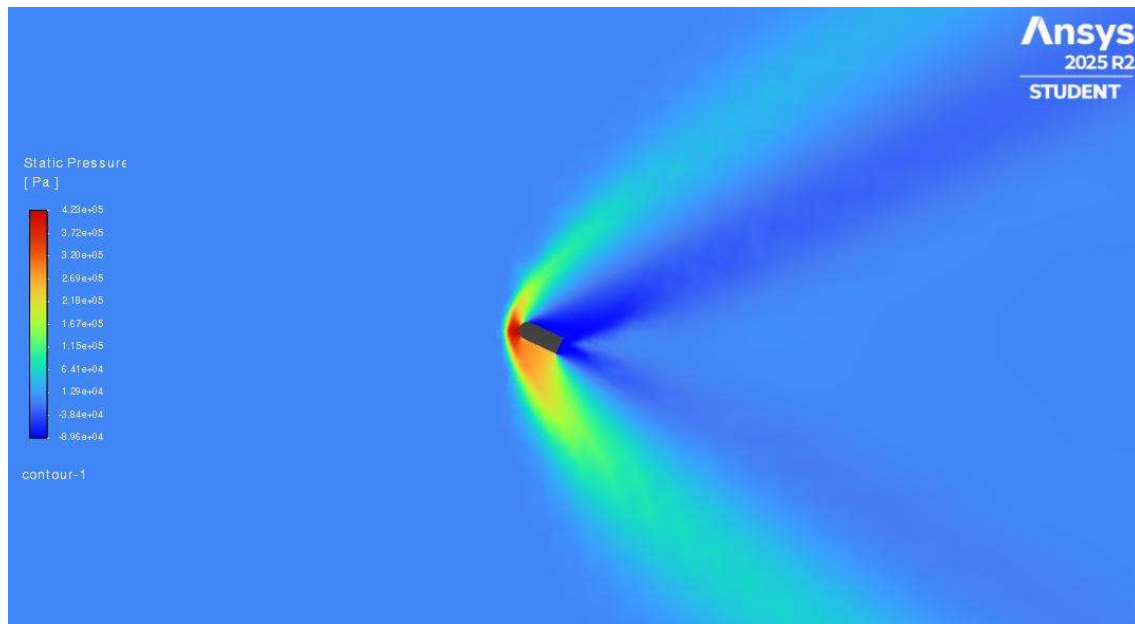
Scaled residuals:



Mach:



Pressure:



30 Degree:

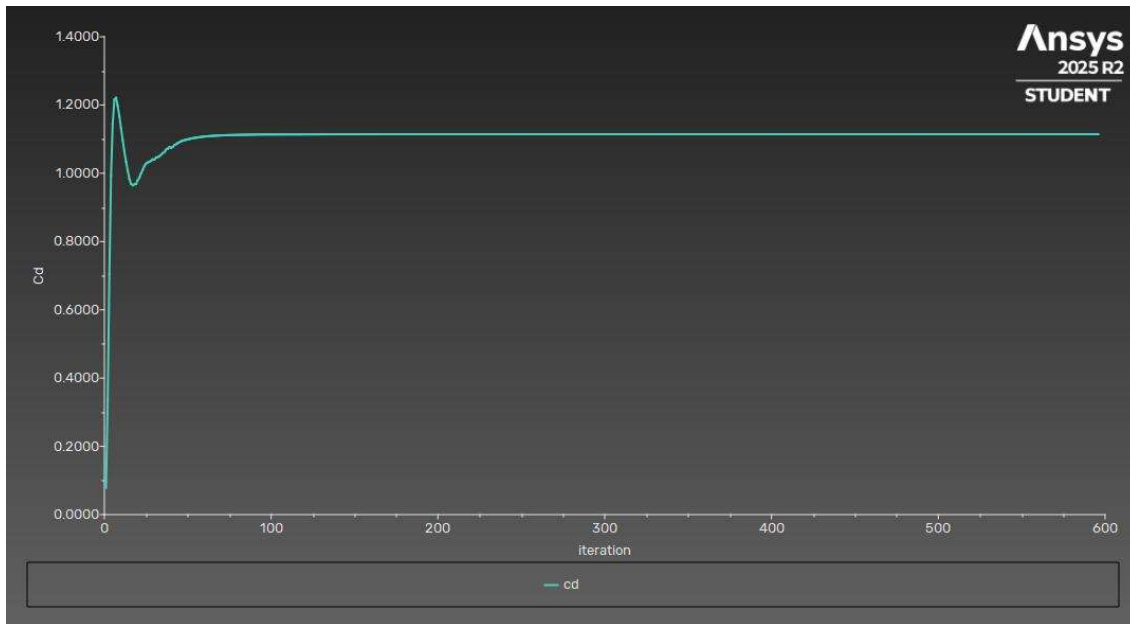
Coefficient of Lift:

Stability: 0.937

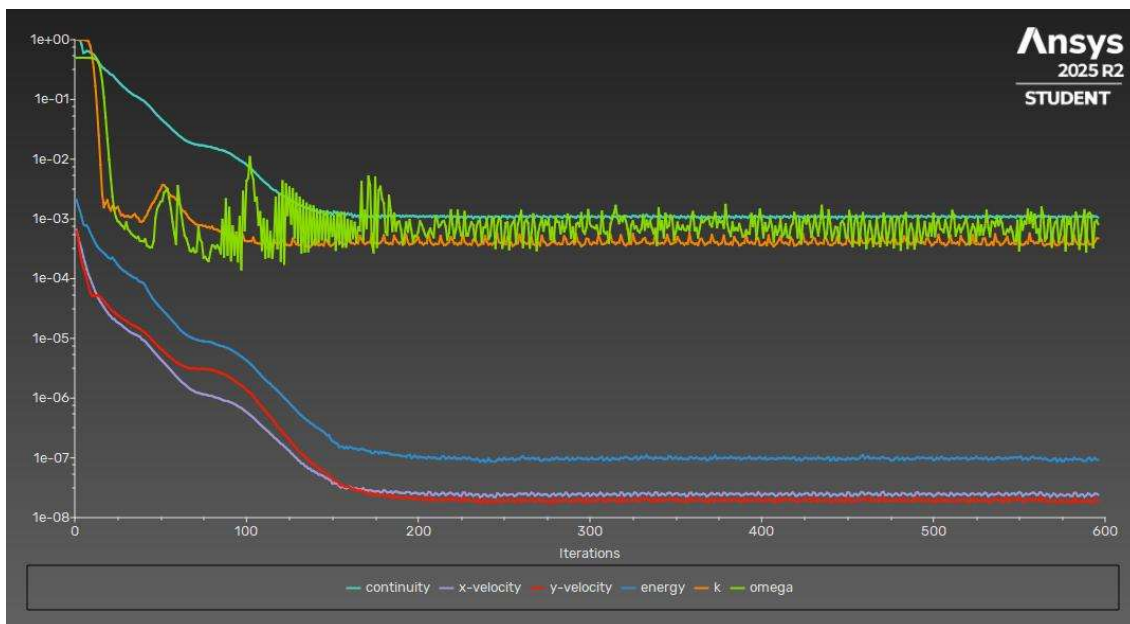


Coefficient of Drag:

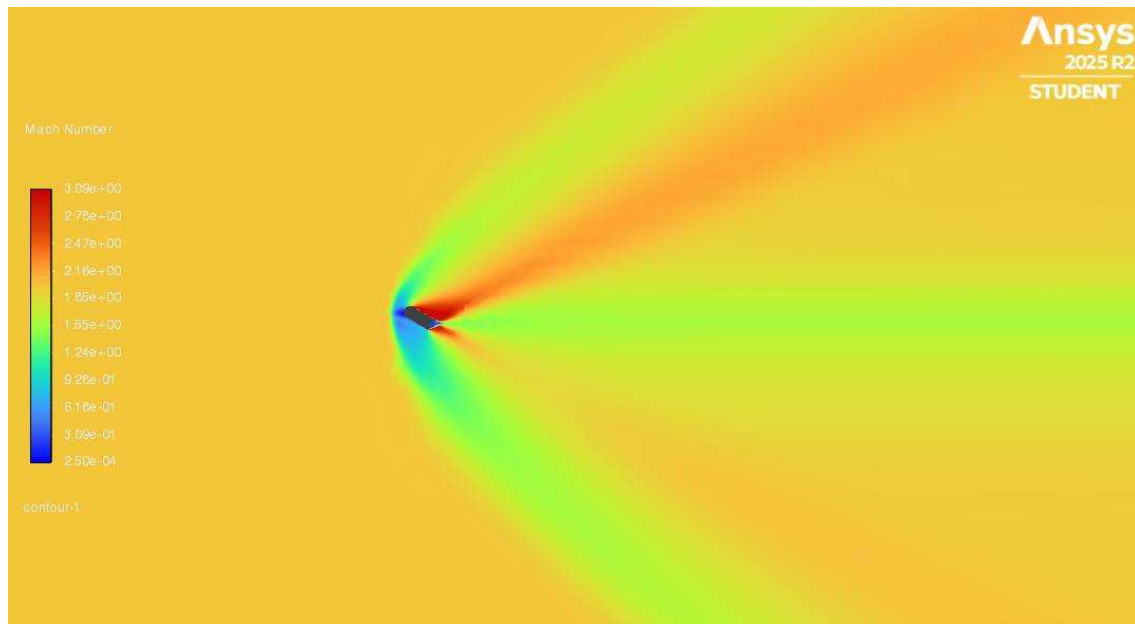
Stability: 1.114



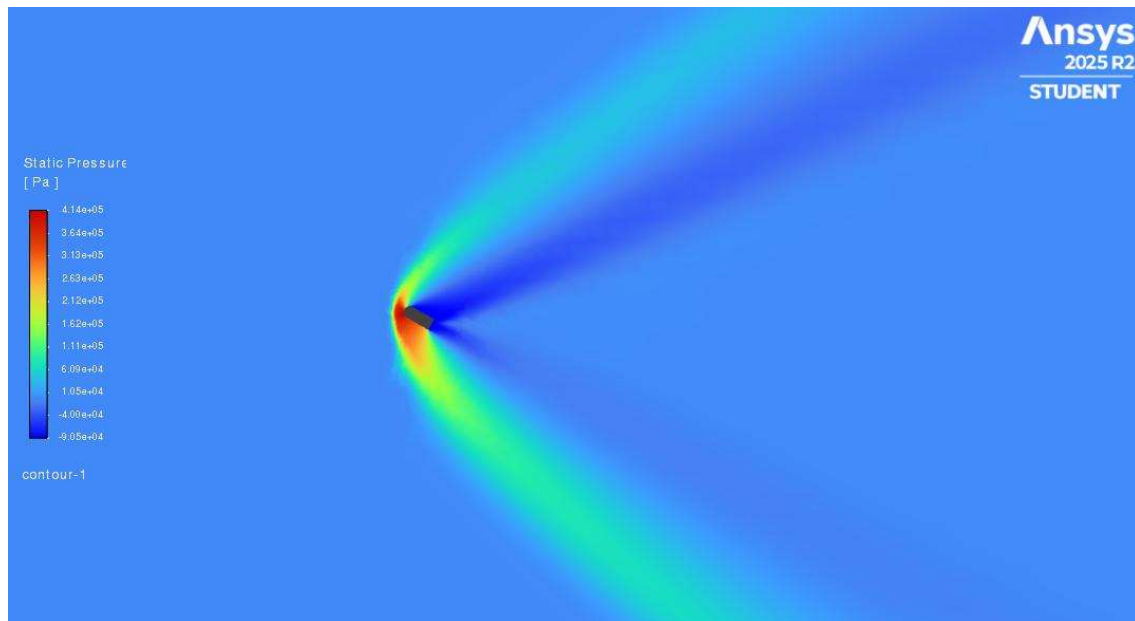
Scaled residuals:



Mach:



Pressure:

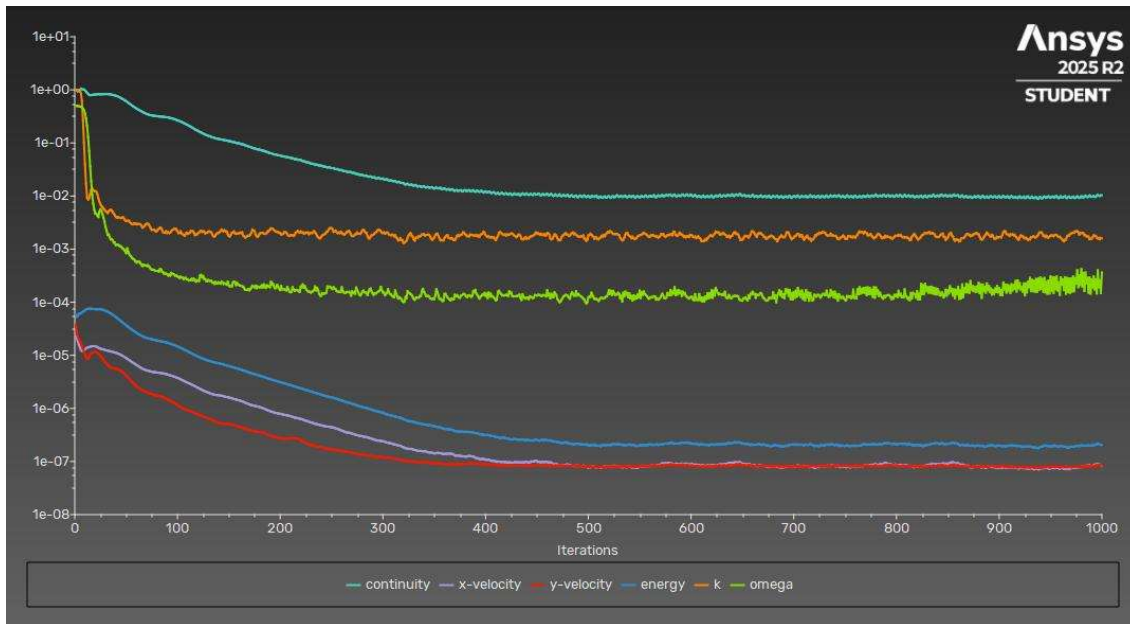


Question D

Figure 2. A

Scaled residuals:

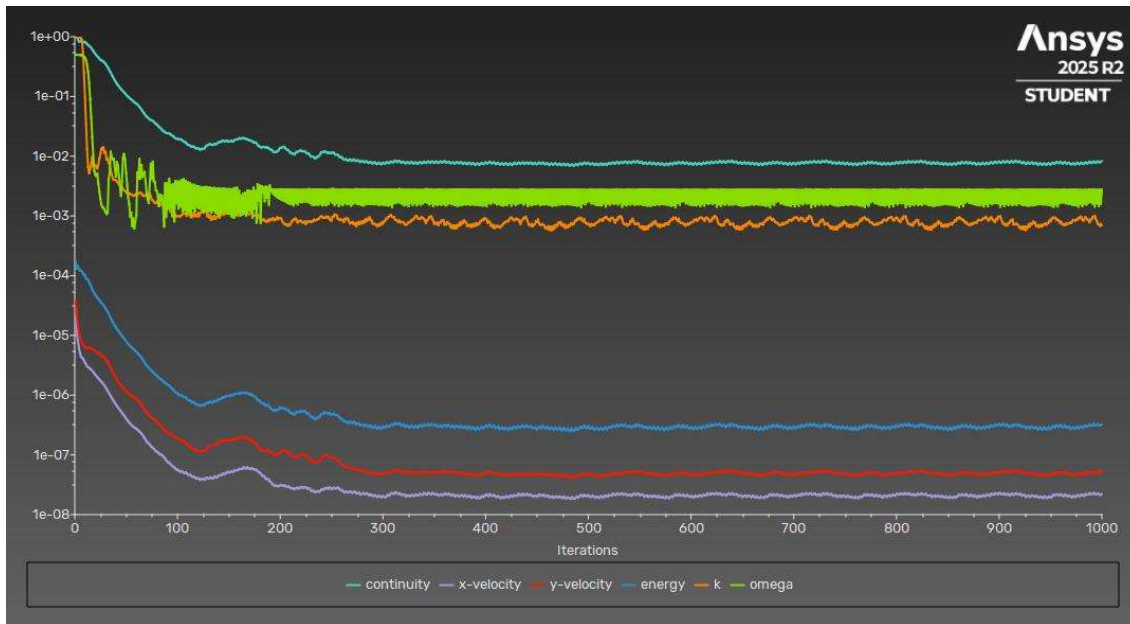
Mach1



Mach2



Mach3



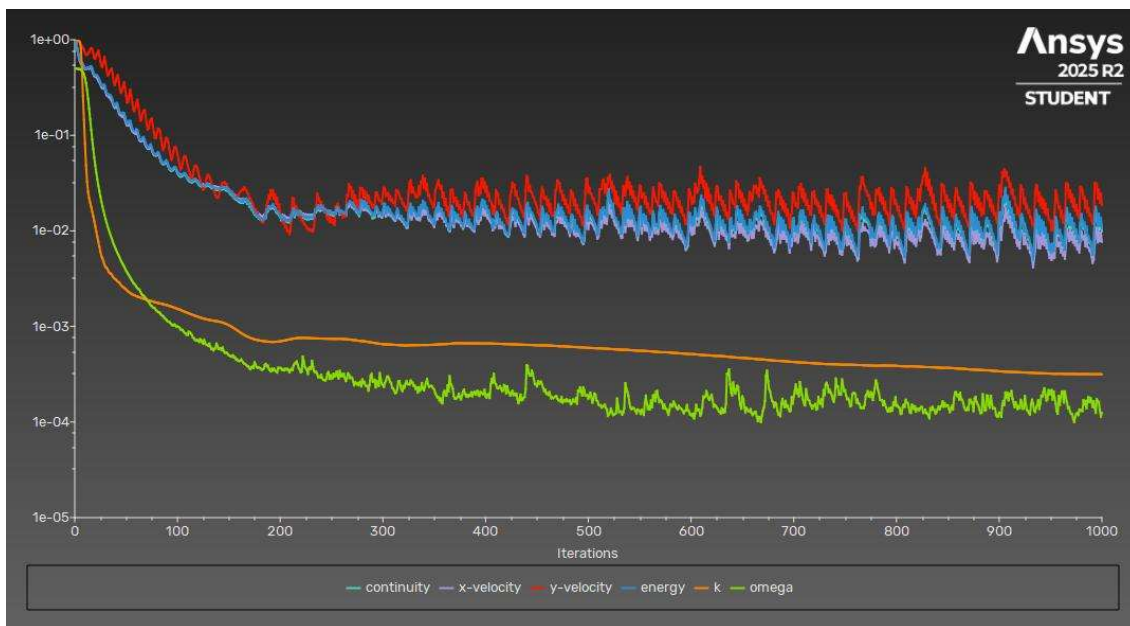
Mach4



Mach5



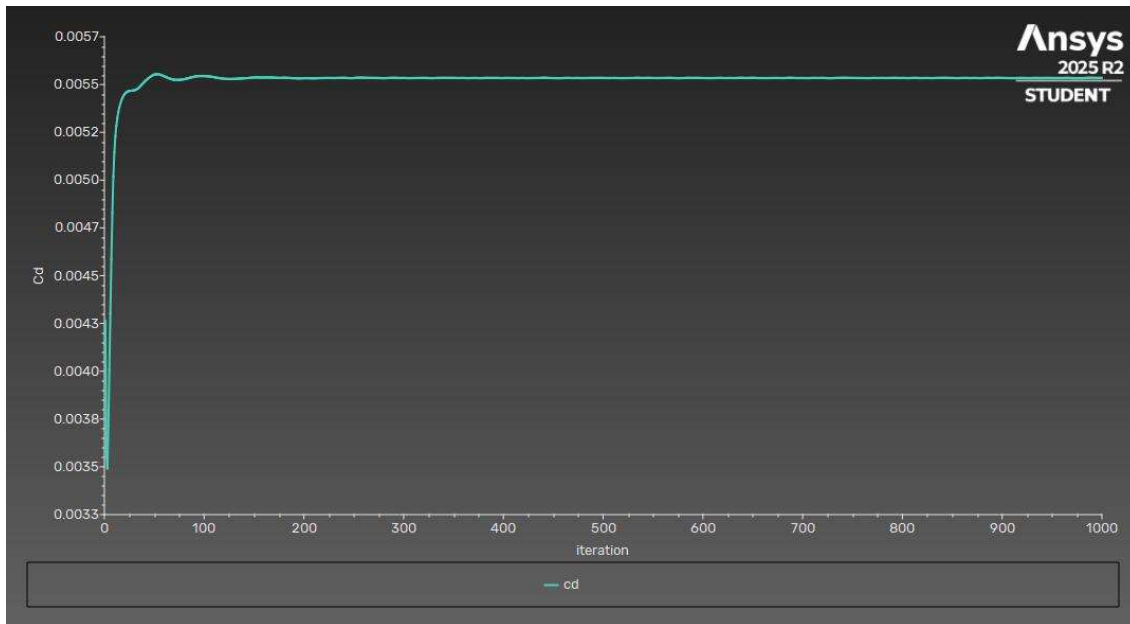
Mach6



Coefficient of Drag:

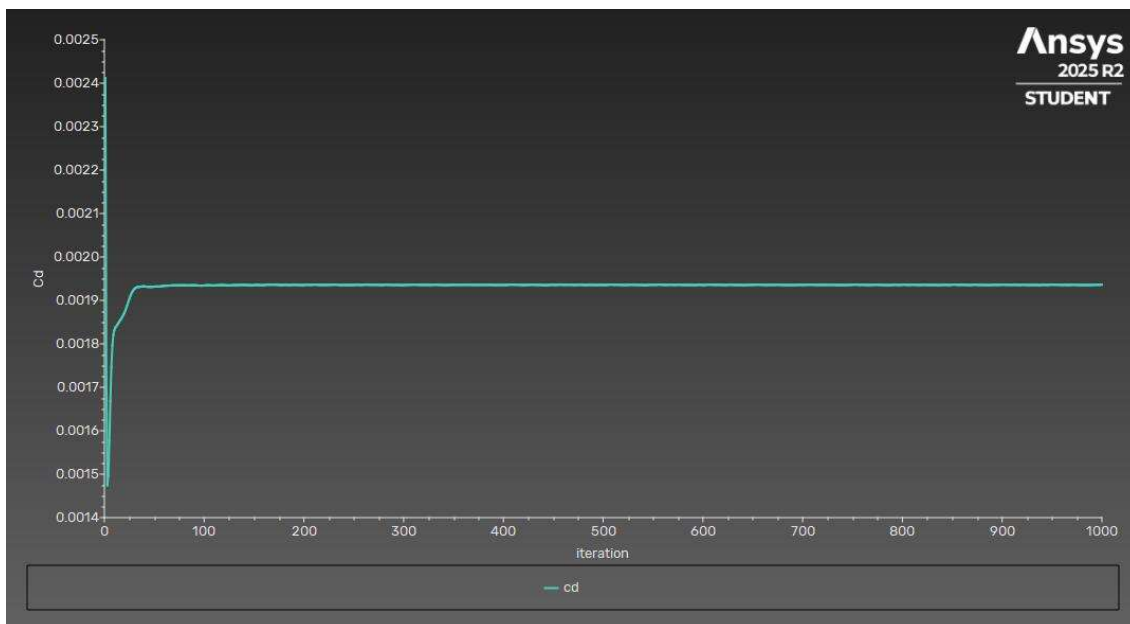
Mach1

Stable: 0.00553



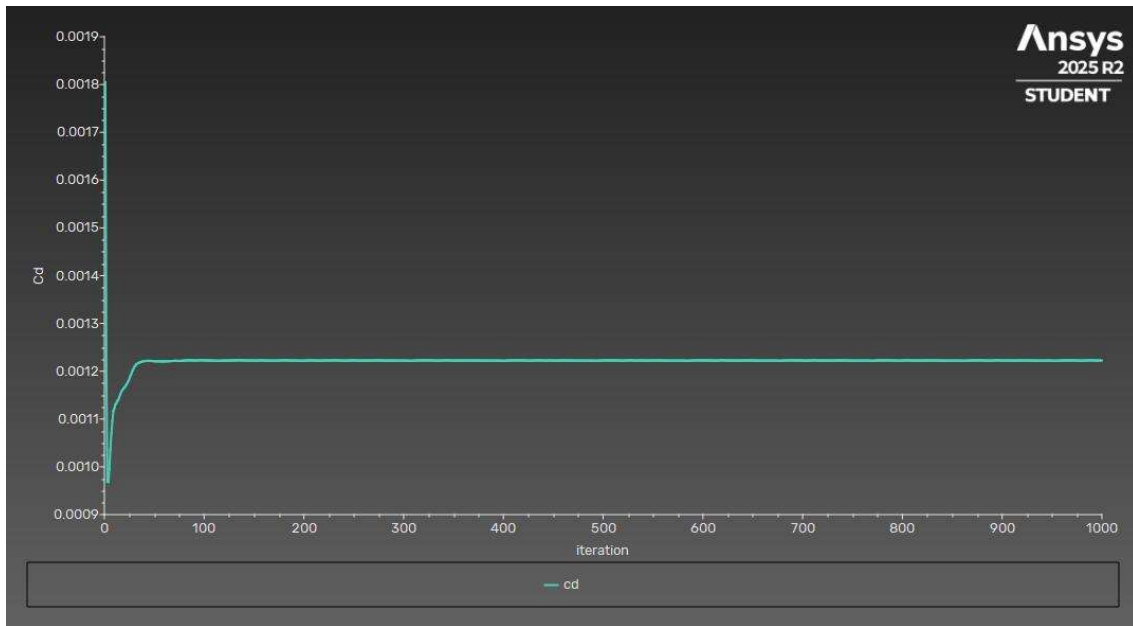
Mach2

Stable: 0.00193



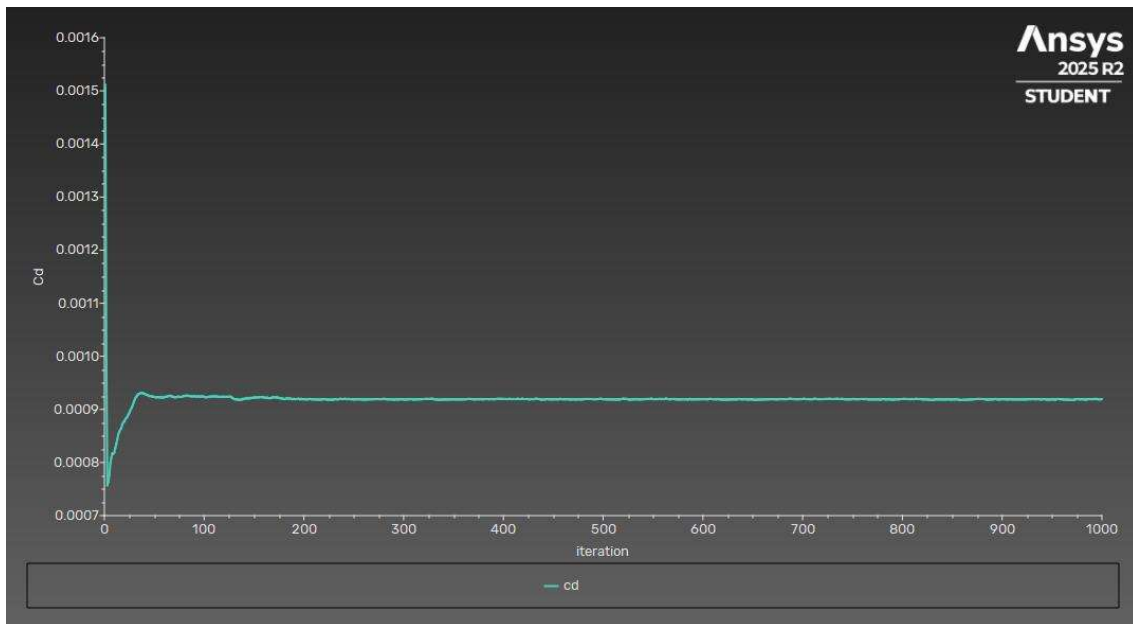
Mach3

Stable: 0.00122



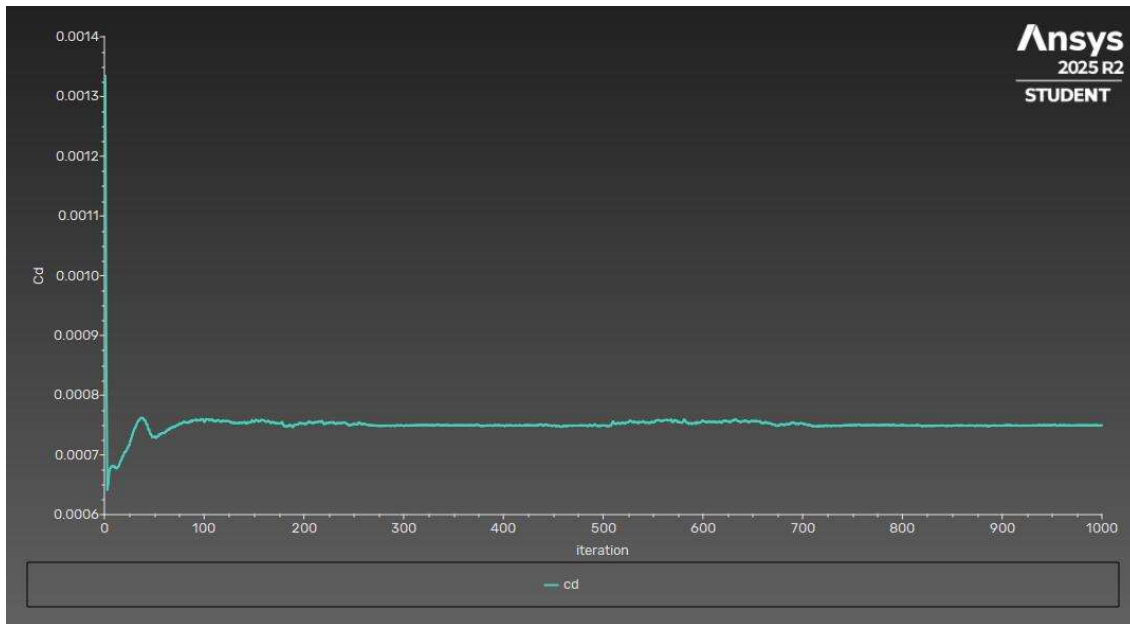
Mach4

Stable: 0.000919



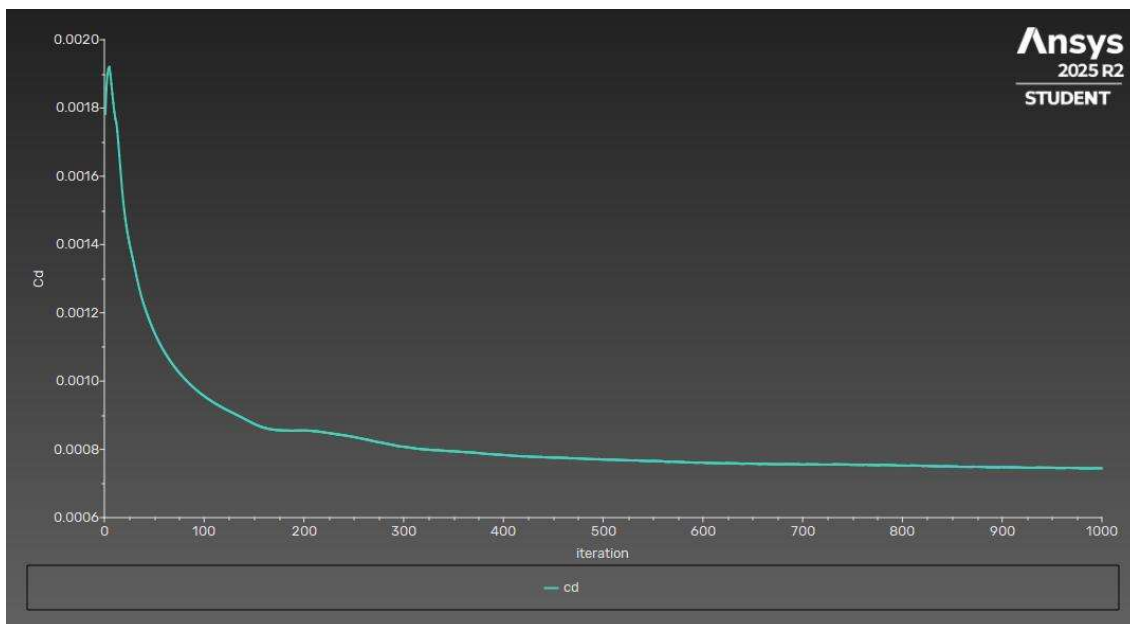
Mach5

Stable: 0.000749



Mach6

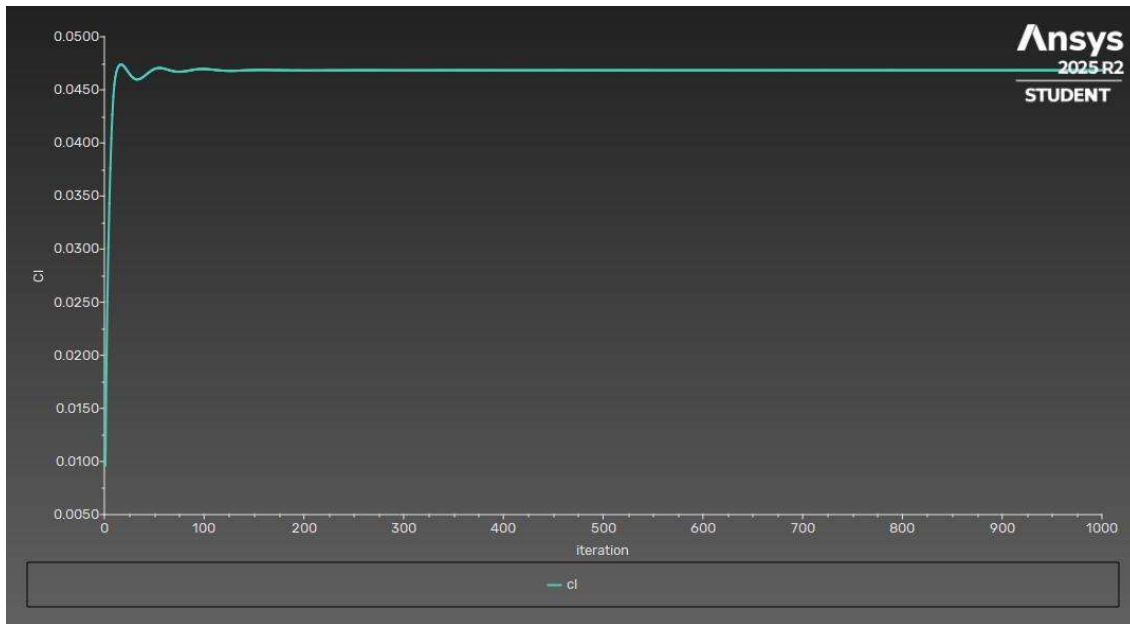
Stable: 0.000744



Coefficient of Lift:

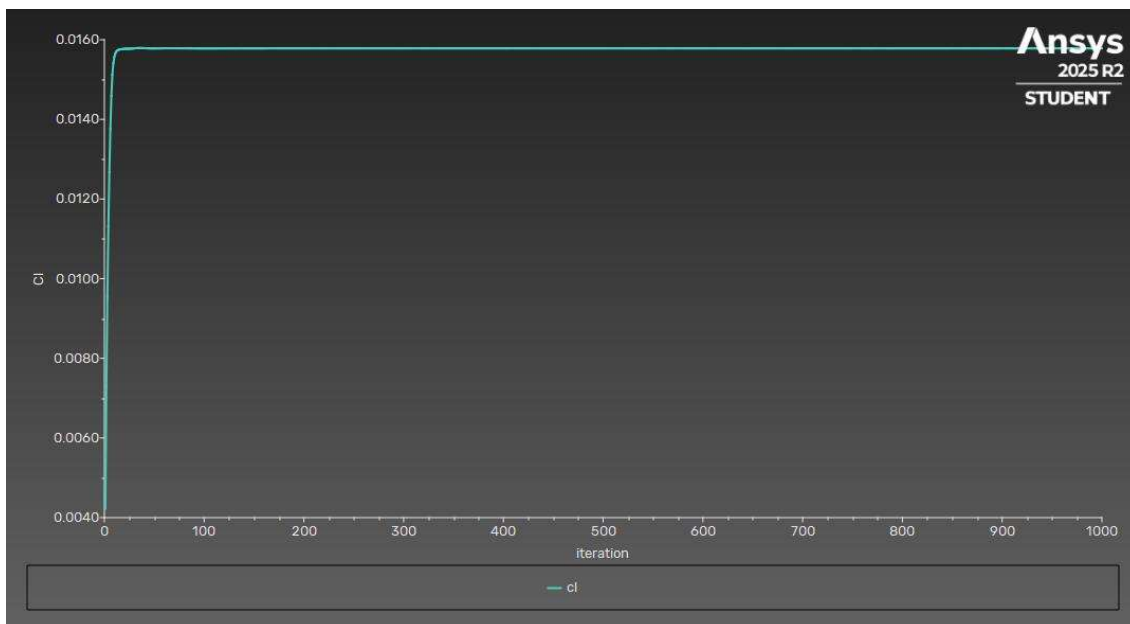
Mach1

Stable: 0.0468



Mach2

Stable: 0.0157



Mach3

Stable: 0.00971



Mach4

Stable: 0.00716



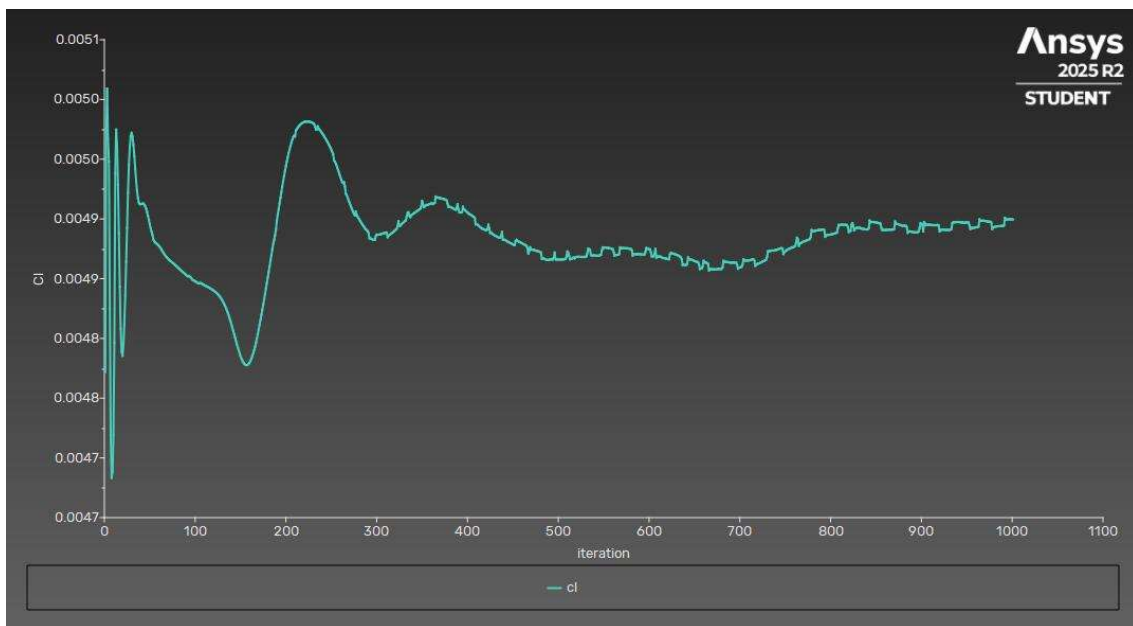
Mach5

Stable: 0.00573



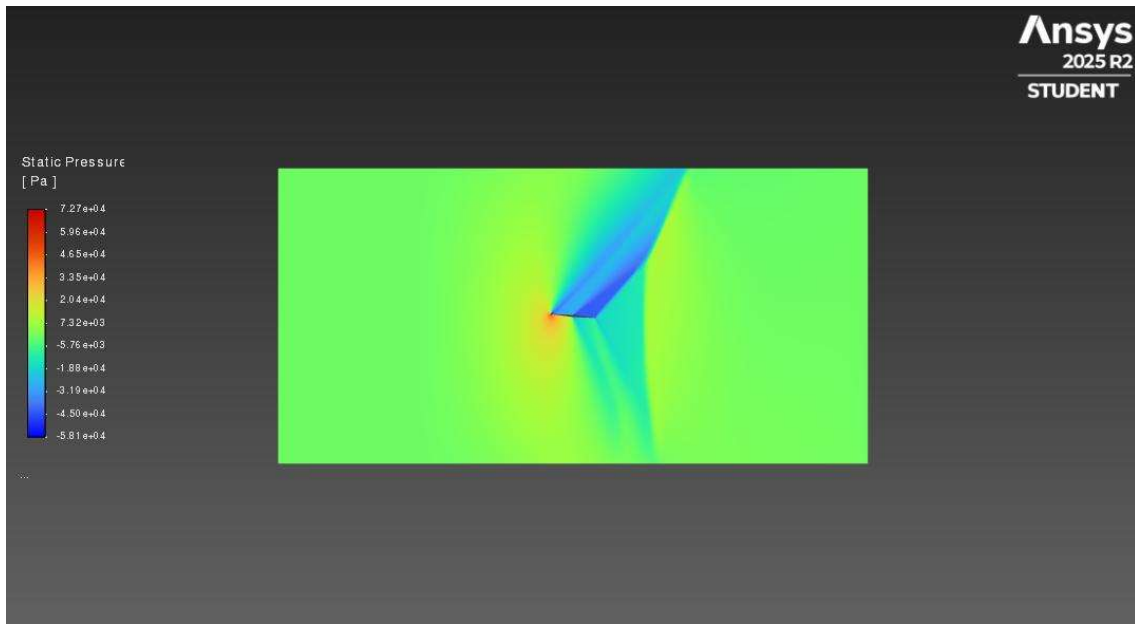
Mach6

Stable: 0.00489

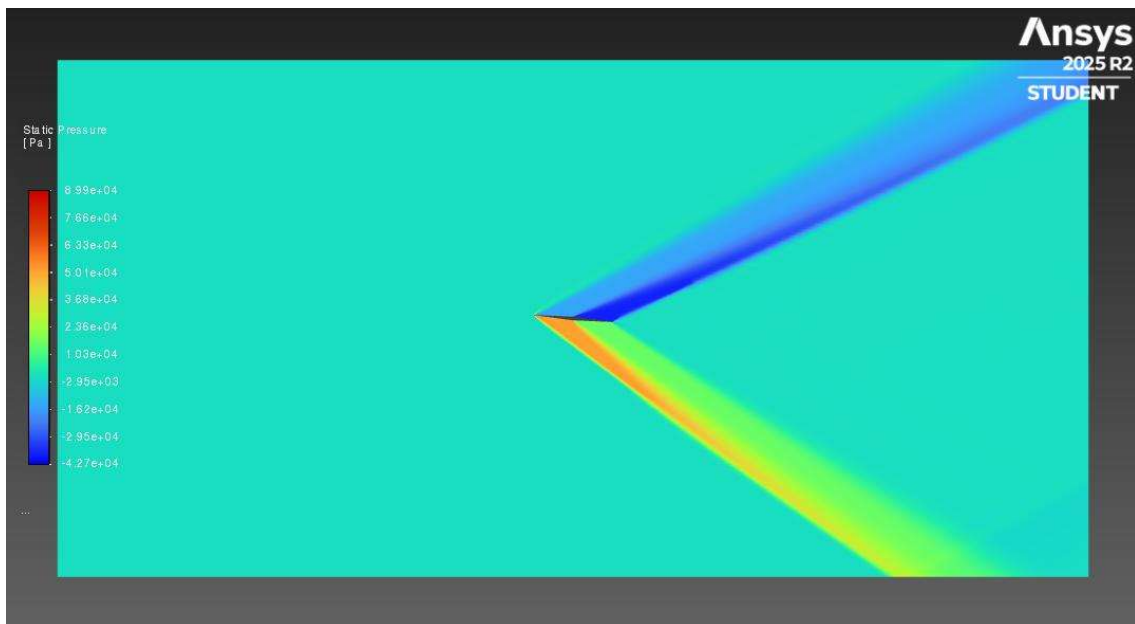


Pressure:

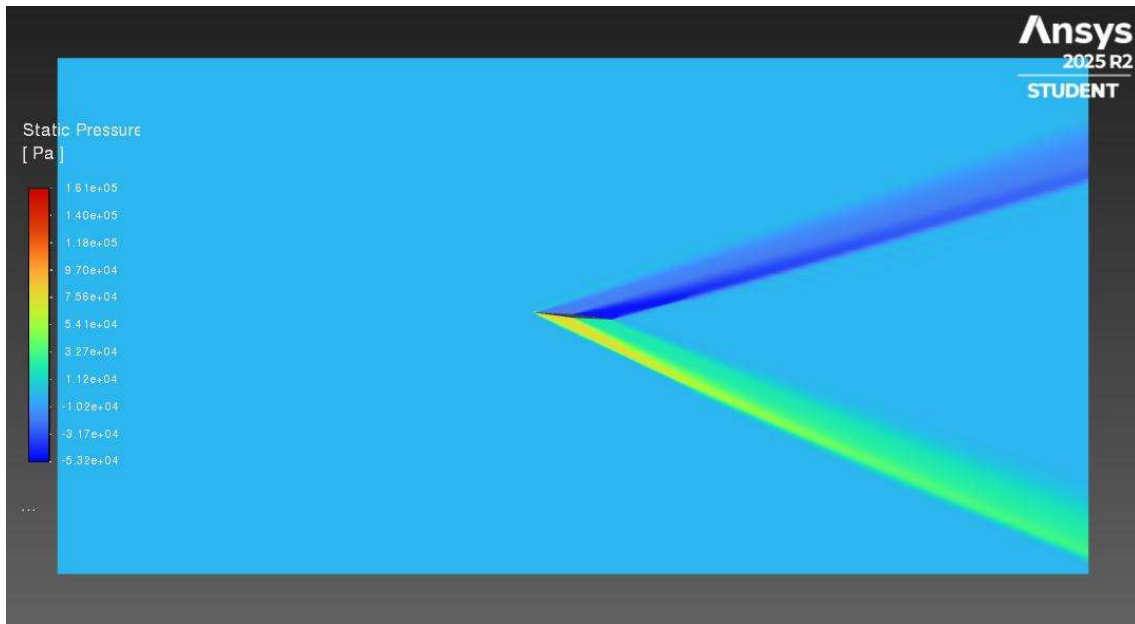
Mach1



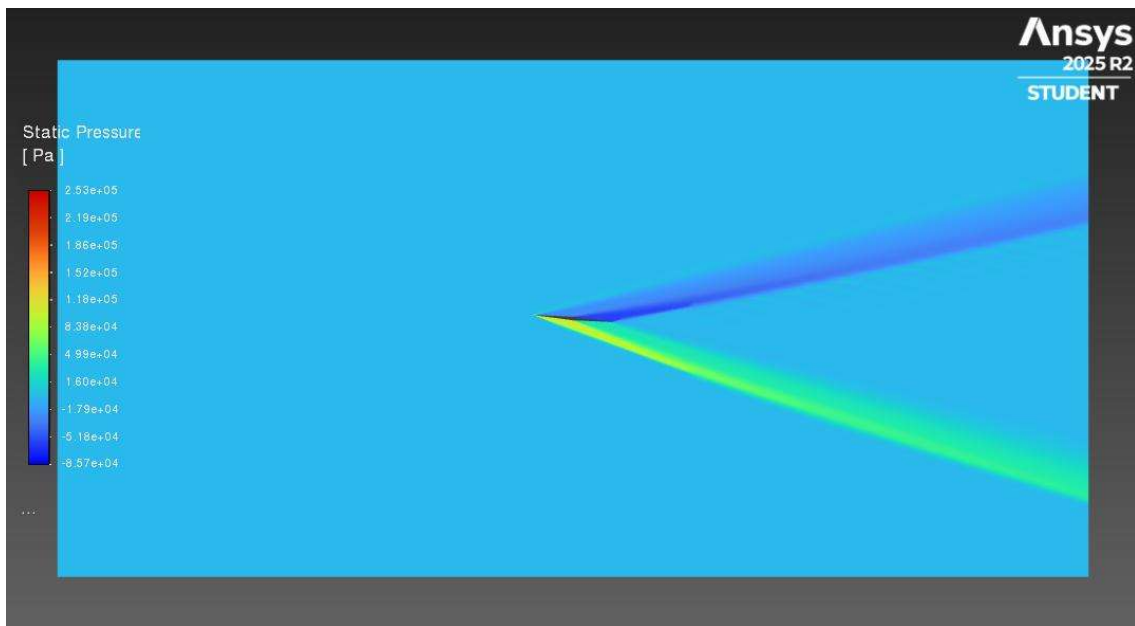
Mach2



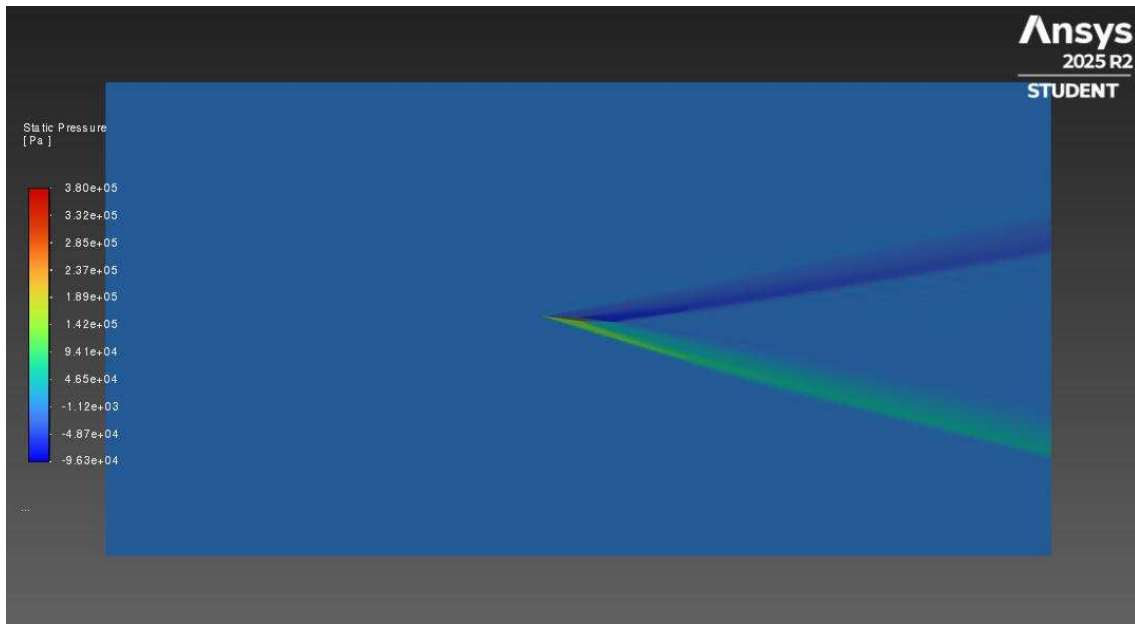
Mach3



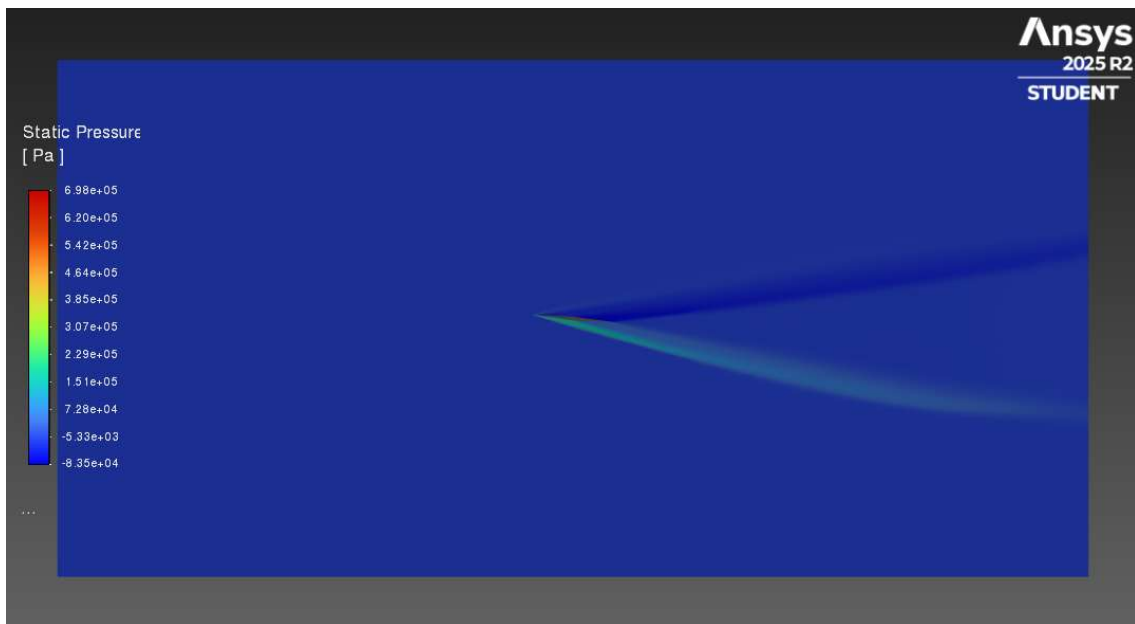
Mach4



Mach5

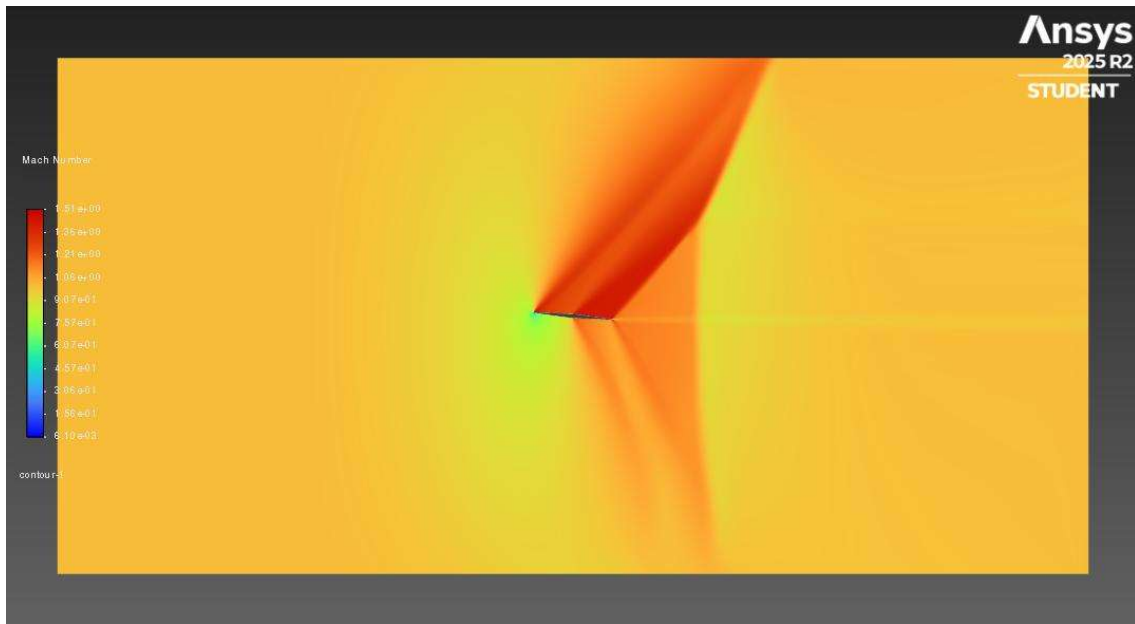


Mach6

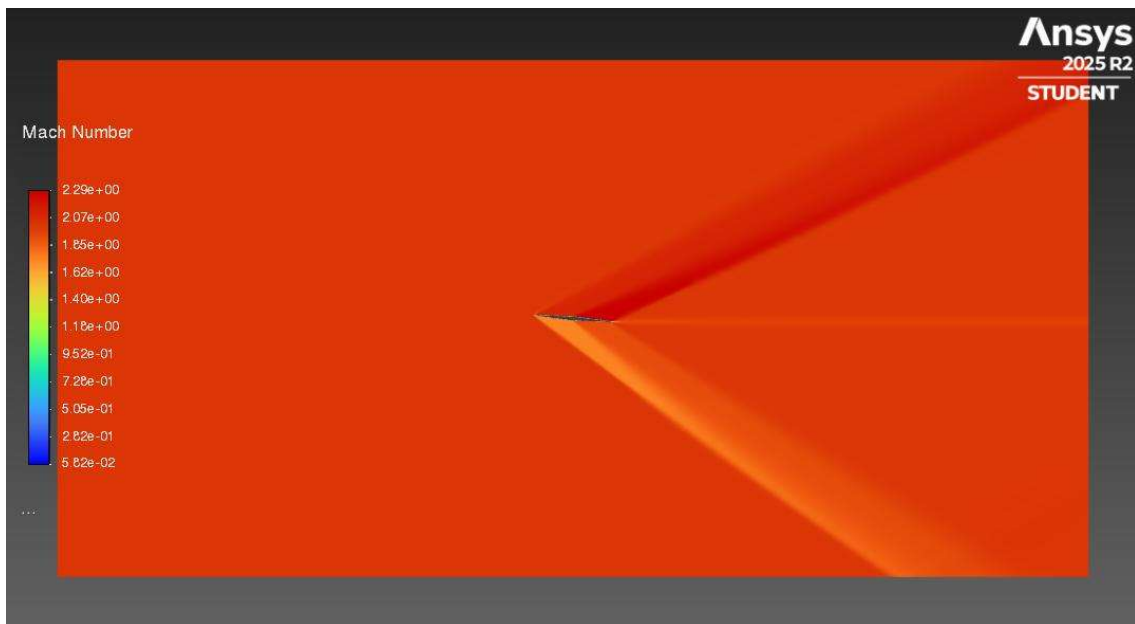


Mach:

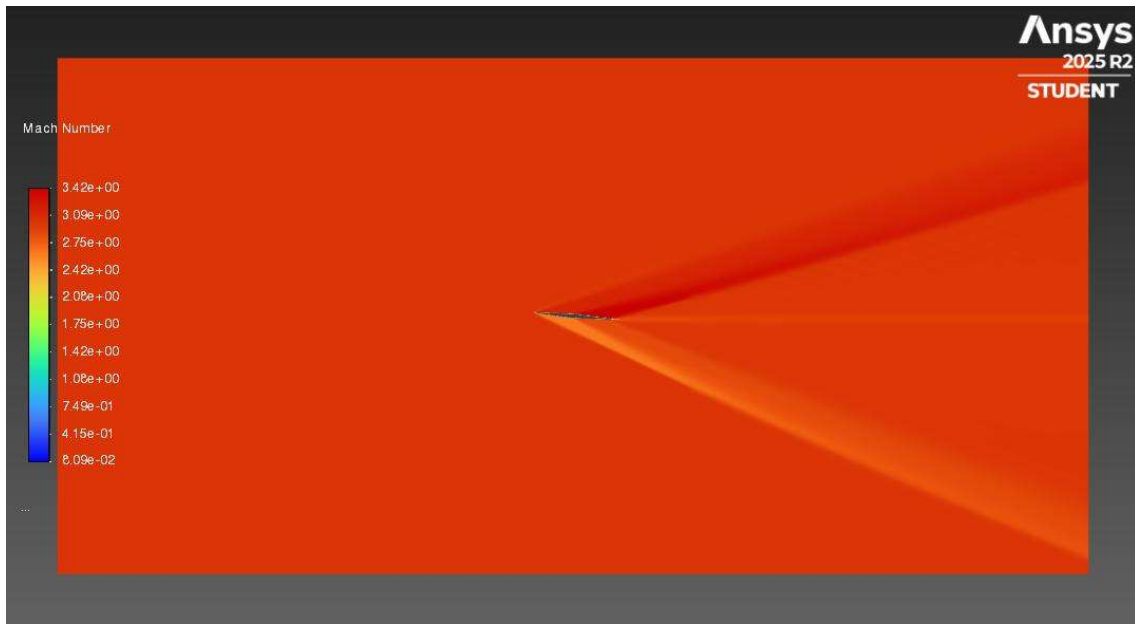
Mach1



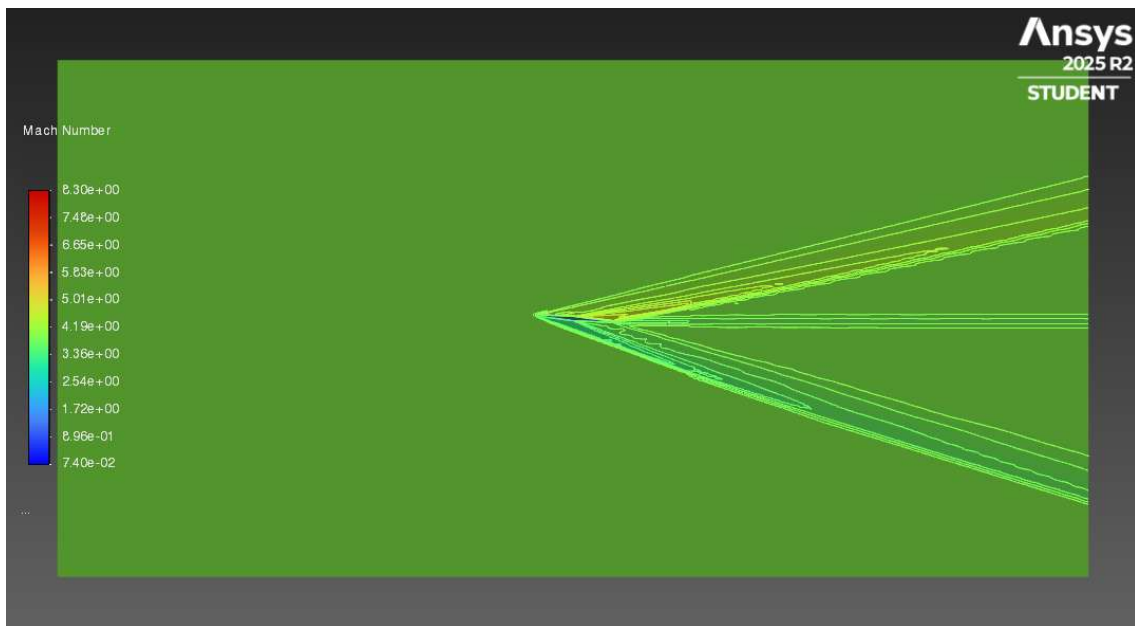
Mach2



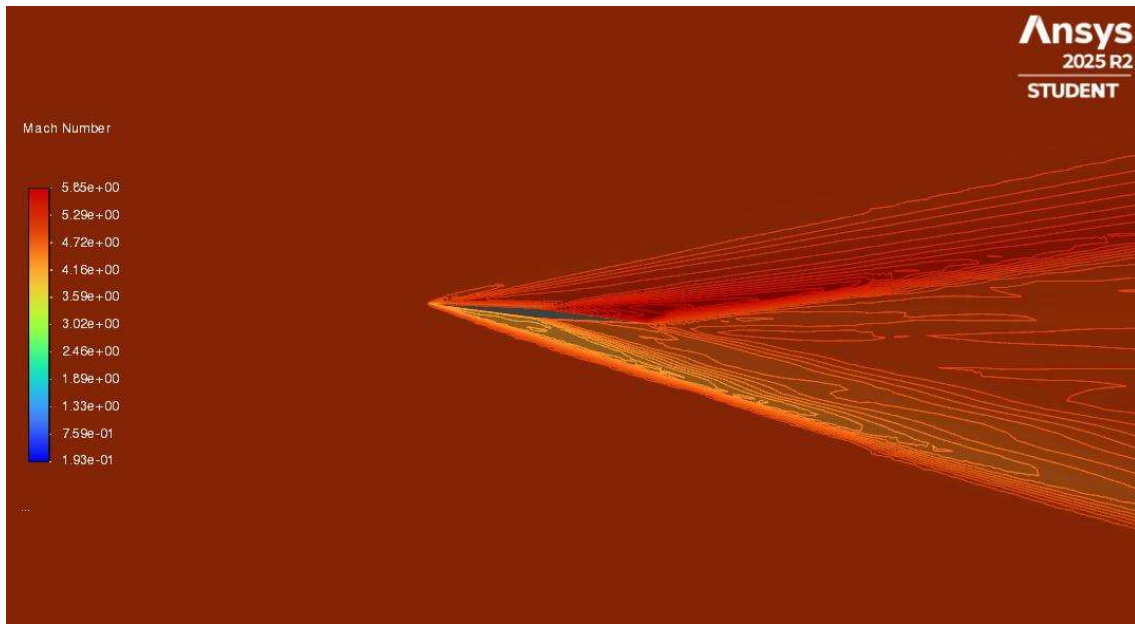
Mach3



Mach4



Mach5



Mach6

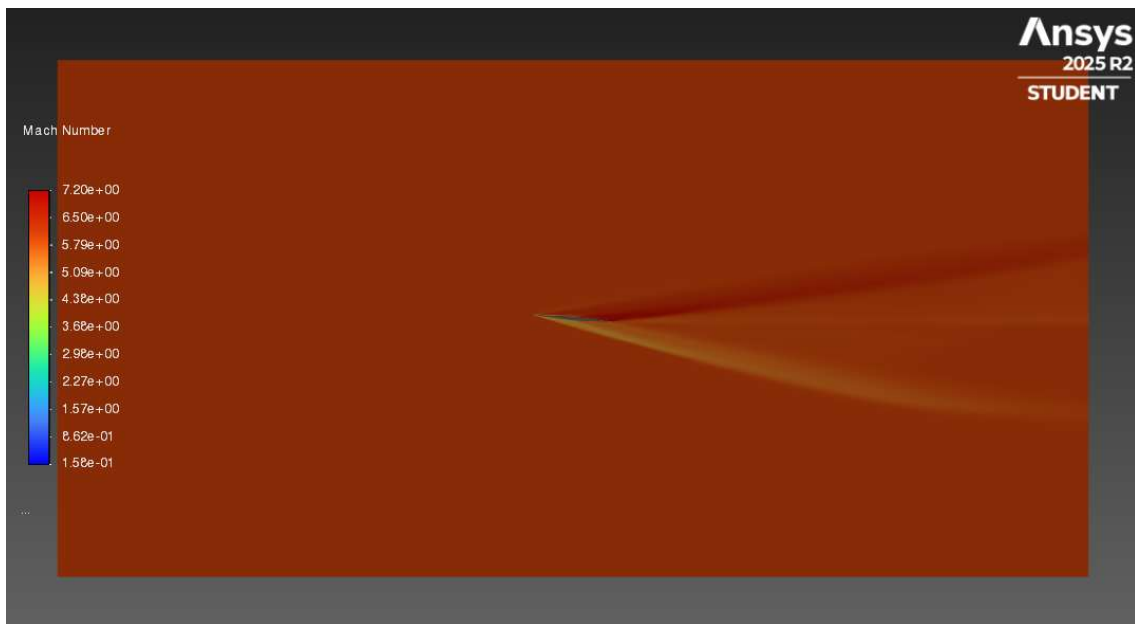
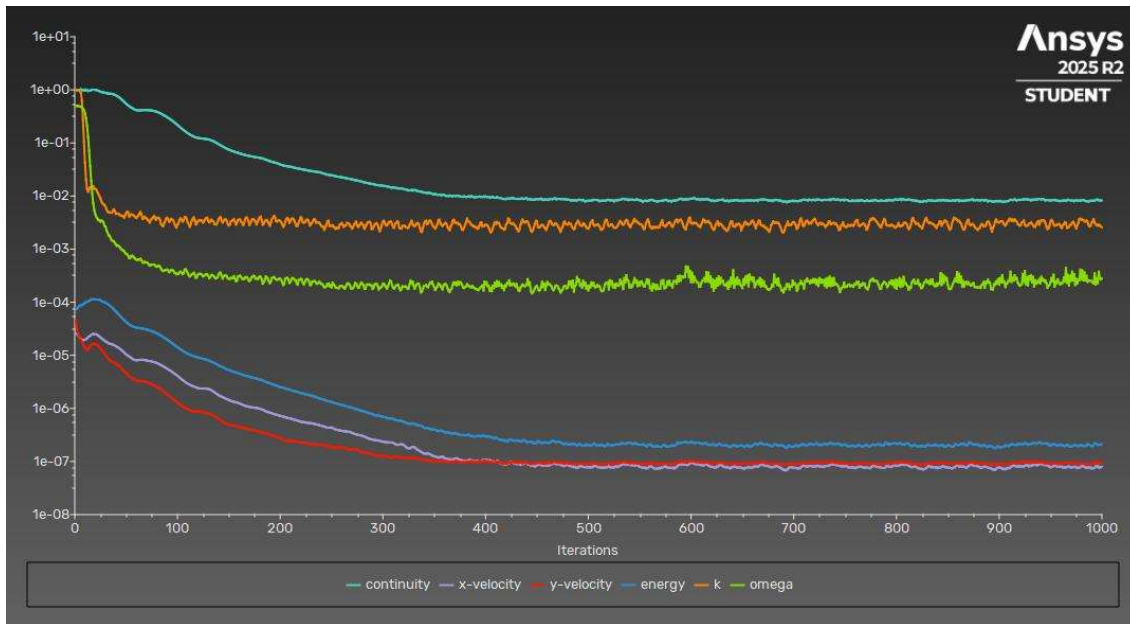


Figure 2. B

Scaled residuals:

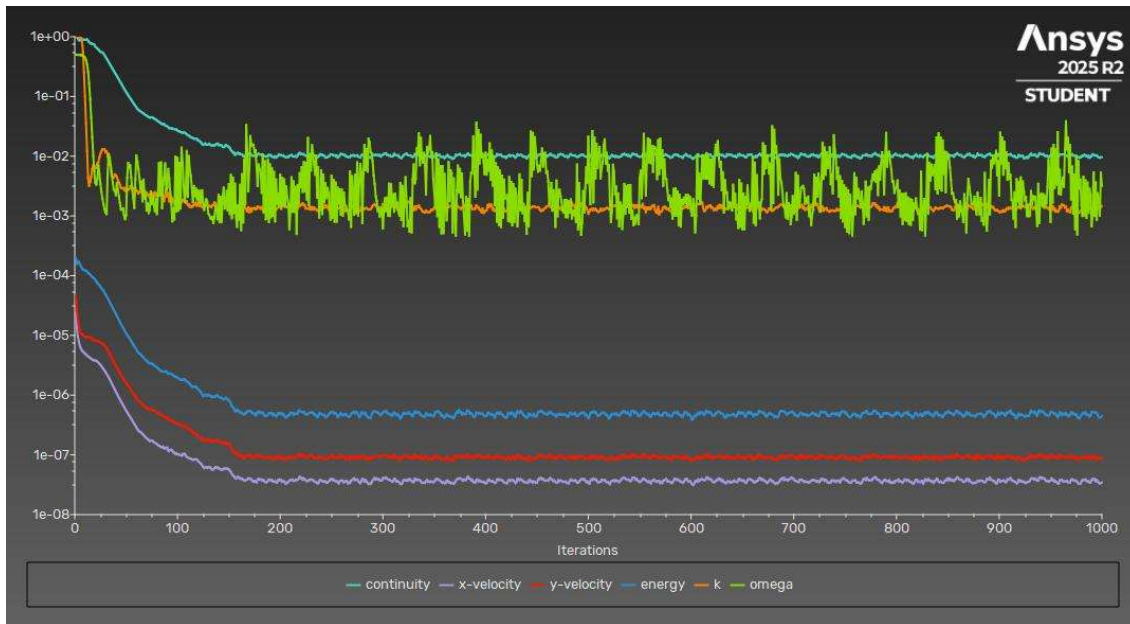
Mach1



Mach2



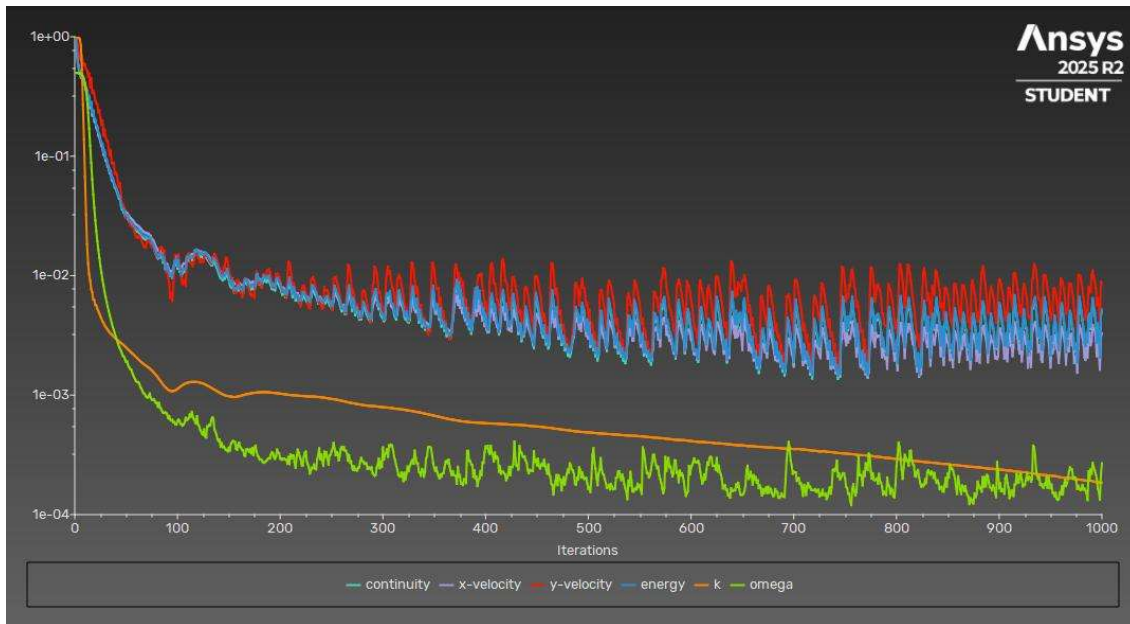
Mach3



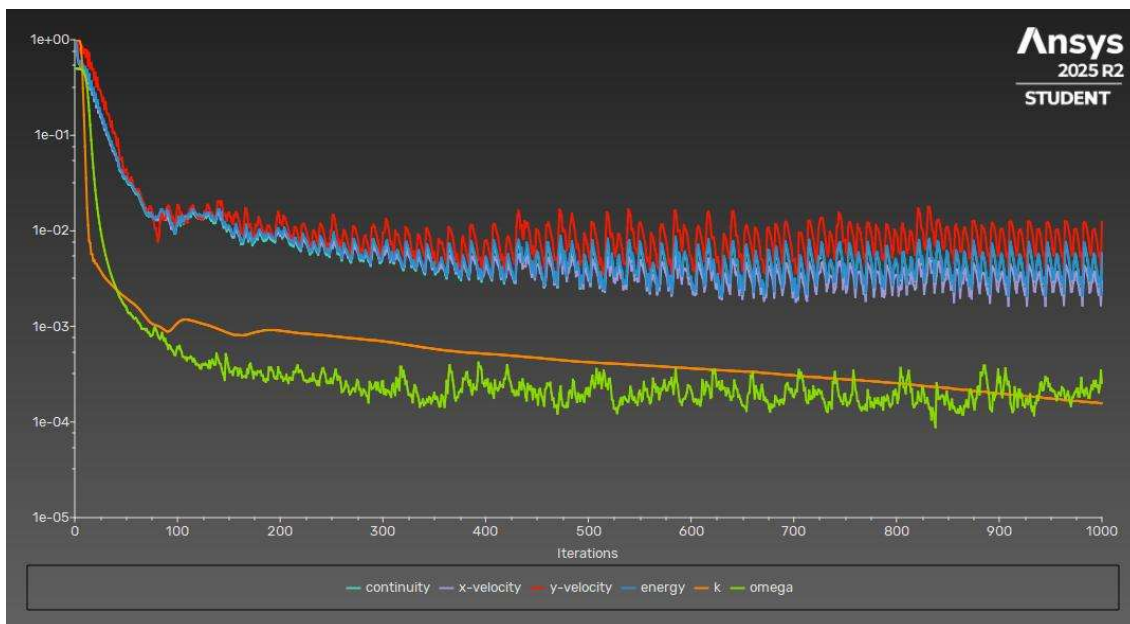
Mach4



Mach5



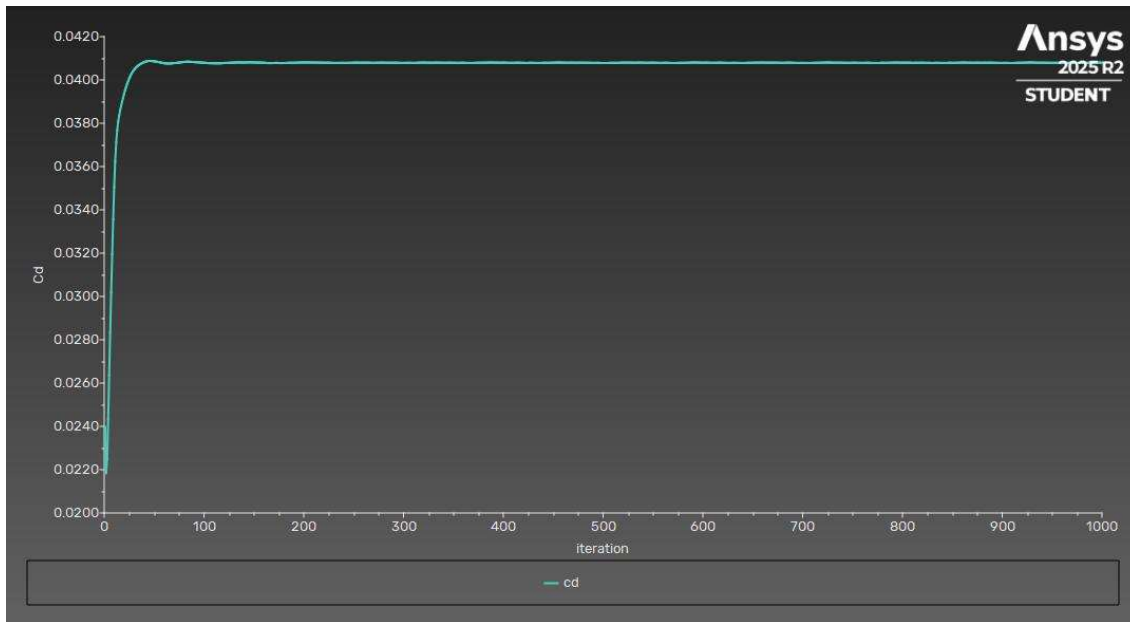
Mach6



Coefficient of Drag:

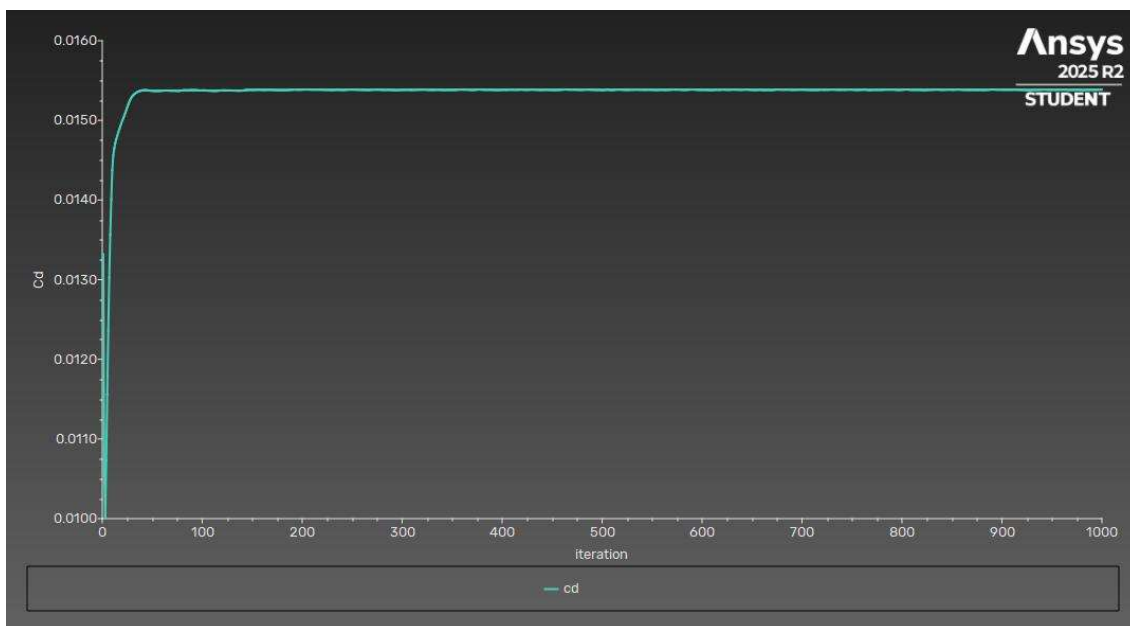
Mach1

Stable: 0.0407



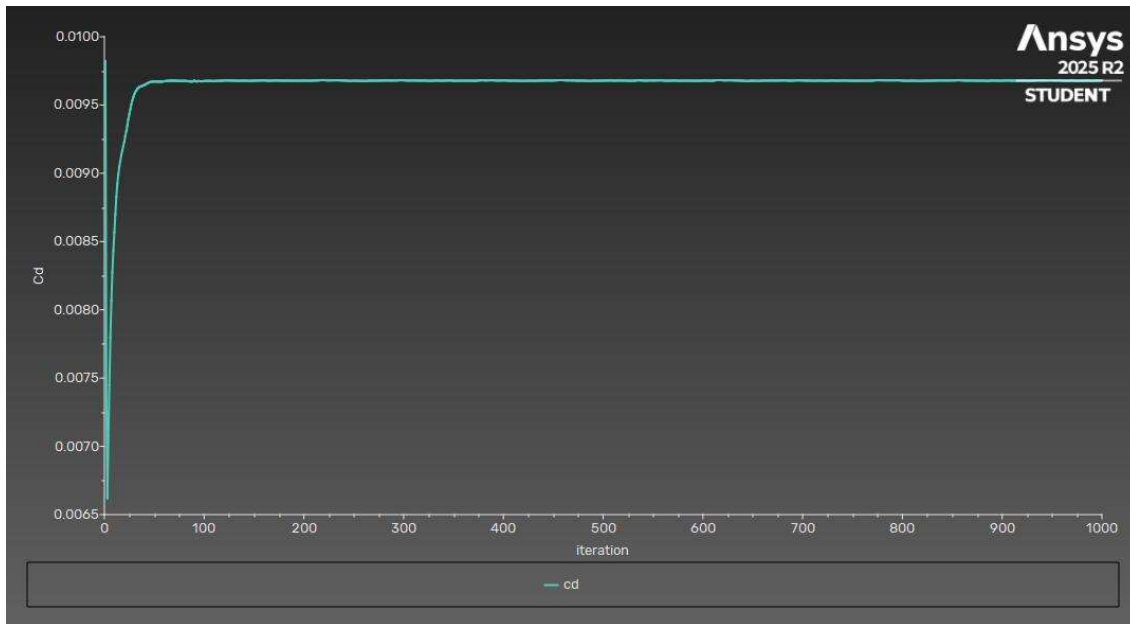
Mach2

Stable: 0.0153



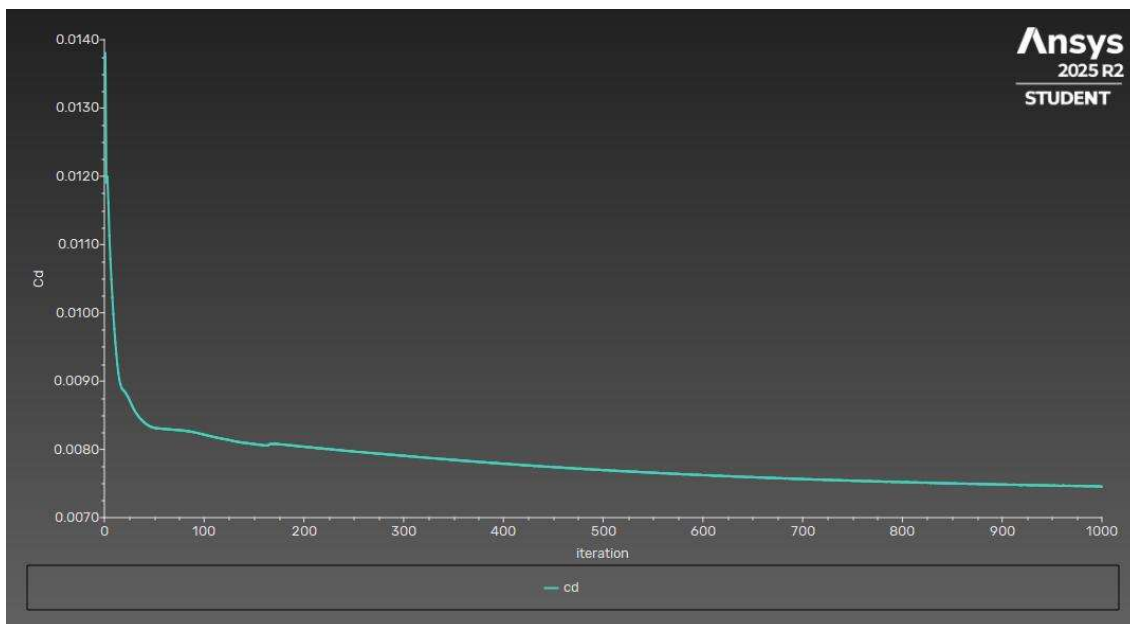
Mach3

Stable: 0.00967



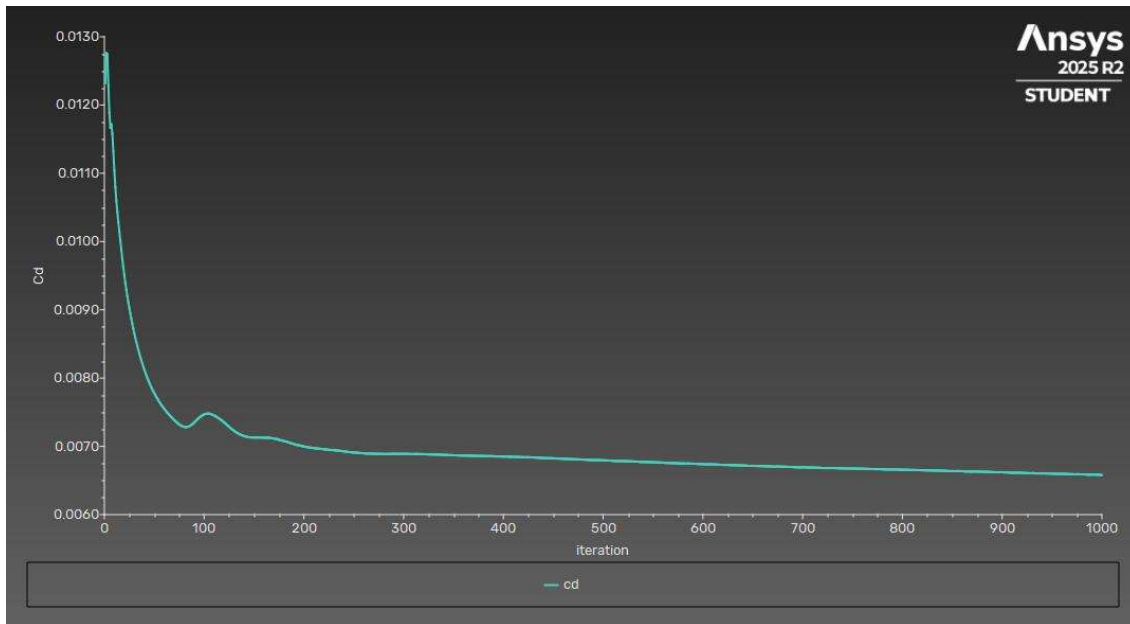
Mach4

Stable: 0.00746



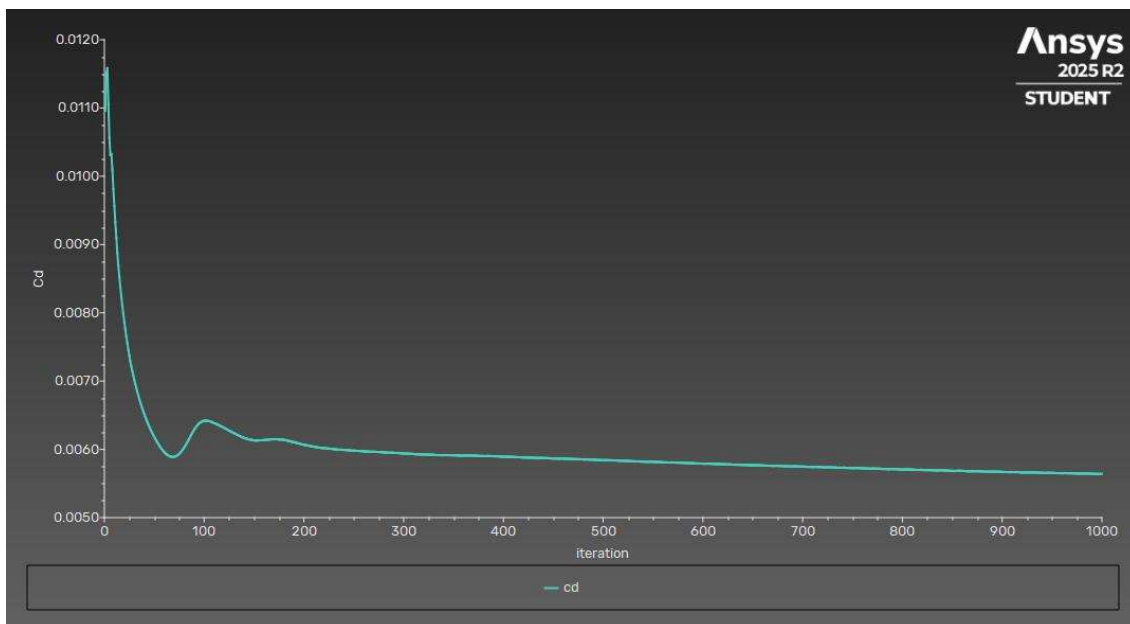
Mach5

Stable: 0.00658



Mach6

Stable: 0.00564



Coefficient of Lift:

Mach1

Stable: 0.213



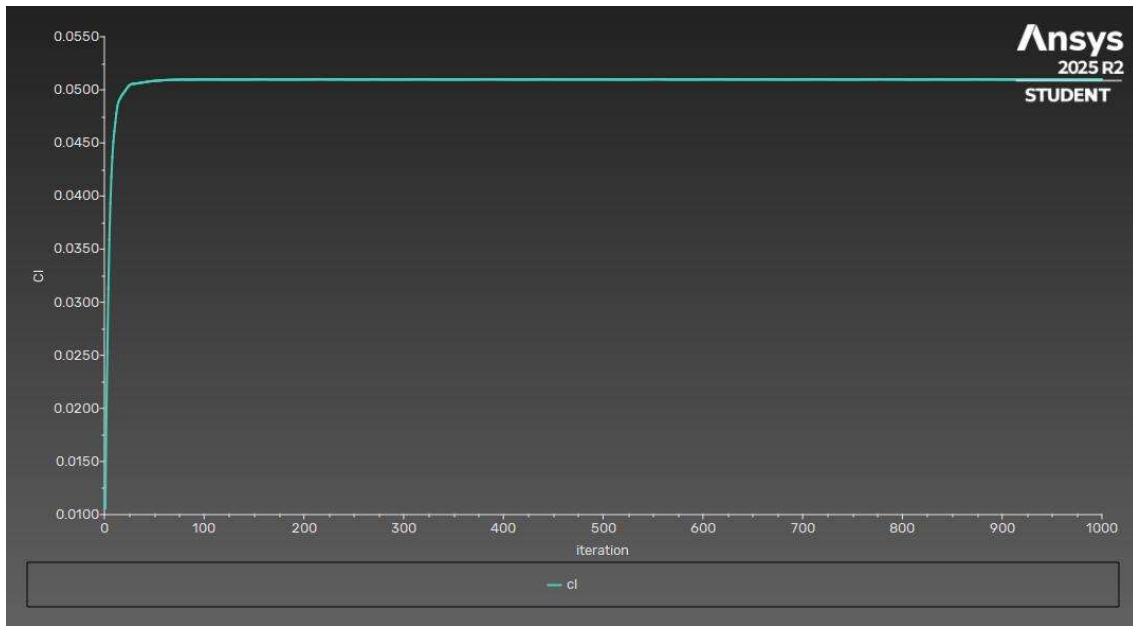
Mach2

Stable: 0.082



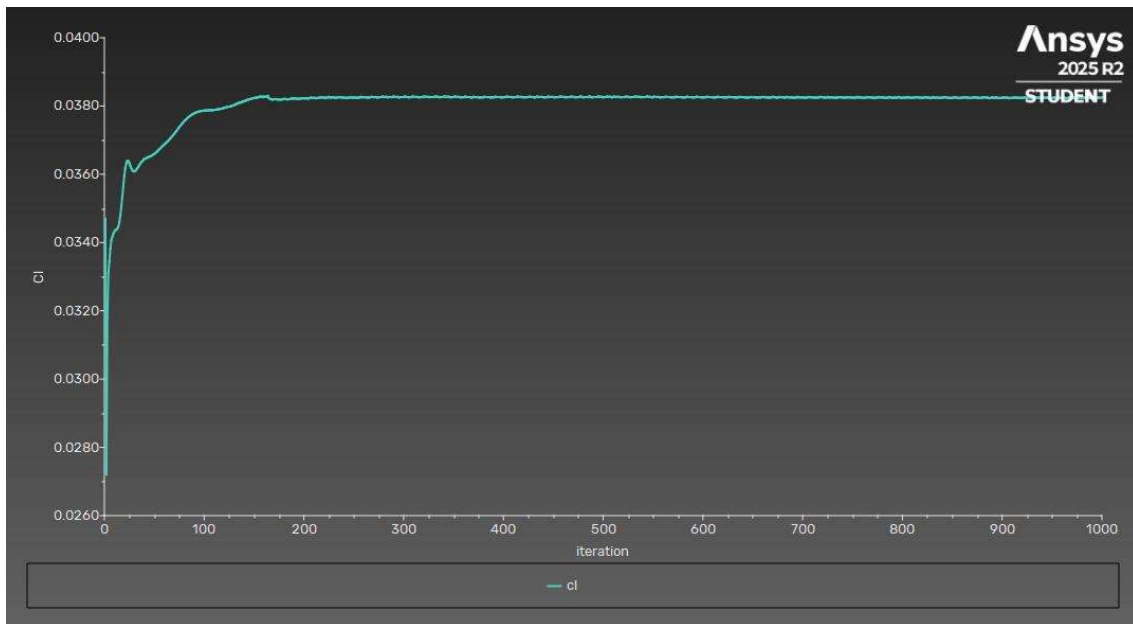
Mach3

Stable: 0.0509



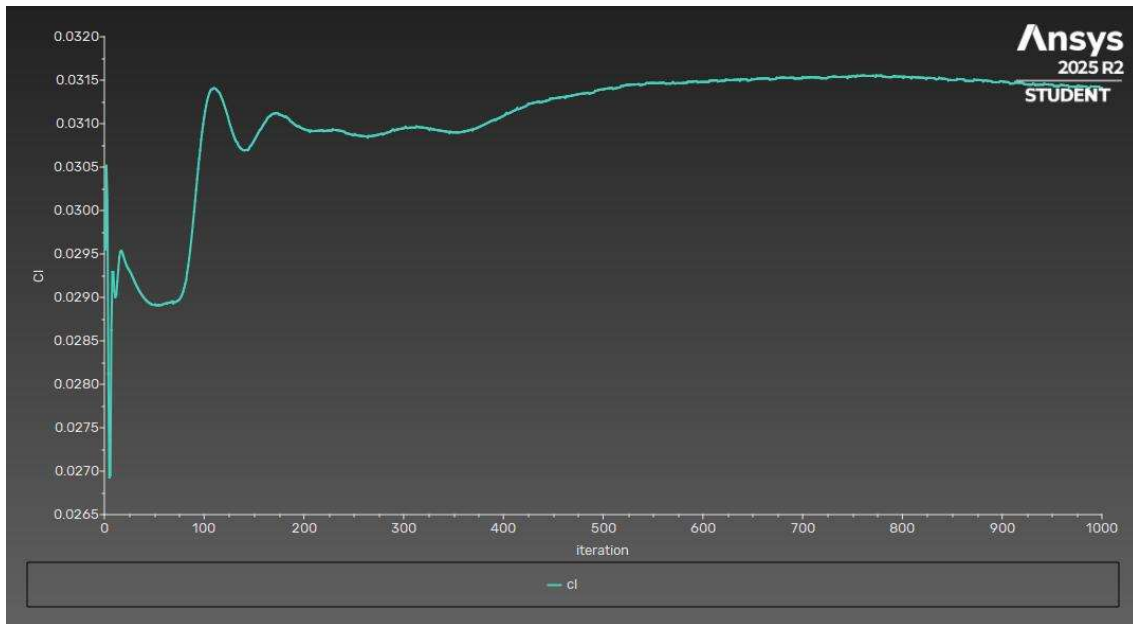
Mach4

Stable: 0.0382



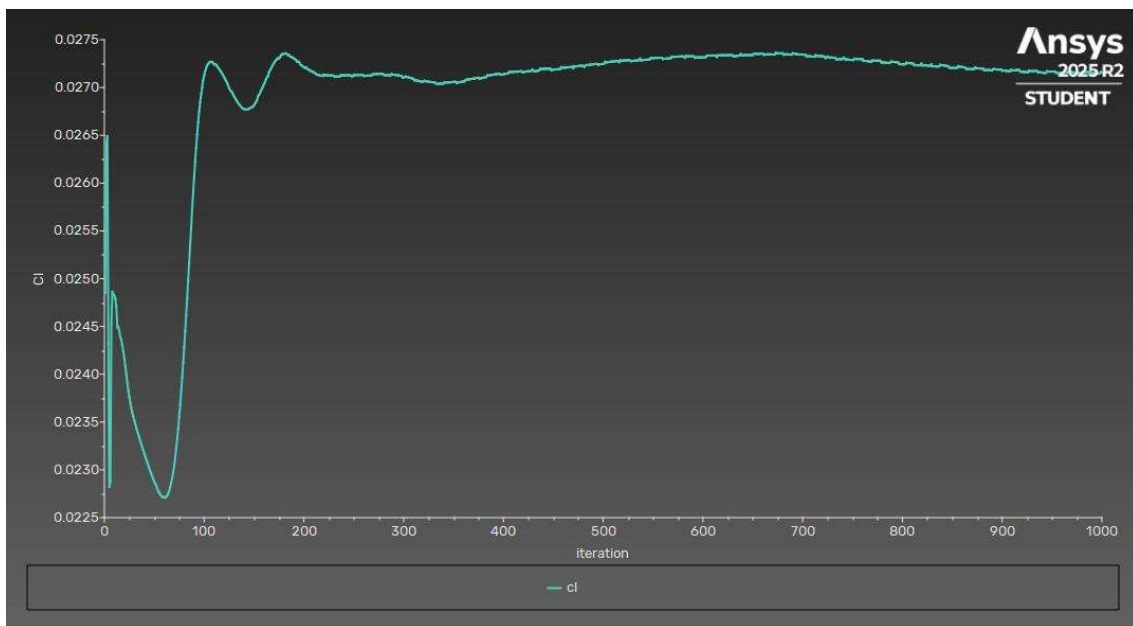
Mach5

Stable: 0.0314



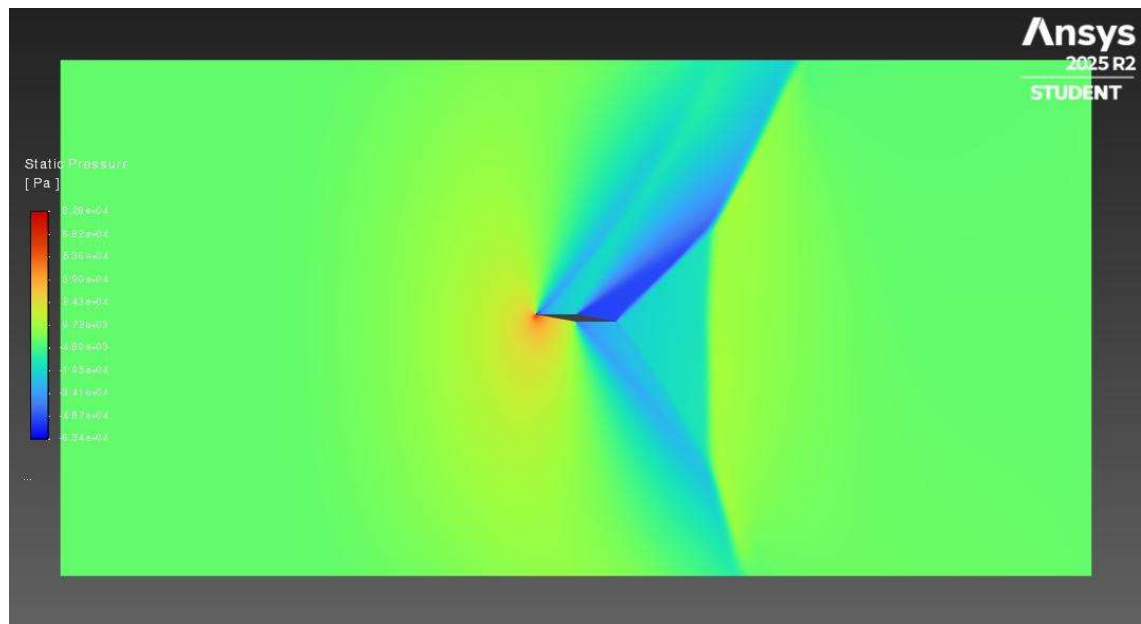
Mach6

Stable: 0.0271

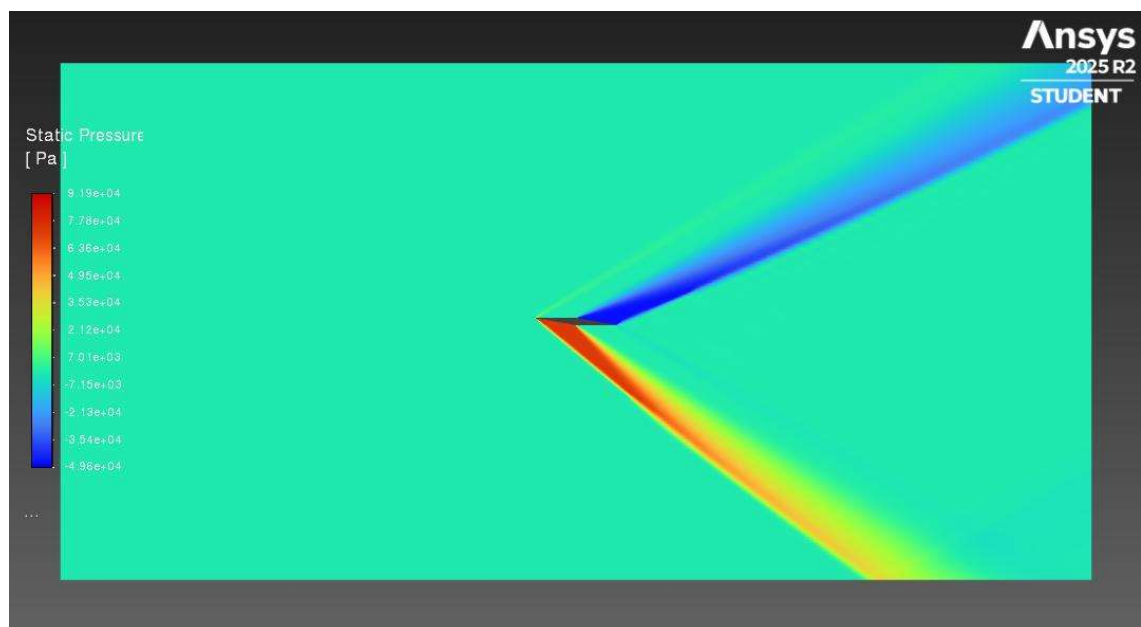


Pressure:

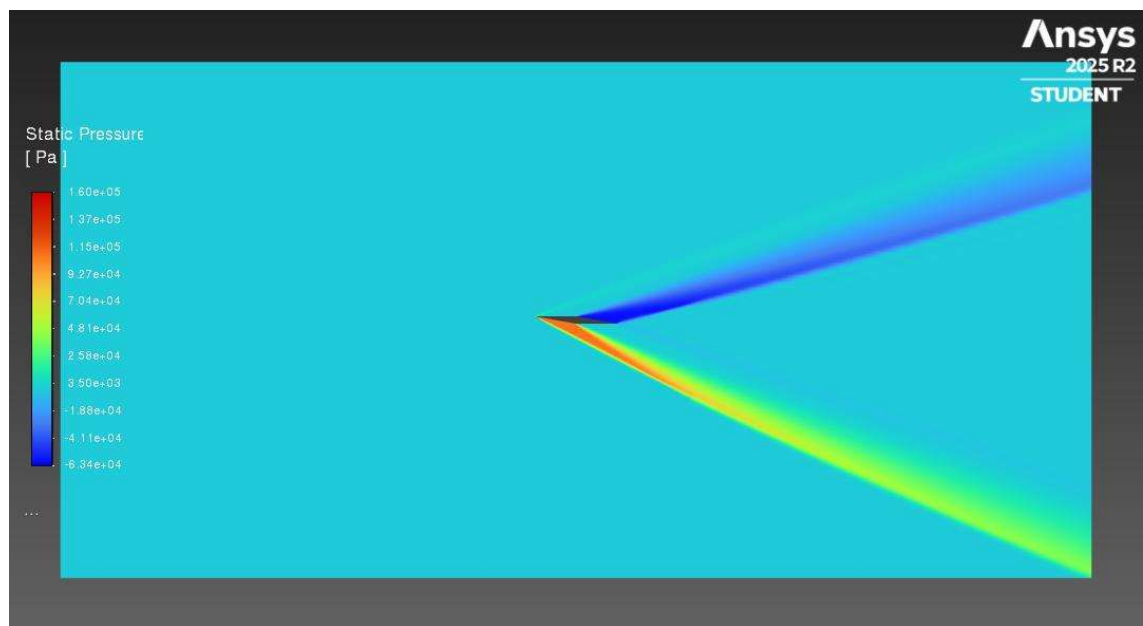
Mach1



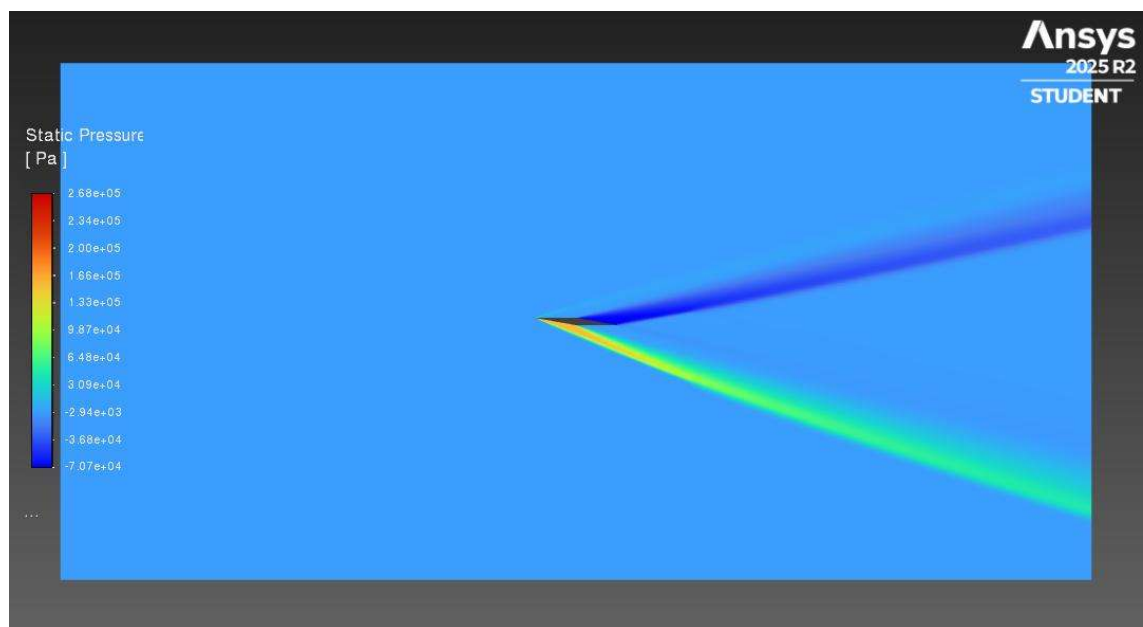
Mach2



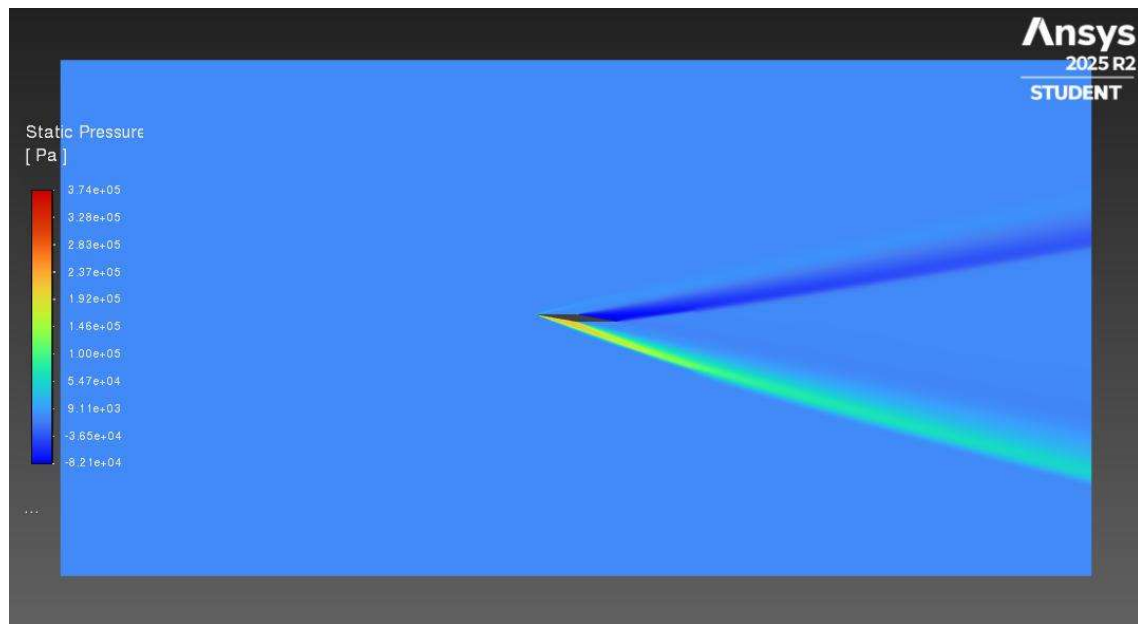
Mach3



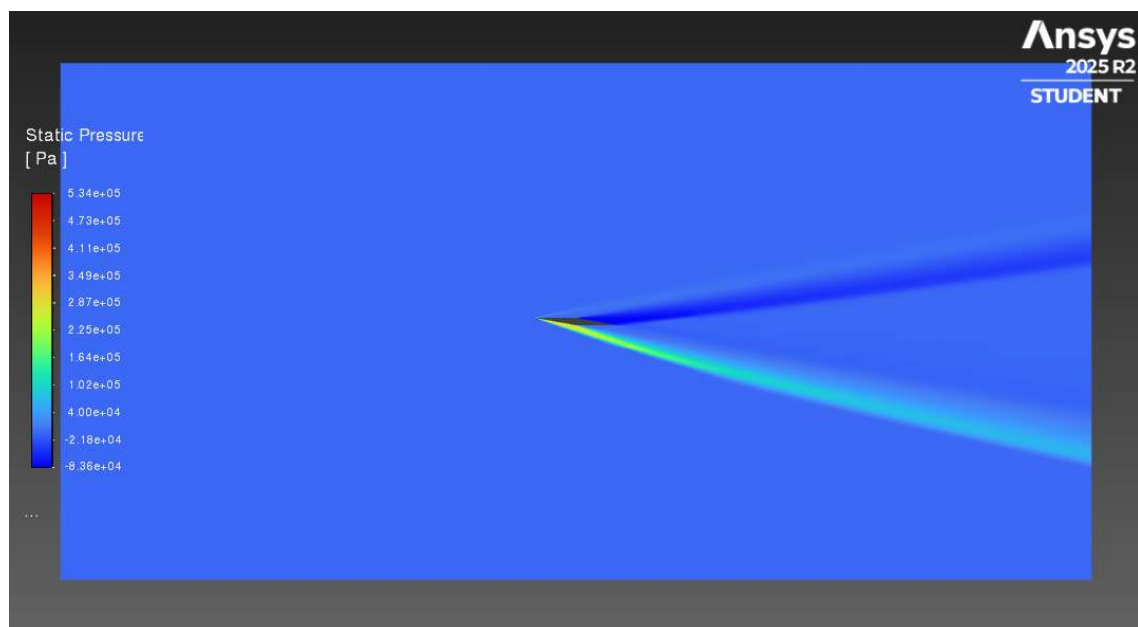
Mach4



Mach5

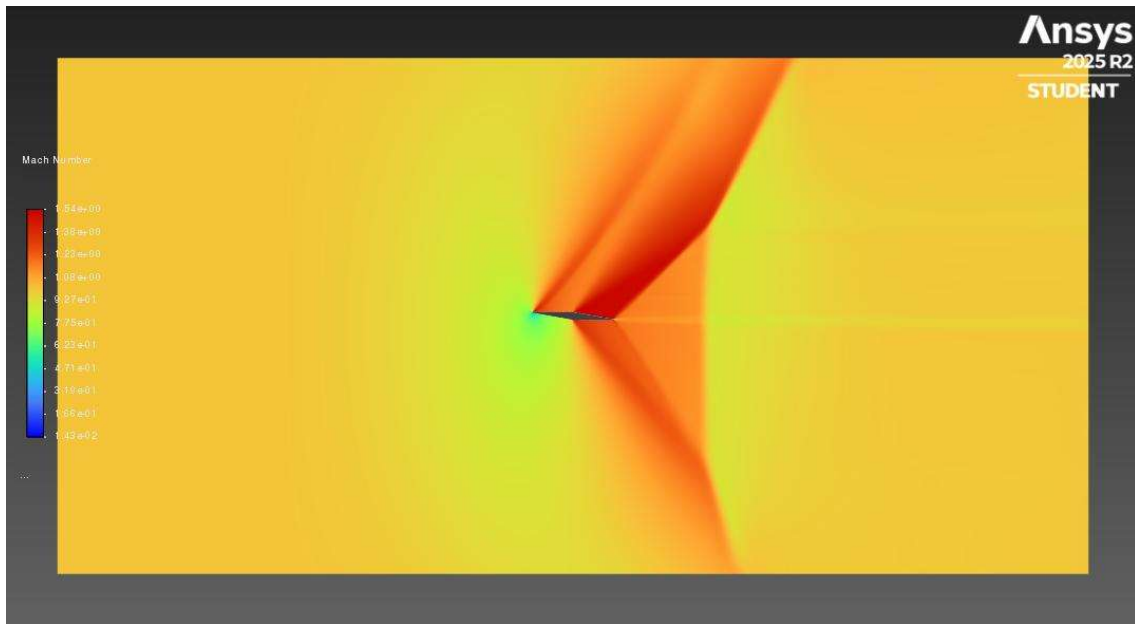


Mach6

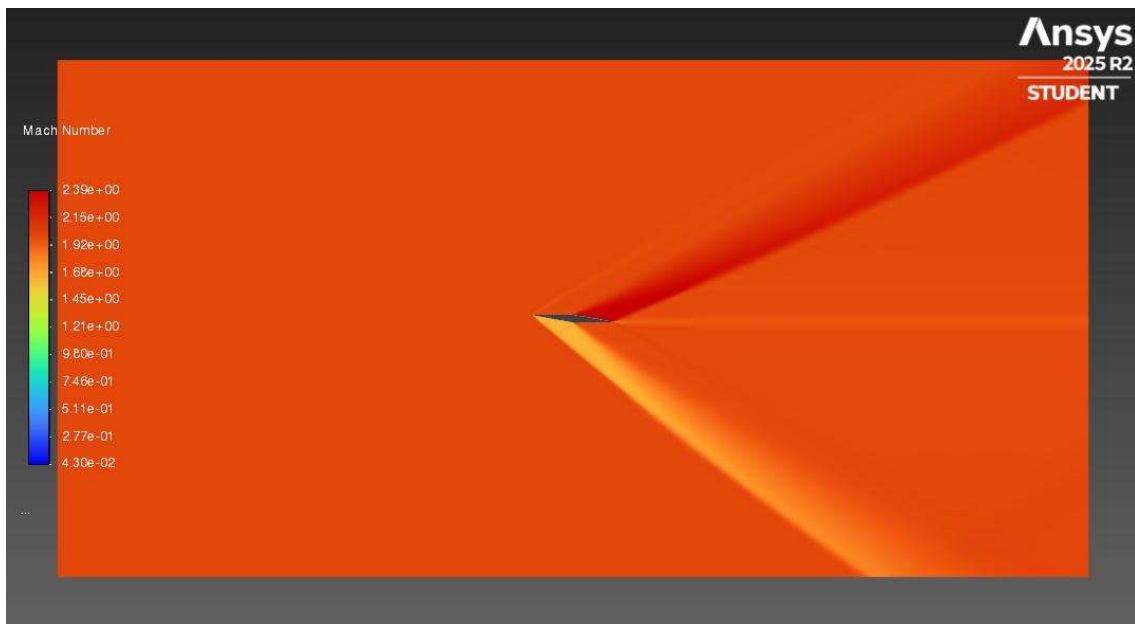


Mach:

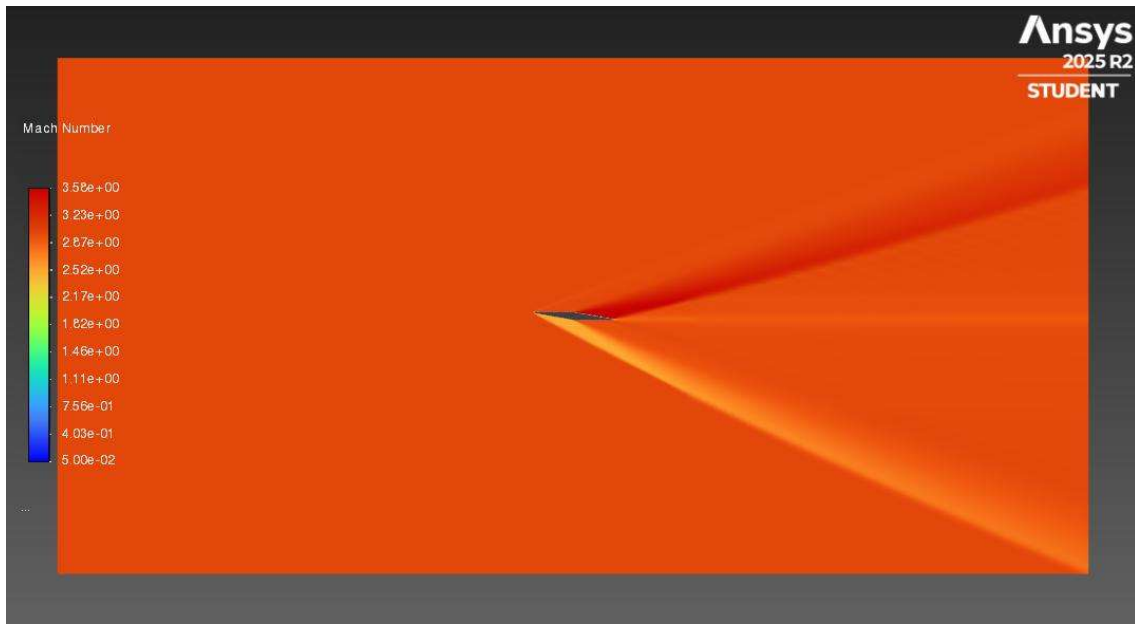
Mach1



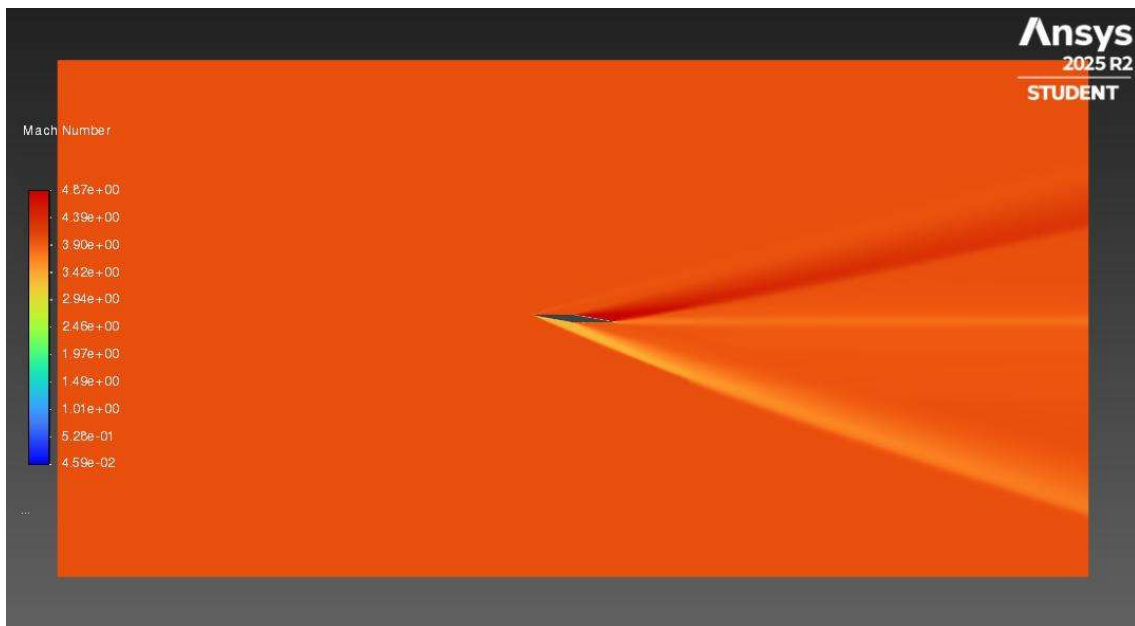
Mach2



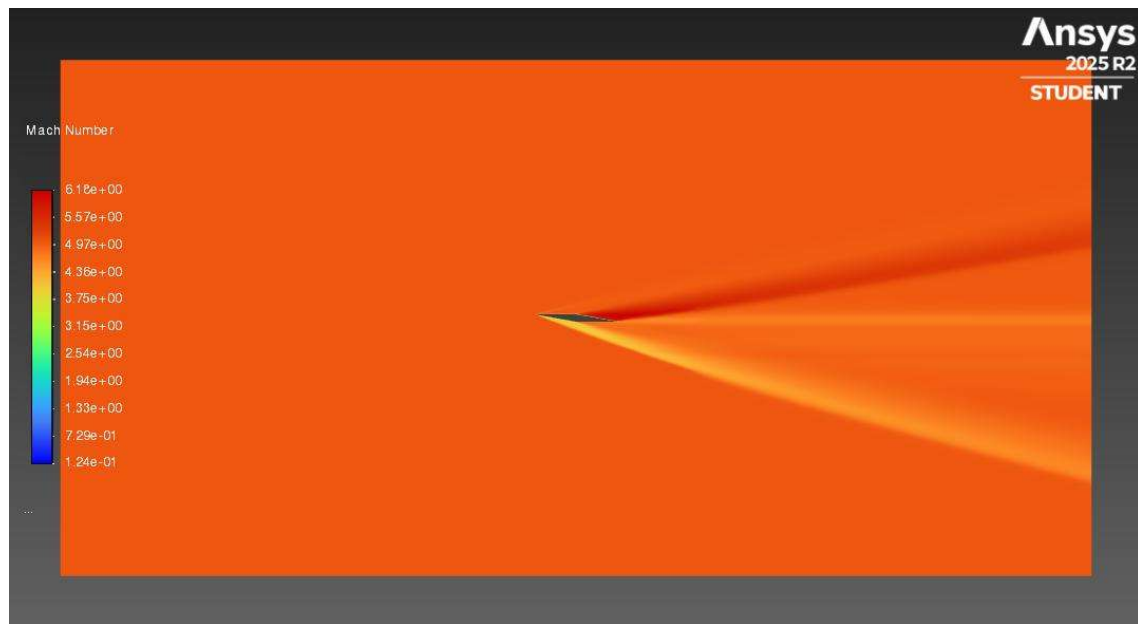
Mach3



Mach4



Mach5



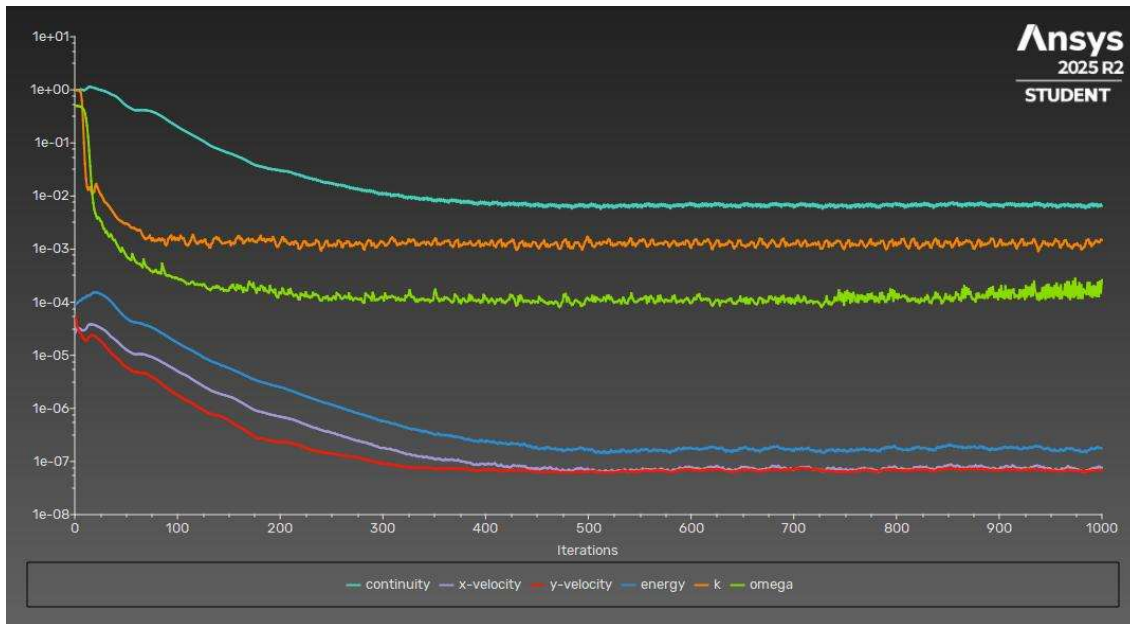
Mach6



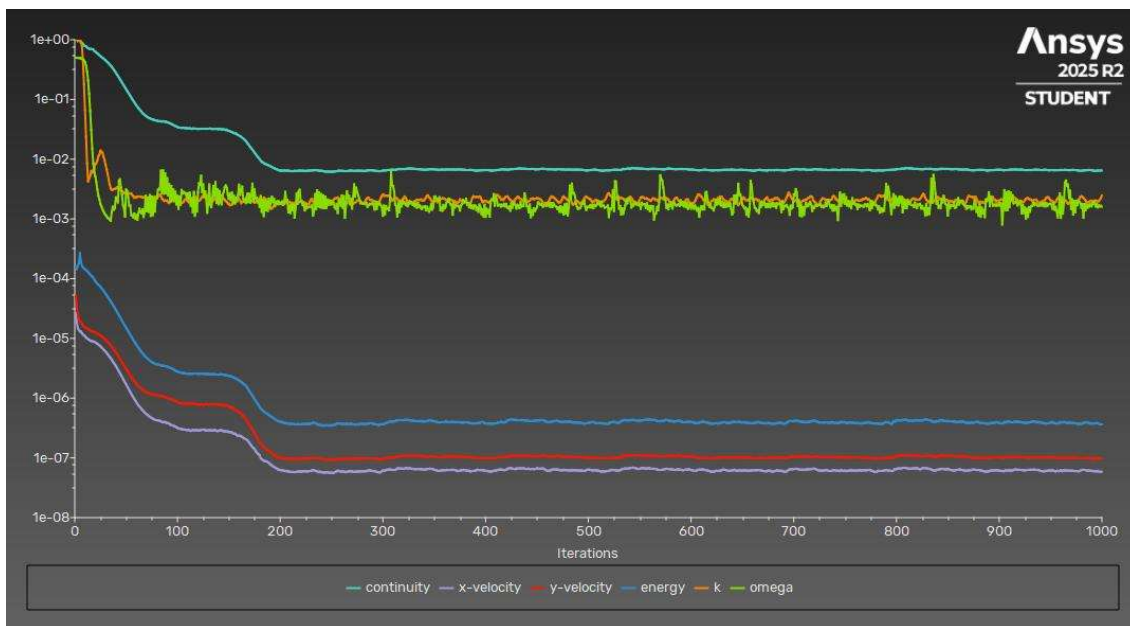
Figure 2. C

Scaled residuals:

Mach1



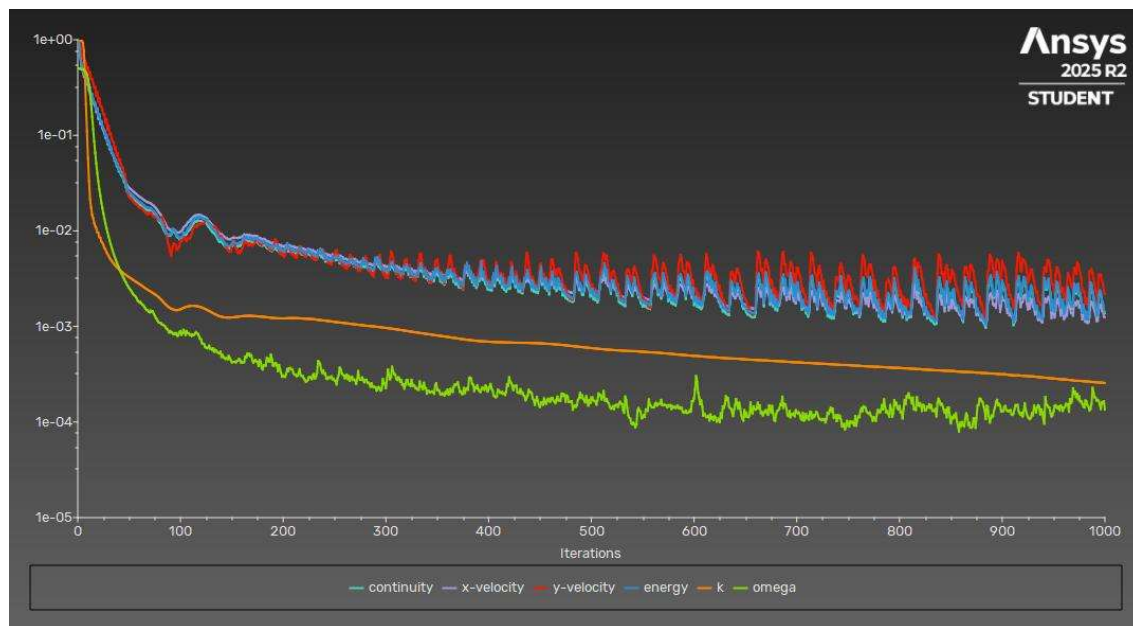
Mach2



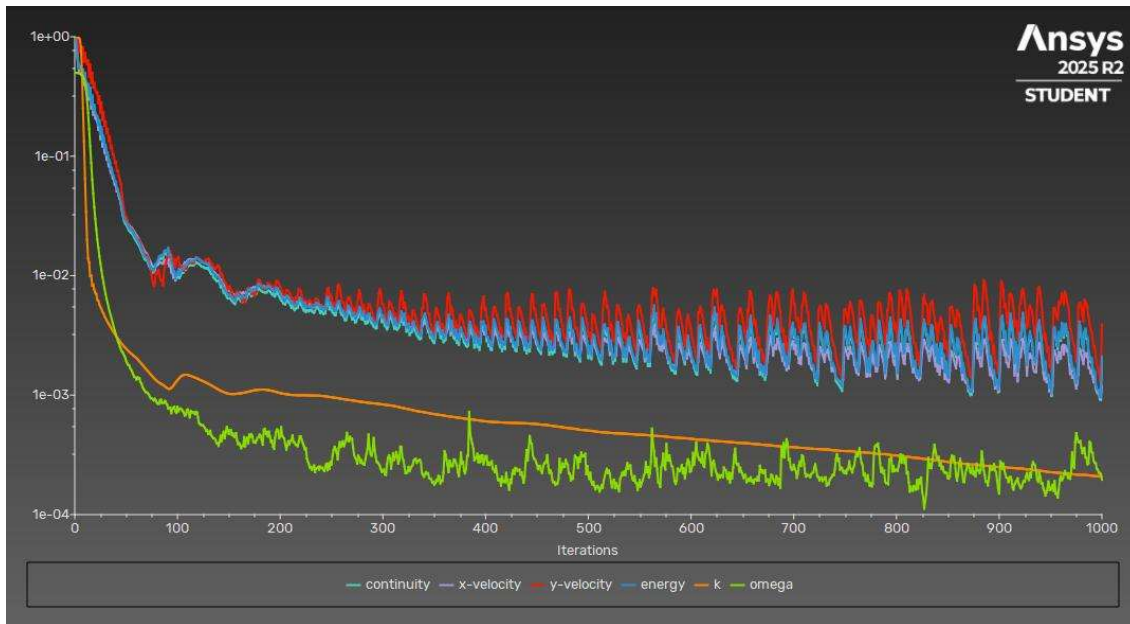
Mach3



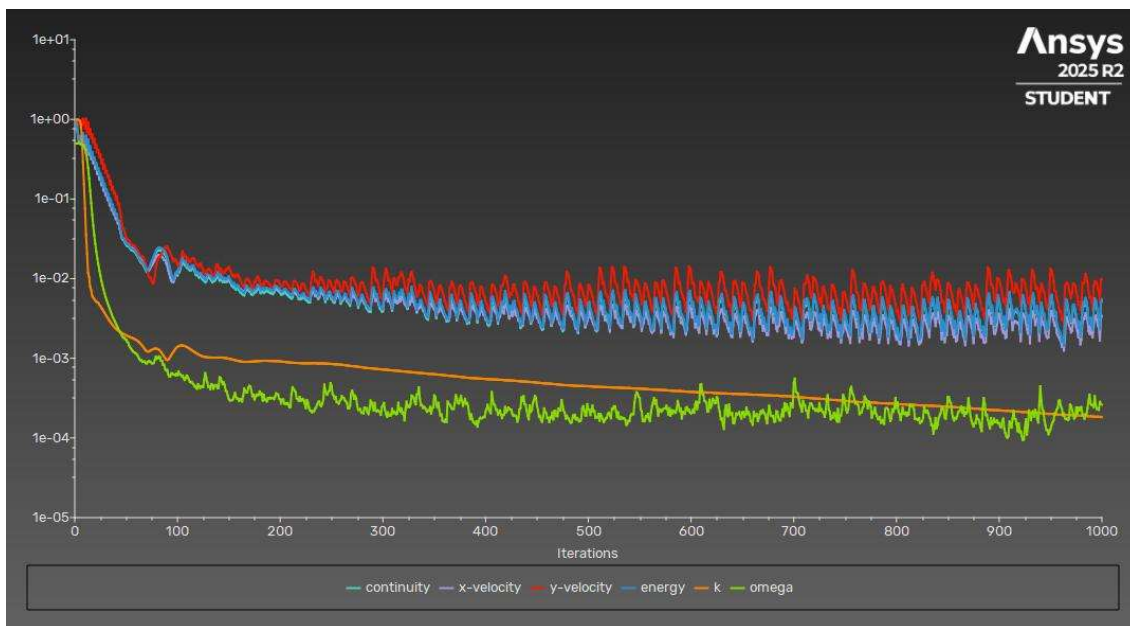
Mach4



Mach5



Mach6



Coefficient of Drag:

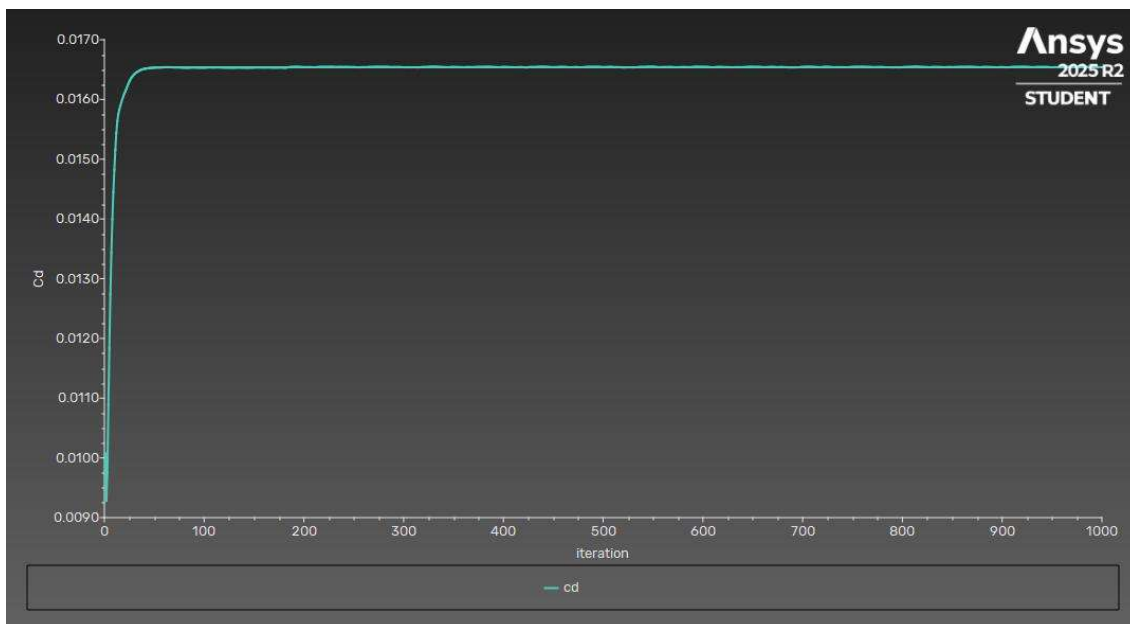
Mach1

Stable: 0.0101



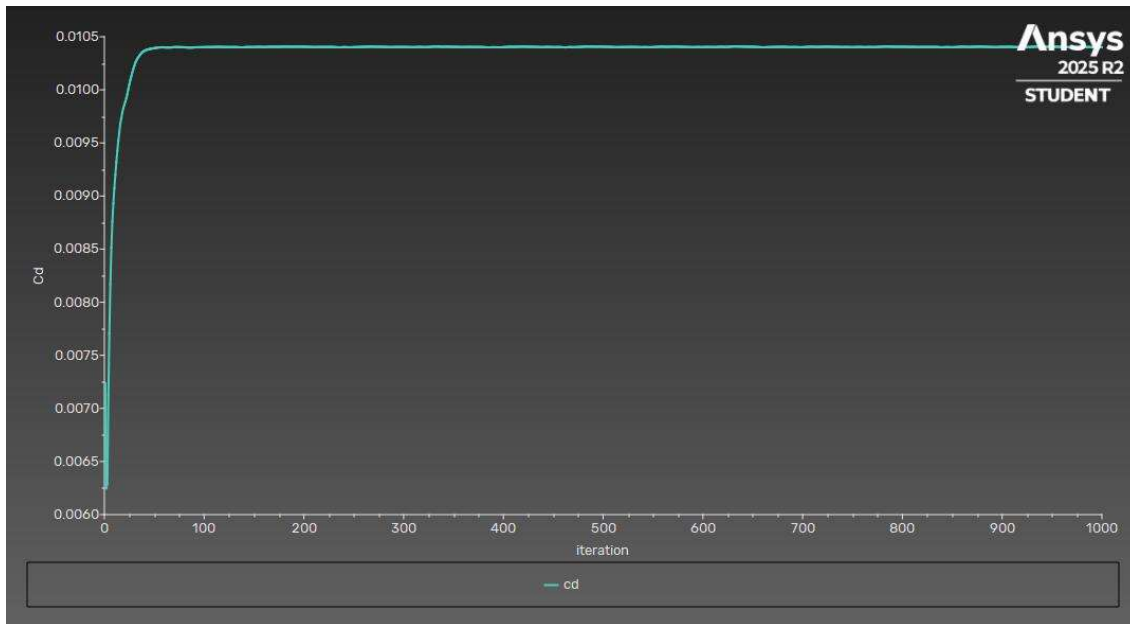
Mach2

Stable: 0.0165



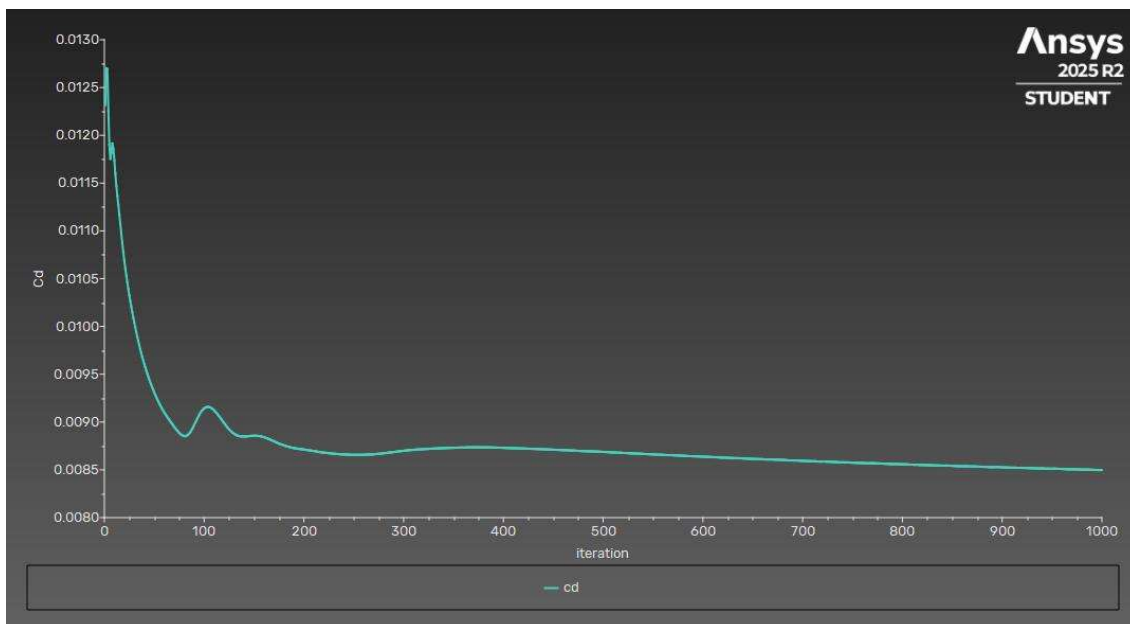
Mach3

Stable: 0.0104



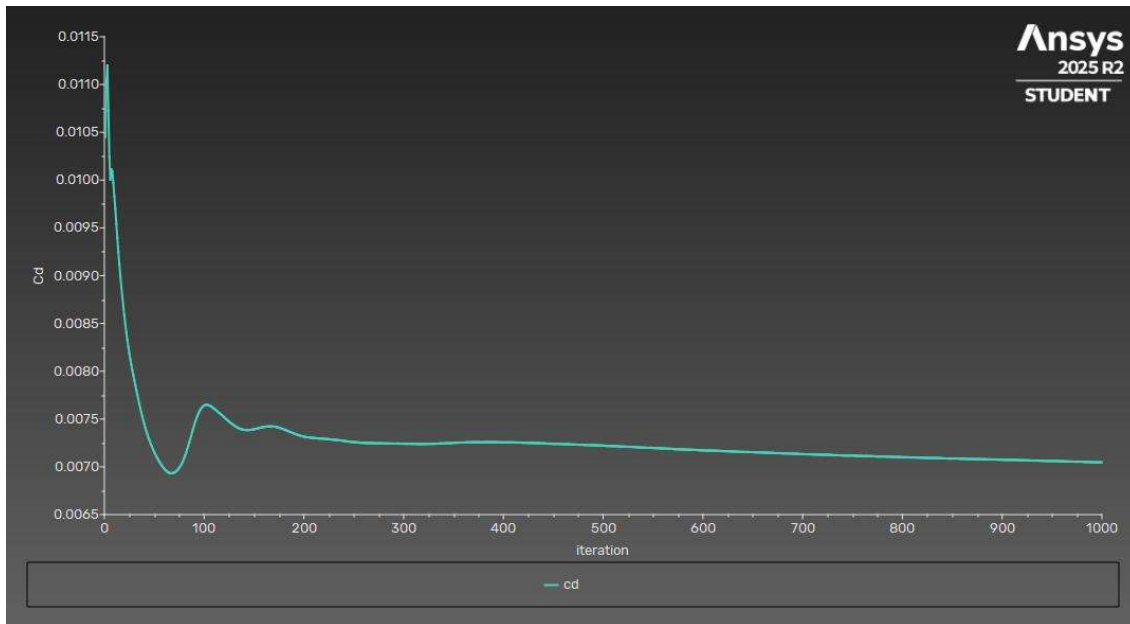
Mach4

Stable: 0.0085



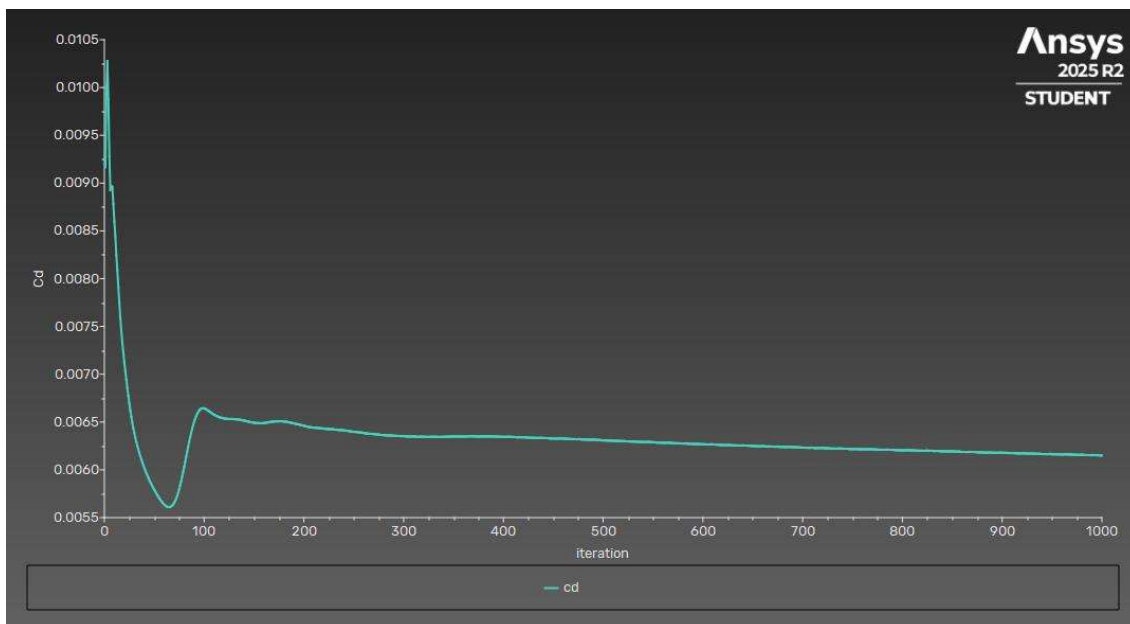
Mach5

Stable: 0.00704



Mach6

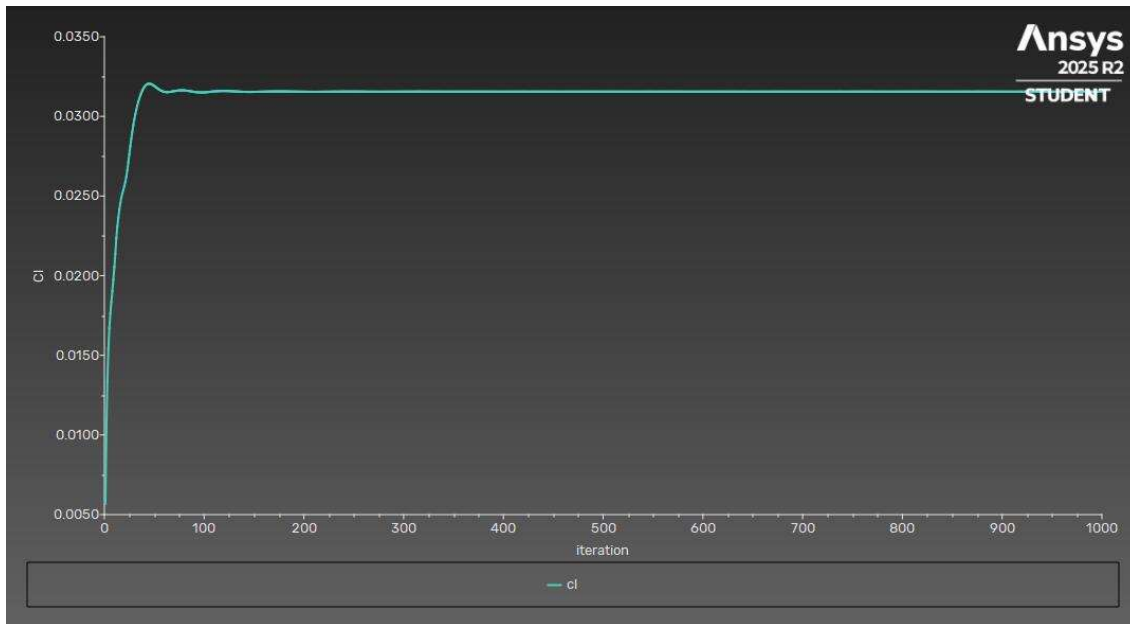
Stable: 0.00615



Coefficient of Lift:

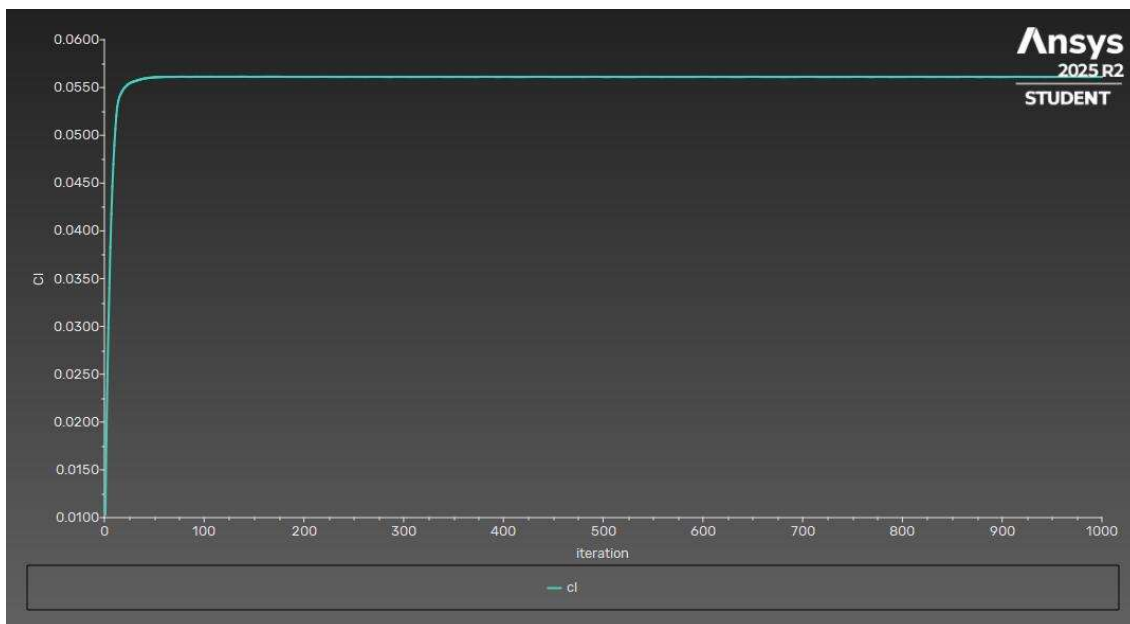
Mach1

Stable: 0.0315



Mach2

Stable: 0.0561



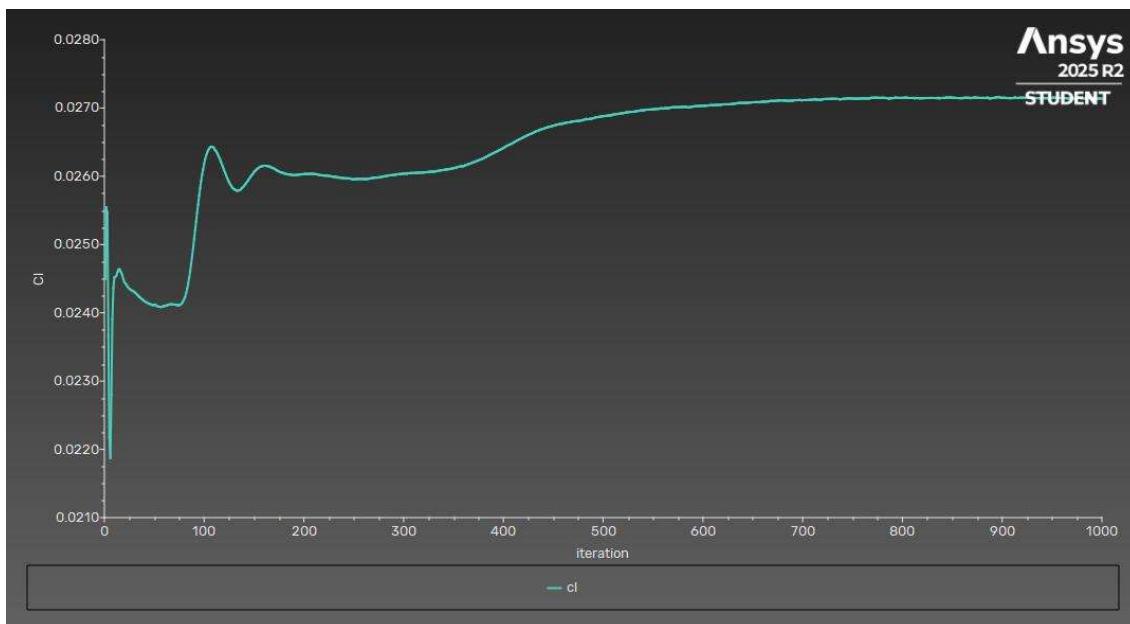
Mach3

Stable: 0.0352



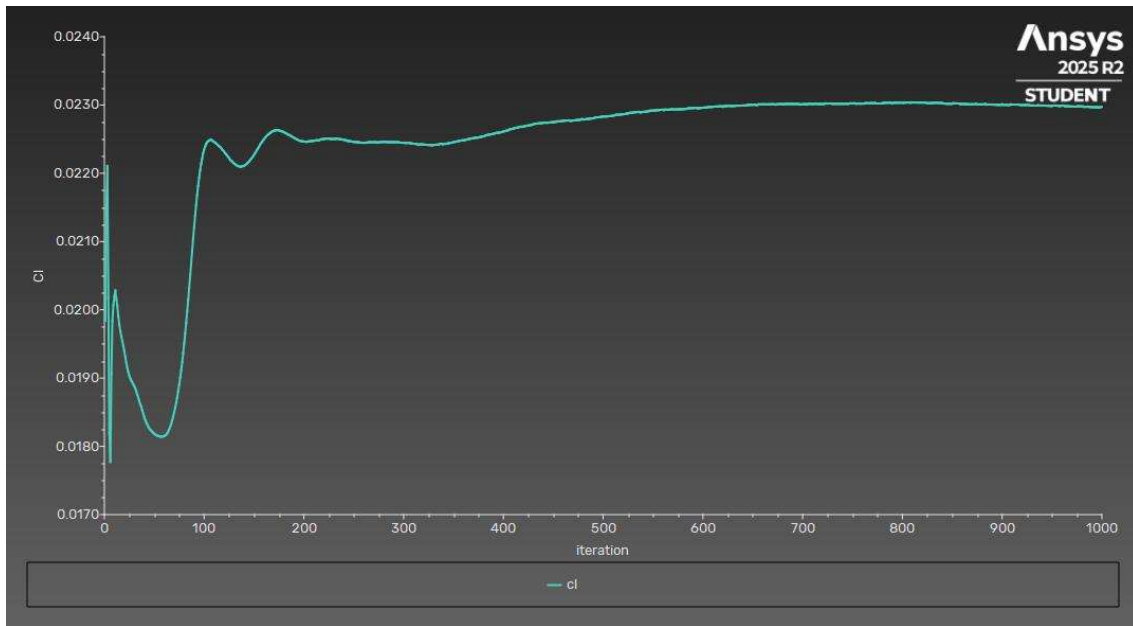
Mach4

Stable: 0.0271



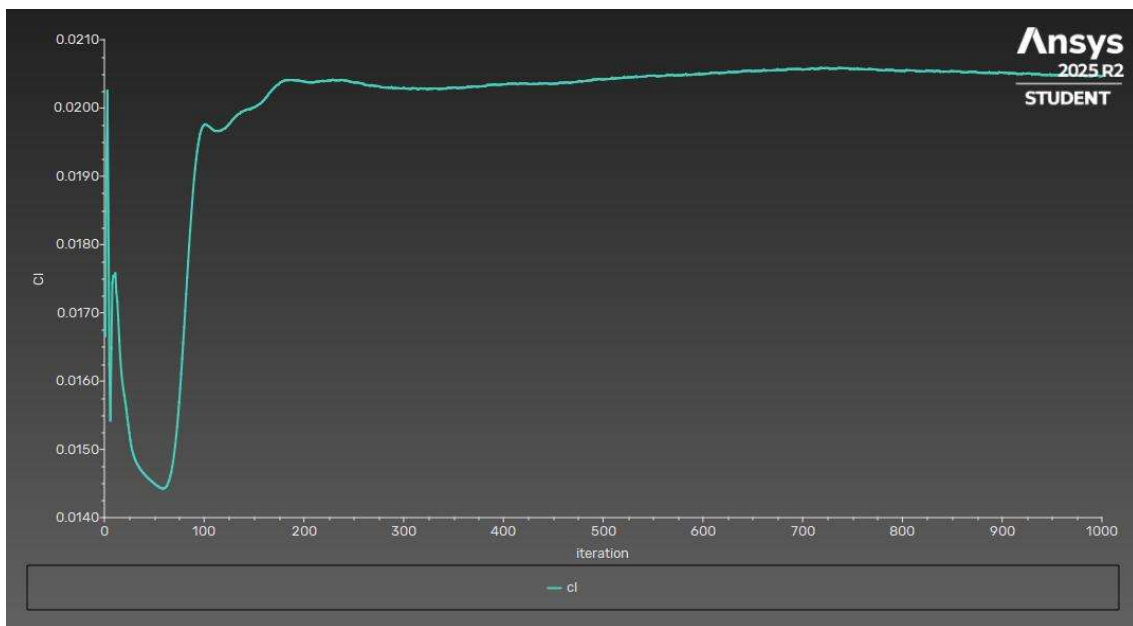
Mach5

Stable: 0.0229



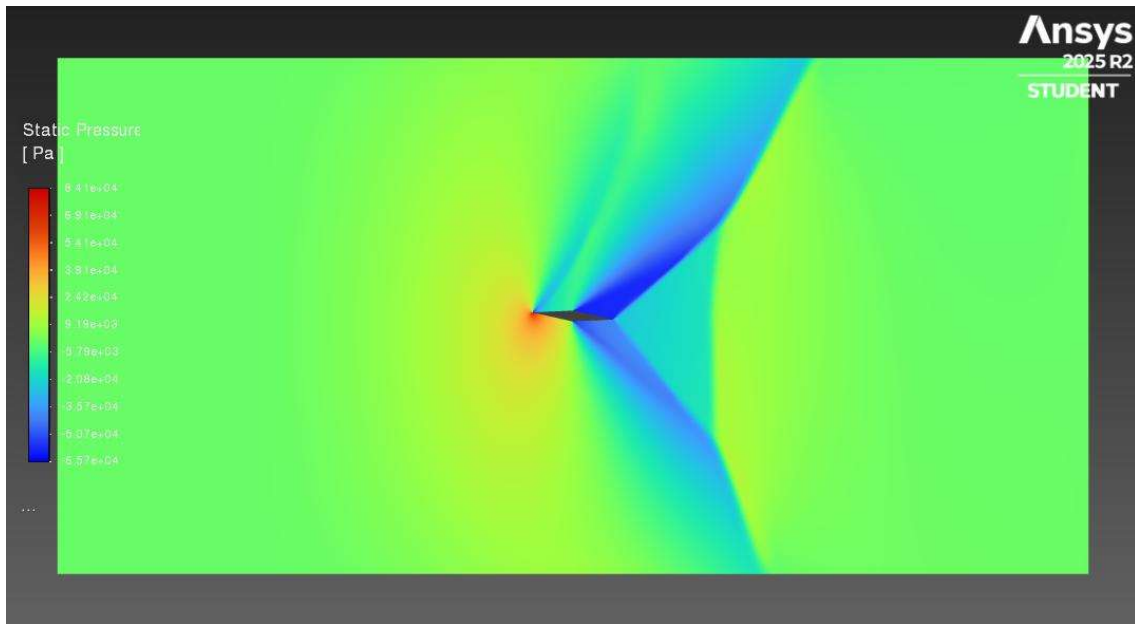
Mach6

Stable: 0.0204

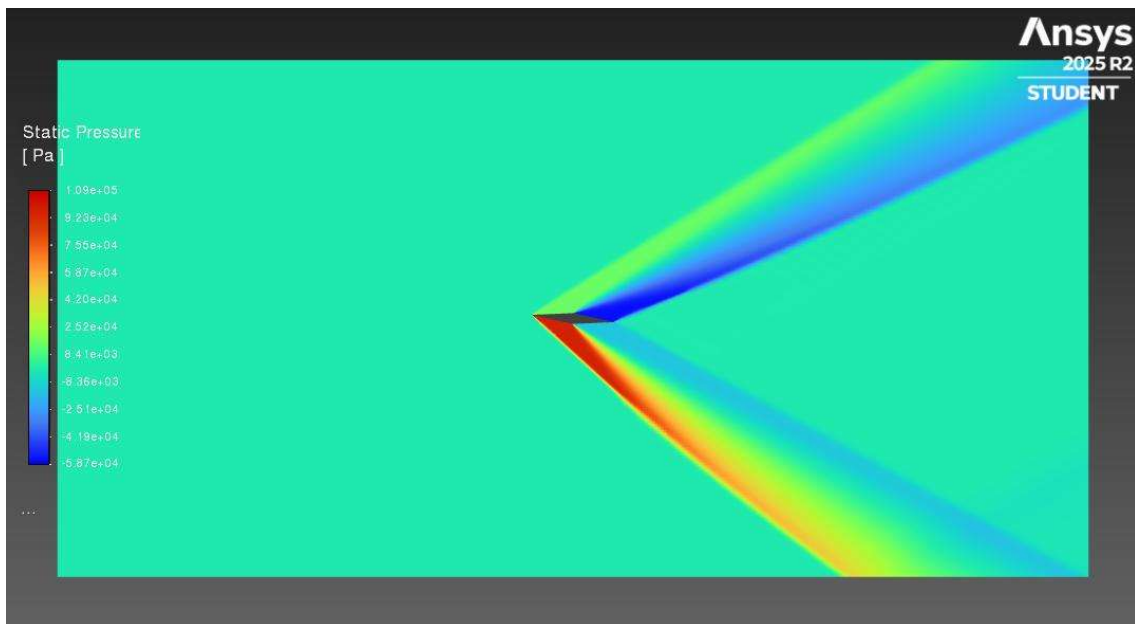


Pressure:

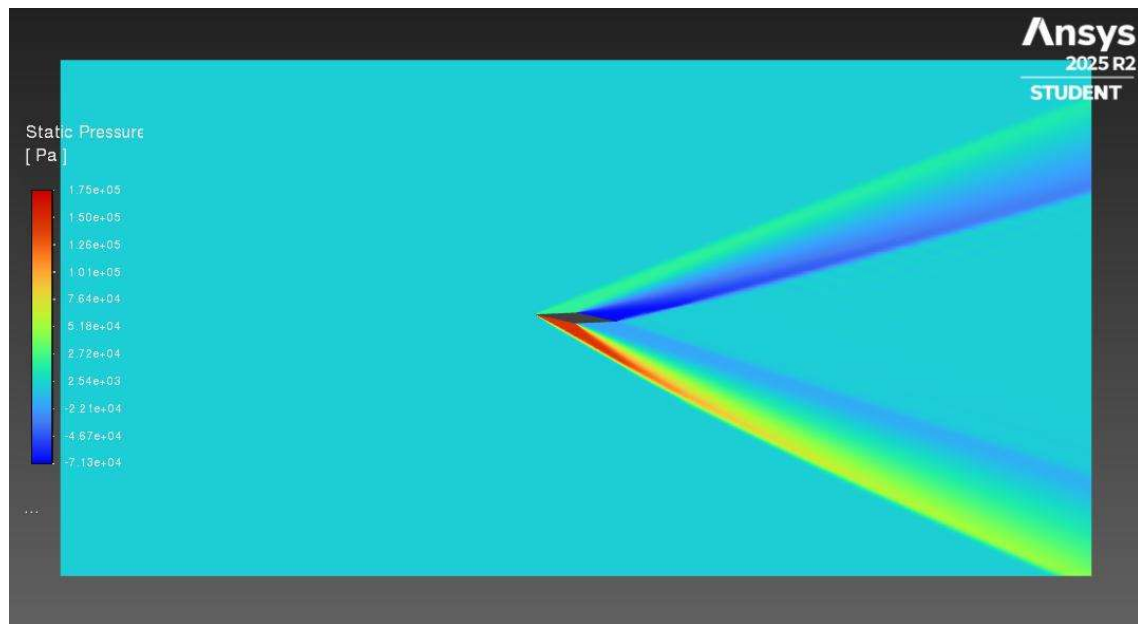
Mach1



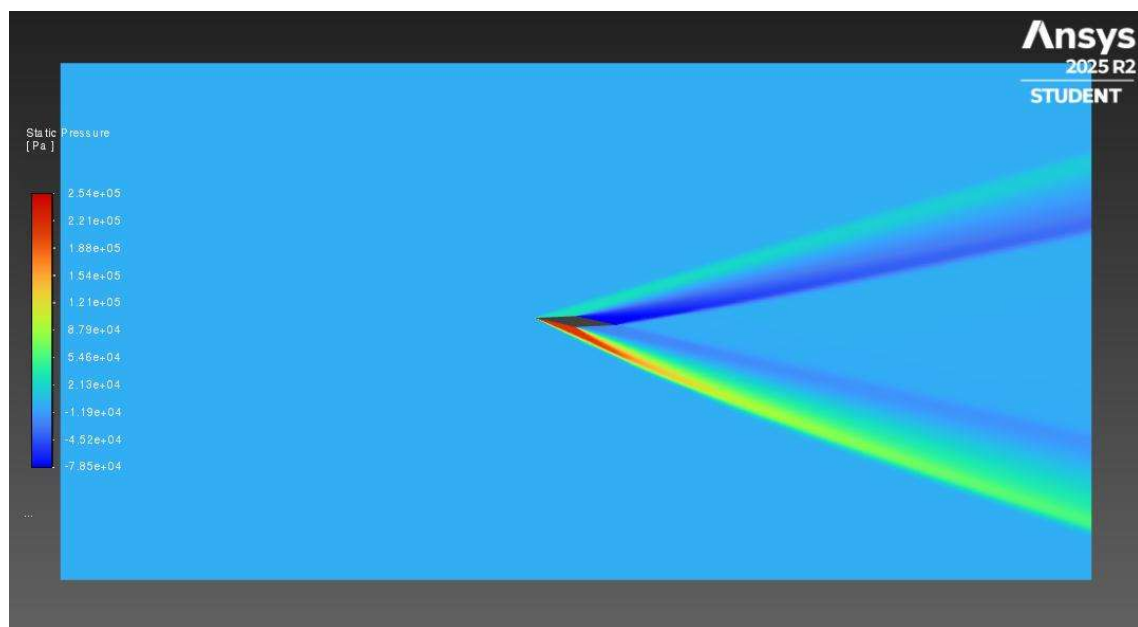
Mach2



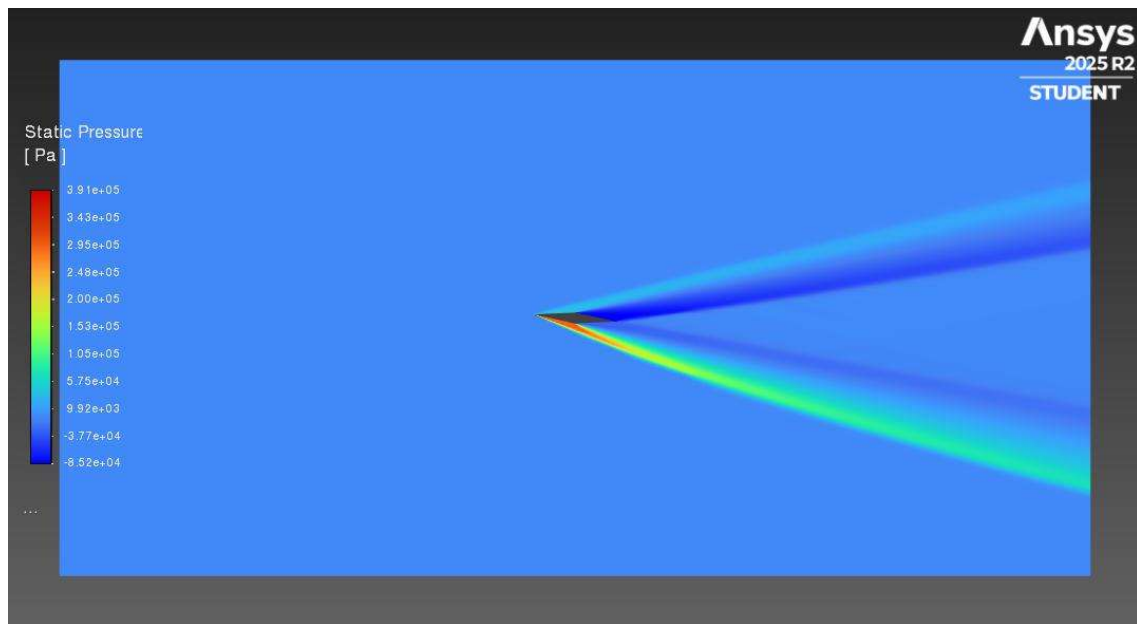
Mach3



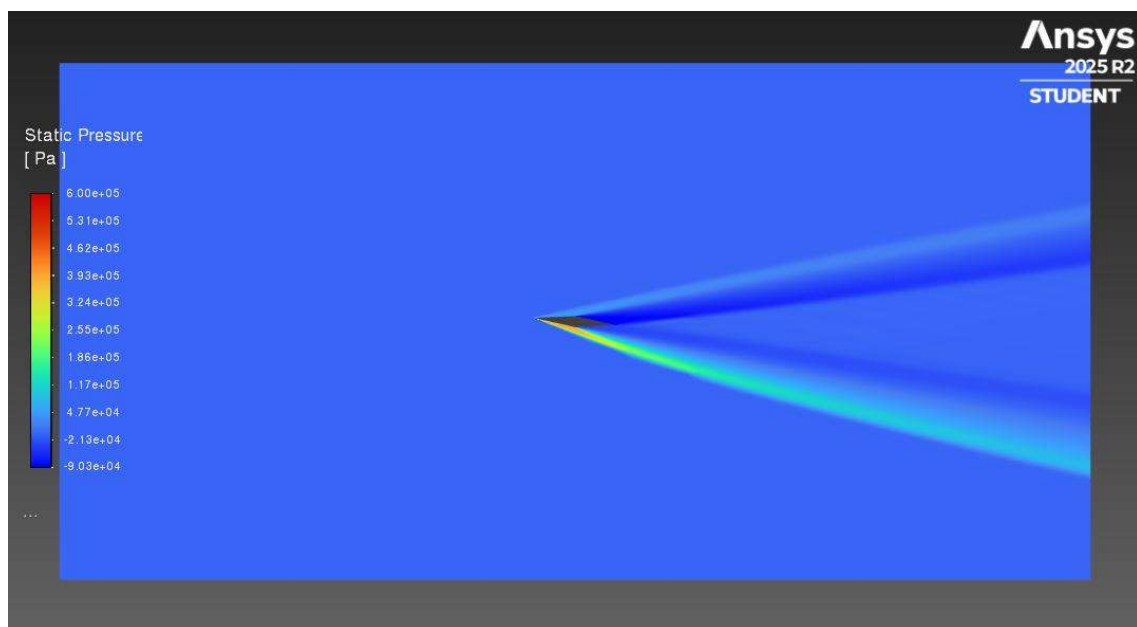
Mach4



Mach5

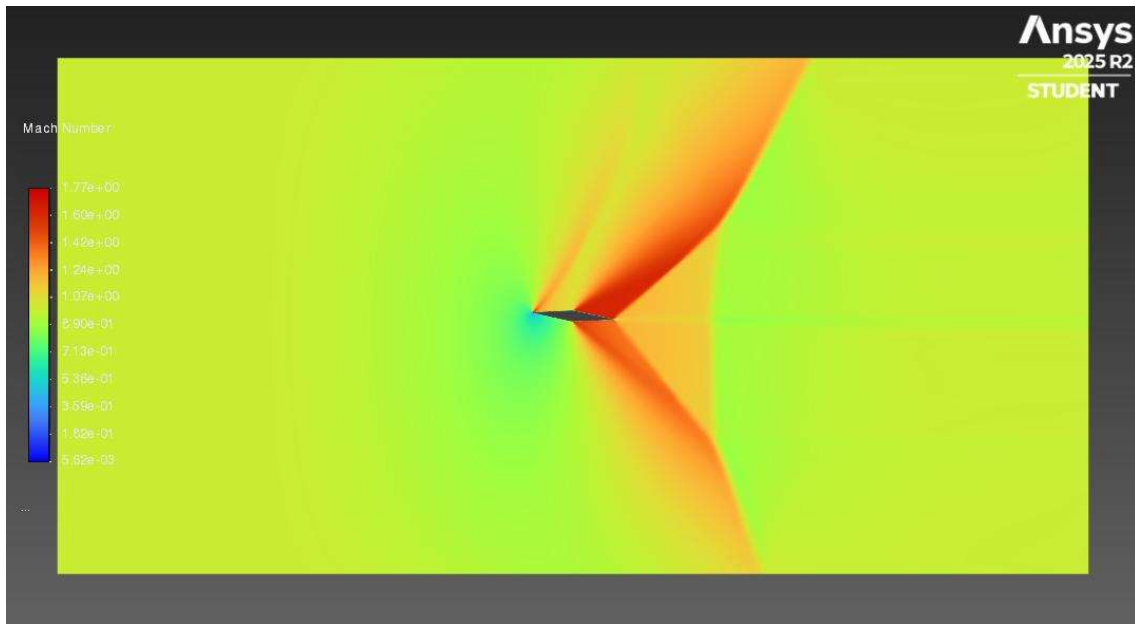


Mach6

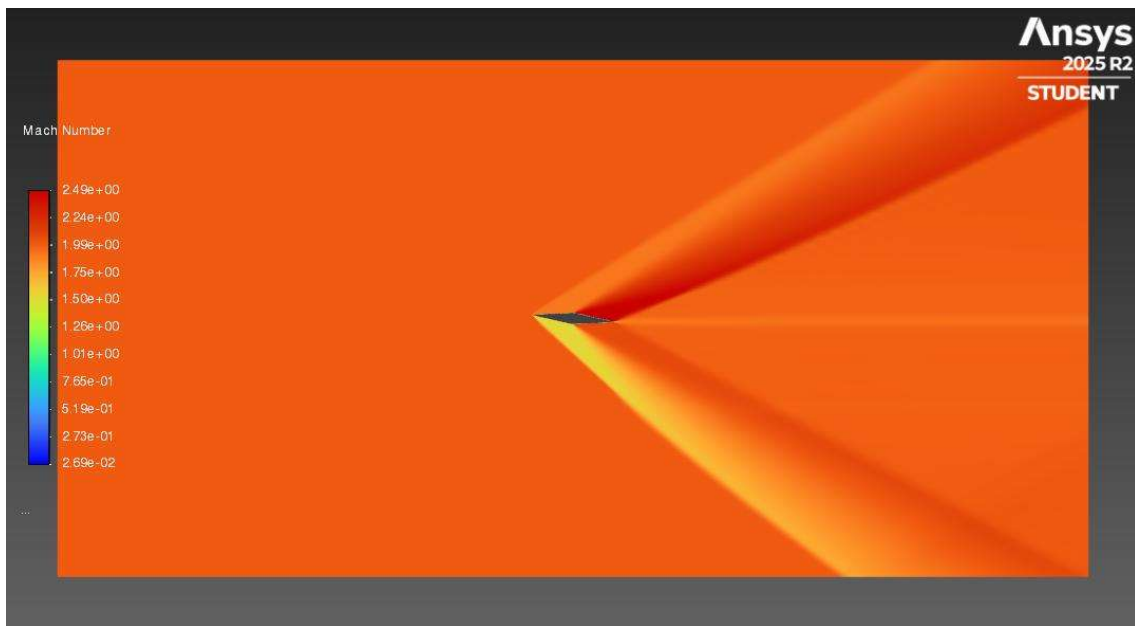


Mach:

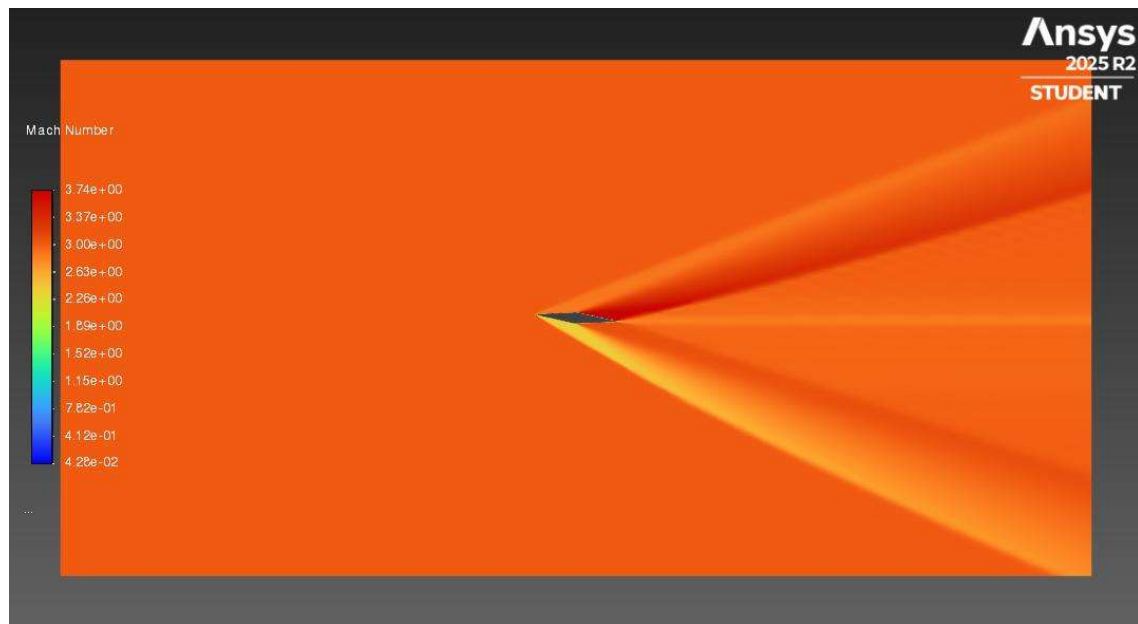
Mach1



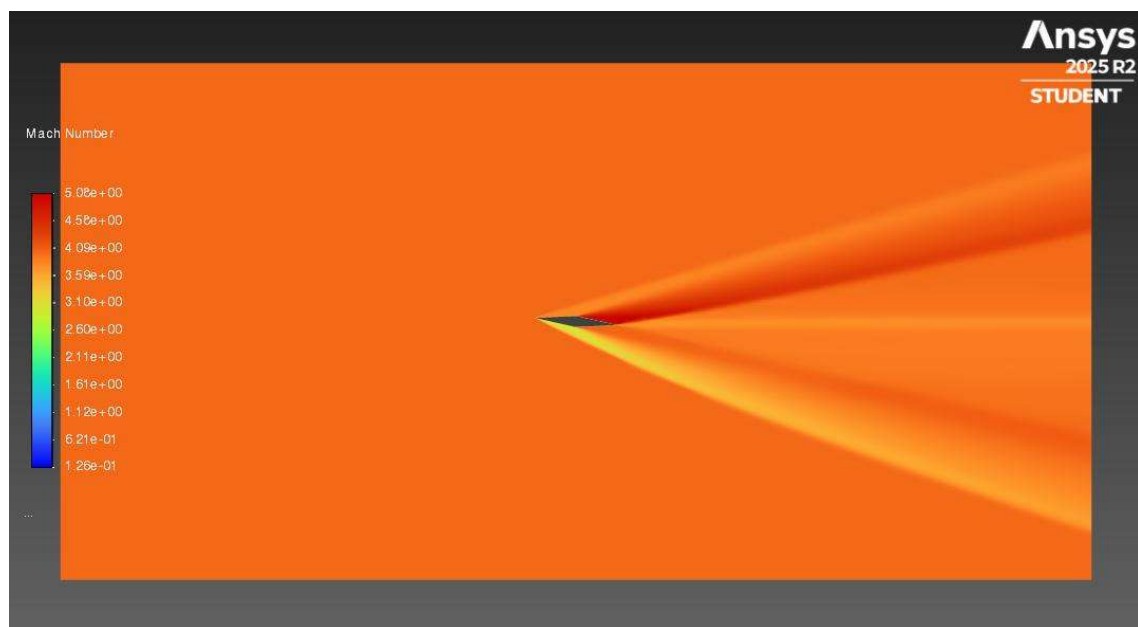
Mach2



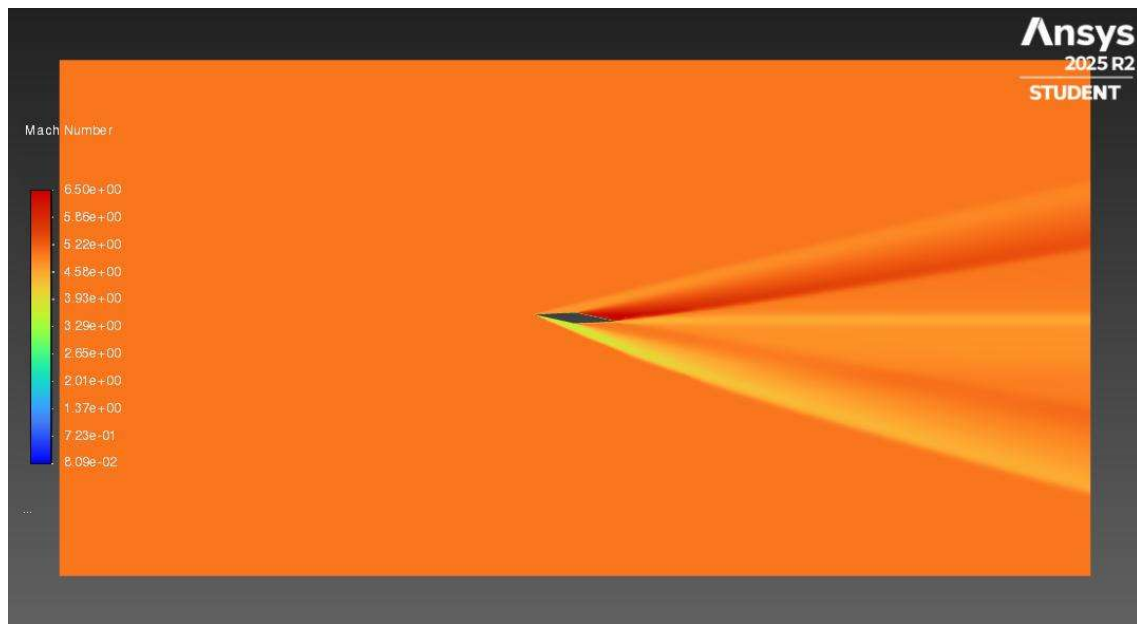
Mach3



Mach4



Mach5



Mach6

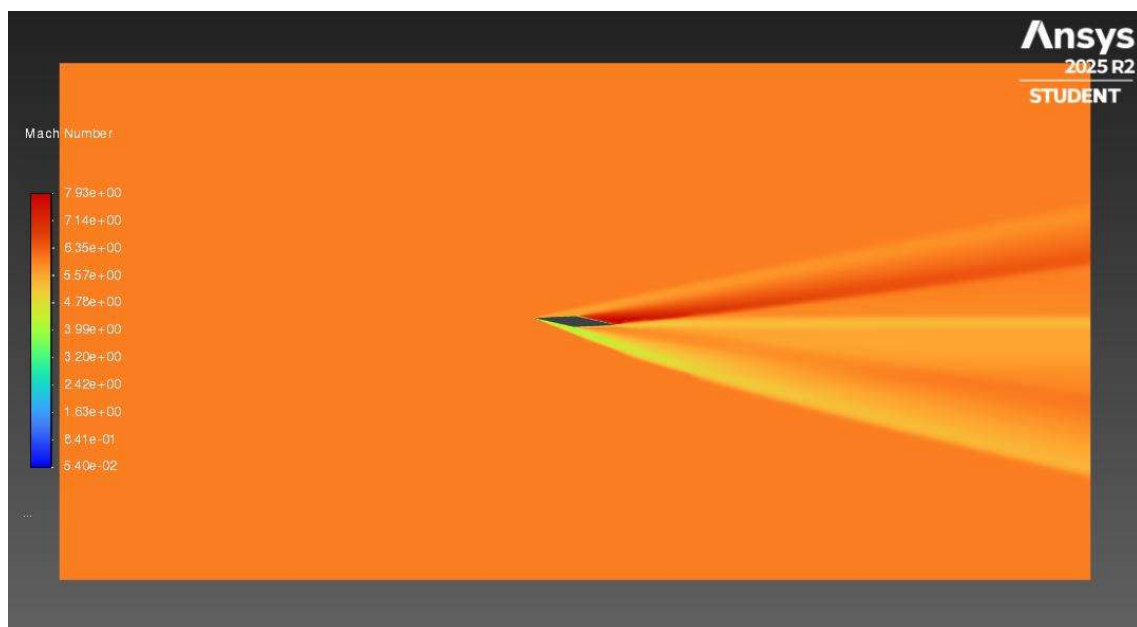
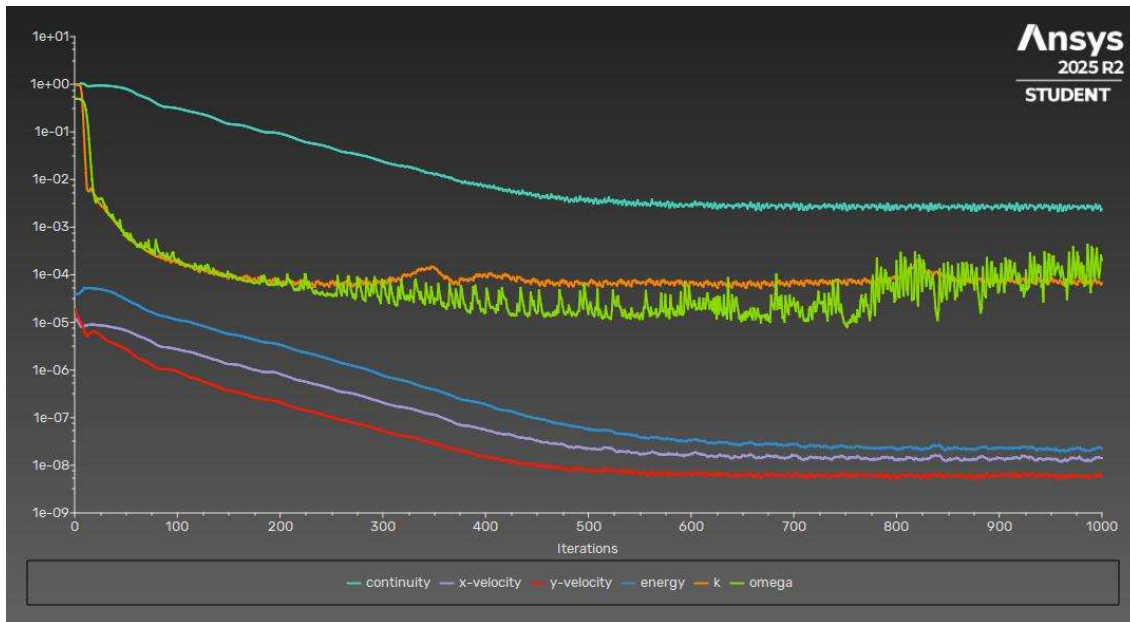


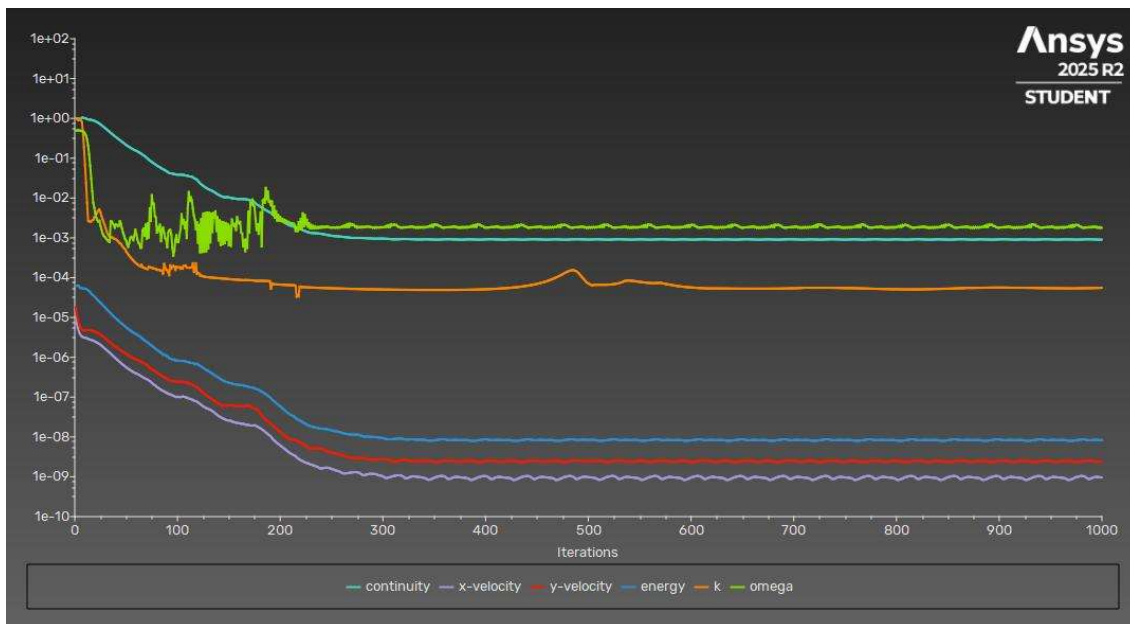
Figure 2. D

Scaled residuals:

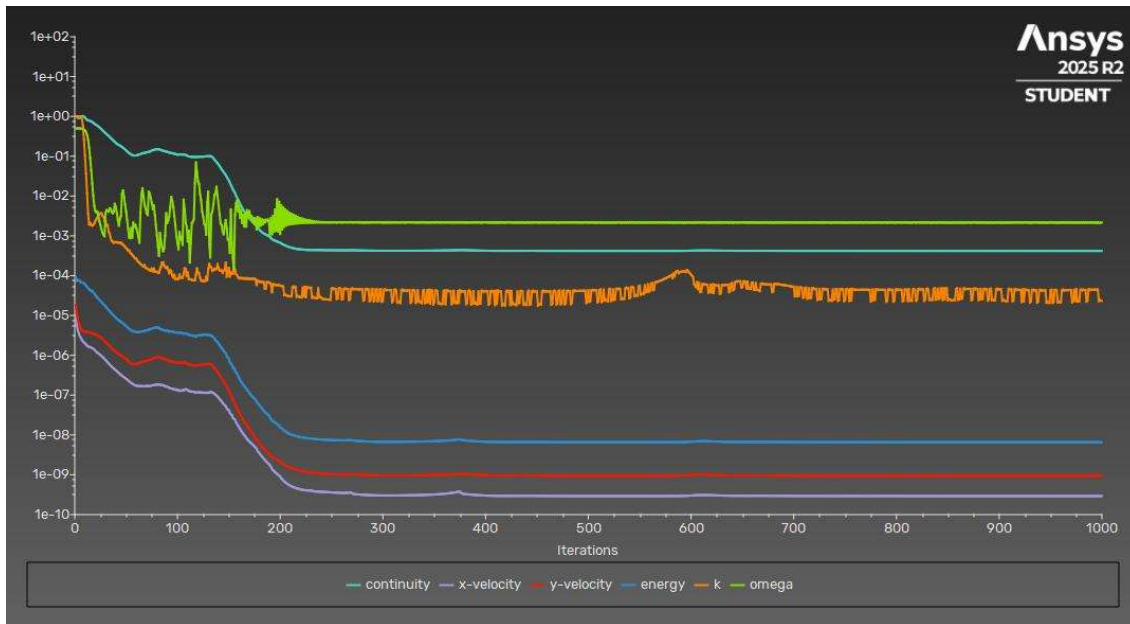
Mach1



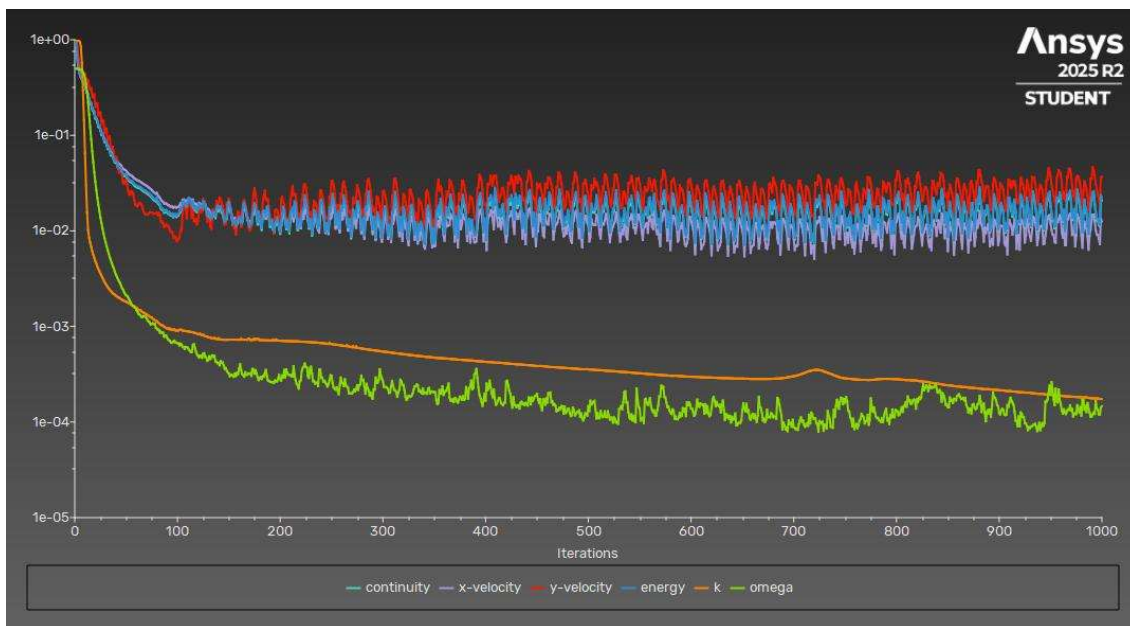
Mach2



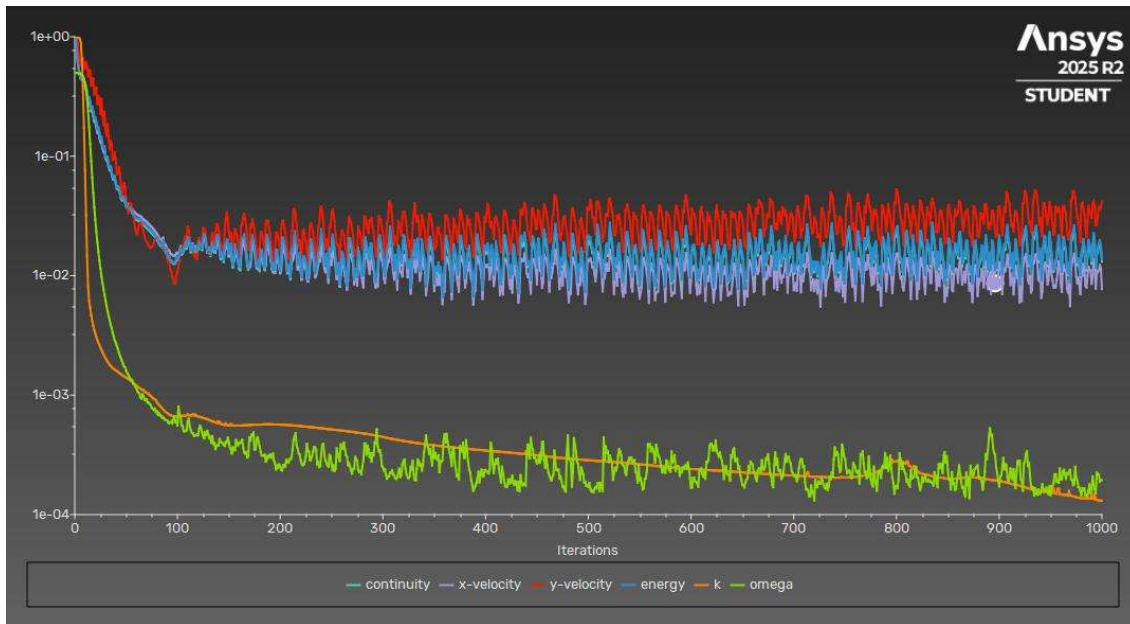
Mach3



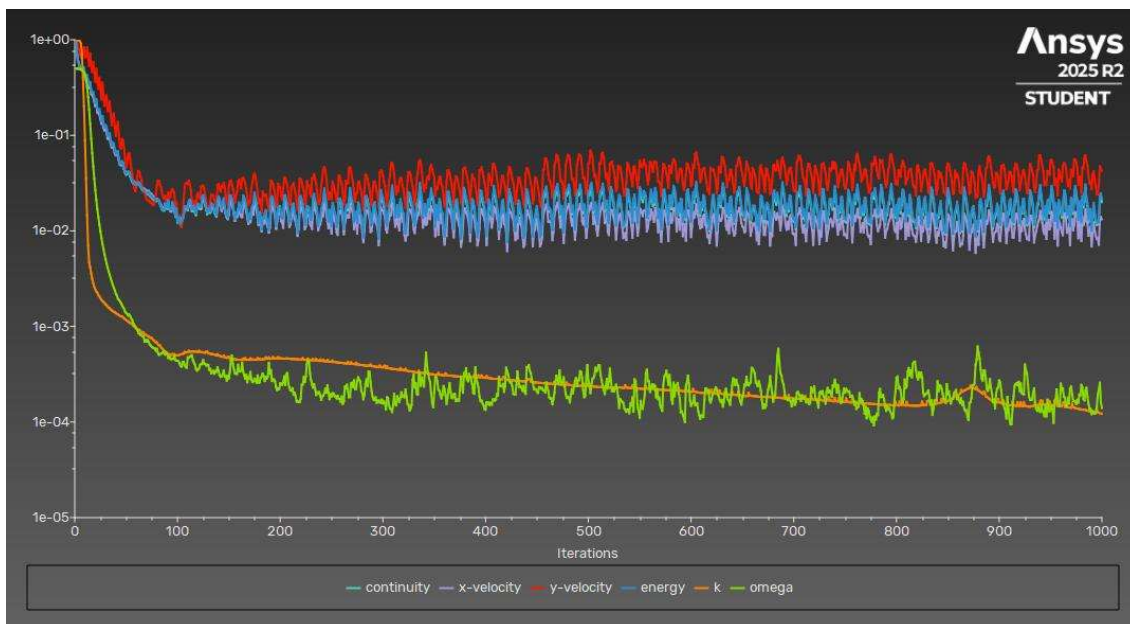
Mach4



Mach5



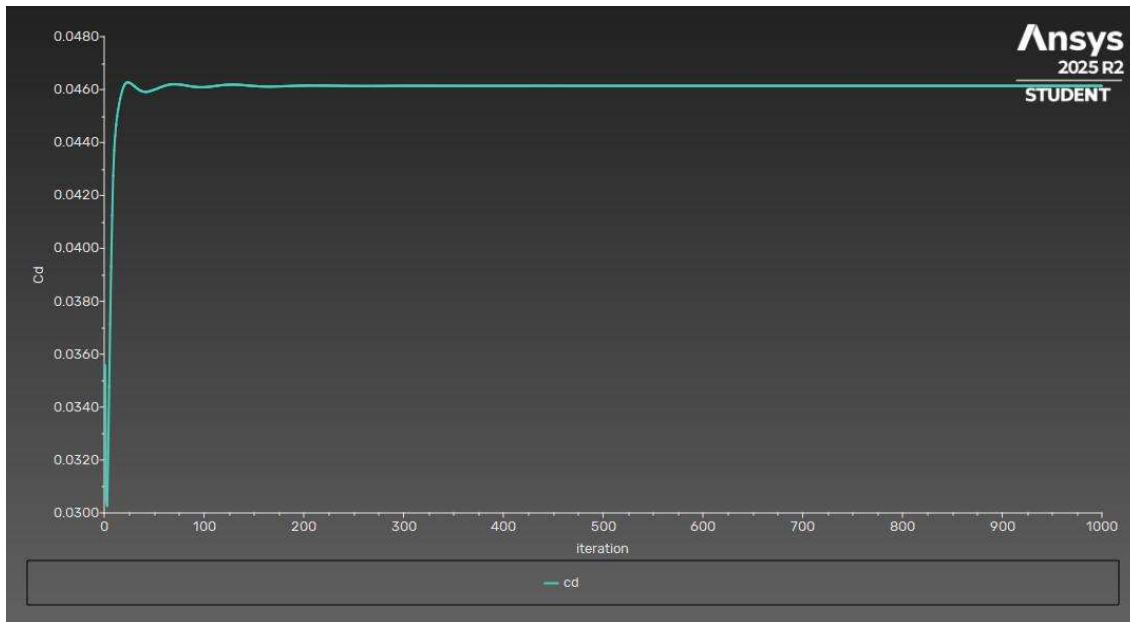
Mach6



Coefficient of Drag:

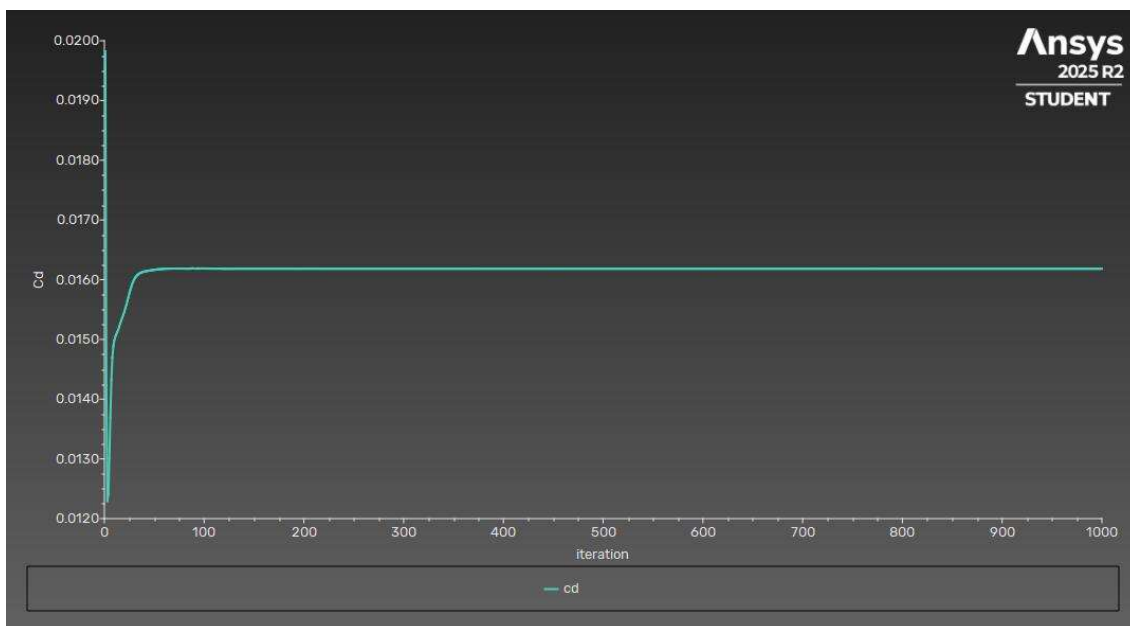
Mach1

Stable:0.0461



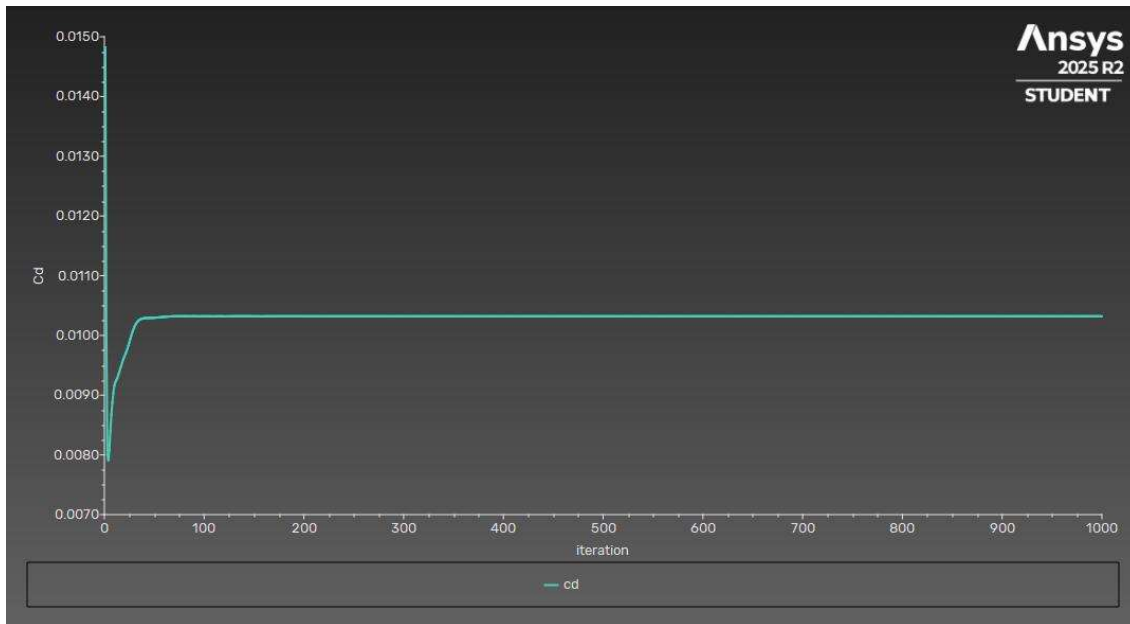
Mach2

Stable: 0.0161



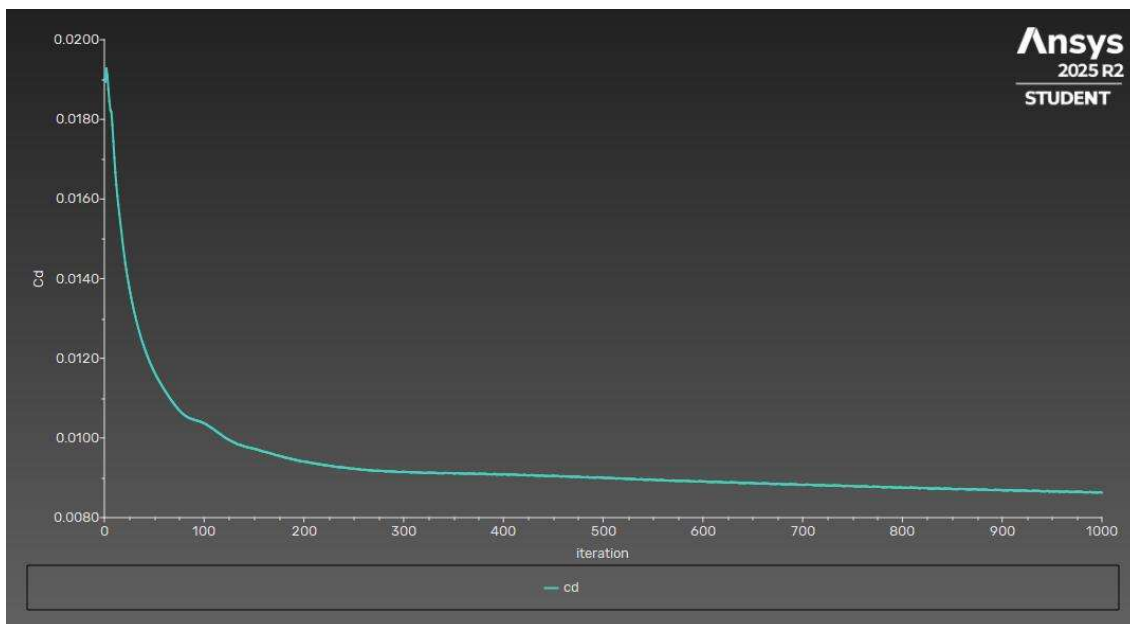
Mach3

Stable: 0.0103



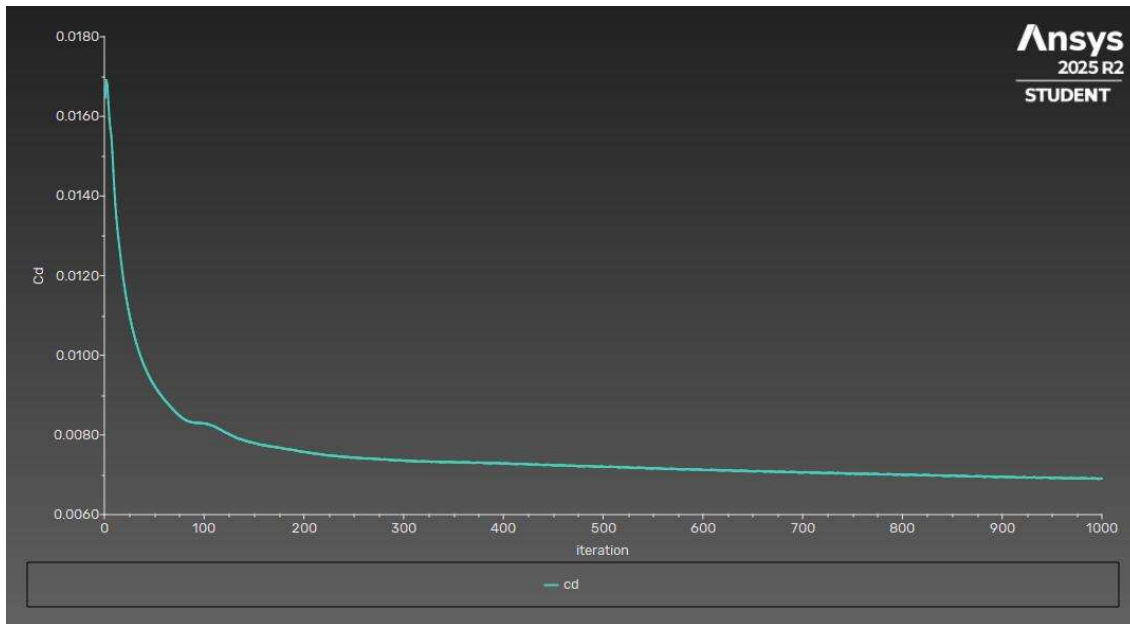
Mach4

Stable: 0.00863



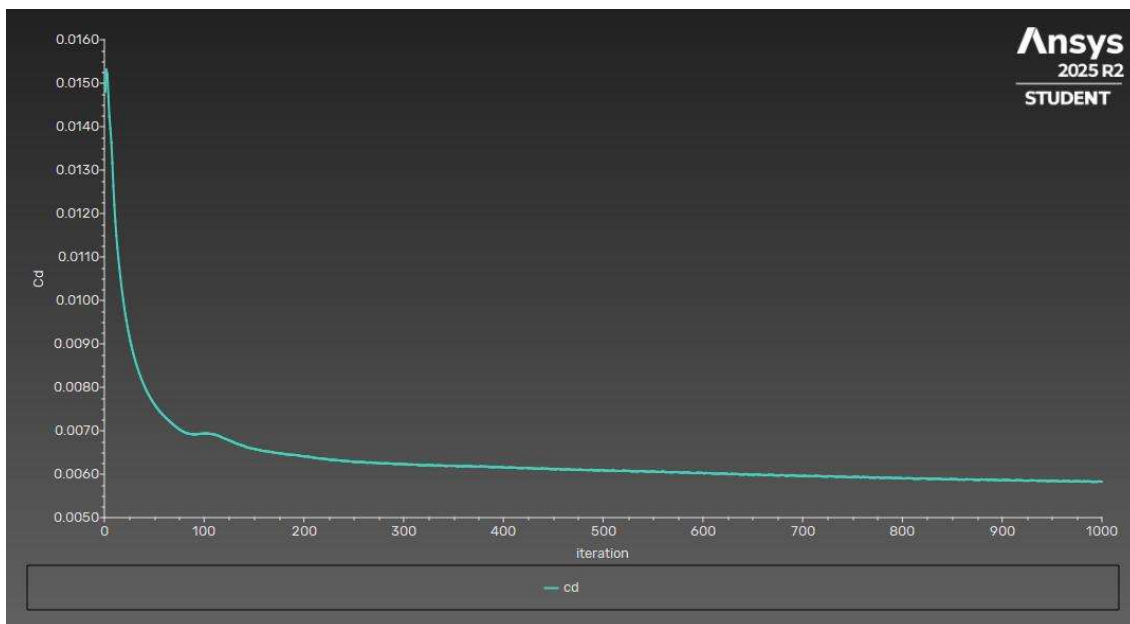
Mach5

Stable: 0.00691



Mach6

Stable: 0.00582



Coefficient of Lift:

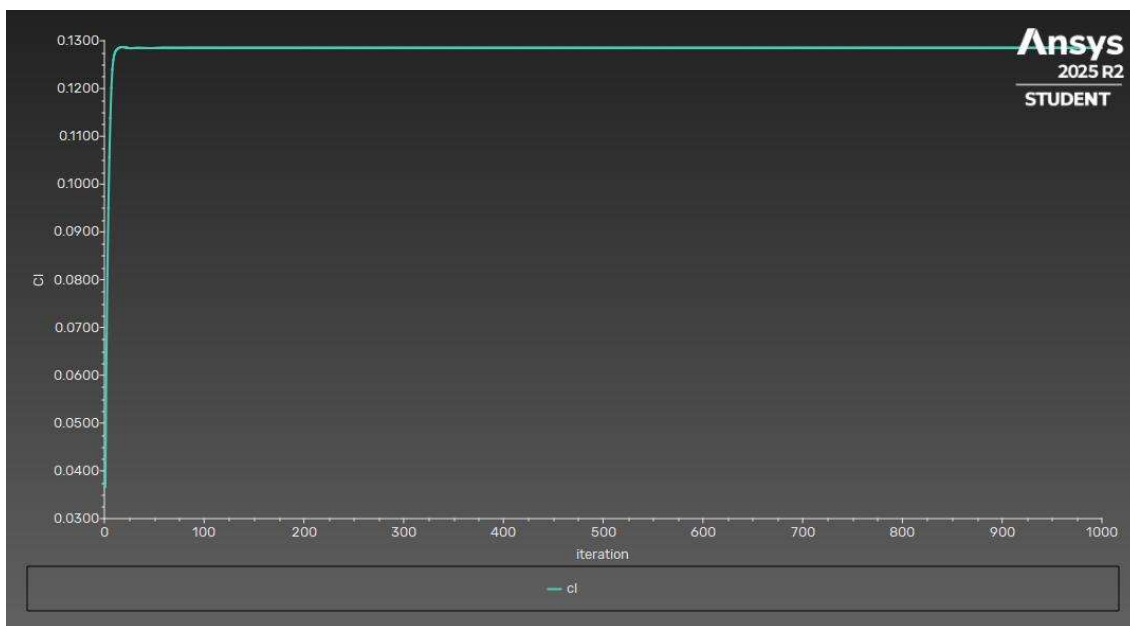
Mach1

Stable: 0.391



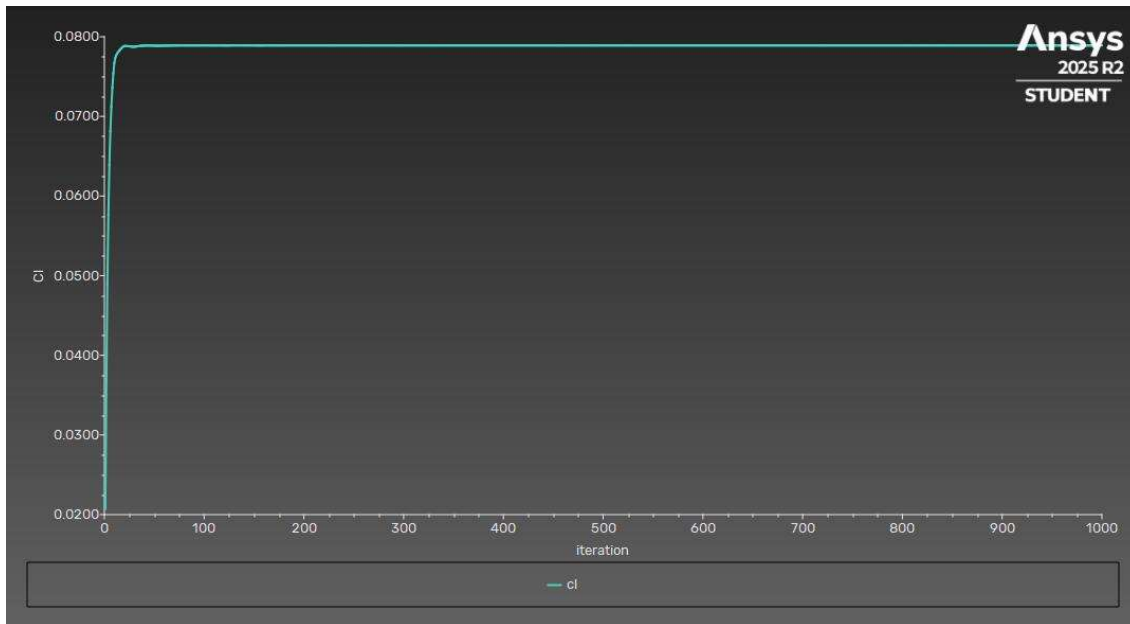
Mach2

Stable: 0.128



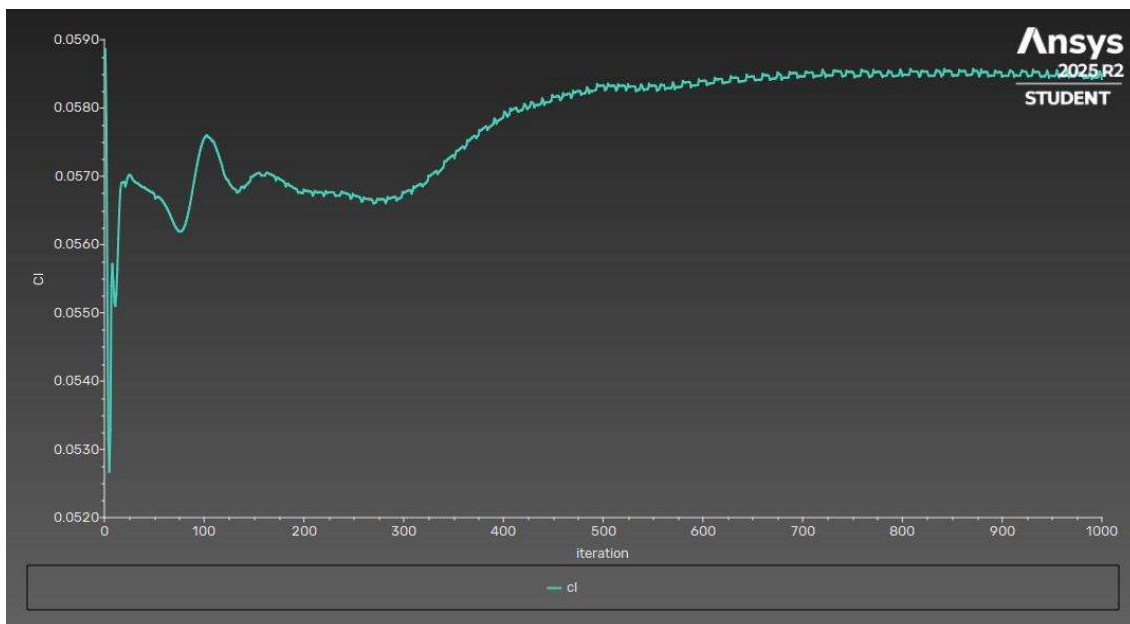
Mach3

Stable: 0.0789



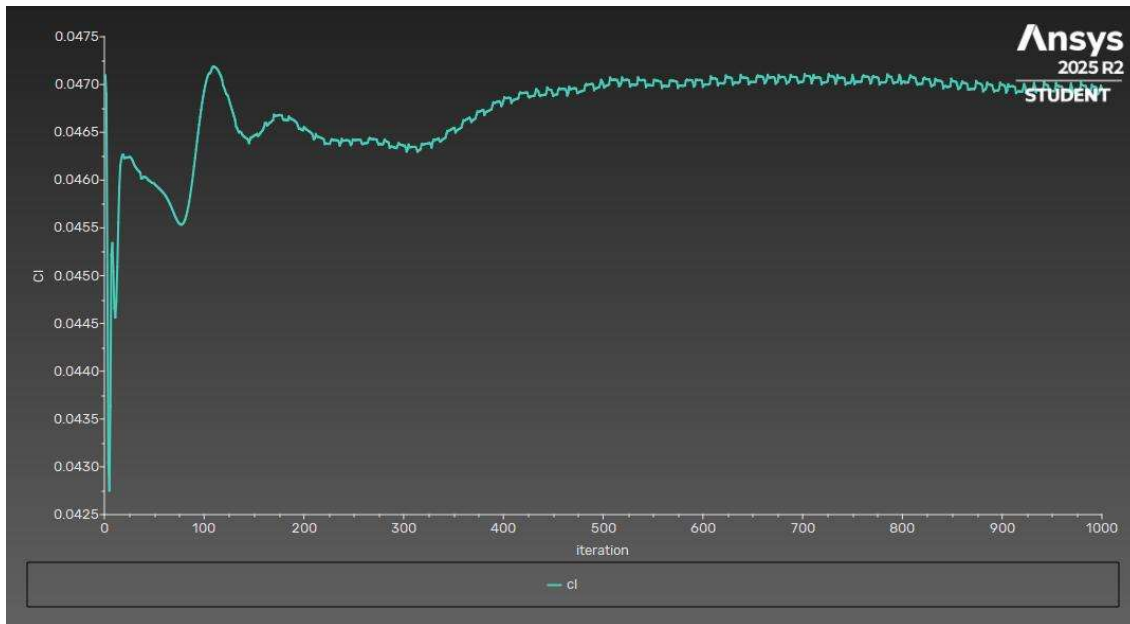
Mach4

Stable: 0.0584



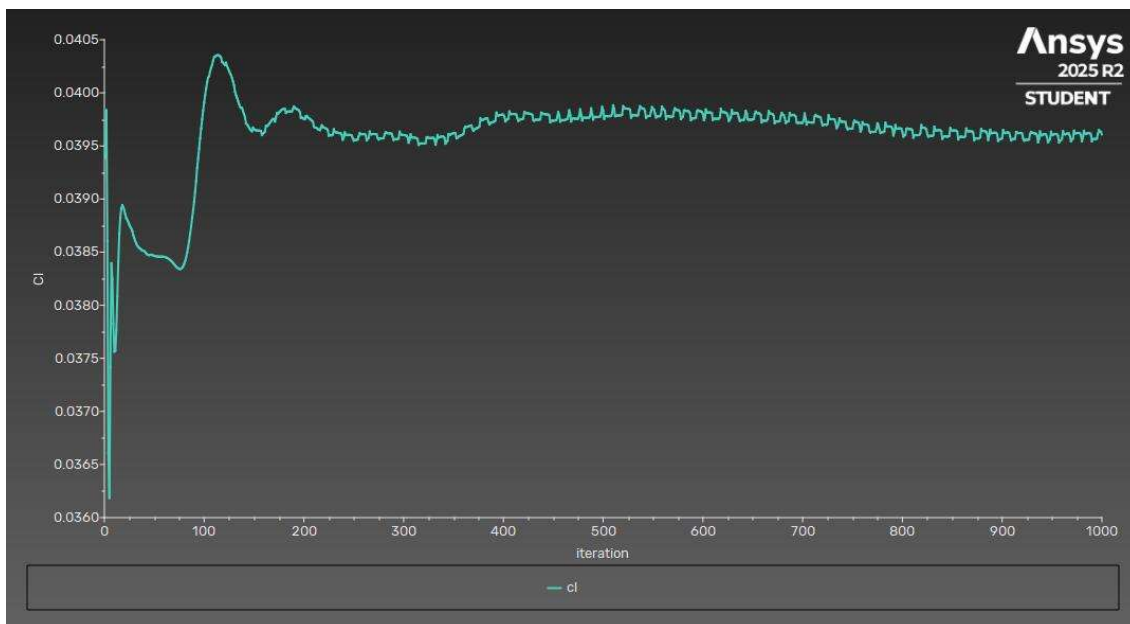
Mach5

Stable: 0.0469



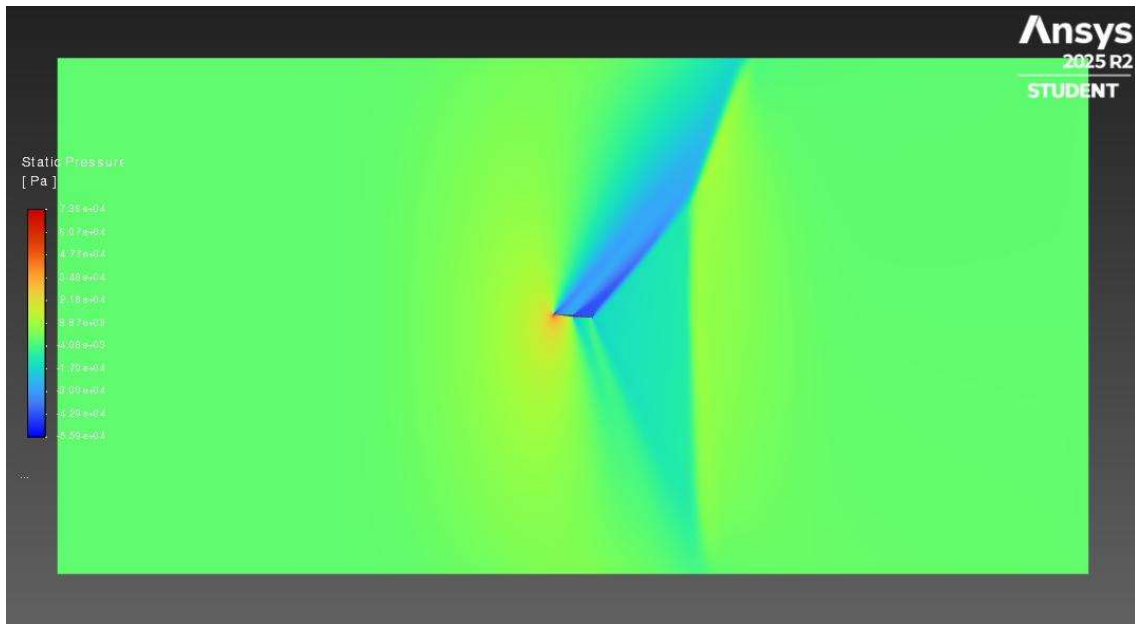
Mach6

Stable: 0.0395

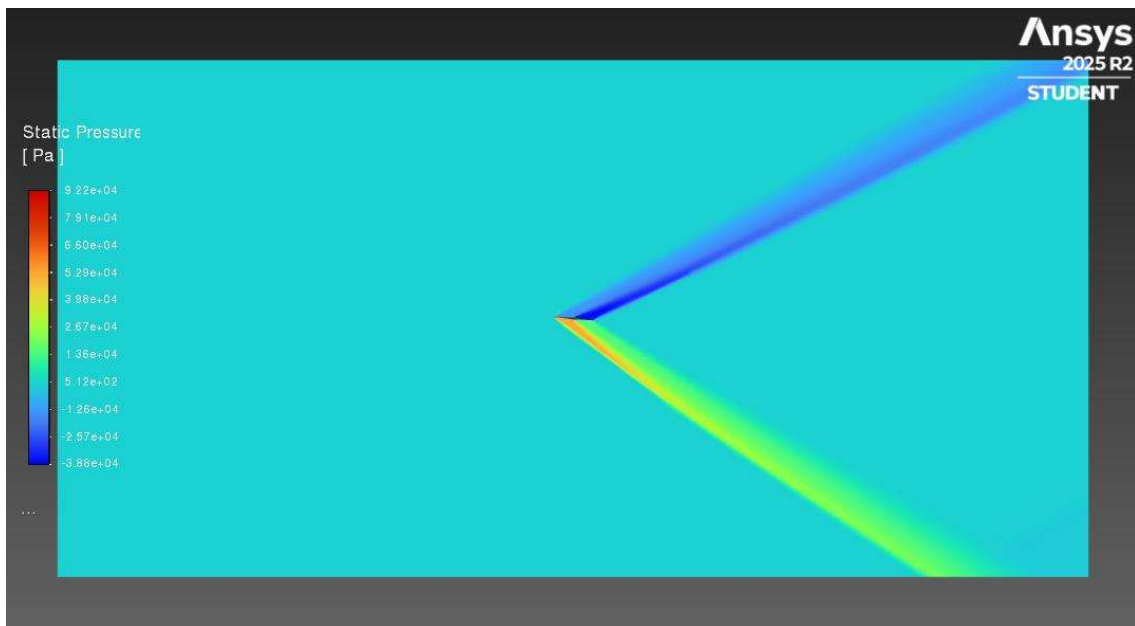


Pressure:

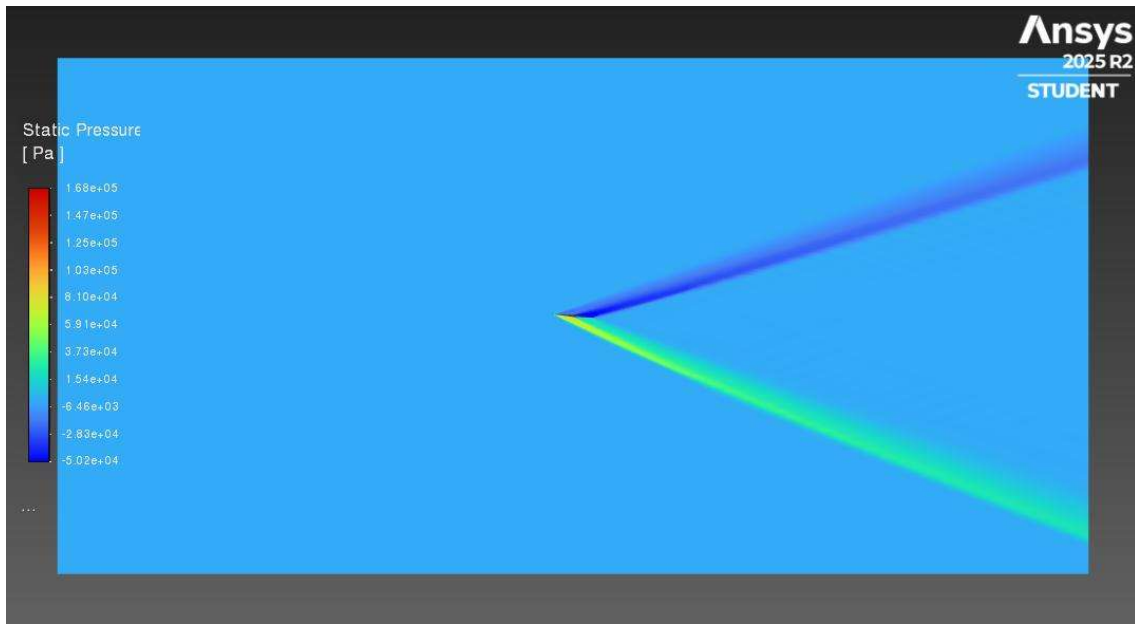
Mach1



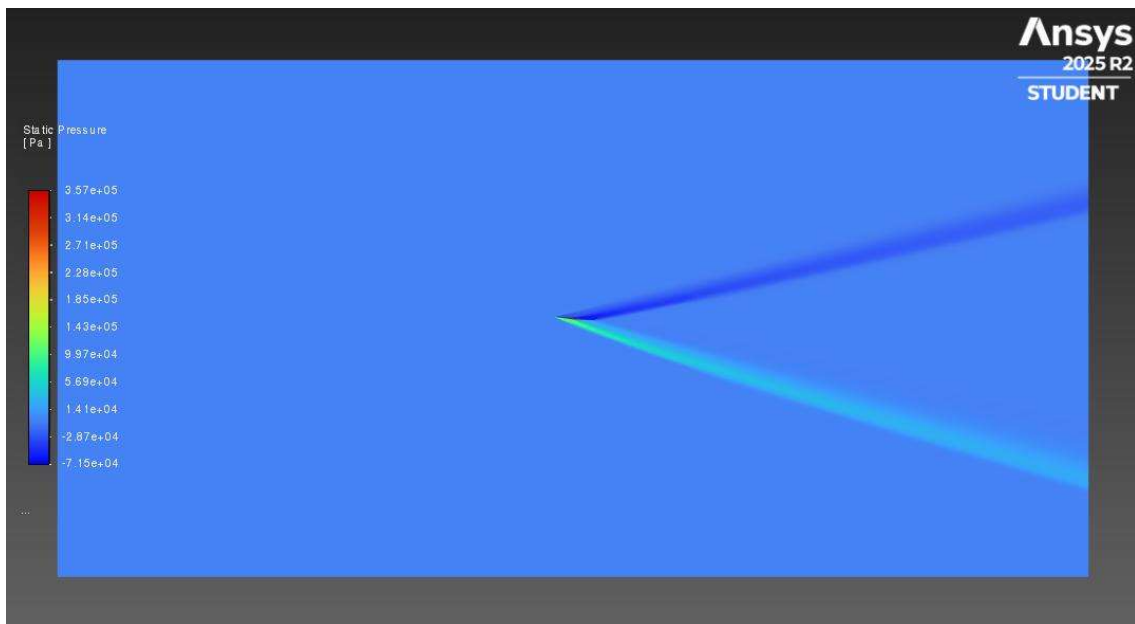
Mach2



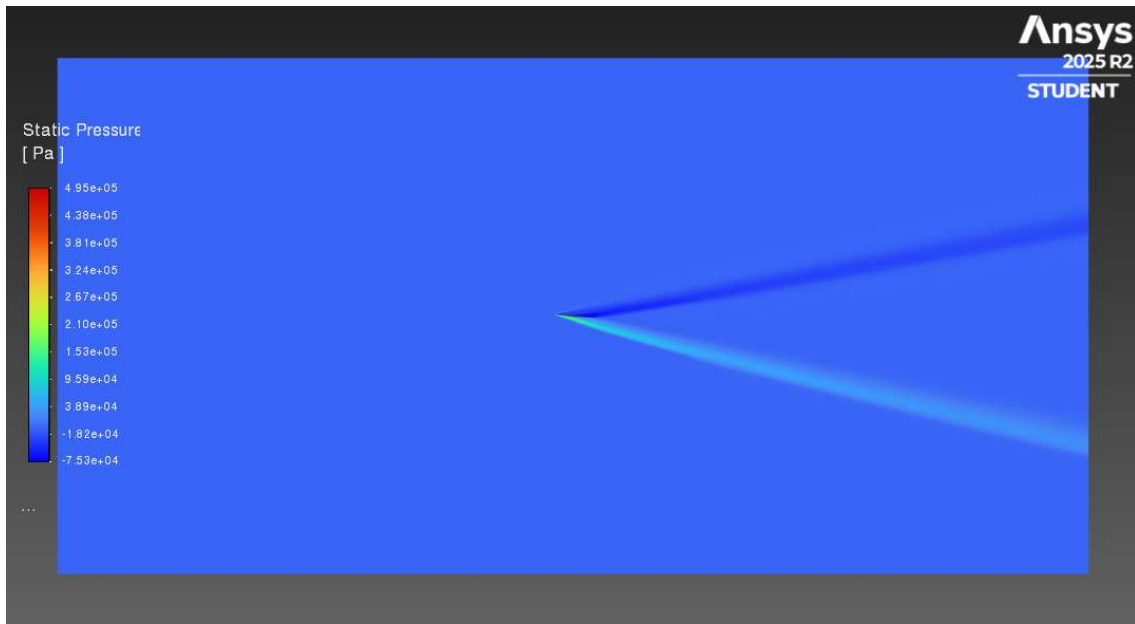
Mach3



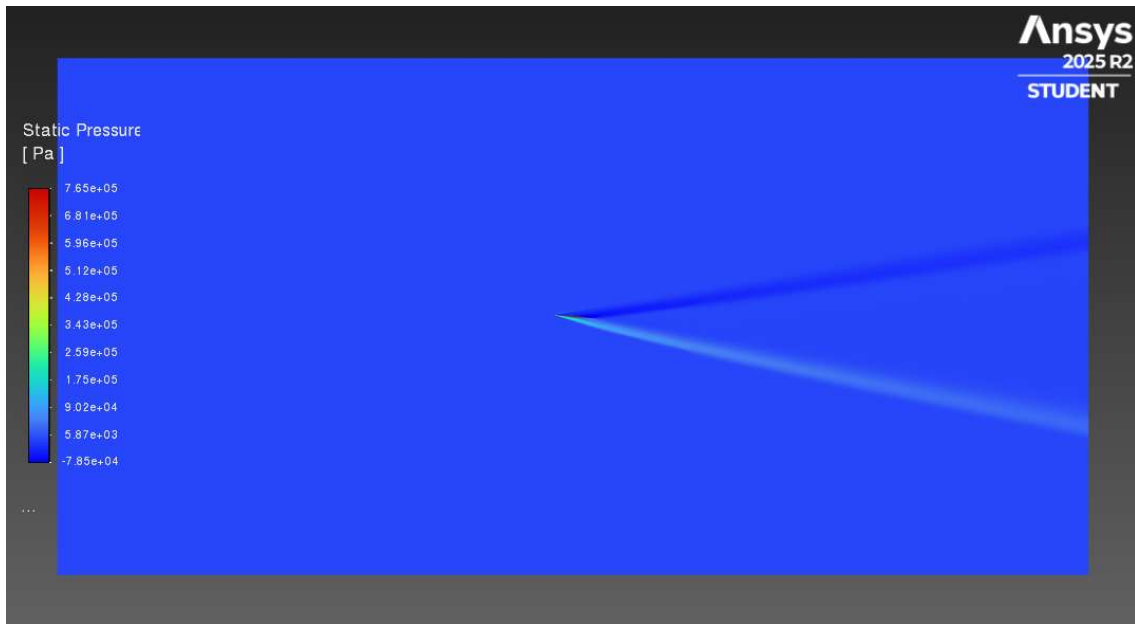
Mach4



Mach5

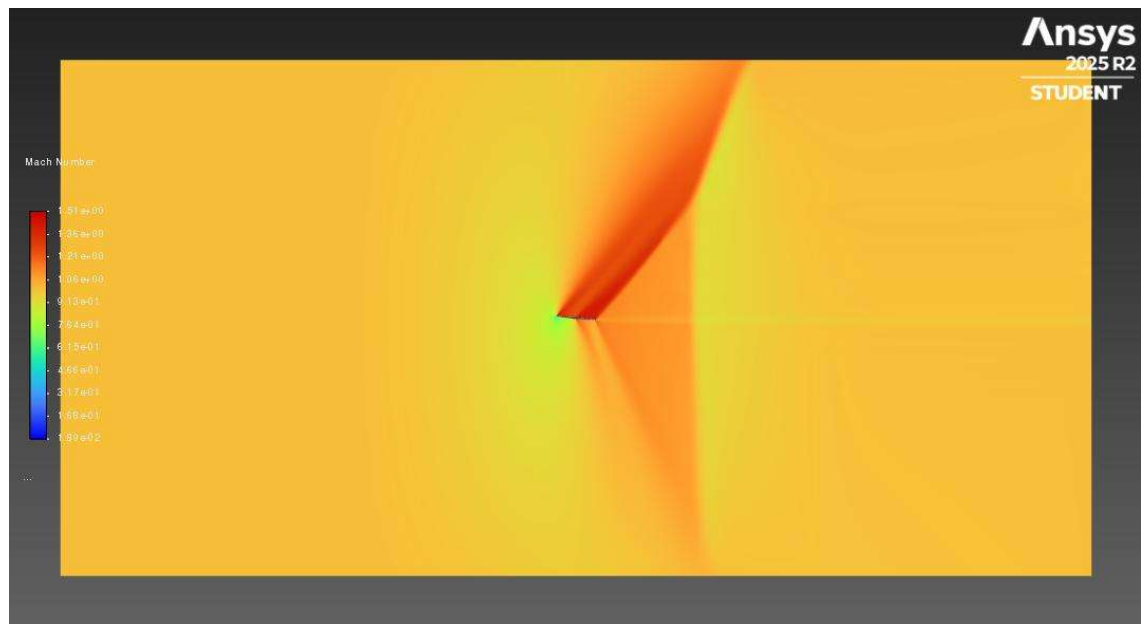


Mach6

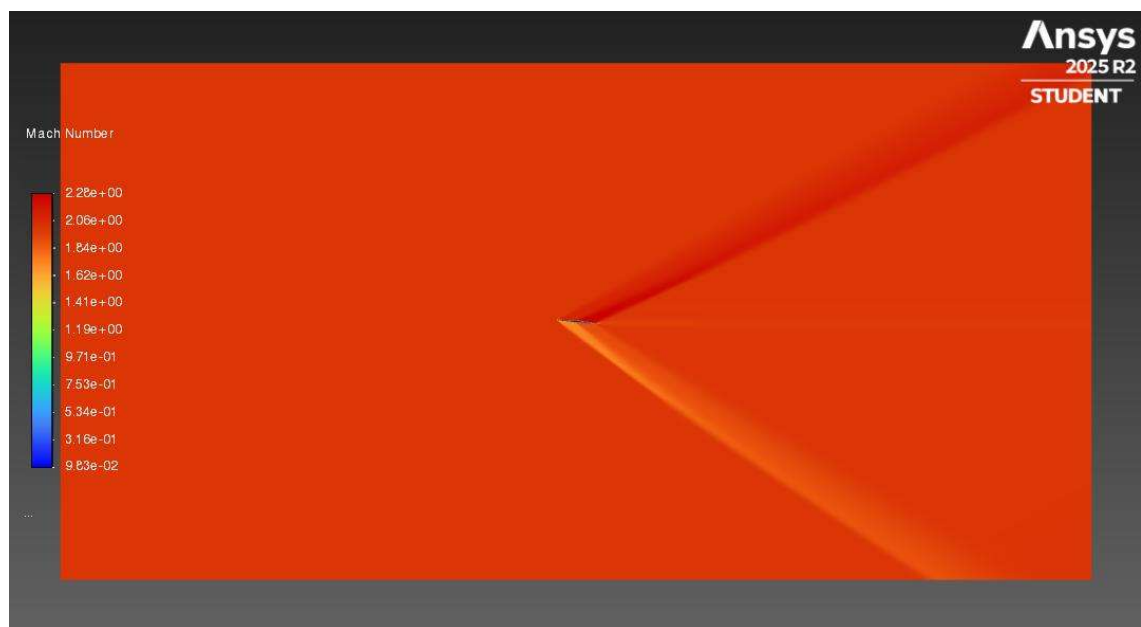


Mach:

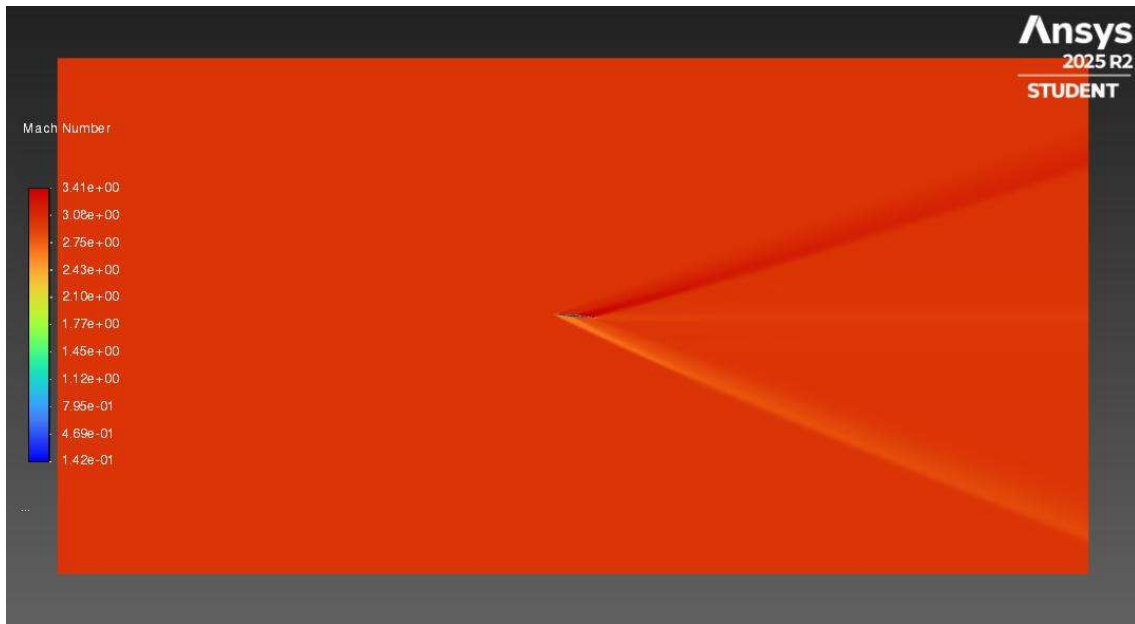
Mach1



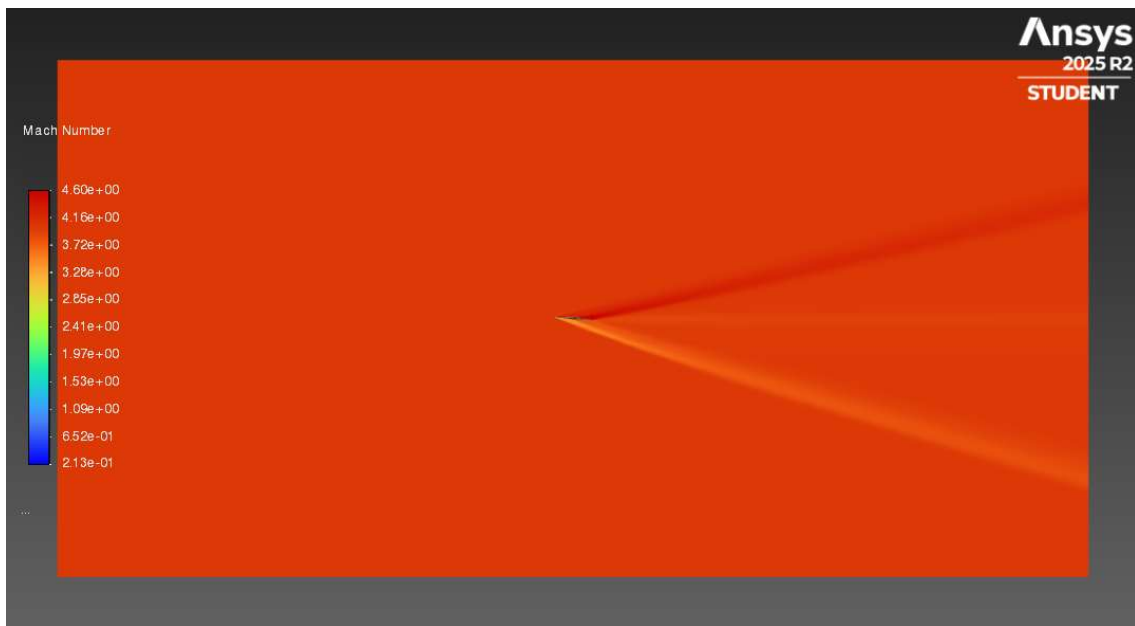
Mach2



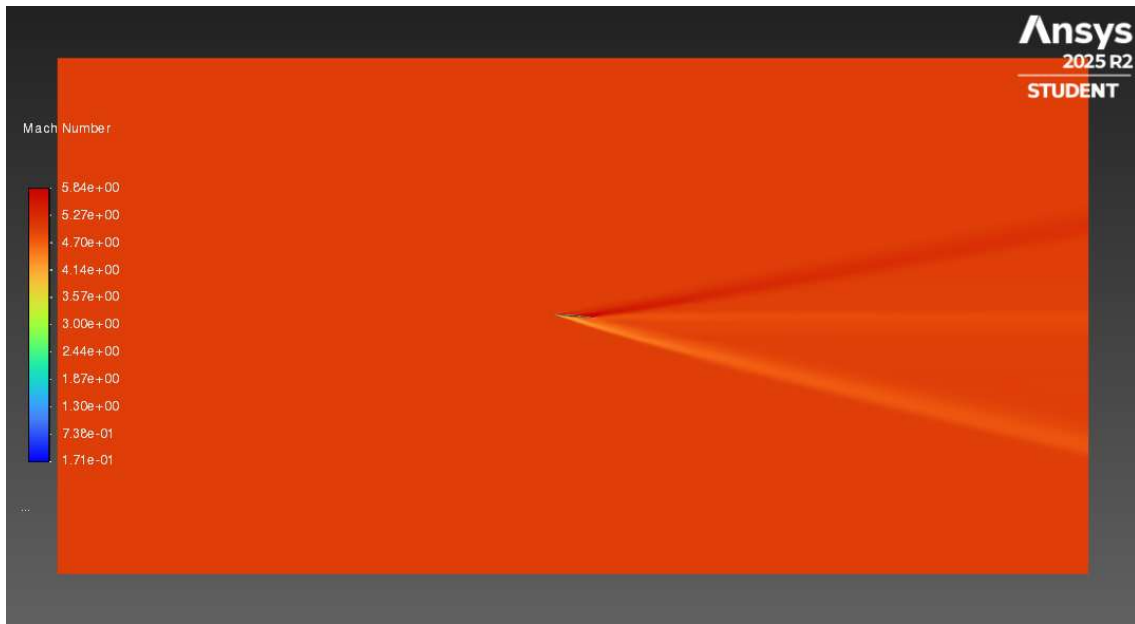
Mach3



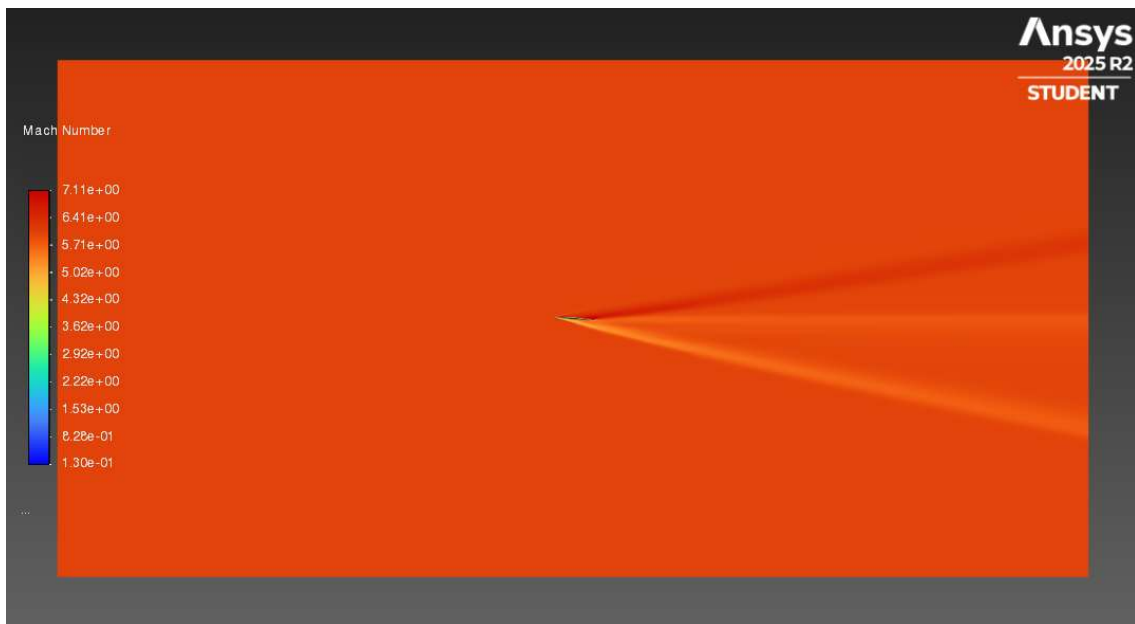
Mach4



Mach5

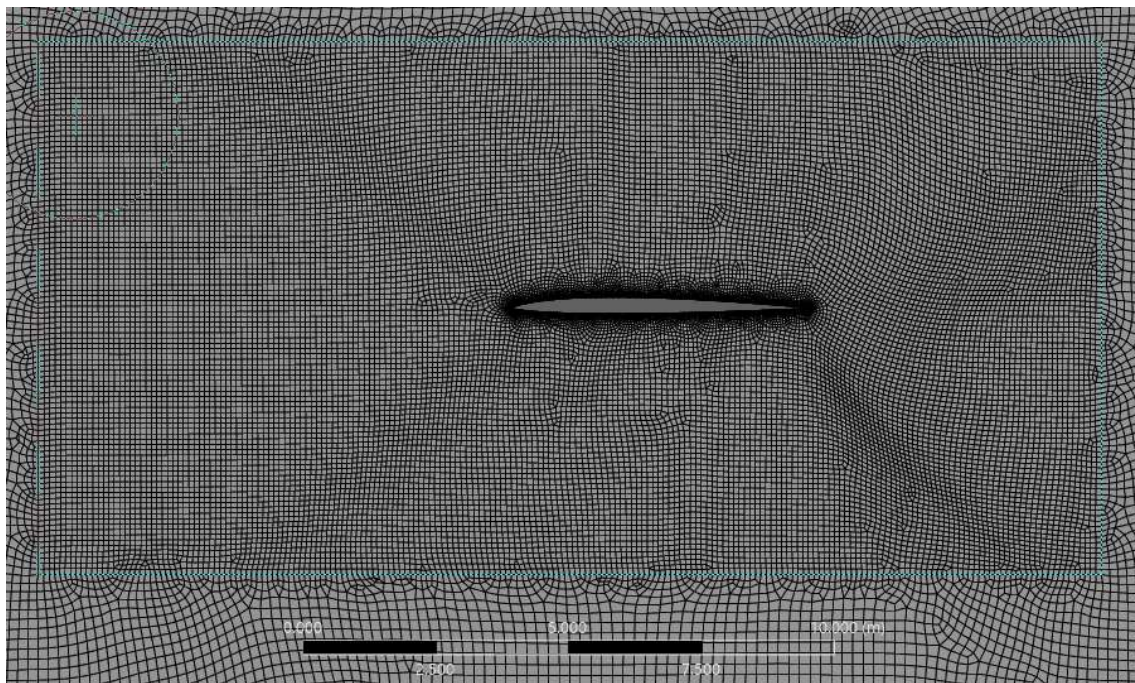
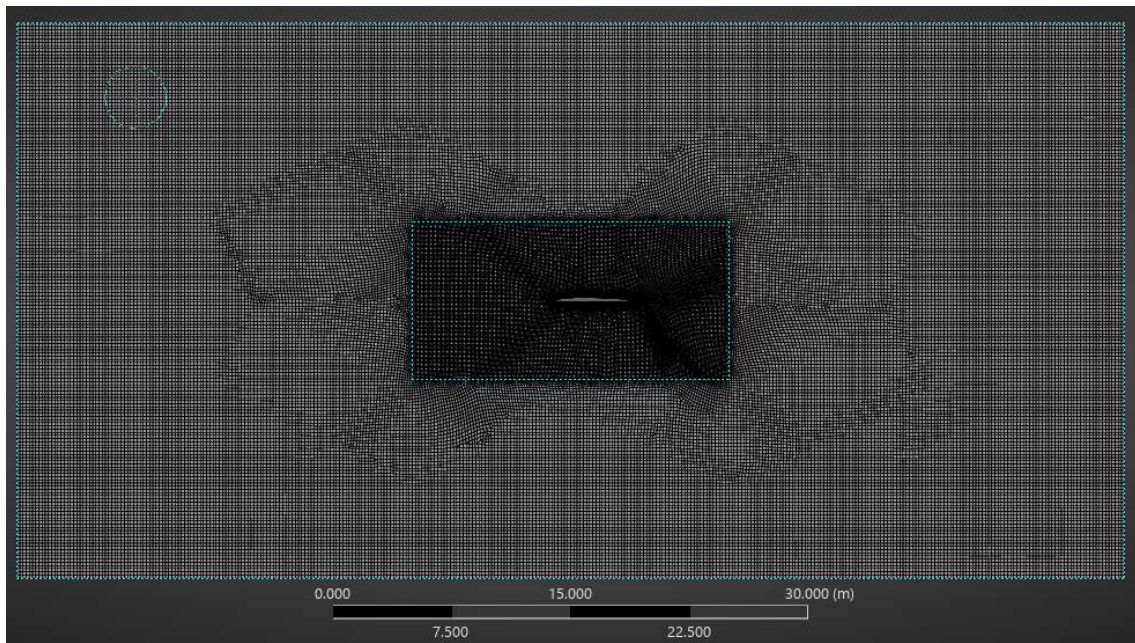


Mach6

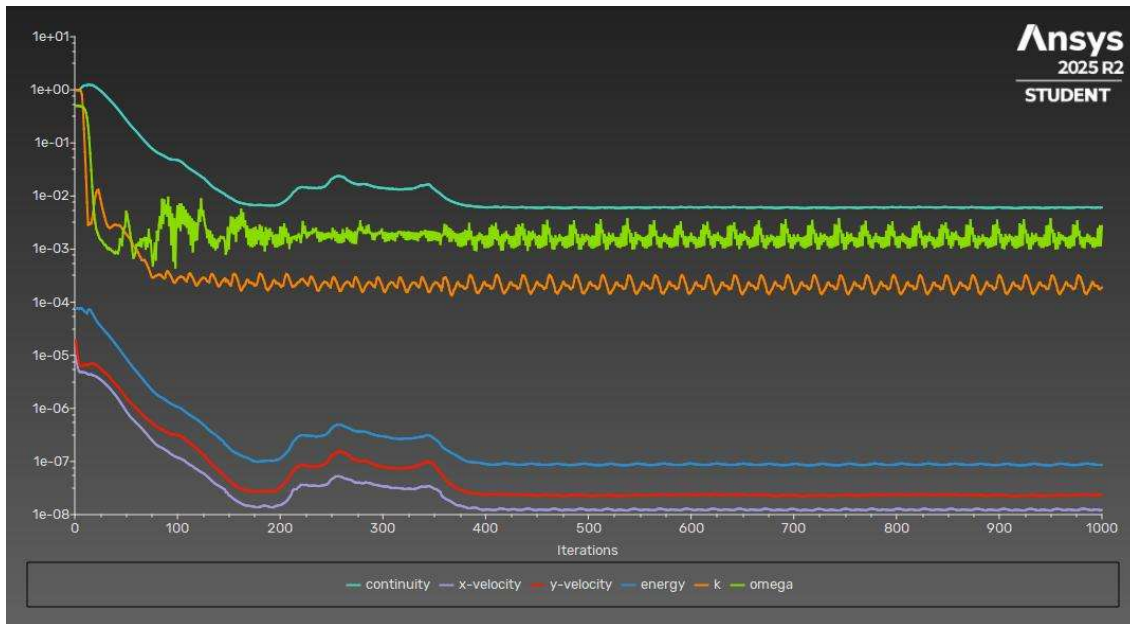


Question E

Mesh



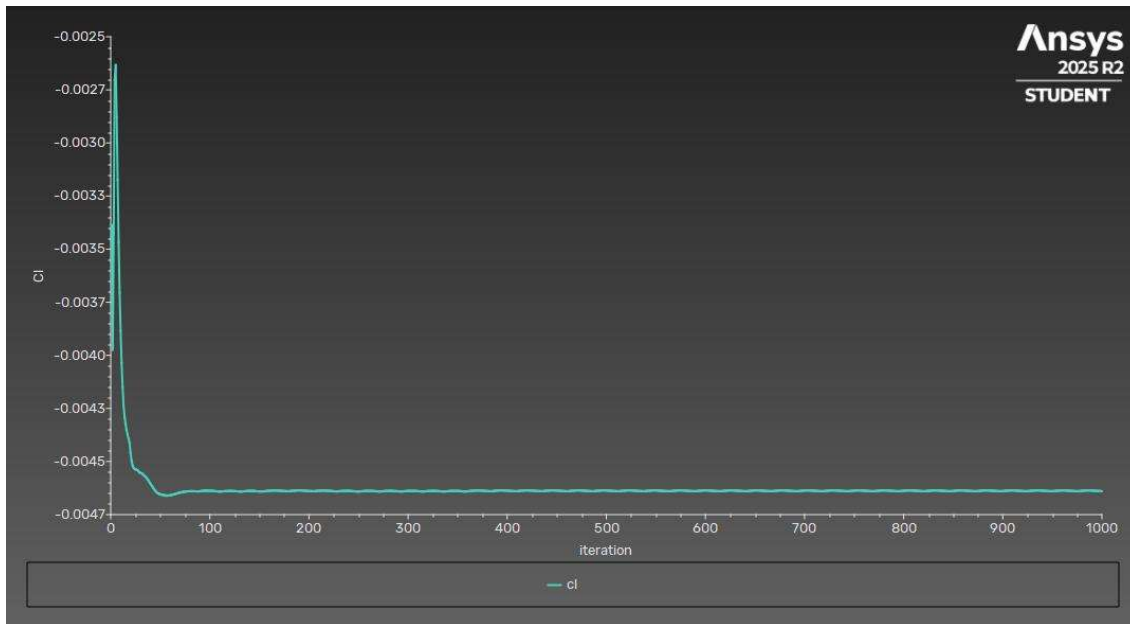
Scaled Residual



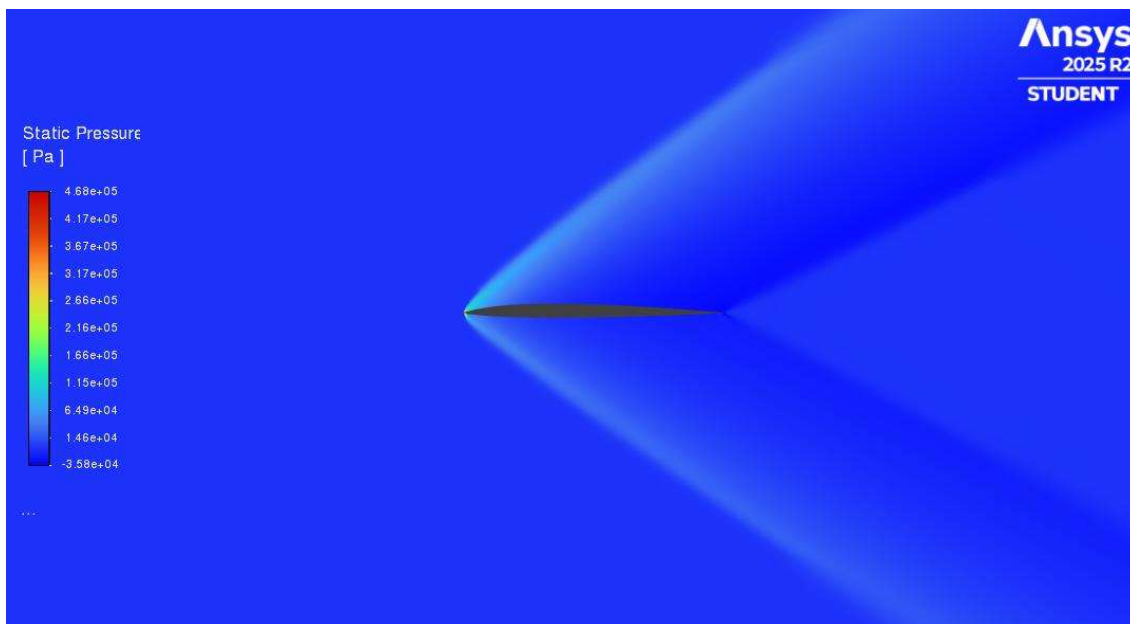
Coefficient of Drag

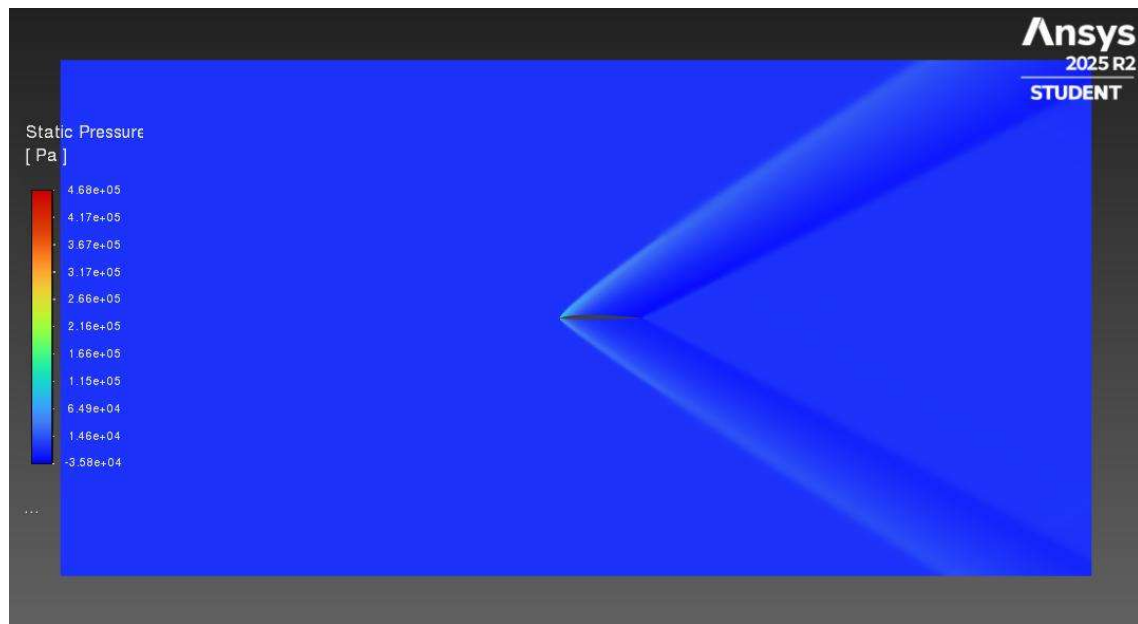


Coefficient of Lift

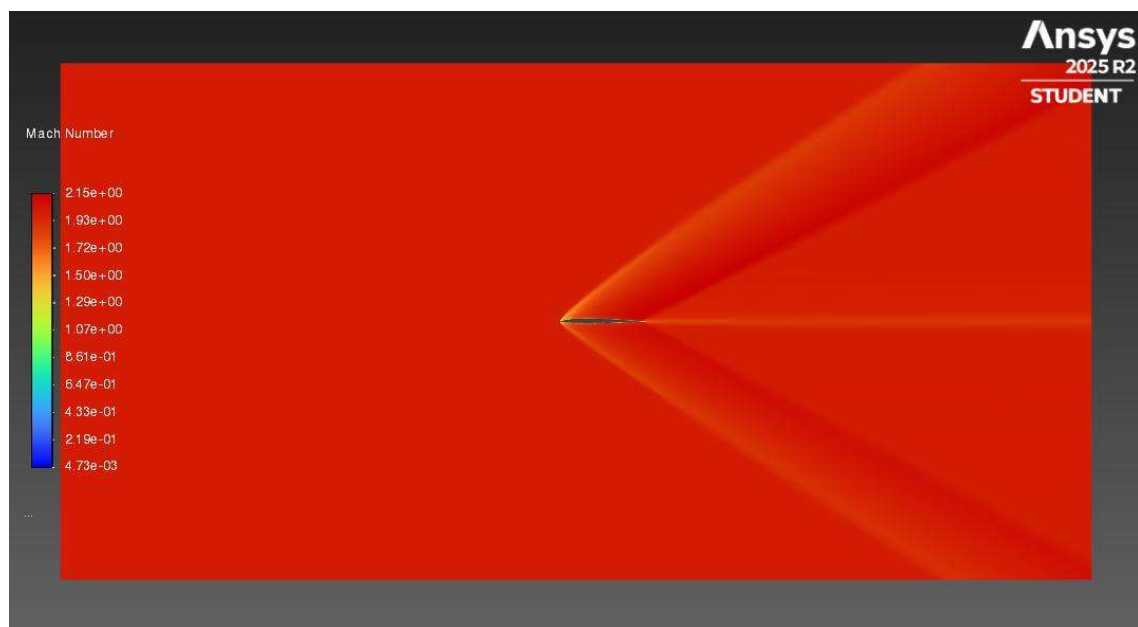


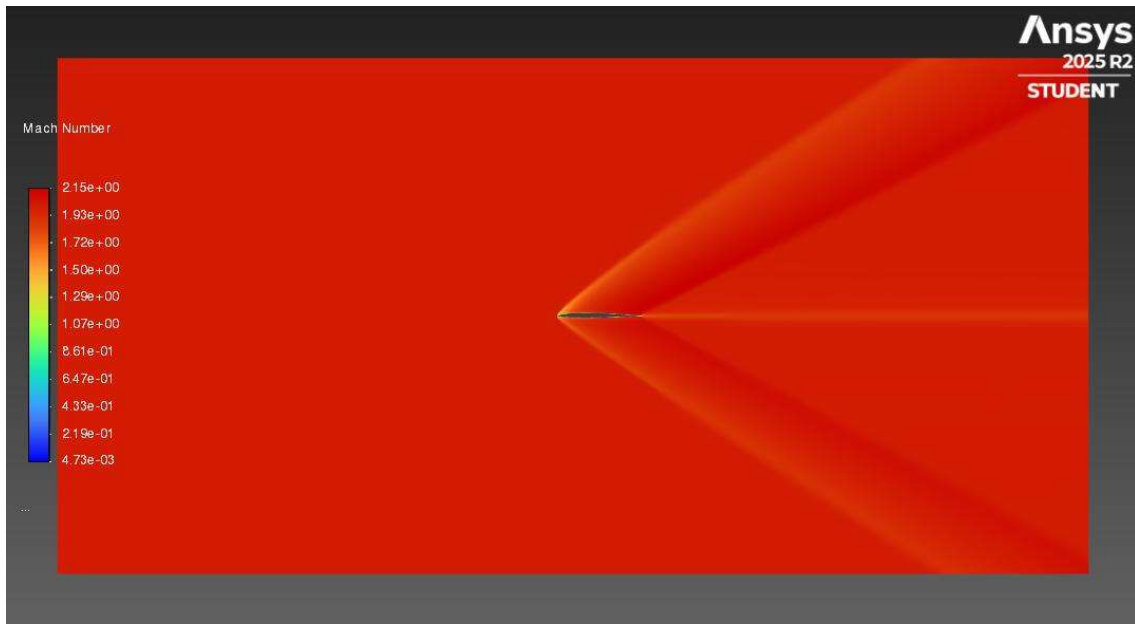
Pressure



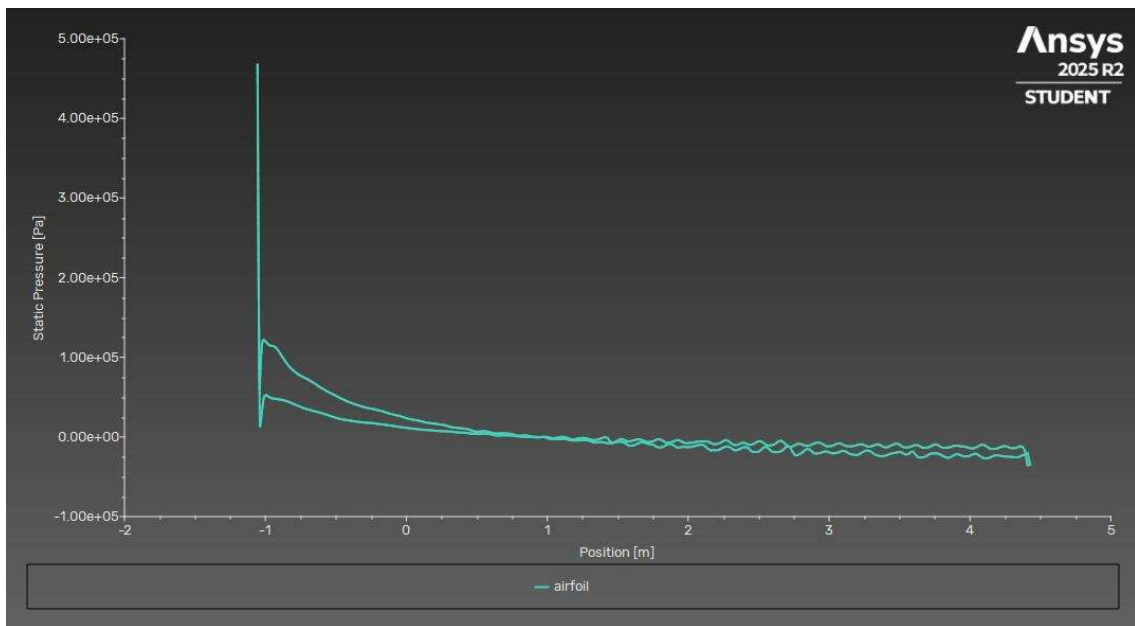


Mach





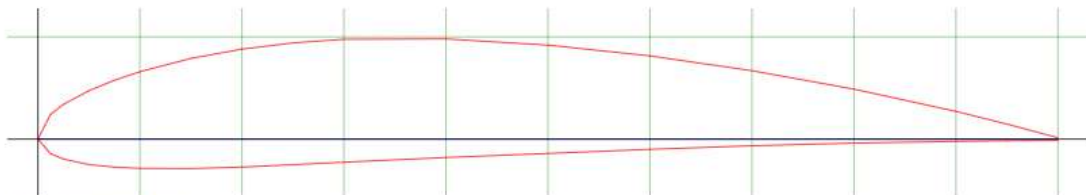
Pressure plot around Airfoil



PART - E FINAL TEST on NACA 4412 @5Degree

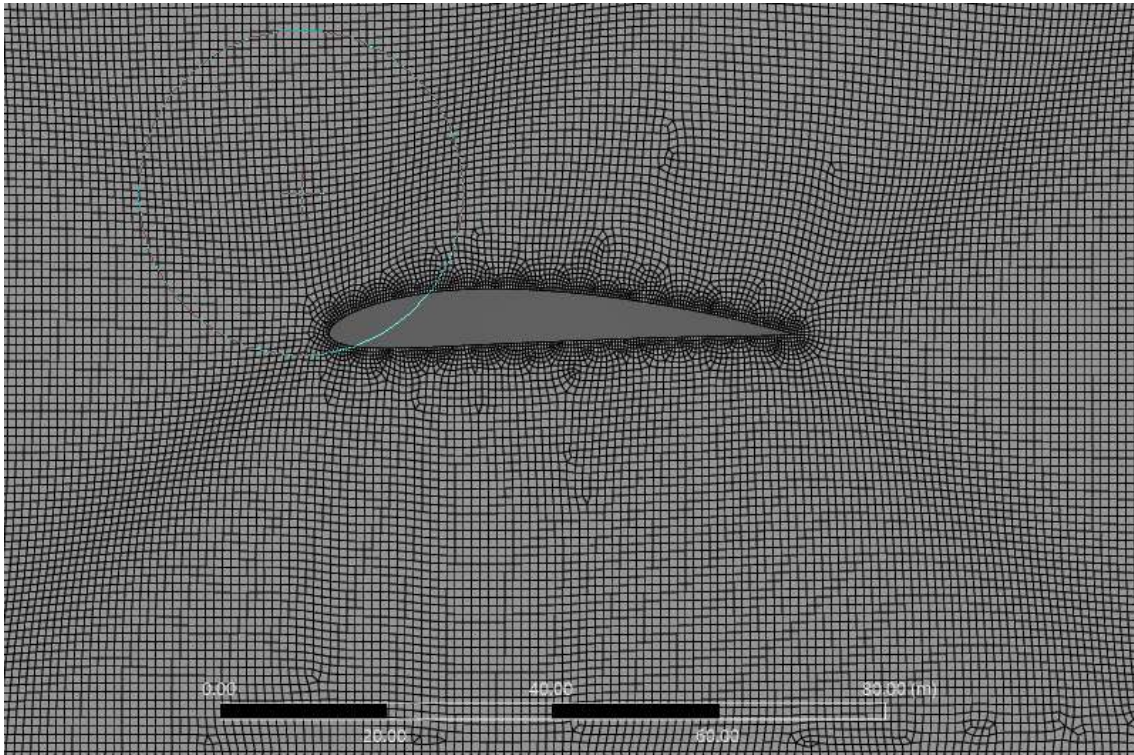
Dimensions

Airfoil	naca4412-il - NACA 4412
Chord (mm)	5596
Radius (mm)	0
Thickness (%)	100
Origin (%)	0
Pitch (degrees)	0
Halo (mm)	0
Halo (mm)	0
Colour	Colour
Line thickness (%)	100
Reverse	☐
Data box	☒
Camber line	☐
X grid (mm)	559.6
Y grid (mm)	559.6
Paper width (mm)	15668.8
Paper height (mm)	10072.8

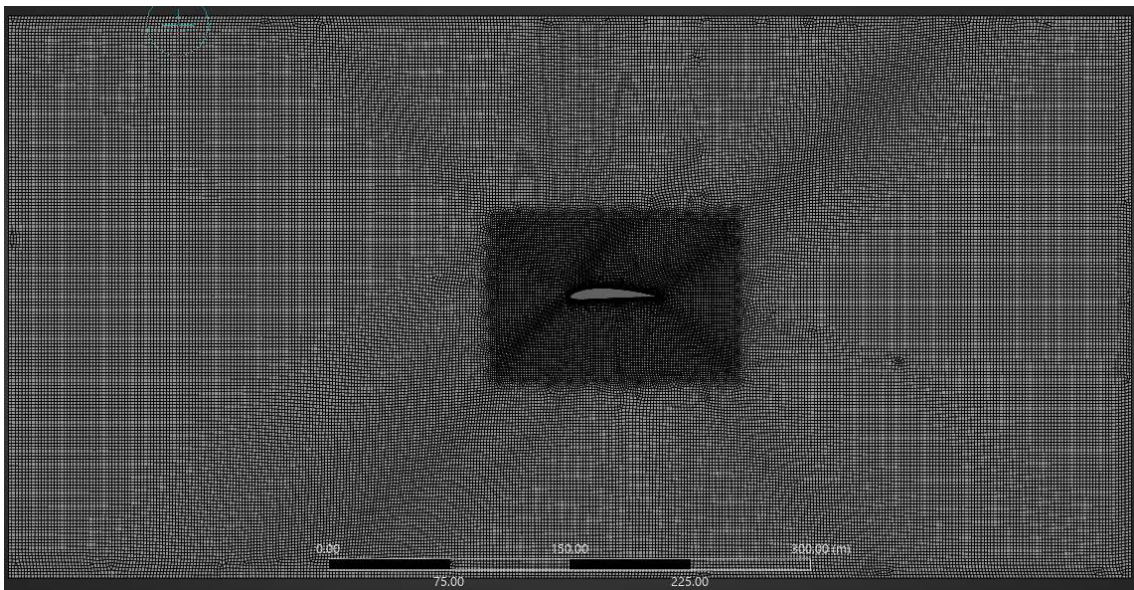


Name = NACA 4412
Chord = 5596mm Radius = 0mm Thickness = 100% Origin = 0% Pitch = 0°

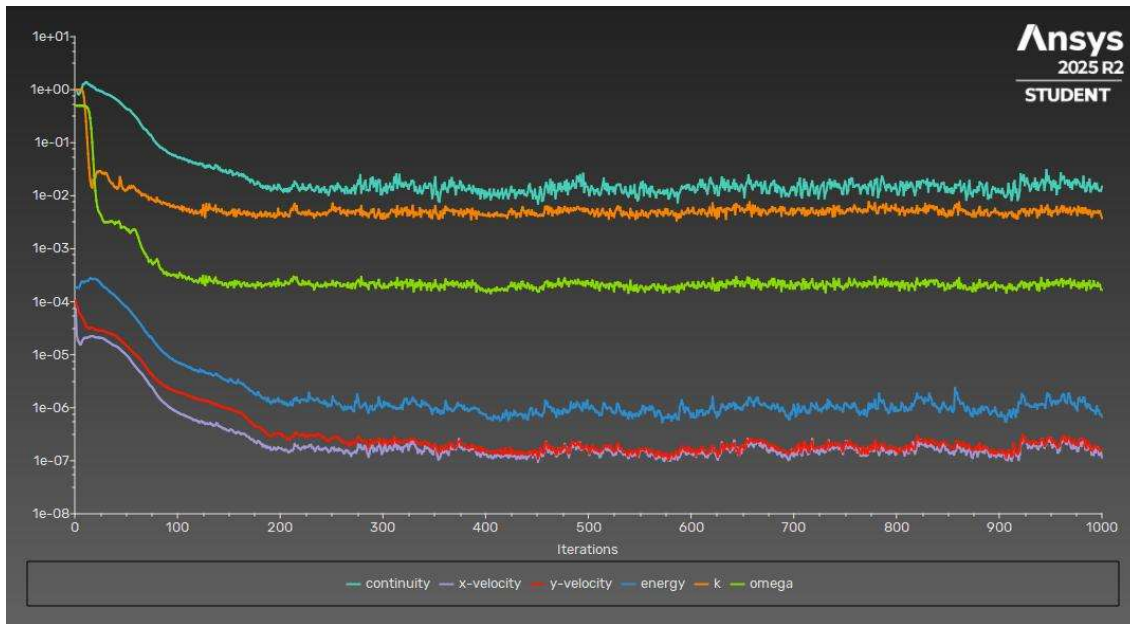
Mesh



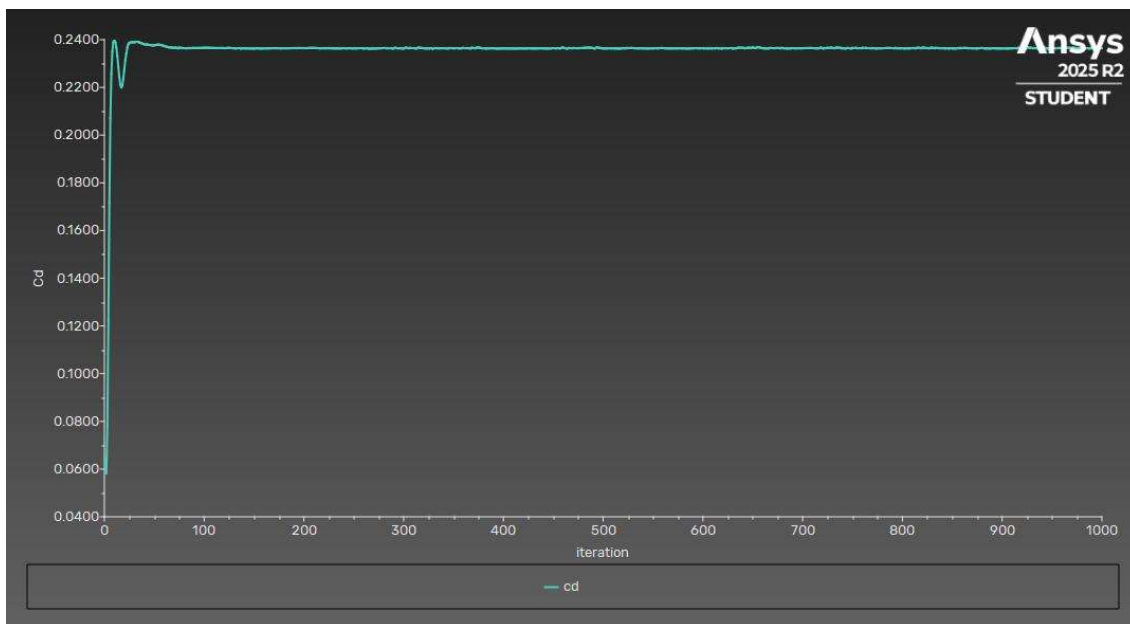
Scaled residual



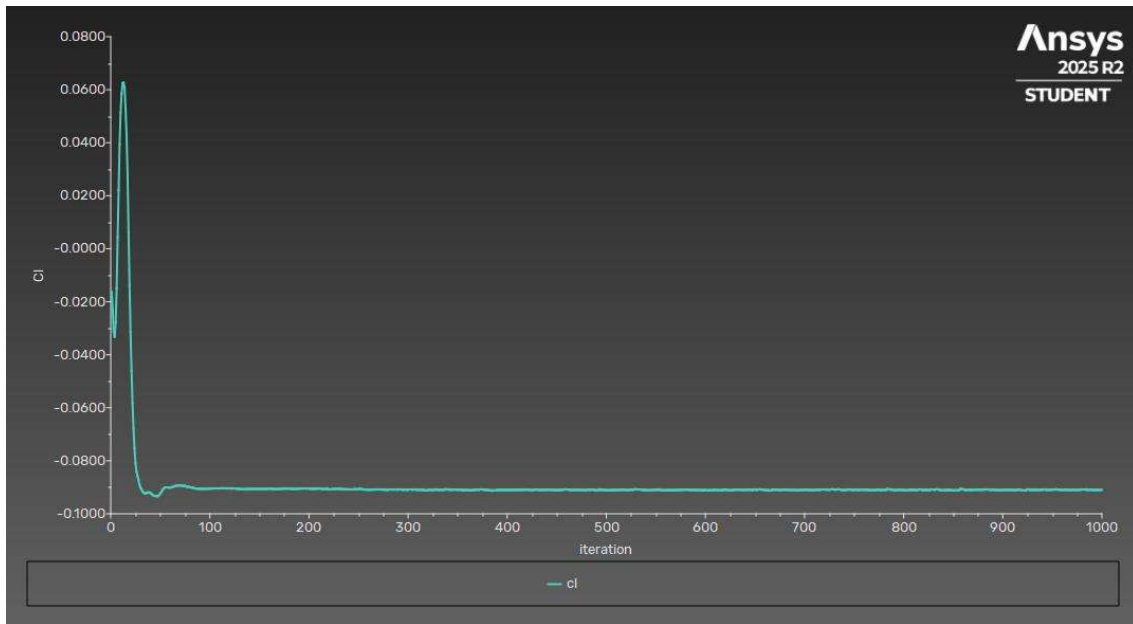
Scaled residuals



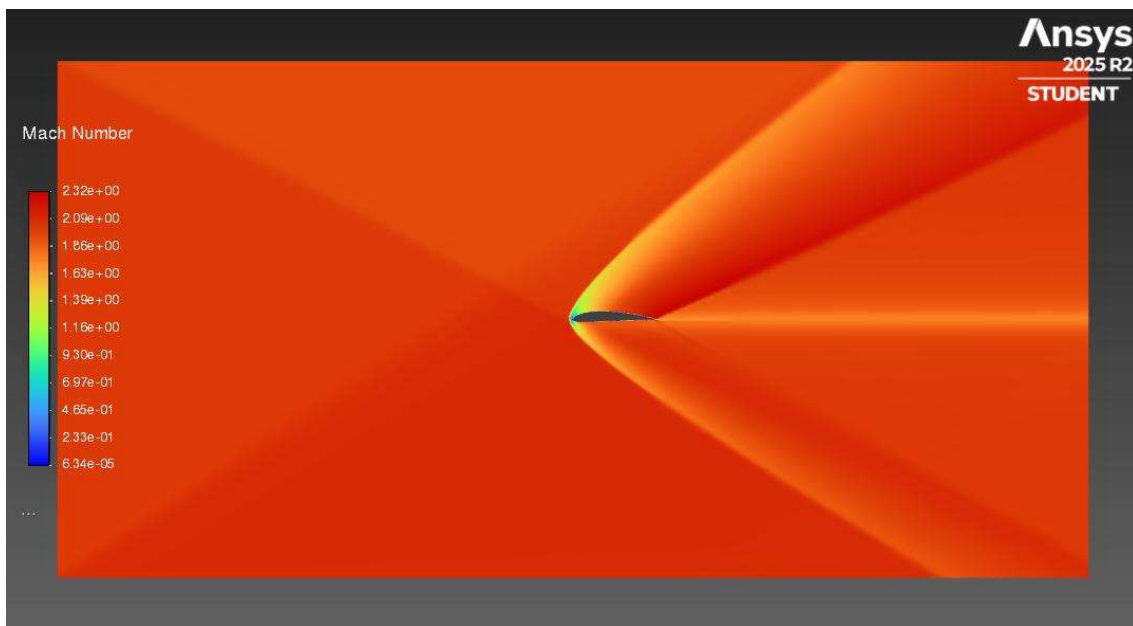
Coefficient of drag



Coefficient of lift



Mach



Pressure

